Class Field Theory

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Preface to the New Edition

The original preface which follows tells about the history of these notes and the missing chapters. This book is a slightly revised edition. Some footnotes and historical comments have been added in an attempt to compensate for the lack of references and attribution of credit in the original. There are two mathematical additions. One is a sketch of the analytic proof of the second inequality in Chapter VI. The other is several additional pages on Weil groups at the end of Chapter XV. They explain that what is there called a Weil group for a finite Galois extension K/F lacks an essential feature of a Weil group in Weil's sense, namely the homomorphism $W_{K,F} \to \mathrm{Gal}(K^{\mathrm{ab}}/F)$, but that we recover this once we construct a Weil group for $\bar{F}/F$ by passing to an inverse limit. There is also a sketch of an abstract version of Weil's proof of the existence and uniqueness of his $W_{K,F}$ for number fields.

I have not renumbered the chapters. After some preliminaries, the book still starts with Chapter V, but the mysterious references to the missing chapters have been eliminated. The book is now in TeX. The handwritten German letters are gone, and many typographical errors have been corrected. I thank Mike Rosen for his help with that effort. For the typos we've missed and other mistakes in the text, the AMS maintains a Web page with a list of errata at

`http://www.ams.org/bookpages/chel-366/`

I would like to thank the AMS for republishing this book, and especially Sergei Gelfand for his patience and help with the preparation of the manuscript.

For those unacquainted with the book, it is a quite complete account of the algebraic (as opposed to analytic) aspects of classical class field theory. The first four chapters, V–VIII, cover the basics of global class field theory, the cohomology of idèle classes, the reciprocity law and existence theorem, for both number fields and function fields. Chapters IX and X cover two more special topics, the structure and cohomology of the connected component of 1 in the idèle class group of a number field, and questions of local vs. global behavior surrounding the Grunwald–Wang theorem. Then there are two chapters on higher ramification theory, generalized local classfield theory, and explicit reciprocity laws. This material is beautifully covered also in [**21**]. For a recent report, see [**8**]. There is a nice generalization of our classical explicit formula in [**13**]. The last three chapters of the book cover abstract class field theory. The cohomological algebra behind the reciprocity law is common to both the local and global class field theory of number fields and function fields. Abstracting it led to the definition of a new algebraic structure, 'class formation', which embodies the common features of the four theories. The difference is in the proofs that the idèle classes globally, and the multiplicative groups locally, satisfy the axioms of a class formation. Chapter XIV concludes with a discussion of the reciprocity law and existence theorem for an abstract class formation. In the last

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chapter XV, Weil groups are defined for finite layers of an arbitrary class formation, and then, for topological class formations satisfying certain axioms which hold in the classical cases, a Weil group for the whole formation is constructed, by passage to an inverse limit, The class formation can be recovered from its Weil group, and the topological groups which occur as Weil groups are characterized by axioms.

The mathematics in this book is the result of a century of developement, roughly 1850–1950. Some history is discussed by Hasse in [**5**] and in several of the papers in [**18**]. The high point came in the 1920's with Takagi's proof that the finite abelian extensions of a number field are in natural one-to-one correspondence with the quotients of the generalized ideal class groups of that field, and Artin's proof several years later that an abelian Galois group and the corresponding ideal class group are canonically isomorphic, by an isomorphism which implied all known reciprocity laws. The flavor of this book is strongly influenced by the last steps in that history. Around 1950, the systematic use of the cohomology of groups by Hochschild, Nakayama and the authors shed new light. It enabled many theorems of the local class field theory of the 1930's to be transferred to the global theory, and led to the notion of class formation embodying the common features of both theories. At about the same time, Weil conceived the idea of Weil groups and proved their existence. With those two developments it is fair to say that the classical one-dimensional abelian class field theory had reached full maturity. There were still a few things to be worked out, such as the local and global duality theories, and the cohomology of algebraic tori, but it was time for new directions.

They soon came. For example:

- Higher dimensional class field theory;
- Non-abelian reciprocity laws and the Langlands program;
- Iwasawa theory;
- Leopold's conjecture;
- Abelian (and non-abelian) ℓ-adic representations;
- Lubin-Tate local theory, Hayes explicit theory for function fields, Drinfeld modules;
- Stark conjectures;
- Serre conjectures (now theorems).

Rather than say more or give references for these, I simply recommend what has become a universal reference, the internet. Searching any of the above topics is rewarding.

John Tate
September 2008

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Preface

This is a chunk of the notes of the Artin–Tate seminar on class field theory given at Princeton University in 1951–1952, namely the part dealing with global class field theory (Chapters V through XII) and the part dealing with the abstract theory of class formations and Weil groups (Chapters XIII–XV). The first four chapters, which are not included, covered the cohomology theory of groups, the fundamentals of algebraic number theory, a preliminary discussion of class formations, and local class field theory. In view of these missing sections, the reader will encounter missing references and other minor flaws of an editorial nature, and also some unexplained notations. We have written a few pages below recalling some of these notations and outlining the local class field theory, in an attempt to reduce the "prerequisites" for reading these notes to a basic knowledge of the cohomology of groups and of algebraic theory, together with patience.

The reason for the long delay in publication was the ambition to publish a revised and improved version of the notes. This new version was to incorporate the advances in the cohomology theory of finite groups which grew out of the seminar and which led to the determination of the higher cohomology groups and to a complete picture of the cohomological aspects of the situation, as outlined in Tate's talk at the Amsterdam Congress in 1954. However this project was never completed and thus served only to prevent the publication of the most important part of the seminar, namely Chapters V through XII of these notes. That this material finally appears is due to the energies of Serge Lang, who took the original notes, continued to urge their publication, and has now made the arrangements for printing. It is a pleasure to express here our appreciation to him for these efforts.

Two excellent general treatments of class field theory, which complement these notes, have appeared during the past year, namely:

Cassels and Fröhlich, *Algebraic Number Theory*, Academic Press, London, 1967. (Distributed in the U.S. by the Thompson Publishing Company, Washington, D.C.).

Weil, *Basic Number Theory*, Springer-Verlag, Berlin/Heidelberg/New York, 1967.

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Preliminaries

1. Idèles and Idèle Classes

A *global field* is either a number field of finite degree over the rational field $\mathbb{Q}$, or a function field in one variable over a finite constant field. Such fields have primes $\mathfrak{p}$, and corresponding canonical absolute values $|\ |_{\mathfrak{p}}$, for which the product formula holds. A *local field* is the completion $k_{\mathfrak{p}}$ of a global field k at a prime $\mathfrak{p}$. Thus a local field is either the real field $\mathbb{R}$, the complex field $\mathbb{C}$, a finite extension of the rational p-adic field $\mathbb{Q}_p$, for some prime number p, or a field of formal power series in one variable over a finite constant field. In other words, a local field is a locally compact non discrete topological field.

The *idèle group* and the *idèle class group* of a global field k are denoted by $J = J_k$ and $C = C_k$ respectively. The *absolute value* (or volume) of an idèle $\mathfrak{a}$ is the product of the absolute values of its components: $|\mathfrak{a}| = \prod_{\mathfrak{p}} |\mathfrak{a}_{\mathfrak{p}}|_{\mathfrak{p}}$. In view of the product formula, the absolute value of an idèle depends only on its class. We denote by J^0 (resp. C^0) the group of idèles (resp. idèle classes) of absolute value 1.

By a *finite set of primes* S of k we always mean a non-empty set, containing the archimedean primes in case k is a number field. The group of S-idèles,

$$J_{k,S} = J_S = \prod_{\mathfrak{p}\in S} k_{\mathfrak{p}}^* \times \prod_{\mathfrak{p}\notin S} U_{\mathfrak{p}}$$

consists of the idèles whose components are units for $\mathfrak{p} \notin S$. The group of S-idèle classes,

$$C_{k,S} = C_S = k^* J_S/k^* \approx J_S/k^* \cap J_S \approx J_S/k_S^*$$

consists of the idèle classes which are represented by S-idèles. Notice that $k_S^* = k^* \cap J_S$ is the group of *S-units* of k, i.e. elements of k which are units for all $\mathfrak{p} \notin S$, and that $C/C_S \approx J/k^* J_S$ is isomorphic to the group of divisor classes in the Dedekind ring formed by the elements of k which are integral for $\mathfrak{p} \notin S$.

We give J the unique topology such that, for each S, J_S is open in J and the topology induced on J_S is the product topology. Uniqueness because $J = \bigcup_S J_S$. Existence because the group of units $U_{\mathfrak{p}}$ is open in $k_{\mathfrak{p}}$ for each non-archimedean $\mathfrak{p}$. Note that J is locally compact because each $U_{\mathfrak{p}}$ is compact, and each $k_{\mathfrak{p}}^*$ is locally compact.

Applying the product formula to $a - 1$ for $a \in k$, $a \neq 1$, one proves that k is a discrete subgroup of J. The theorems on finiteness of class number and the existence of units can be used to show that $C^0 = J^0/k^*$ is compact. Of course, $C/C^0 \approx J/J^0$ is isomorphic to $\mathbb{R}^+$ or to $\mathbb{Z}$ in the number field and function field cases respectively.

If K is a finite extension of k, then there are injections $J_k \to J_K$ and $C_k \to C_K$, and in each case the smaller group maps homeomorphically onto a closed

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subgroup of the larger one. If K/k is Galois, the Galois group G operates on K, J_K, and C_K, and we have $k \approx K^G$, $J_k \approx J_K^G$, and $C_k \approx C_K^G$. The first of these isomorphisms follows from Galois theory, the second from definitions of idèles and of the imbedding $J_k \subset J_K$, and the last follows from the first two in view of "Hilbert's Theory 90": $H^1(G, K^*) = 0$.

If S is a finite set of primes $\mathfrak{p}$ of k, we let the same symbol S stand also for the set of primes $\mathfrak{P}$ of K which divide some prime $\mathfrak{p} \in S$. With this understanding we have

$$J_{K,S} = \prod_{\mathfrak{P}\in S} K_{\mathfrak{P}}^* \times \prod_{\mathfrak{P}\notin S} U_{\mathfrak{P}} = \prod_{\mathfrak{p}\in S}\Big(\prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}^*\Big) \times \prod_{\mathfrak{P}\notin S}\Big(\prod_{\mathfrak{P}|\mathfrak{p}} U_{\mathfrak{P}}\Big)$$

and consequently

$$H^r(G, J_{K,S}) \approx \prod_{\mathfrak{p}\in S} H^r\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}^*\Big) \times \prod_{\mathfrak{p}\notin S} H^r\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} U_{\mathfrak{P}}\Big).$$

Now the operation of G on $\prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}^*$ permutes the factors, and the subgroup of G consisting of the elements which carry a given factor $K_{\mathfrak{P}}^*$, into itself is the decomposition group $G_{\mathfrak{P}}$ of $\mathfrak{P}$. It follows that $\prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}^*$ is the G-module "induced" by the $G_{\mathfrak{P}}$-module $K_{\mathfrak{P}}^*$ and the cohomological theory of induced modules (some time referred to as Shapiro's Lemma, referred to in these notes by the catchword *semilocal theory*) shows that we have isomorphisms

$$H^r\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}^*\Big) \approx H^r(G_{\mathfrak{P}}, K_{\mathfrak{P}}^*)$$

for any fixed prime $\mathfrak{P}$ dividing $\mathfrak{p}$, and similarly

$$H^r\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} U_{\mathfrak{P}}\Big) \approx H^r(G_{\mathfrak{P}}, U_{\mathfrak{P}}).$$

These isomorphisms are canonical, coming from the restriction from G to $G_{\mathfrak{P}}$ and the projection of the $\mathfrak{P}$-factor. By the theory of local fields, we have $H^r(G_{\mathfrak{P}}, U_{\mathfrak{P}}) = 0$ for $r > 0$ if $K_{\mathfrak{P}}$ is unramified over $k_{\mathfrak{p}}$, and therefore if our set S contains all primes $\mathfrak{p}$ of k which ramify in K we have

$$H^r\big(G, J_{K,S}\big) \approx \prod_{\mathfrak{p}\in S} H^r\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}^*\Big) \approx \prod_{\mathfrak{p}\in S} H^r(G_{\mathfrak{P}}, K_{\mathfrak{P}}^*).$$

Passing to the direct limit over larger and larger S, we find for $r > 0$:

$$H^r(G, J_K) \approx \varinjlim_{S} H^r(G, J_{K,S}) \approx \coprod_{\mathfrak{p}} H^r(G_{\mathfrak{P}}, K_{\mathfrak{P}}^*)$$

where $\coprod$ denotes direct sum, and for each $\mathfrak{p}$ of k, $\mathfrak{P}$ denotes a selected prime of K above $\mathfrak{p}$. In this way, the Galois cohomology of the idèles is reduced to the cohomology of the local fields. The isomorphism is of course functorial and commutes with restriction, transfer, and inflation in the $k \subset L \subset K$ situation in the obvious way: A global restriction from $G_{K/k}$ to $G_{K/L}$ is reflected in the local restrictions from $G_{K_{\mathfrak{P}}/k_{\mathfrak{p}}}$ to $G_{K_{\mathfrak{P}}/L_{\mathfrak{q}}}$. The global transfer in the other direction is reflected, for each $\mathfrak{p}$, in the sum over the primes $\mathfrak{q}$ of L dividing $\mathfrak{p}$, of the local transfers. The global inflation corresponds to the totality of all local inflations.

Let Ω be an infinite extension of k. The idèle group J_Ω is by definition the union of the idèle groups J_K of the finite subextensions K/k of Ω/k, and the same goes for the idèle class group C_Ω. If Ω is a Galois extension then its Galois group $G = G_{\Omega/k}$

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operates on J_Ω, C_Ω, and Ω^*. For each K, we have $J_K = J_\Omega^{G_K}$, where $G_K = G_{\Omega/K}$ is the subgroup of G_Ω corresponding to the field K, and the same is true for the idèle class groups and for the multiplicative groups of the fields. Thus for $A = J_\Omega$, C_Ω, or Ω^*, the mathematical system $(G, \{G_K\}, A)$ represents a "formation" in the sense of the beginning paragraphs of Chapter XIV. The reader might well begin his study of these notes by reading the first three sections of Chapter XIV, where the notion of class formation is defined in abstracto, because the first three chapters (Chapters V–VII) of these notes are devoted to the proof that the idèle classes of global fields do constitute a class formation.

Our notation and terminology for formations is rather naive. For all intents and purposes, the notion of a formation is equivalent to a pair (G, A) consisting of a compact totally disconnected topological group G and a G-module A on which G acts continuously in the sense that the stabilizer of every element $a \in A$ is an open subgroup of G. Given such a pair one defines cohomology groups $H^r(G, A)$ for $r \geq 0$ either by using *continuous* standard cochains, or equivalently, as the direct limits under the inflation maps of the cohomology groups $H^r(G/U, A^U)$ of the finite quotients of G by its open normal subgroups U. As functors of (G, A) these cohomology groups have all the properties of ordinary cohomology groups of groups, and the freedom to vary the "formation" module A and "Galois group" G is a considerable technical advantage. For more details the reader can consult [**21**, Ch. X] or for still more details, [**22**].

2. Cohomology

In the proof of the main theorem, p. 154, we refer to the following result in the cohomology of finite groups.

THEOREM A. *Let G be a finite group, and let $A \times B \to C$ be a G-pairing of two G-modules into a third. Let $a \in H^p(G, A)$. Then for each $q \in \mathbb{Z}$ and each subgroup $S \subset G$ the cup product with the restriction of α to S yields a homomorphism*

$$\alpha_{q,S} \colon H^q(S, B) \to H^{p+q}(S, C).$$

Suppose for some q_0 that the maps $\alpha_{q_0-1,S}$ are surjective, the maps $\alpha_{q_0,S}$ are bijective, and the maps $\alpha_{q_0+1,S}$ are injective for all subgroups S. Then the maps $\alpha_{q,S}$ are bijective for all q and all S.

The proof is not difficult. By dimension shifting, one reduces to the case $p = 0$, in which case $\alpha_{q,S}$ is just the map induced by some G-homomorphism $f \colon B \to C$. Adding to C a cohomologically trivial module into which B injects, we see that it is no loss of generality to assume that f is an injection. We then consider the cohomology sequence associated with the exact sequence

$$0 \to B \xrightarrow{f} C \to D \to 0$$

and see from our hypotheses that

$$H^{q_0-1}(S, D) = H^{q_0}(S, D) = 0$$

for all subgroups S of G. It follows now from the theory of cohomological triviality that $H^q(S, D) = 0$ for all q and all S, whence the result.

In discussing cohomological triviality one first reduces to the Sylow groups and then, by induction in cyclic towers, to cyclic groups, where the matter is trivial. These methods have been refined in recent works of Nakayama and Rim. The main

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theorem itself is proved directly in Serre's Bourbaki seminar report of February 1953 ([**23**], see also [**21**, Chap. IX]).

At the time when Chapters V–XII of these notes were written, the isomorphism of the main theorem were not known except for $q = -2, -1, 0, 1, 2$, nor was it seen that these constituted part of a general pattern. Indeed, the possibility of extending the cohomology of finite groups to negative dimensions was not known at that time. Therefore the all-important norm-residue isomorphisms

$$(*) \qquad G_{K/F}/G^c_{K/F} \approx H^{-2}(G_{K/F}, \mathbb{Z}) \xrightarrow[\approx]{\alpha_{-2}} H^0(G_{K/F}, A_K) \approx A_F/N_{K/F}A_K,$$

(here G^c denotes the commutator subgroup of a group G) for a normal layer K/F in a class formation was treated separately by itself in the missing Chapters I–IV. Two descriptions of it were given.

The first method was to map $\sigma \in G_{K/F}$ onto the element $\prod_\tau a_{\tau,\sigma} \in A_F \pmod{N_{K/F}A_K}$, where $\{a_{\tau,\sigma}\}$ is a fundamental 2-cocycle for the layer K/F. Indeed, this Nakayama map[1] is just an explicit formula for the cup product of the fundamental class with the class $\zeta_\sigma \in H^{-2}(G, \mathbb{Z})$ corresponding to σ, so that this method amounts to giving an explicit description of the map α_{-2} of the main theorem without recognizing it as a cup product.

However in Chapters V–XII, the emphasis is placed on a dual description of the isomorphism $(*)$, namely that of Propositions 6 and 6′ of Chapter XIV, which involves no negative dimensional cohomology, even implicitly. Before explaining this method, we must introduce a formalism with characters which is also useful in other connections. Let G be a finite group. Since $\mathbb{Q}$ is infinitely and uniquely divisible it has trivial cohomology, and the exact sequence

$$0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$$

gives an isomorphism

$$\delta\colon H^1(G, \mathbb{Q}/\mathbb{Z}) \to H^2(G, \mathbb{Z}).$$

We can identify $H^1(G, \mathbb{Q}/\mathbb{Z})$ with the character group $\widehat{G} = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ and we then denote the image of a character χ under the isomorphism δ by $\delta\chi \in H^2(G, \mathbb{Z})$. Thus if for each $\sigma \in G$ we let $\bar\chi(\sigma)$ be a rational number such that $\chi(\sigma) \equiv \bar\chi(\sigma) \pmod 1$, then

$$\delta\bar\chi(\tau, \sigma) = \bar\chi(\tau) + \bar\chi(\sigma) - \bar\chi(\tau\sigma)$$

is a 2-cocycle with values in $\mathbb{Z}$ representing the class $\delta\chi$. Summing over τ we obtain

$$\sum_{\tau \in G} \delta\bar\chi(\tau, \sigma) = n\bar\chi(\sigma)$$

where $n = (G : 1)$ is the order of G. Thus passing to cohomology, it follows that the pairing

$$H^{-2}(G, \mathbb{Z}) \times H^2(G, \mathbb{Z}) \to H^0(G, \mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$$

[1] The map $\sigma \mapsto \prod_{\tau \in G} a_{\tau,\sigma}$ was introduced by Tadasi Nakayama in 1935 in his paper [**19**]. He showed that if K/k is a Galois extension with group G, and $a_{\sigma,\tau}$ is a 2-cocycle of G in K^*, then: (1) The map $\sigma \mapsto \prod_{\tau \in G} a_{\tau,\sigma}$ induces a homomorphism $G \to k^*/NK^*$; (2) If the cohomology class of $a_{\tau,\sigma}$ has the maximum possible order $n = |G|$, and G is abelian, that homomorphism is injective; and (3) If k is a $\mathfrak{p}$-adic field, and $a_{\sigma,\tau}$ corresponds to a division algebra with Hasse invariant $1/n$, then the homomorphism is the inverse of the norm residue homomorphism of local class field theory. A few months later, Yasuo Akizuki showed that for $a_{\sigma,\tau}$ as in (2), then for arbitrary G the kernel of the homomorphism is G', by reducing the general case to the abelian one (see footnote after (XIII, Sect. 3, Theorem 5)).

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is given by

$$\boxed{\zeta_\sigma \cup \delta\chi = n\bar{\chi}(\sigma) \pmod{n\mathbb{Z}}}.$$

From this formula we obtain a conceptual proof for the periodicity of the cohomology of cyclic groups:

THEOREM B. *Let G be a finite cyclic group of order n, and A a G-module, and φ a generator of G. Let χ be the character of G such that $\chi(\varphi) = 1/n \pmod 1$. Then the cup products with $\delta\chi$ and with ζ_φ give mutually inverse isomorphisms*

$$H^p(G, A) \underset{\cup\zeta_\varphi}{\overset{\cup\delta\chi}{\rightleftarrows}} H^{p+2}(G, A).$$

Indeed, we have $\zeta_\varphi \cup \delta\chi = \delta\chi \cup \zeta_\varphi = n\bar{\chi}(\varphi) \equiv 1 \pmod n$ in this case.

Now let K/F be a normal layer in a class formation. For each $a \in A_F$, we let $(a, K/F)$ denote the element of $G_{K/F}/G^c_{K/F}$ corresponding to the residue class of $a \mod N_{K/F}A_K$ under the norm residue isomorphism $(*)$. Then $(a, K/F)$ is characterized by the fact that

$$\chi((a, K/F)) = \mathrm{inv}_F(\varkappa a \cup \delta\chi) \qquad \text{for all } \chi \in \hat{G}_{K/F}$$

where $\varkappa a$ denotes the 0-dimensional cohomology class corresponding to a. Indeed, if $(a, K/F) = \sigma \mod G^c$ then by definition, $\varkappa a = \alpha \cup \zeta_\sigma$ where α is the fundamental class of the layer, hence

$$\varkappa a \cup \delta\chi = \alpha \cup \zeta_\sigma \cup \delta\chi = n\bar{\chi}(\sigma)\alpha$$

and this 2-dimensional class does have invariant $\chi(\sigma)$ because α has invariant $1/n$.[2]

3. The Herbrand Quotient

The Herbrand quotient is used so frequently in class field theory that we recall here its definition and properties. If f is an endomorphism of an abelian group A, we shall denote its kernel and image by A_f and A^f respectively.

Let f and g be endomorphisms of an abelian group A such that $fg = 0 = gf$. Then the *Herbrand quotient* is defined by the expression

$$q(A) = q_{f,g}(A) = \frac{(A_f : A^g)}{(A_g : A^f)}$$

provided both indices are finite.

Special Case: G is a finite cyclic group of order n. A is a G-module, $f = 1 - \varphi$ and $g = 1 + \varphi + \cdots + \varphi^{n-1}$ where φ is a generator of G. We have

$$A_f/A^g \approx H^0(G, A) \approx H^2(G, A)$$
$$A_g/A^f \approx H^{-1}(G, A) \approx H^1(G, A)$$

and thus the Herbrand quotient is denoted in this case by $h_{2/1}(G, A)$ or by $h_{2/1}(A)$, because it is the ratio of the orders of the 2- and 1-dimensional cohomology groups, and so appears in many applications.

[2]The interpretation of the Nakayama map as the cup product with ζ_σ gives a quick proof of the theorem of Akizuki–Nakayama mentioned in the previous footnote. Indeed, denoting the class of $a_{\sigma.\tau}$ by α and the class of $\delta\bar{\chi}$ by $\delta\chi$ we have

$$(\zeta_\sigma \cdot \alpha) \cdot \delta\chi = (\alpha \cdot \zeta_\sigma) \cdot \delta\chi = \alpha \cdot (\zeta_\sigma \cdot \delta\chi) = n\bar{\chi}(\sigma)\alpha.$$

Hence, if α is of order $n = |G|$ and $\zeta_\sigma \cdot \alpha = 0$, then $\bar{\chi}(\sigma)$ is an integer, i.e., $\chi(\sigma) = 0$, for all χ, so $\sigma \in G'$. (More generally, if α is of order m, then $\sigma^{n/m} \in G'$.)

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The basic properties of the Herbrand quotient result from the fact that it is the "multiplicative" Euler–Poincaré characteristic of the circular complex

$$\begin{tikzcd} A \arrow[r, bend left, "f"] & A \arrow[l, bend left, "g"] \end{tikzcd}$$

i.e. is the ratio of the orders of the two cohomology groups of the complex. Thus, by usual arguments, we find:

THEOREM q.1. *Suppose in the following diagram the horizontal rows are exact and the horizontal arrows commute with the vertical arrows*:

$$\begin{tikzcd} 0 \arrow[r] & A' \arrow[r] \arrow[d, shift right, "f'"'] & A \arrow[r] \arrow[d, shift right, "f"'] & A'' \arrow[r] \arrow[d, shift right, "f''"'] & 0 \\ 0 \arrow[r] & A' \arrow[r] \arrow[u, shift right, "g'"'] & A \arrow[r] \arrow[u, shift right, "g"'] & A'' \arrow[r] \arrow[u, shift right, "g''"'] & 0. \end{tikzcd}$$

Then $q(A) = q(A')q(A'')$ in the sense that if any two of the three quotients are defined then the third is also and the above equality holds.

THEOREM q.2. *If A is finite, then $q(A) = 1$.*

It is an elementary exercise to prove:

THEOREM q.3. *If g and h are commuting endomorphisms of an abelian group A, then*

$$q_{0,gh}(A) = q_{0,g}(A)q_{0,h}(A)$$

in the sense that if either side is defined then the other is also, and equality holds.

We sketch a proof of the following theorem which generalizes a theorem of Chevalley (Class Field Theory, Nagoya, 1953–54, Theorem 10.3).

THEOREM q.4. *Let G be a cyclic group of prime order p, and let A be a G-module such that $q_{0,p}(A)$ is defined. Then $q_{0,p}(A^G)$ and $h_{2/1}(A)$ are defined and we have*:

$$(h_{2/1}(G, A))^{p-1} = \frac{q_{0,p}(A^G)^p}{q_{0,p}(A)}.$$

Let φ be a generator of G and consider the exact sequence

$$0 \to A^G = A_{1-\varphi} \to A \xrightarrow{1-\varphi} A^{1-\varphi} \to 0.$$

Since $A^{1-\varphi}$ is at the same time a quotient group and subgroup of A, we see that $q_{0,p}(A^{1-\varphi})$ is defined, because $q_{0,p}(A)$ is so by hypothesis. Since both of these are defined, so also is $q_{0,p}(A^G)$ by Theorem q.1. But $h_{2/1}(A^G) = q_{0,p}(A^G)$ because G operates trivially on A^G. Hence

$$h_{2/1}(A) = q_{0,p}(A^G)h_{2/1}(A^{1-\varphi})$$
$$q_{0,p}(A) = q_{0,p}(A^G)q_{0,p}(A^{1-\varphi}).$$

Substituting these equations in the statement of the theorem, we see that we are reduced to proving that $h_{2/1}(A^{1-\varphi})$ is defined, and satisfies

$$h_{2/1}(A^{1-\varphi})^{p-1} = \frac{1}{q_{0,p}(A^{1-\varphi})}.$$

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But the endomorphism $1+\varphi+\cdots+\varphi^{p-1}$ annihilates $A^{1-\varphi}$ so that we can view $A^{1-\varphi}$ as a module over the ring $\mathbb{Z}[X]/(1+X+\cdots+X^{p-1})$, i.e. we can treat φ as a primitive p-th root of unity, and we are to prove

$$q_{0,p}(A^{1-\varphi}) = \big(q_{0,1-\varphi}(A^{1-\varphi})\big)^{p-1}.$$

But this follows from Theorem q.3 because in the ring of integers in the field of p-th roots of unity, the factorization of p is given by $p=(1-\varphi)^{p-1}\varepsilon$ where ε is a unit.

Another proof can be found in notes of Serre's course "Homologie des groupes, applications arithmetiques", Collège de France, 1958–59.

Theorem q.4 gives the fastest known proof of the basic lemma (Chapter V, § 3) in the proof of the global first inequality in case of cyclic extension of prime degree p, which is the only case necessary. To compute the Herbrand quotient $h_{2/1}(K_S)$ of the S-units one needs only know the ranks of the finitely generated groups K_S and $k_S = K_S^G$, because $q_{0,p}$ depends obviously on the rank. The details can be found in Chevalley's Nagoya book.

Another application of Herbrand's quotient is in the computation of the power index

$$(k^* : k^{*n})$$

in a local field k. Since this index is used at the end of the proof of Lemma 2, Chapter VI, § 3 we outline the computation here. Let k be a non-archimedean local field, let U be the group of units in k, and for each integer $i \geq 1$, let $U_i = 1+\mathfrak{p}^i$ be the group of units congruent to 1 mod $\mathfrak{p}^i$. We let f be the map $f(x)=1$ for all $x \in k^*$, and $g(x)=x^n$. Then

$$q_{0,n}(k^*) = \frac{(k^* : k^{*n})}{(k_n^* : 1)}.$$

We observe that the denominator is the order of the group of n-th roots of unity in k. Now for any i,

$$\begin{aligned} q_{0,n}(k^*) &= q_{0,n}(k^*/U)q_{0,n}(U/U_i)q_{0,n}(U_i) \\ &= q_{0,n}(\mathbb{Z})q_{0,n}(\text{finite group})q_{0,n}(U_i) \\ &= n\, q_{0,n}(U_i). \end{aligned}$$

If i is large and n prime to the characteristic of k, then the map $x \to x^n$ maps U_i isomorphically onto U_{i+m} where $m = \mathrm{ord}_k(n)$, as one sees for example by the binomial series for $(1+t)^n$ and $(1+t)^{1/n}$. Hence

$$q_{0,n}(U_i) = (U_i : U_{i+m}) = (\mathfrak{o} : \mathfrak{p})^m = (\mathfrak{o} : n\mathfrak{o}) = 1/|n|_k$$

where $|n|_k$ denotes the normed absolute value of n in k. Thus finally:

$$(k^* : k^{*n}) = \frac{n}{|n|_k}(k_n^* : 1)$$

and in particular, if the n-th roots of unity are contained in k, then

$$(k^* : k^{*n}) = \frac{n^2}{|n|_k}.$$

These formulaa can be checked directly in the archimedean cases, $k=\mathbb{R}$ and $k=\mathbb{C}$ (recall that in the latter case, the normed absolute value is the square of the ordinary absolute value). They hold formally if the characteristic of k divides n, because then $|n|_k = 0$ and $(k^* : k^{*n}) = \infty$.

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4. Local Class Field Theory

We now turn to local class field theory. Let k be a local field, and Ω its algebraic closure. We are to show that the formation $(G_{\Omega/k}, \Omega^*)$ is a class formation. If k is archimedean this is completely trivial, so we assume k non-archimedean. For any normal layer K/F we have $H^1(G_{K/F}, K^*) = 0$ by Hilbert's Theorem 90, i.e. our formation is trivially a field formation in the terminology of Chapter XIV, because it is a formation of (multiplicative groups of) fields.

Probably the best way to prove the Second Inequality

$$(F^* : N_{K/F}K^*) \leqslant [K : F]$$

is the method of Chapter XI, because the detailed study of the norm mapping carried out there is necessary for the ramification theory, and must be done sometime anyhow. On the other hand, there are short cuts if one wants only the inequality. As explained in Chapter XIV, § 3, one needs only establish our inequality for cyclic layers of prime degree.

Let K/F be cyclic of degree n. Then for any submodule V of finite index in the group of units U of K, we have

$$\begin{aligned} h_{2/1}(G_{K/F}, K^*) &= h_{2/1}(G_{K/F}, \mathbb{Z}) h_{2/1}(G_{K/F}, U/V) h_{2/1}(G_{K/F}, V) \\ &= n\, h_{2/1}(G_{K/F}, V) \end{aligned}$$

and we shall prove this is equal to n by constructing in any normal layer K/F, cyclic or not, a subgroup V such that $H^r(G, V) = 0$ for all r. Indeed, let $\{\theta^\sigma\}$ be a normal basis for K/F. Replace θ by $\pi^i\theta$ where π is a prime element in F, and where i is sufficiently large so that if we put

$$M = \sum_{\sigma \in G} \mathfrak{o}_F \theta^\sigma$$

we have $M^2 \subset \pi M$ and $M \subset \pi \mathfrak{o}_K$. Finally, put $V = 1 + M$. Then it is easy to see that V is an open G-submodule of U, and moreover V is filtered by subgroups $V_i = 1 + \pi^i M$ such that for each i, the module

$$V_i/V_{i+1} \approx M/\pi M$$

is G-regular, and hence has vanishing cohomology. Now we can apply the following elementary lemma whose proof we leave to the reader.

Lemma. *Let A be a complete topological group and G a finite group operating continuously on A. Let*

$$A = A_0 \supset A_1 \supset A_2 \supset \dots$$

be a decreasing sequence of subgroups invariant under G, and which shrink to the identity in the sense that for each neighborhood U of 1 in A, there is an index i such that $A_i \subset U$. If $H^r(G, A_i/A_{i+1}) = 0$ for all i and some r, then $H^r(G, A) = 0$.

(In characteristic 0, one could avoid the preceding construction by taking a sufficiently small neighborhood of 0 in the additive group of K, and mapping it onto a neighborhood of 1 in K^* by means of the exponential function.)

For cyclic layers K/F of prime degree p different from the characteristic, one can also deduce $h_{2/1}(G_{K/F}, K^*) = p$ from Theorem q.4 above, using the equations

$$q_{0,p}(K^*) = \frac{p}{|p|_k} \qquad \text{and} \qquad q_{0,p}(F^*) = \frac{p}{|p|_k}$$

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obtained in our computation of the power index $(K^* : K^{*p})$ above.

Finally, one could ignore the second inequality completely by proving directly that every 2-dimensional class has an unramified splitting field, or what is the same, that the Brauer group of the maximal unramified extension of k is trivial (cf. for example [**15**]).

To complete the proof that our formation of multiplicative groups of local fields is a class formation, we must establish Axiom II′ of Chapter XIV. For this, we consider the *unramified* extension K/F of degree n. Since the residue class field is finite, the Galois group $G_{K/F}$ is cyclic, with a canonical generator, the Frobenius automorphism $\varphi = \varphi_{K/F}$.

For any normal layer K/F, unramified or not, the exact sequence

$$0 \to U_K \to K^* \to \mathbb{Z} \to 0$$

yields, on passage to cohomology,

$$F^* = K^{*G} \to \mathbb{Z} \to H^1(G_{K/F}, U_K) \to H^1(G_{K/F}, K^*) = 0,$$

from which we see that $H^1(G_{K/F}, U_K)$ is isomorphic to the cokernel of $F^* \to \mathbb{Z}$, i.e. is cyclic of order equal to the ramification index $e_{K/F}$, because $\mathbb{Z}$ here represents the value group of K. Thus, for our unramified K/F, we have $H^1(G_{K/F}, U_K) = 0$.

On the other hand, we have $H^0(G_{K/F}, U_K) = U_F/N_{K/F}U_K = 0$ also. This follows in various ways: either a direct refinement process showing that every unit in F is a norm of a unit in K, or from the fact that $h_{2/1}(U_K) = 1$, as was shown in course of proving $h_{2/1}(K^*) = n$ above. Thus for unramified K/F we have $H^r(G_{K/F}, U_K) = 0$ for all r, a fact of importance for the idèle cohomology, as mentioned in the first paragraphs of this introduction.

From our exact sequence we obtain canonical isomorphisms

$$H^2(G_{K/F}, K^*) \approx H^2(G_{K/F}, \mathbb{Z}) \approx \widehat{G}_{K/F}.$$

On the other hand, $\chi \to \chi(\varphi_{K/F})$ gives an isomorphism

$$\widehat{G}_{K/F} \approx \frac{1}{n}\mathbb{Z}/\mathbb{Z}.$$

Composing these two we obtain an injection

$$\mathrm{inv}_{K/F} \colon H^2(G_{K/F}, K^*) \to \mathbb{Q}/\mathbb{Z}.$$

For any $a \in F^*$, and $\chi \in \widehat{G}$, the class $\varkappa a \cup \delta\chi \in H^2(G_{K/F}, K^*)$ is represented by the 2-cocycle $a^{\delta\bar{\chi}(\sigma,\tau)}$. Taking ordinals, i.e. applying the map $K^* \to \mathbb{Z}$, we get the 2-cocycle

$$(\mathrm{ord}_K\, a)\delta\bar{\chi}(\sigma, \tau)$$

which represents the class $\mathrm{ord}_K\, a \cup \delta\chi = \delta(\chi^{\mathrm{ord}_K\, a})$ in $H^2(G_{K/F}, \mathbb{Z})$. Thus we have

$$(**) \qquad \mathrm{inv}_{K/F}(\varkappa a \cup \delta\chi) = \chi^{\mathrm{ord}_K\, a}(\varphi) = \chi(\varphi^{\mathrm{ord}_F\, a})$$

since $\mathrm{ord}_F\, a = \mathrm{ord}_K\, a$, our extension being unramified. This shows that for an unramified extension K/F, we will have

$$(a, K/F) = \varphi_{K/F}^{\mathrm{ord}_F\, a}$$

(In some classical texts, the opposite sign is chosen.) Furthermore, since $G_{K/F}$ is cyclic, every 2-dimensional class is of the form $\varkappa a \cup \delta\chi$ and we can use the rule $(**)$ to establish the required properties of $\mathrm{inv}_{K/F}$. First of all, the invariant does not change under inflation to a bigger unramified extension L/F with $L \supset K \supset F$,

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because $\varphi_{K/F}$ is the image of $\varphi_{L/F}$ under the canonical map $G_{L/F} \to G_{K/F}$, and inflation of $\varkappa a \cup \delta\chi$ amounts to viewing a character χ of $G_{K/F}$, as a character of $G_{L/F}$ by this same canonical map. Hence

$$\bar{H}^2(*/F) = \bigcup_{K/F \text{ unramified}} H^2(G_{K/F}, K^*)$$

the subgroup of the Brauer group $H^2(*/F)$ consisting of the elements coming from unramified layers. We obtain an isomorphism

$$\mathrm{inv}_F \colon \bar{H}^2(*/F) \to \mathbb{Q}/\mathbb{Z}$$

(surjectivity because there exist unramified extension of arbitrary degree).

To complete the proof of Axiom II′, we must show that the invariant multiplies by the degree $[E : F]$ under restriction from F to E. This follows from $(**)$ when one takes into account that $\mathrm{ord}_E = e\,\mathrm{ord}_F$, where e is the ramification index, and that, under the canonical map $G_{KE/E} \to G_{K/F}$ the image of $\varphi_{KE/E}$ is $\varphi^f_{K/F}$ where f is the residue class degree. Hence the invariant multiplies by $ef = [E : F]$.

This just about completes our introductory comments. Concerning the existence theorem, we have given in Chapter XIV, §6 an abstract discussion which shows that the existence theorem follows in abstracto from Axioms IIIa–IIIe. In both global and local class field theory, these axioms are all trivial to verify except for IIId. The proof of this axiom in the global case is carried out in Chapter VI, § 5. In the local case, it is not covered in these notes, but would follow readily from the theory of the norm residue symbol in Kummer fields.

Chapter XIII and Chapter XV are not needed for the remaining parts, but note that there is a proof of the principal ideal theorem in Chapter XIII.

We hope that the preceding remarks will to some extent reduce the inconvenience which the reader will suffer from the missing portions of the notes, and other imperfections occurring in them.

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CHAPTER V

The First Fundamental Inequality

1. Statement of the First Inequality

In this entire chapter, k is a global field and K/k a *cyclic* extension of degree n with Galois Group G. We let $J = J_K$ be the idèles of K, and $C = C_K$ be the idèle classes of K. Then G acts on J and C, and the fixed groups are $J^G = J_k$, $C^G = C_k$.

We let h_1 and h_2 denote the orders of the first and second cohomology groups. $h_{2/1}$ abbreviates h_2/h_1. We wish to determine the order $h_2(G, C)$ of $\mathcal{H}^2(G, C)$, and it will be shown in this chapter that $h_2(G, C) \geqslant n$. In fact, we prove

THEOREM 1. *Let k be a global field and let K/k be a cyclic extension of degree n with group G. Then*

$$h_2(G, C_K) = n \cdot h_1(G, C_K)$$

or in other words,

$$h_{2/1}(G, C_K) = n.$$

To simplify the notation we omit G and write $h_i(C)$ instead of $h_i(G, C)$ whenever G is the group of operators.

We shall prove this inequality first in function fields, because considerable simplifications occur in this special case. Afterwards, we shall give a unified proof for all global fields.

We shall make constant use of the properties of the index $h_{2/1}$ developed on pages 5–7 (Section 3 of "Preliminaries"), and recall here the three most important properties for the convenience of the reader.

PROPERTY 1. The index $h_{2/1}$ is multiplicative. In other words, if A is an abelian group on which G acts, and A_0 is a subgroup invariant under G we have

$$h_{2/1}(A) = h_{2/1}(A/A_0)h_{2/1}(A_0),$$

in the sense that if two of these quotients are finite, then so is the third, and the relation holds.

PROPERTY 2. If A_0 is a finite group, then $h_{2/1}(A_0) = 1$, and hence $h_{2/1}(A) = h_{2/1}(A/A_0)$.

PROPERTY 3. If $A \approx \mathbb{Z}$ is infinite cyclic and G operates trivially, then $h_{2/1}(\mathbb{Z}) = n$ is the order of G.

2. First Inequality in Function Fields

We suppose here that k is a function field. We let $U = \prod_{\mathfrak{P}} U_{\mathfrak{P}}$ be the unit idèles of K, and $J_0 = J_K^0$ the idèles of volume 1 of K, i.e. those idèles $\mathfrak{a}$ such that

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$\prod_{\mathfrak{P}} |\mathfrak{a}|_{\mathfrak{P}} = 1$. Then $J_0 \supset U$ obviously, and $J_0 \supset K^*$ by the product formula. Hence $J_0 \supset UK^*$.

The multiplicativity of $h_{2/1}$ gives

$$h_{2/1}(J/K^*) = h_{2/1}(J/J_0)h_{2/1}(J_0/UK^*)h_{2/1}(UK^*/K^*),$$

and it will come out that all three quotients on the right are finite.

To begin with, J/J_0 is G-isomorphic to the additive group of integers $\mathbb{Z}$ with trivial action under G, via the degree map. Hence

$$h_{2/1}(J/J_0) = n.$$

Since the number of divisor classes of degree zero is finite, J_0/UK^* is a finite group. Hence

$$h_{2/1}(J_0/UK^*) = 1.$$

The factor group UK^*/K^* is G-isomorphic to $U/(U \cap K^*)$ and hence

$$h_{2/1}(UK^*/K^*) = h_{2/1}(U/(U \cap K^*)) = h_{2/1}(U)\big(h_{2/1}(U \cap K^*)\big)^{-1}.$$

Here we use the multiplicativity in reverse, and it will be shown that both $h_{2/1}(U)$ and $h_{2/1}(U \cap K^*)$ are 1.

We know that $U \cap K^* = K_0^*$ is the multiplicative group of the constant field of K, and is finite. Hence $h_{2/1}(U \cap K^*) = 1$. We contend finally that $h_{2/1}(U) = 1$. Indeed, we can express U as a direct product,

$$U = \prod_{\mathfrak{p}} \Big(\prod_{\mathfrak{P}|\mathfrak{p}} U_{\mathfrak{P}}\Big)$$

where each component $\prod_{\mathfrak{P}|\mathfrak{p}} U_{\mathfrak{P}}$ is semilocal, and invariant under G. For each $\mathfrak{p}$ let $U_{\mathfrak{p}}$ be one of the groups $U_{\mathfrak{P}}$, and let $G_{\mathfrak{p}}$ be the local group, leaving $U_{\mathfrak{p}}$ invariant. The semilocal theory states that

$$\mathcal{H}^r\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} U_{\mathfrak{P}}\Big) \approx \mathcal{H}^r(G_{\mathfrak{p}}, U_{\mathfrak{p}})$$

and we have

$$\mathcal{H}^r(U) \approx \prod_{\mathfrak{p}} \mathcal{H}^r(G_{\mathfrak{p}}, U_{\mathfrak{p}}).$$

We know from the local class field theory that

$$h_1(G_{\mathfrak{p}}, U_{\mathfrak{p}}) = h_2(G_{\mathfrak{p}}, U_{\mathfrak{p}}) = e_{\mathfrak{p}}$$

where $e_{\mathfrak{p}}$ is the ramification index. But $e_{\mathfrak{p}} = 1$ at almost all $\mathfrak{p}$. This shows that $h_2(U)$ and $h_1(U)$ are both equal to $\prod_{\mathfrak{p}} e_{\mathfrak{p}}$ and therefore that $h_{2/1}(U) = 1$, as was to be shown.

If we piece together the information just derived, we get the desired result:

$$h_{2/1}(C) = h_{2/1}(J/K^*) = n.$$

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3. First Inequality in Global Fields

We treat now the two cases simultaneously. The existence of archimedean primes prevents us from giving the same proof for number fields that was given for function fields in the preceding section. Using Haar measure, and a generalized Herbrand quotient for Haar measure, one could indeed give an argument in number fields which parallels completely that of function fields. Since we wish to avoid the use of Haar measure, we give below a modified version of our preceding proof.

We let k be a global field and S a finite non-empty set of primes of k including all archimedean primes. The subgroups K^*, J_S, K^*J_S of J are invariant under G. (Recall that S also stands for the set of all the primes of K dividing those of k which are in S.) We have therefore

$$h_{2/1}(J/K^*) = h_{2/1}(J/J_SK^*)h_{2/1}(J_SK^*/K^*),$$

and it will be shown that both indices on the right are finite.

From the finiteness of class number theorem we know that J/J_SK^* is a finite group, and consequently $h_{2/1}(J/J_SK^*) = 1$.

J_SK^*/K^* is G-isomorphic to $J_S/J_S \cap K^* = J_S/K_S^*$. Consequently

$$h_{2/1}(J/K^*) = h_{2/1}(J_S)(h_{2/1}(K_S^*))^{-1},$$

and it will be shown that both quotients on the right are finite. We can write

$$J_S = \Big(\prod_{\mathfrak{P}\notin S} U_{\mathfrak{P}}\Big) \times \Big(\prod_{\mathfrak{P}\in S} K_{\mathfrak{P}}^*\Big)$$

where each factor is invariant under G. By an argument similar to that used in function fields, we have $h_{2/1}\Big(\prod_{\mathfrak{P}\notin S} U_{\mathfrak{P}}\Big) = 1$. We can decompose the finite product

$$\prod_{\mathfrak{P}\in S} K_{\mathfrak{P}}^* = \prod_{\mathfrak{p}\in S}\Big(\prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}^*\Big).$$

Let $K_{\mathfrak{p}}^*$ be any one of the fields $K_{\mathfrak{P}}^*$, and let $G_{\mathfrak{p}}$ be its local group. The semilocal theory shows that

$$h_{2/1}\Big(G, \prod_{\mathfrak{P}\in S} K_{\mathfrak{P}}^*\Big) = \prod_{\mathfrak{p}\in S} h_{2/1}\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}^*\Big) = \prod_{\mathfrak{p}\in S} h_{2/1}\left(G_{\mathfrak{p}}, K_{\mathfrak{p}}^*\right).$$

From the local theory we know that $h_2(G_{\mathfrak{p}}, K_{\mathfrak{p}}^*) = n_{\mathfrak{p}}$ is the local degree, and $h_1(G_{\mathfrak{p}}, K_{\mathfrak{p}}^*) = 1$ is trivial. Hence

$$h_{2/1}(J_S) = \prod_{\mathfrak{p}\in S} n_{\mathfrak{p}}.$$

We shall therefore have completed the proof of the first inequality if we succeed in proving the following

Lemma. *Let S be a finite set of primes of k including all archimedean primes and let K/k be a cyclic extension of degree n with group G. Then*

$$h_{2/1}(K_S^*) = \Big(\prod_{\mathfrak{p}\in S} n_{\mathfrak{p}}\Big)/n$$

where $n_{\mathfrak{p}}$ is the local degree $[K_{\mathfrak{P}} : k_{\mathfrak{p}}]$.

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PROOF. [1] Let s be the number of primes $\mathfrak{P}$ of the set S in K, and let $\mathbb{R}^s$ be the additive group of Euclidean s-space. Let $\{X_{\mathfrak{P}}\}_{\mathfrak{P}\in S}$ be a basis of $\mathbb{R}^s$ and map the S-idèles of K into $\mathbb{R}^s$ as follows:

$$\mathfrak{a} \to L(\mathfrak{a}) = \sum_{\mathfrak{P}\in S} \log |\mathfrak{a}|_{\mathfrak{P}} X_{\mathfrak{P}}.$$

This map is an algebraic homomorphism of J_S into $\mathbb{R}^s$. We shall make it a G-homomorphism by defining a suitable action of G on $\mathbb{R}^s$. Namely we let

$$X_{\mathfrak{P}}^{\sigma} = X_{\mathfrak{P}^{\sigma}}, \qquad \sigma \in G,$$

and extend σ to $\mathbb{R}^s$ by linearity. Then G acts on $\mathbb{R}^s$, permutes the vectors $X_{\mathfrak{P}}$ but not necessarily transitively. We recall that $|\mathfrak{a}^{\sigma}|_{\mathfrak{P}^{\sigma}} = |\mathfrak{a}|_{\mathfrak{P}}$ and using these facts we have

$$\begin{aligned} L(\mathfrak{a}^{\sigma}) &= \sum_{\mathfrak{P}\in S} \log |\mathfrak{a}^{\sigma}|_{\mathfrak{P}} X_{\mathfrak{P}} \\ &= \sum_{\mathfrak{P}\in S} \log |\mathfrak{a}^{\sigma}|_{\mathfrak{P}^{\sigma}} X_{\mathfrak{P}^{\sigma}} \\ &= \sum_{\mathfrak{P}\in S} \log |\mathfrak{a}|_{\mathfrak{P}} X_{\mathfrak{P}^{\sigma}} \\ &= L(\mathfrak{a})^{\sigma}. \end{aligned}$$

This proves that L is a G-homomorphism of J_S into $\mathbb{R}^s$.

The image $L(K_S)$ is a lattice of dimension $s-1$ in $\mathbb{R}^s$, according to the Unit Theorem (see [**16**, p. 104] or [**5**, proof of theorem, p. 72]). This lattice is contained in the hyperplane $\mathbb{R}^{s-1}$ of all elements $\sum x_{\mathfrak{P}} X_{\mathfrak{P}}$ such that $\sum x_{\mathfrak{P}} = 0$ because of the product formula, and consequently the lattice $L(K_S)$ spans this hyperplane.

The kernel of L in K_S consists of all roots of unity and is a finite group. The map

$$K_S \to L(K_S)$$

is a G-homomorphism and $L(K_S)$ is G-isomorphic to $K_S/$(roots of unity). Hence

$$h_{2/1}(L(K_S)) = h_{2/1}(K_S)$$

because $h_{2/1}$ of a finite group is 1.

We now face the task of determining $h_{2/1}(L(K_S))$. We first extend the $s-1$ dimensional lattice $L(K_S)$ to an s-dimensional lattice M as follows. We let $X = \sum_{\mathfrak{P}} X_{\mathfrak{P}}$. Then the vector X does not lie in the hyperplane $\mathbb{R}^{s-1}$ spanned by $L(K_S)$. We let M be the lattice generated by $L(K_S)$ and by X, i.e.

$$M = L(K_S) + \mathbb{Z}X \qquad (\mathbb{Z} \text{ are the integers}).$$

Then M is s-dimensional, and spans $\mathbb{R}^s$. Since $X^{\sigma} = X$ for all $\sigma \in G$, M is invariant under G and both $L(K_S)$ and $\mathbb{Z}X$ are G-modules. The module $\mathbb{Z}X$ is

[1] The key idea in this proof is that for a finite cyclic group G and a finitely generated G-module M, the Herbrand quotient $h_{2/1}(G, M)$ is defined and depends only on the $\mathbb{R}(G)$-module $M \otimes_{\mathbb{Z}} \mathbb{R}$ (see, for example, [**5**, Ch. IV, Sect. 8, Prop. 12]). In fact, if G is of order n, generated by σ, and the characteristic polynomial of σ acting on the vector space $M \otimes \mathbb{R}$ is $\Phi(x) = (x-1)^n \Psi(x)$ with $\Psi(1) \neq 0$, then

$$h_{2/1}(G, M) = \frac{n^r}{\Psi(1)}.$$

For example, if $\Phi(x) = x^n - 1$, then $h_{2/1} = n/r$.

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G-isomorphic to $\mathbb{Z}$ with trivial action. Hence $h_{2/1}(\mathbb{Z}X) = n$. Furthermore, the above sum is clearly direct, and therefore

$$h_{2/1}(M) = h_{2/1}(L(K_S)) \cdot n.$$

The proof of our lemma will therefore be complete if we prove that

$$h_{2/1}(M) = \prod_{\mathfrak{p} \in S} n_{\mathfrak{p}}.$$

For this purpose, we prove the following proposition.

PROPOSITION. *Given any s-dimensional lattice M in our $\mathbb{R}^s$ that is invariant under G, there exists a sublattice of finite index $\widetilde{M}$ which is invariant under G, and generated by basis elements $Y_{\mathfrak{P}}$ ($\mathfrak{P} \in S$) on which the action of G is given by*

$$Y_{\mathfrak{P}}^{\sigma} = Y_{\mathfrak{P}^{\sigma}} \qquad \sigma \in G.$$

Before proving the proposition, we show how our lemma follows from it. Suppose we have found a sublattice $\widetilde{M}$ of M with the action described in the proposition. For each prime $\mathfrak{p}$, let $\widetilde{M}_{\mathfrak{p}}$ be the sublattice of $\widetilde{M}$ generated by all elements $Y_{\mathfrak{P}}$, $\mathfrak{P} \mid \mathfrak{p}$. (In other words, break up $\widetilde{M}$ into transitivity domains under G.) Then each $\widetilde{M}_{\mathfrak{p}}$ is invariant under G and

$$\widetilde{M} = \sum \widetilde{M}_{\mathfrak{p}}$$

is a direct sum. Furthermore, each pair $(G, \widetilde{M}_{\mathfrak{p}})$ is semilocal. By the semilocal theory, we have for each

$$h_{2/1}(G, \widetilde{M}_{\mathfrak{p}}) = h_{2/1}(G_{\mathfrak{P}}, \mathbb{Z}Y_{\mathfrak{P}})$$

and $G_{\mathfrak{P}}$ acts trivially on the infinite cyclic group generated by $Y_{\mathfrak{P}}$. Consequently $h_{2/1}(\widetilde{M}_{\mathfrak{p}}) = n_{\mathfrak{p}}$ and $h_{2/1}(\widetilde{M}) = \prod n_{\mathfrak{p}}$. Since $\widetilde{M}$ is of finite index in M, it follows that

$$h_{2/1}(M) = h_{2/1}(\widetilde{M}) = \prod n_{\mathfrak{p}}.$$

This proves our lemma. □

PROOF OF PROPOSITION. For convenience we define a norm on $\mathbb{R}^s$ by putting

$$\Big|\sum_{\mathfrak{P}} x_{\mathfrak{P}} X_{\mathfrak{P}}\Big| = \max_{\mathfrak{P}} |x_{\mathfrak{P}}| \qquad x_{\mathfrak{P}} \in \mathbb{R}.$$

This norm is clearly invariant under the effect of $\sigma \in G$.

Let b be a constant such that for any vector $A \in \mathbb{R}^s$ there exists $Y \in M$ such that $|A - Y| < b$. Such a b exists since M is s-dimensional.

For each prime $\mathfrak{p} \in S$ let $\bar{\mathfrak{p}}$ be one of the primes $\mathfrak{P} \mid \mathfrak{p}$. Let $Y_{\bar{\mathfrak{p}}} \in M$ be such that

$$|Y_{\bar{\mathfrak{p}}} - bX_{\bar{\mathfrak{p}}}| < b. \tag{1}$$

Let $Y_{\mathfrak{P}} = \sum_{\substack{\sigma \in G \\ \bar{\mathfrak{p}}^{\sigma} = \mathfrak{P}}} Y_{\bar{\mathfrak{p}}}^{\sigma}$. We contend that the vectors $Y_{\mathfrak{P}}$ generate a sublattice of the desired kind.

We first verify that G has the proper effect, i.e. $Y_{\mathfrak{P}}^{\tau} = Y_{\mathfrak{P}^{\tau}}$. Indeed

$$Y_{\mathfrak{P}}^{\tau} = \sum_{\bar{\mathfrak{p}}^{\sigma} = \mathfrak{P}} Y_{\bar{\mathfrak{p}}}^{\tau\sigma} = \sum_{\substack{\sigma \in G \\ \bar{\mathfrak{p}}^{\tau^{-1}\sigma} = \mathfrak{P}}} Y_{\bar{\mathfrak{p}}}^{\sigma} = \sum_{\substack{\sigma \in G \\ \bar{\mathfrak{p}}^{\sigma} = \mathfrak{P}^{\tau}}} Y_{\bar{\mathfrak{p}}}^{\sigma} = Y_{\mathfrak{P}^{\tau}}$$

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as was to be shown.

All that remains to be done is to show that the vectors $Y_{\mathfrak{P}}$ are linearly independent. Given a relation

$$\sum x_{\mathfrak{P}} Y_{\mathfrak{P}} = 0$$

we shall prove that all $x_{\mathfrak{P}} = 0$.

We note that the number of $\sigma \in G$ such that $\bar{\mathfrak{p}}^{\sigma} = \mathfrak{P}$ is exactly the local degree $n_{\mathfrak{p}}$. Because of (1) there exist vectors $B_{\bar{\mathfrak{p}}}$ such that

$$Y_{\bar{\mathfrak{p}}} = bX_{\bar{\mathfrak{p}}} + B_{\bar{\mathfrak{p}}}$$

where $|B_{\bar{\mathfrak{p}}}| < b$.

From the definition of $Y_{\mathfrak{P}}$ we get

$$\begin{aligned} Y_{\mathfrak{P}} &= \sum_{\bar{\mathfrak{p}}^{\sigma}=\mathfrak{P}} bX_{\bar{\mathfrak{p}}}^{\sigma} + \sum_{\bar{\mathfrak{p}}^{\sigma}=\mathfrak{P}} B_{\bar{\mathfrak{p}}}^{\sigma} \\ &= \sum_{\bar{\mathfrak{p}}^{\sigma}=\mathfrak{P}} bX_{\mathfrak{P}} + C_{\mathfrak{P}} \\ &= n_{\mathfrak{p}} bX_{\mathfrak{P}} + C_{\mathfrak{P}} \end{aligned}$$

where $C_{\mathfrak{P}}$ is a vector such that $|C_{\mathfrak{P}}| < n_{\mathfrak{p}} b$. Substituting in the relation yields

$$0 = \sum_{\mathfrak{P}} x_{\mathfrak{P}} n_{\mathfrak{p}} bX_{\mathfrak{P}} + \sum_{\mathfrak{P}} x_{\mathfrak{P}} C_{\mathfrak{P}}. \tag{2}$$

Let $(C_{\mathfrak{P}}, X_{\mathfrak{P}})$ be the component of $C_{\mathfrak{P}}$ along $X_{\mathfrak{P}}$. Since the vectors $X_{\mathfrak{P}}$ are independent, the component of each $X_{\mathfrak{P}}$ in (2) must be 0. It is

$$x_{\mathfrak{P}}\left[n_{\mathfrak{p}} b + (C_{\mathfrak{P}}, X_{\mathfrak{P}})\right]$$

and the inequality $|C_{\mathfrak{P}}| < n_{\mathfrak{p}} b$ shows that $n_{\mathfrak{p}} b + (C_{\mathfrak{P}}, X_{\mathfrak{P}})$ cannot be 0. Hence $x_{\mathfrak{P}} = 0$, and the $Y_{\mathfrak{P}}$ are linearly independent.

This concludes the proof of the first inequality. □

4. Consequences of the First Inequality

We note that in Theorem 1, we have proved more than an inequality. Namely, we have actually proved that $h_2(G, C)$ is equal to $n \cdot h_1(G, C)$. This will be used at a later stage of the theory. However, at this point we shall only give applications of the inequality itself.

Since G is cyclic, we have

$$h_2(G, C) = (C^G : NC) = (C_k : N_{K/k} C_K)$$

or in terms of idèles,

$$h_2(G, C) = (J_k : k^* N_{K/k} J_K).$$

To simplify the notation, we abbreviate $N_{K/k}$ by N, and locally write $N_{\mathfrak{P}}$ for $N_{K_{\mathfrak{P}}/k_{\mathfrak{p}}}$.

The first inequality implies therefore that the norm index $(J_k : kNJ_K)$ of a cyclic extension is greater or equal to its degree. In particular, if this index is 1, then the cyclic extension is trivial, and $[K : k] = 1$ also. This remark is used to prove an important consequence of the first inequality.

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THEOREM 2. *Let K be a normal extension of a global field k. If $K \neq k$ then there exists infinitely many primes $\mathfrak{p}$ of k which do not split completely in K. Phrased another way: If $K_{\mathfrak{P}} = k_{\mathfrak{p}}$ for all but a finite number of primes $\mathfrak{p}$, then $K = k$.*

PROOF. First let K/k be cyclic. Suppose $K_{\mathfrak{P}} = k_{\mathfrak{p}}$ at almost all primes. We shall prove that any idèle $\mathfrak{a} \in J_k$ can be written

$$\mathfrak{a} = \alpha N \mathfrak{a}_K \qquad \mathfrak{a}_K \in J_K.$$

Let S be the set of primes where $K_{\mathfrak{P}} \neq k_{\mathfrak{p}}$. We can find $\alpha \in k$ such that $\alpha^{-1}\mathfrak{a}$ is so close to 1 at all $\mathfrak{p} \in S$ that $\alpha^{-1}\mathfrak{a}_{\mathfrak{p}}$ is a local norm: $\alpha^{-1}\mathfrak{a}_{\mathfrak{p}} = N_{\mathfrak{P}}\alpha_{\mathfrak{P}}$. At all other primes, $\alpha^{-1}\mathfrak{a}$ is a fortiori a local norm of an $\alpha_{\mathfrak{P}} = \alpha_{\mathfrak{p}}$ because $K_{\mathfrak{P}} = k_{\mathfrak{p}}$. We make up an idèle $\mathfrak{a}_K \in J_K$ by giving it the component $\alpha_{\mathfrak{P}}$ at one $\mathfrak{P} \mid \mathfrak{p}$, and 1 at the other $\mathfrak{P} \mid \mathfrak{p}$. Then

$$\mathfrak{a} = \alpha \cdot N\,\mathfrak{a}_K$$

as was to be shown. Therefore $(J_k : kNJ_K) = 1$, $K = k$

If K/k is normal, and $K \neq k$, then there is a *non trivial* cyclic subgroup of G which leaves fixed a subfield F of K over which K is cyclic. The theorem is now obvious since there are infinitely many primes in F that do not split completely in K. □

In the following special cases, one can use the Hilbert Theory to sharpen Theorem 2.

THEOREM 3. *Let K/k be a cyclic extension of prime power degree p^r. Then there exist infinitely many primes of k which do not split at all in K, i.e. which remain prime in K.*

PROOF. K contains exactly one subfield F of degree p over k. Suppose that almost all primes split in some way. Then for such primes $\mathfrak{p}$,

$$[K_{\mathfrak{p}} : k_{\mathfrak{p}}] = [Kk_{\mathfrak{p}} : k_{\mathfrak{p}}] = [K : (K \cap k_{\mathfrak{p}})] = \frac{[K : k]}{[(K \cap k_{\mathfrak{p}}) : k]}$$

is the local degree and is strictly less than p^r. Hence $[(K \cap k_{\mathfrak{p}}) : k] > 1$ and therefore $F \subset k_{\mathfrak{p}}$. This would mean that all but a finite number of primes split completely in F, and this is impossible. □

The next theorem will be used in the proof of the second inequality.

THEOREM 4. *Let $K_1, \ldots, K_r/k$ be r cyclic extensions of prime degree p which are mutually disjoint over k, i.e. $K_\nu \cap (K_1 \ldots \widehat{K}_\nu \ldots K_r) = k$. Then there exist infinitely many primes $\mathfrak{p}$ that split completely in K_i $(i > 1)$ and remain prime in K_1.*

PROOF. Let $K = K_1 \ldots K_r$ be the compositum of all K_ν. Then $K/(K_2 \ldots K_r)$ is cyclic. Let $\mathfrak{q}$ be a prime in $(K_2 \ldots K_r)$ which remains prime in K and which divides a prime $\mathfrak{p}$ of k which is unramified in K. (There exist infinitely many such primes.) Then $K_{\mathfrak{q}}/(K_2 \ldots K_r)_{\mathfrak{q}}$ is cyclic of degree p. But $K_{\mathfrak{q}}/k_{\mathfrak{p}}$ is also cyclic because $\mathfrak{p}$ is unramified in K. The Galois group of K/k is of type $(p, p, \ldots, p)$ and that of $K_{\mathfrak{q}}/k_{\mathfrak{p}}$ a cyclic subgroup. This means that $[K_{\mathfrak{q}} : k_{\mathfrak{p}}] \leqslant p$. Together with $[K_{\mathfrak{q}} : (K_2 \ldots K_r)_{\mathfrak{q}}] = p$ this shows $(K_2 \ldots K_r)_{\mathfrak{q}} = k_{\mathfrak{p}}$. It follows that $\mathfrak{p}$ splits completely in $(K_2 \ldots K_r)$. It remains prime in K_1, or else $K_{\mathfrak{q}} = k_{\mathfrak{p}}$ which is not the case. □

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The previous theorems concerning the existence of primes splitting in a certain way are weak instances of more general results concerning the statistical behavior of primes in normal extension, which can be proved using analysis. What we have proved in Theorems 2, 3, and 4 will suffice for our development of class field theory.

We shall obtain one more consequence of the first inequality in function fields.

THEOREM 5. *Let k be a function field in one variable over a finite constant field k_0. Then there exists a divisor of degree* 1 *in k.*

PROOF. We have already observed (Section 3, Theorem 8) that the degrees of the divisors of k form an ideal of the ordinary integers. This ideal is principal, generated by the positive integer δ, which is the greatest common divisor of all degrees of the divisors. It is also the g.c.d. of the degrees of the primes, and we shall now prove that $\delta = 1$.

Let k_1/k_0 be the extension of degrees δ and let $K = kk_1$ be the corresponding constant field extension of K. For each prime $\mathfrak{p}$ of k, the residue class field $\bar{k}_{\mathfrak{p}}$ contains k_1, and therefore for any $\mathfrak{P} \mid \mathfrak{p}$, $K_{\mathfrak{P}} = k_{\mathfrak{p}}$. By the first inequality (Theorem 2) it follows that $K = k$, i.e. $\delta = 1$. $\square$

It may be convenient for the reader to recapitulate the essential ingredients of the preceding proof:

The first inequality in function fields is actually an immediate consequence of the Riemann–Roch Theorem and of the fact that the constant field is finite. The only other facts used in its proof are the elementary properties of the index $h_{2/1}$, and some elementary local properties.

We see therefore that the existence of a divisor of degree 1 is fairly shallow. The existence of a prime of degree 1 is a much more serious question. Such primes do not always exist, and conditions under which they exist are obtained from the Riemann Hypothesis in function fields.

We see that in function fields, the value group of the idèle classes is the integers, and is generated by an idèle class having volume exactly q. Denoting such an idèle class by a_1, so that $|a_1| = q$, we see that we can write any idèle class a uniquely in the form $a = a_1^{\nu} \cdot a_0$ where a_0 is an idèle class of absolute value 1. We may therefore assign ordinals to idèle classes, just as we did in the local theory, and we shall see later that from the point of view of class field theory, the group of idèle classes behaves in exactly the same fashion as does the local group.

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CHAPTER VI

Second Fundamental Inequality

1. Statement and Consequences of the Inequality

Let k be a global field, of characteristic p ($= 0$ or > 0). K/k denotes a normal extension of degree n with group G. We abbreviate $N_{K/k}$ by N and $N_{K_{\mathfrak{P}}/k_{\mathfrak{p}}}$ by $N_{\mathfrak{P}}$.

The proof of the first inequality showed that if K/k is cyclic,

$$h_2(C_K) = (C_k : NC_K) = h_1(C_K) \cdot n.$$

In this chapter, we shall prove the

Second Fundamental Inequality. *For any normal extension K/k the norm index $(C_k : NC_K)$ divides the degree $[K : k]$.*

Coupling the two statements together, we obtain immediately $h_1(C_K) = 1$ for all cyclic layers of the formation of idèle classes. In Chapter XIV, Section 2, Lemma 1 it is shown that the triviality of $\mathcal{H}^1$ in all cyclic layers implies the triviality of $\mathcal{H}^1$ in all layers. Consequently we get

Theorem 1. *Let k be a global field and let K/k be a normal extension with group G. Then $\mathcal{H}^1(G, C_K)$ is trivial. The formation of idèle classes of the algebraic closure of k is a field formation.*

If we consider the exact sequence

$$\mathcal{H}^1(J_K/K^*) \to \mathcal{H}^2(K^*) \to \mathcal{H}^2(J_K)$$

we notice that the natural imbedding of $\mathcal{H}^2(K^*)$ into $\mathcal{H}^2(J_K)$ must be injective. We have already discussed the structure of $\mathcal{H}^2(J_K)$ on page 2. It is a weak direct product of the local groups $\mathcal{H}^2(G_{\mathfrak{p}}, K^*_{\mathfrak{p}})$. A cocycle class $c \in \mathcal{H}^2(J_K)$ is completely determined by its local components $c_{\mathfrak{p}}$. We obtain therefore

Theorem 2. *Let K/k be a normal extension. Let $c \in \mathcal{H}^2(G, K^*)$. Then $c = 1$ if and only if $c_{\mathfrak{p}} = 1$ for all primes $\mathfrak{p}$. In other words, a 2-cocycle of K^* splits in K if and only if it splits everywhere locally.*

Let $c \in \mathcal{H}^2(J_K)$. According to the local class field theory, each local component $c_{\mathfrak{p}}$ of c can be assigned canonically an invariant $\operatorname{inv}_{\mathfrak{p}} c$. We shall prove later (Reciprocity Law) that the cocycle classes of K^* are characterized as those cocycle classes $c \in \mathcal{H}^2(J_K)$ having $\sum_{\mathfrak{p}} \operatorname{inv}_{\mathfrak{p}} c = 0$.

In order to prove the second inequality, we have to distinguish two cases depending on the characteristic of k. These will be handled respectively in Sections 3 and 4. The Kummer Theory of abelian extension will be used in our proofs and is given in Section 2. In Section 5 we point out that the infinite divisibility of the group of universal norms in idèle classes is an immediate consequence of the procedure used to prove the second inequality.

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We shall now perform certain trivial reduction steps showing that it suffices to prove the inequality in cyclic fields of prime degree ℓ. If $\ell \neq p$, we may even assume that k contains the ℓ-th roots of unity.

LEMMA. *If E/k is any finite extension of degree m, then $(C_k : N_{E/k}C_E)$ is finite and divides a power of m.*

PROOF. To prove the norm index finite it suffices to prove it for normal extensions (because if $K \supset E$ and K/k is normal, $NC_K \subset N_{E/k}C_E$). In the later case, let S be a finite set of primes including all archimedean primes, all ramified primes, and enough primes such that[1]

$$J_k = k^* J_k^S, \qquad J_K = K^* J_K^S.$$

Then $k^* N J_K = k^* \cdot N K^* \cdot N J_K^S = k^* N J_K^S$ whence

$$(J_k : k^* N J_K) = (k^* J_k^S : k^* N J_K^S) \leqslant (J_k^S : N J_K^S) \leqslant \prod_{\mathfrak{p} \in S} n_{\mathfrak{p}}$$

by an argument similar to that used in the first inequality (V, 3). This proves the finiteness. □

The norm index divides a power of the degree because for any $a \in C_k$, $a^m \in N_{E/k}C_E$.

LEMMA. *Let $E \supset F \supset k$ be two finite extensions. Then:*

1. *$(C_k : N_{F/k}(C_F))$ divides $(C_k : N_{E/k}(C_E))$.*
2. *$(C_k : N_{E/k}(C_E))$ divides $(C_k : N_{F/k}(C_F)) \cdot (C_F : N_{E/F}(C_E))$.*

Consequently if the inequality holds in the steps of a tower, it holds in the tower itself.

PROOF. We note that

$$(*) \qquad (C_k : N_{E/k}(C_E)) = (C_k : N_{F/k}(C_F))\big(N_{F/k}(C_F) : N_{F/k}(N_{E/F}(C_E))\big).$$

This already proves 1. The map $C_F \to N_{F/k}(C_F)$ is a homomorphism so that the second factor of the right side of $(*)$ divides $(C_F : N_{E/F}(C_E))$. This proves 2. □

From these lemmas we obtain

REDUCTION 1. *If the second inequality holds in all cyclic extensions of prime degree, then it holds in all normal extensions.*

PROOF. Let K/k be normal and let ℓ be a prime. Let E be the fixed field of an ℓ-Sylow subgroup of the Galois group G. K/E is a tower of cyclic fields of degree ℓ and by the lemma we may assume that the inequality holds in K/E:

$(C_E : N_{K/E}(C_K))$ divides $[K : E]$. On the other hand, $(C_k : N_{E/k}(C_E))$ divides a power of $[E : k]$ and is therefore prime to ℓ. From the fact that $(C_k : N_{K/k}(C_K))$ divides $(C_k : N_{E/k}(C_E)) \cdot (C_E : N_{K/E}(C_K))$ it follows now that for each prime ℓ the ℓ-contribution to $(C_k : N_{K/k}(C_K))$ divides $[K : E]$ and consequently $[K : k]$. The inequality for K/k follows. □

REDUCTION 2. *If $\ell \neq p$ it suffices to prove the second inequality for cyclic fields of prime degree over fields k which contain a primitive ℓ-th root of unity ζ.*

[1] Often in this chapter we write J_k^S instead of $(J_k)_S$.

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PROOF. If K/k is cyclic of degree ℓ then the norm index of K/k divides that of $K(\zeta)/k$ which in turn divides the product of the one of $k(\zeta)/k$ and the one of $K(\zeta)/k(\zeta)$. The norm index of $k(\zeta)/k$ is prime to ℓ since the degree is prime to ℓ. The norm index of K/k is a power of ℓ and divides therefore the norm index of $K(\zeta)/k(\zeta)$, a cyclic extension of prime degree ℓ of a field containing ζ. $\square$

2. Kummer Theory

Let k be any field with any characteristic p, K/k an abelian extension of k (finite or infinite) and G its Galois group with the Krull topology. We shall consider only cases in which G is of finite exponent n, meaning by this that $\sigma^n = 1$ for all $\sigma \in G$.

We shall give an algebraic characterization of these extension fields K/k in two certain special cases.

a. $p \nmid n$ and k contains the primitive n-th roots of unity. The n-th roots of unity form a *multiplicative* group of order n in k and we use it as value group for the characters of G.

If χ is a character, its value is in k, so that it is invariant under the action of G. This implies $\chi(\sigma\tau) = \chi(\sigma)\chi(\tau) = \chi(\sigma)(\chi(\tau))^\sigma$ and shows that the function $\chi(\sigma)$ is a continuous 1-cocycle of (G, K^*). $\mathcal{H}^1(G, K^*)$ is trivial. Consequently there exists an $\alpha \in K^*$ such that $\chi(\sigma) = \alpha^{1-\sigma}$. This α is not arbitrary, since $(\alpha^n)^{1-\sigma} = (\chi(\sigma))^n = 1$ for all $\sigma \in G$, so that α^n is in k.

This suggests that we introduce the discrete multiplicative group A of all $\alpha \in K^*$ such that $\alpha^n \in k$. We have then $\chi(\sigma) = \alpha^{1-\sigma}$ with $\alpha \in A$. If we form conversely with any $\alpha \in A$ the function $\chi(\sigma) = \alpha^{1-\sigma}$ then this function is continuous on G since its value is 1 on the subgroup of G that has the field $k(\alpha)$ as fixed field. It satisfies $(\chi(\sigma))^n = (\alpha^n)^{1-\sigma} = 1$. $\chi(\sigma)$ is therefore an n-th root of unity in k and consequently invariant under the action of G. Finally $\chi(\sigma)\chi(\tau) = \chi(\sigma)(\chi(\tau))^\sigma = \alpha^{1-\sigma+\sigma-\sigma\tau} = \alpha^{1-\sigma\tau} = \chi(\sigma\tau)$, in other words, $\chi(\sigma)$ is a character of G.

We introduce now the symbol:

$$(\alpha, \sigma) = \alpha^{1-\sigma}$$

and see that it defines a pairing of the groups A and G into the roots of unity of k. The kernel of A in this pairing, the set of all $\alpha \in A$ with $\alpha^{1-\sigma} = 1$ for all $\sigma \in G$, is k ($k \subset A$ is trivial). In order to find the kernel of G let σ_0 be an element $\neq 1$. By the Duality Theorem, there exists a character χ such that $\chi(\sigma_0) \neq 1$. We have shown above that $\chi(\sigma) = \alpha^{1-\sigma}$ for some α. This shows that σ_0 is not in the kernel, in other words that the kernel of G is 1.

Let now H be a closed subgroup of G and $H^\perp$ the orthogonal group to H under our pairing. Then $k^* \subset H^\perp \subset A$ and any subgroup of A that contains k will come from precisely one closed subgroup H of G namely its orthogonal group. We adjoin the elements of H^* to k and obtain the subfield $k(H^\perp)$ of K. Which subgroup of G determines this subfield? It consists of those elements σ of G which leave every element of $H^\perp$ invariant. The translation of this statement into the language of the pairing shows that the group in question is orthogonal to $H^\perp$. According to the duality theory it is therefore H. We have therefore a 1–1 correspondence between the subfields of K/k and the subgroups of A that contain k^*.

The connection between the structure of G and that of A is given by

$$A/k^* \simeq \widehat{G}$$

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and the map defined by our pairing.

The group A can now be mapped onto a subgroup of k by raising each element into the n-th power. The kernel of the map $\alpha \to \alpha^n$ consists of the n-th roots of unity and is therefore contained in the previous kernel k. All our statements can therefore be compounded with this further map. We obtain $A^n/k^n \simeq \widehat{G}$, and have a 1–1 correspondence between the subfields of K/k and the subgroups of A^n of k^* which contain k^{*n}.

We must now do these steps in the reverse order. Suppose we start with an arbitrary subgroup Δ of k^* which contains k^{*n} (this Δ is to play the role of the final group A^n). We extract all n-th roots of elements of Δ and obtain a group $\Delta^{1/n}$ in the algebraic closure of k. ($\Delta^{1/n}$ is to play the role of A. It clearly contains k.) Construct the field $K = k(\Delta^{1/n})$. It is a normal extension because it is a splitting field of the separable polynomials $x^n - \delta$ of all $\delta \in \Delta$. Let G be its group. We contend that G is abelian. It suffices to show that the action of G on each generator $\alpha \in \Delta^{1/n}$ of the field K/k is commutative. We have $\alpha^n \in k$ or $(\alpha^n)^\sigma = \alpha^n$ for each $\sigma \in G$. This shows that $\alpha^\sigma = \zeta_\sigma \alpha$ where ζ_σ is an n-th root of unity. If τ in G, then τ leaves ζ_σ fixed and we obtain $(\alpha^\sigma)^\tau = \zeta_\sigma \zeta_\tau \alpha$, an equation, which shows the required commutativity. We also see $\sigma^n = 1$ which shows that the field K/k is of our previous type. We determine now the group A in this field K. Clearly $A \supset \Delta^{1/n} \supset k^*$. Since subgroups of A are in 1–1 correspondence with subfields of K and since $k(\Delta^{1/n}) = K$ it follows, that actually $A = \Delta^{1/n}$. We summarize our main results in:

THEOREM 3. *Let k be a field and suppose that the primitive n-th roots of unity lie in k, $p \nmid n$. There exists a 1–1 correspondence between subgroups Δ of k^* which contain k^{*n} and abelian extensions K/k with groups G of exponent n. The correspondence is given by*

$$\Delta \leftrightarrow K_\Delta = k(\Delta^{1/n}).$$

The character group $\widehat{G}$ is isomorphic to the factor group Δ/k^{*n}.

The following corollaries are obvious:

COROLLARY 1. *If Δ_1, Δ_2 are two subgroups of k^* containing k^{*n} then we have the correspondences*

$$\Delta_1\Delta_2 \leftrightarrow K_{\Delta_1} \cdot K_{\Delta_2}$$
$$\Delta_1 \cap \Delta_2 \leftrightarrow K_{\Delta_1} \cap K_{\Delta_2}.$$

COROLLARY 2. *If K/k is finite then $G \simeq \widehat{G}$ and $[K : k] = (\Delta : k^{*n})$.*

In all our discussion, we have taken $\Delta \supset k^{*n}$. We now remark that if Δ is a subgroup of k^* which does not contain k^{*n}, we can always form the composite group $\Delta \cdot k^{*n}$. Then the Kummer group of the field $K = k(\Delta^{1/n})$ is $\Delta \cdot k^{*n}$. The factor group $\Delta \cdot k^{*n}/k^{*n}$ is isomorphic to $\Delta/(\Delta \cap k^{*n})$ and in case K/k is finite, we get

$$[K : k] = (\Delta : (\Delta \cap k^{*n})).$$

COROLLARY 3. *K/k is cyclic if and only if Δ/k^* is cyclic. If δ generates Δ/k^{*n} then $K = k(\sqrt[n]{\delta})$.*

For global fields these statements have to be supplemented by arithmetical statements:

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THEOREM 4. *Let k be a global field containing the primitive n-th roots of unity, $p \nmid n$. Let $K = k(\Delta^{1/n})$ be a finite Kummer extension and $\mathfrak{p}$ a prime of k.*

1. *$\mathfrak{p}$ splits completely in K if and only if $\Delta \subset k_{\mathfrak{p}}^{*n}$.*

2. *Suppose that $\mathfrak{p}$ is finite and that $\mathfrak{p} \nmid n$. $\mathfrak{p}$ is unramified in K if and only if there exists a set of generators of Δ/k^{*n} which are units at $\mathfrak{p}$. In other words: $\Delta \subset U_{\mathfrak{p}} \cdot k^{*n}$.*

PROOF. 1. The complete splitting of $\mathfrak{p}$ means that the completion of K at a prime dividing $\mathfrak{p}$ is $k_{\mathfrak{p}}$. Since this completion is $k_{\mathfrak{p}}(\Delta^{1/n})$ the contention is obvious.

2. Let $\mathfrak{P} \mid \mathfrak{p}$ in k. $K_{\mathfrak{P}}/k_{\mathfrak{p}}$ is unramified if the value group of the valuation of $K_{\mathfrak{P}}$ is the same as that of $k_{\mathfrak{p}}$. Let δ be one of the generators. Since $\delta = (\delta^{1/n})^n$ in $K_{\mathfrak{P}}$, its value is an n-th power of a value. The same is therefore true in $k_{\mathfrak{p}}$ and this means that the ordinal of δ is divisible by $n : \mu = rn$. If π is an element of k with ordinal 1 at $\mathfrak{p}$ then $\delta(\pi^{-r})^n$ is a unit at $\mathfrak{p}$ and can be used as generator instead of δ.

3. Assume conversely that Δ/k^{*n} has generators δ which are units at $\mathfrak{p}$. It will suffice to show that each field $k(\sqrt[n]{\delta})$ is unramified. $\alpha = \sqrt[n]{\delta}$ satisfies the equation $f(x) = x^n - \delta = 0$ and $f'(\alpha) = n\alpha^{n-1}$ is prime to $\mathfrak{p}$. By an elementary local criterion $\mathfrak{p}$ is unramified. $\square$

b. $n = p$, the characteristic of k. The elements of the prime field of k form an *additive* cyclic group of order $n = p$ and we use it as value group for the characters of G. The theory goes over with very little change, one has only to be careful to replace the multiplicative group A by an additive one.

This time a character satisfies $\chi(\sigma\tau) = \chi(\sigma) + \chi(\tau) = \chi(\sigma) + \sigma\chi(\tau)$ and is an additive 1-cocycle of (G, K). The first cohomology group in the additive setup is trivial again and one obtains $\chi(\sigma) = (1 - \sigma)\alpha$ for some α in K. This α is not arbitrary. To find the group A to which it belongs we introduce the additive operator $\wp$ of K by defining:

$$\wp(\beta) = \beta^p - \beta.$$

If β is in the kernel of this operator then $\beta^p - \beta = 0$ and this equation has as solution exactly the elements of the prime field. The operator $\wp$ commutes with the elements of G. We apply now to the equation $\chi(a) = (1 - \sigma)\alpha$ the operator $\wp$. The left side becomes 0 since $\chi(\sigma)$ is in the prime field and the right side shows now that $\wp(\alpha) \in k$. We define therefore A to be the *additive* subgroup of those elements $\alpha \in K$ for which $\wp(\alpha) \in k$. If we form conversely with such an element $\alpha \in A$ the function $\chi(\sigma) = (\sigma - 1)\alpha$ and apply $\wp$ to it, then the right side becomes 0 and shows that $\chi(\sigma)$ is in the prime field (and consequently left fixed by G). Its continuity together with

$$\chi(\sigma) + \chi(\tau) = \chi(\sigma) + \sigma\chi(\tau) = (\sigma - 1)\alpha + (\sigma\tau - \sigma)\alpha = (\sigma\tau - 1)\alpha = \chi(\sigma\tau)$$

shows that this function is a character of G.

The symbol $(\alpha, \sigma) = (\sigma - 1)\alpha$ defines now a pairing of the additive group A and the multiplicative group G into the additive group of the prime field. The kernel of A is again k and we prove as before that the kernel of G is 1. We find as before a 1–1 correspondence between the subfields of K and the (additive) subgroups of A which contain k.

The mapping of A into the ground field k is done by the operator $\wp$. We obtain $\wp A/\wp k \simeq \widetilde{G}$, and have a 1–1 correspondence between the subfields of K/k and the additive subgroups of B which contain $\wp k$ given by $B = \wp A$.

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If we start with an additive group Δ of k which contains $\wp k$, we form in the algebraic closure of k the group $\frac{1}{\wp}\Delta$ where the symbol $\frac{1}{\wp}(\alpha)$ means all solutions of $\wp(x) = x^p - x = \alpha$. We form the field $K = k\big(\frac{1}{\wp}\Delta\big)$ and arrive by completely similar arguments at the theorem:

THEOREM 5. *Let k be a field of characteristic $p > 0$. There is a 1–1 correspondence between the additive subgroups Δ of k which contain $\wp k$, and abelian extensions K/k with groups of exponent p. The correspondence is given by*

$$\Delta \to K_\Delta = k\left(\frac{1}{\wp}\Delta\right).$$

3. Proof in Kummer Fields of Prime Degree

Let k be a global field, of characteristic p ($= 0$ or > 0). Let n be an integer, $p \nmid n$. ζ_n denotes a primitive n-th root of unity. We assume ζ_n lies in k.

We shall introduce auxiliary groups of idèles in the following way. Let S be a finite set of primes containing at least all archimedean primes and divisors of n. This set S is split up into two *disjoint* sets S_1 and S_2, one of which may be empty: $S = S_1 \cup S_2$. Let i stand for one of the two subscripts 1, 2 and call j the other. We define the group D_i to be the group of all idèles $\mathfrak{a}$ such that

1. $\mathfrak{a}$ is an n-th power at all $\mathfrak{p} \in S_j$,
2. $\mathfrak{a}$ is a unit outside S,

whereas no condition is put on the primes of S_i. We see that D_i can be written naturally as a product:

$$D_i = \prod_{\mathfrak{p}\in S_j} k_{\mathfrak{p}}^{*n} \times \prod_{\mathfrak{p}\in S_i} k_{\mathfrak{p}}^{*} \times \prod_{\mathfrak{p}\notin S} U_{\mathfrak{p}}.$$

An element $\mathfrak{a} \in D_i$ can be written $\mathfrak{a} = \mathfrak{b}^n \cdot \mathfrak{c}$ with $\mathfrak{b} \in \prod_{\mathfrak{p}\in S_j} k_{\mathfrak{p}}^*$, and $\mathfrak{c} \in \prod_{\mathfrak{p}\in S_i} k_{\mathfrak{p}}^* \times \prod_{\mathfrak{p}\notin S} U_{\mathfrak{p}}$.

We put $\Delta_i = D_i \cap k$. It is clear that $k_S^* \supset \Delta_i \supset k_S^{*n}$ and we know from the Unit Theorem that k_S^*/k_S^{*n} is finite. In fact k_S^* is a free abelian group on $s-1$ generators if S consists of s primes, modulo the roots of unity. Since k contains ζ_n, the roots of unity form a cyclic group of an order divisible by n and this shows

$$(k_S : k_S^{*n}) = n^s. \tag{1}$$

The extension $K_i = k(\Delta_i^{1/n})$ is a finite Kummer extension, belonging to $\Delta_i k^{*n}$. We obviously have $\Delta_i \cap k^{*n} = k_S^{*n}$. From this we get for the degree

$$[K_i : k] = (\Delta_i k^{*n} : k^{*n}) = (\Delta_i : k_S^{*n}),$$

by Theorem 3, Corollary 2. In addition, from Theorem 4 we obtain a crude description of the splitting of primes in K_i as follows:

If $\mathfrak{p} \notin S$, then $\mathfrak{p}$ is unramified in K_i.

If $\mathfrak{p} \in S_j$, then $\mathfrak{p}$ splits completely in K_i.

In a similar way we define K_j. We denote the idèles of K_i and K_j by J_i resp. J_j and norms from these fields to k by N_i resp. N_j.

The proof of the second inequality is contained in the following lemmas.

LEMMA 1. *Let K/k be a finite abelian extension of exponent n (i.e. $\sigma^n = 1$ for all $\sigma \in G$). Let $\mathfrak{a} \in J_k$. Then $\mathfrak{a}^n \in k^* N_{K/k} J_K$.*

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PROOF. We first note that even though G has exponent n, it does not mean that G is of order n. If G has order n, then the lemma is of course trivial.

If we wish to use local class field theory, we can even prove that $\mathfrak{a}^n \in NJ_K$. Indeed, $\mathfrak{a}_\mathfrak{p}^n \in N_\mathfrak{p} K_\mathfrak{p}^*$, so the lemma is obvious. We can give an elementary proof, however. Let $\alpha \in k^*$ be such that $\alpha\mathfrak{a}$ is close to 1 at all ramified primes. Then $(\alpha\mathfrak{a})^n$ is norm at these primes.[2] At the other primes, the local extension is cyclic, and therefore has a degree dividing n. Hence $(\alpha\mathfrak{a})^n$ is a norm at the unramified primes also. This proves the lemma. □

The next lemma is fundamental, and shows how the Kummer theory of the two fields K_i intermingles with their class field theory.

LEMMA 2. *Let k contain the n-th roots of unity, $p \nmid n$. Let S be a finite set of primes including*

- *all $\mathfrak{p} \mid n$ and all archimedean primes,*
- *enough primes so that $J_k = k^* J_k^S$.*

Let $S = S_1 \cup S_2$ as above. Let $K_i = k(\Delta_i^{1/n})$. Then

1. $k^* D_i \subset k^* N_j J_j$.
2. $(J_k : k^* D_1)(J_k : k^* D_2) = [K_1 : k][K_2 : k]$

and consequently

$$(J_k : k^* N_1 J_1)(J_k : k^* N_2 J_2) \leqslant [K_1 : k][K_2 : k].$$

PROOF. We begin with the first statement. Let $\mathfrak{a} \in D_i$. Write $\mathfrak{a} = \mathfrak{b}^n \cdot \mathfrak{c}$ as above. Then $\mathfrak{b}^n \in k^* N_j J_j$ by Lemma 1. $\mathfrak{c}$ has component 1 at all $\mathfrak{p} \in S_j$. Furthermore, if $\mathfrak{p} \in S_i$ then $\mathfrak{p}$ splits completely in K_j, and hence $\mathfrak{c}_\mathfrak{p}$ is a local norm. If $\mathfrak{p} \notin S$, $\mathfrak{p}$ is unramified, $\mathfrak{c}_\mathfrak{p}$ is a unit and consequently a local norm. Thus $\mathfrak{c}_\mathfrak{p}$ is local norm at all primes, and hence $\mathfrak{c} \in N_j J_j$. This proves that $\mathfrak{a} \in k^* N_j J_j$.

The second statement will follow from a brief index computation. We first remind the reader of the general rule: $(A : B) = (AC : BC)\big((A \cap C) : (B \cap C)\big)$. We have therefore

$$\begin{aligned}
(J_k : k^* D_i) &= (k^* \cdot J_k^S : k^* \cdot D_i) \\
&= \frac{(J_k^S : D_i)}{\big((J_k^S \cap k^*) : (D_i \cap k^*)\big)} \\
&= \frac{(J_k^S : D_i)}{(k_S^* : \Delta_i)} \\
&= \frac{\prod_{\mathfrak{p} \in S_j} (k_\mathfrak{p}^* : k_\mathfrak{p}^{*n})}{(k_S^* : k_S^{*n})} (\Delta_i : k_S^{*n}) \\
&= \frac{\prod_{\mathfrak{p} \in S_j} (k_\mathfrak{p}^* : k_\mathfrak{p}^{*n})}{n^s} [K_i : k].
\end{aligned}$$

We form now the analogous expression for $(J_k : kD_j)$ and multiply the two formulas. This yields

$$(J_k : k^* D_1)(J_k : k^* D_2) = \frac{\prod_{\mathfrak{p} \in S} (k_\mathfrak{p}^* : k_\mathfrak{p}^{*n})}{n^{2s}} [K_1 : k][K_2 : k].$$

[2]See Ch. 11, Sections 4 and 5, for the details of just how close to 1 $(\alpha\mathfrak{a})^n$ needs to be. But the fact that $N_\mathfrak{p} K_\mathfrak{p} * n$ is open is more elementary. For example, it contains $U_\mathfrak{p}^{[K_\mathfrak{p} : k_\mathfrak{p}]}$ which is compact of finite index in $U_\mathfrak{p}$, hence open.

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In the remarks on pages 6 and 7 we computed the local index

$$(k_{\mathfrak{p}} : k_{\mathfrak{p}}^n) = n^2/|n|_{\mathfrak{p}}.$$

Since $|n|_{\mathfrak{p}} = 1$ for all $\mathfrak{p} \notin S$ we get from the product formula

$$\prod_{\mathfrak{p}\in S} |n|_{\mathfrak{p}} = \prod_{\mathfrak{p}} |n|_{\mathfrak{p}} = 1.$$

and consequently

$$\prod_{\mathfrak{p}\in S} (k_{\mathfrak{p}}^* : k_{\mathfrak{p}}^{*n}) = n^{2s}$$

which leads to the desired formula

$$(J_k : k^* D_1)(J_k : k^* D_2) = [K_1 : k][K_2 : k].$$

The second inequality will now be an immediate consequence of the next lemma. Note how the Kummer theory relates K_i with D_i whereas the class field theory relates K_i with D_j. □

Lemma 3. *Let n be a prime. Let $k(\sqrt[n]{\alpha}) = K$ be any given Kummer extension of degree n. Then there exist two disjoint sets of primes S_1 and S_2 whose union S satisfies the condition of Lemma 2, for which $J_k = kD_1$, and for which the associated field K_1 happens to be K.*

Before proving this lemma we indicate right away how the second inequality follows from it. Since $J_k = k^* D_1 \subset k^* \cdot N_2 J_2$ we have $(J_k : k^* \cdot N_2 J_2) = 1$. By the first inequality this implies that $K_2 = k$. The last line of Lemma 2 reads now $(J_k : k^* N_1 J_1) \leqslant [K_1 : k]$ and this is the second inequality since $K = K_1$. (The divisibility follows from the first lemma of this chapter.)

Proof of Lemma 3. Let S_1 be a finite set of s_1 primes containing

- all $\mathfrak{p} \mid n$ and all archimedean primes,
- all $\mathfrak{p} \mid \alpha$ (which makes α an S_1-unit),
- enough primes so as to have $J_k = kJ_k^{S_1}$.

To simplify the notation we put $J = J_k$, and $J_{S_1} = J_k^{S_1}$.

$k_{S_1}/k_{S_1}^n$ is of type $(n, n, \ldots, n)$ since n is a prime. α is not an n-th power and may therefore be extended to a basis of $k_{S_1}/k_{S_1}^n$. Let $\alpha_1, \ldots, \alpha_{s_1}$ be such a basis, $\alpha = \alpha_1$.

Put $E_\mu = k(\sqrt[n]{\alpha_\mu})$. Then $K = E_1$. The full compositum of these fields has degree n^{s_1}, the compositum of all but one field n^{s_1-1}. This follows from the preceding section. Our fields satisfy therefore the condition of Theorem 4 of Chapter V so that there exist primes $\mathfrak{q}_i$ outside the set S_1 such that $\mathfrak{q}_i$ remains prime in the field E_i but splits completely in all the other fields E_μ. Of these primes we utilize only the primes $\mathfrak{q}_2, \mathfrak{q}_3, \ldots, \mathfrak{q}_{s_1}$ and form with them the set S_2 (so without the prime $\mathfrak{q}_1$). Then α_1 is an n-th power at all the primes of S_2 and α_i (for $i \neq 1$) is an n-th power at any $\mathfrak{q}_j \neq \mathfrak{q}_i$ but not an n-th power at $\mathfrak{q}_i$.

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We look now at the structure of the groups J_{S_1}, D_1 and $J_{S_1} \cap D_1$:

$$J_{S_1} = \prod_{\mathfrak{p} \in S_1} k_{\mathfrak{p}}^* \times \prod_{i=2}^{s_1} U_{\mathfrak{q}_i} \times \prod_{\mathfrak{p} \notin S} U_{\mathfrak{p}},$$

$$D_1 = \prod_{\mathfrak{p} \in S_1} k_{\mathfrak{p}}^* \times \prod_{i=2}^{s_1} k^{*n}_{\mathfrak{q}_i} \times \prod_{\mathfrak{p} \notin S} U_{\mathfrak{p}},$$

$$J_{S_1} \cap D_1 = \prod_{\mathfrak{p} \in S_1} k_{\mathfrak{p}}^* \times \prod_{i=2}^{s_1} U^n_{\mathfrak{q}_i} \times \prod_{\mathfrak{p} \notin S} U_{\mathfrak{p}}.$$

The local index $(U_{\mathfrak{p}} : U^n_{\mathfrak{p}})$ has been computed on p. 7. Its value is $n/|n|_{\mathfrak{p}}$. For a prime outside S_1 this is simply n since all $\mathfrak{p} \mid n$ are in S_1. n is a prime and so the factor groups $U_{\mathfrak{q}_i}/U^n_{\mathfrak{q}_i}$ are cyclic of order n. If we can find idèles $\mathfrak{a}_i \in J_{S_1}$ which are n-th powers at each $\mathfrak{a}_j \neq \mathfrak{a}_i$ and not an n-th power at $\mathfrak{a}_i$ then these idèles $\mathfrak{a}_i$ will generate the factor group $J_{S_1}/J_{S_1} \cap D_1$. The peculiar properties of $\alpha_2, \ldots, \alpha_{s_1}$ show that these principal idèles (they are in k_{S_1}) will serve. Since they are principal idèles we see that $J_{S_1} \subset k^* \cdot (J_{S_1} \cap D_1) \subset k^* D_1$ and consequently $J = k^* J_{S_1} \subset k^* D_1$ which proves the first part of our lemma.

We have still to show that $K = K_1 = k(\Delta_1^{1/n})$. We remark to this effect that the group D_1 (see the display of its structure above) is obviously a subgroup of $J_{S_1} \cdot J^n = J_{S_1}(k^* J_{S_1})^n = k^{*n} J_{S_1}$. A generator δ of the Kummer group $\Delta_1 k^{*n}$ mod k^{*n} of the field K_1 can therefore be assumed in J_{S_1}, i.e. in k_{S_1}. It has then the form

$$\delta = \alpha_1^{\nu_1} \alpha_2^{\nu_2} \ldots \alpha_{s_1}^{\nu_{s_1}}.$$

But δ has to lie in $\Delta_1 k^n$ which implies that δ must be an n-th power at every prime $\mathfrak{q}_i \in S_2$ $(i \neq 1)$. At $\mathfrak{q}_i$ every α_j with $j \neq i$ is an n-th power and α_i is not. This shows that $\nu_2, \nu_3, \ldots, \nu_{s_1}$ are divisible by n. Since n-th powers can be absorbed into k^n we may now assume $\delta = \alpha_1^{\nu_1}$. Such a δ is indeed in Δ_1 since it is an n-th power at every prime of S_2 and a unit outside S. Since $\alpha_1 = \alpha$ this proves the contention. □

4. Proof in p-extensions

a. A lemma on derivations. In the course of the proof a certain lemma on derivations will be needed and we prove it separately because it is of independent interest.

By a derivation D of a field E one means an additive map of E into itself which satisfies the usual rule

$$D(xy) = xD(y) + yD(x).$$

Putting $x = y = 1$ one obtains $D(1) = 0$. Letting now $y = x^{-1}$ one sees

$$D(x^{-1}) = x^{-2} D(x).$$

The additive group of all $x \in E$ for which $D^n(x) = 0$ shall be denoted by P_n. ($D^0 =$ identity, hence $P_0 = \{0\}$.)

As usual, we denote the additive operator that multiplies the elements of E by a given element y of E also by y. A confusion between y as element and y as operator is avoided by the use of parenthesis: Dy means the operator product y followed by D whereas $D(y)$ means the effect D has on the element y.

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We say that an element $x \in E$ is a logarithmic derivative in E if there exists an element $y \in E^*$ such that $x = \frac{D(y)}{y}$. We say then that x is the logarithmic derivative of y.

LEMMA 4. *An element x of E is the logarithmic derivative of an element $y \in P_n$ which is not in P_{n-1} $(n > 0)$ if and only if the n-th power of the operator $D + x$ applied to 1 is 0 and the $(n-1)$-st power applied to 1 is different from 0: $(D+x)^n(1) = 0$, $(D+x)^{n-1}(1) \neq 0$.*

PROOF. Let $x = \frac{D(y)}{y}$ for some $y \in E^*$. For all $z \in E$ we have $(D+x)(z) = D(z) + \frac{D(y)}{y} z = y^{-1}D(yz) = y^{-1}Dy(z)$. This means in operators that $D + x = y^{-1}Dy$ and consequently for any $n \geqslant 0$ that

$$(D+x)^n = y^{-1}D^n y.$$

Assume now in addition that $y \in P_n$ and $y \notin P_{n-1}$. Then $(D+x)^n(1) = y^{-1}D^n y(1) = y^{-1}D^n(y) = 0$ and similarly $(D+x)^{n-1}(1) \neq 0$. Assume conversely that $(D+x)^n(1) = 0$ and $(D+x)^{n-1}(1) \neq 0$. Put

$$(D+x)^{n-1}(1) = y^{-1}.$$

Then $(D+x)(y^{-1}) = 0 = -y^{-2}D(y) + xy^{-1}$ and consequently $x = \frac{D(y)}{y}$. Therefore $D + x = y^{-1}Dy$, $(D+x)^{n-1}(1) = y^{-1}D^{n-1}(y) \neq 0$ and similarly $y^{-1}D^n(y) = 0$. Hence $y \in P_n$ but $y \notin P_{n-1}$. □

Frequently the derivation D has the additional property that for every x in E we have $D^n(x) = 0$ for some n or in other words that E is the union of all P_n. Then our lemma shows that an element $x \in E$ is a logarithmic derivative in E if and only if $(D+x)^n(1) = 0$ for some n. If x is now an element of a subfield F of E that is stable under D, $(D(F) \subset F)$, and if x is an element of F, then x is a logarithmic derivative in F if and only if it is a logarithmic derivative. in E.

This situation occurs in the following case: Let k be a field of characteristic $p > 0$ and $E = k\{t\}$ the field of formal power series in t with coefficients in k. If D is the ordinary differentiation with respect to t then $D^p = 0$. We obtain therefore

COROLLARY. *Let $E = k((t))$ be a power series field of characteristic $p > 0$ and F a subfield stable under the ordinary derivation $D = \frac{d}{dt}$. An element $x \in F$ is logarithmic derivative in F if and only if it is a logarithmic derivative in E.*

b. A pairing connected with function fields. Notations:

k a global function field of characteristic $p > 0$.

k_0 its constant field, $\mathbb{Z}_p$ the prime field.

$k_{\mathfrak{p}}$ the completion of k at a prime $\mathfrak{p}$ and also the additive group of $k_{\mathfrak{p}}$.

$k_{\mathfrak{p}}^*$ the multiplicative group of $k_{\mathfrak{p}}$.

k_1 the residue class field of $k_{\mathfrak{p}}$. k_1 contains naturally k_0, and $k_{\mathfrak{p}}$ may be viewed naturally as the power series field $k_1((t))$, where t is a local uniformizing parameter, i.e. an element of ordinal 1. If necessary, we can select t from k and even assume that it is separating, i.e. that k is separable over $k_0(t)$. (Every $\mathfrak{p}$ is regular in the sense of [**2**, XVII, 4] because k_0 is perfect.)

S the trace from k_1 to $\mathbb{Z}_p$.

$\operatorname{res}(x\,dy)$ the residue of a local differential $x\,dy$ computed with any local uniformizing parameter (it does not depend on the choice of the parameter). The reader will find the necessary explanations and proofs in [**2**, X, 3].

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α. *A local pairing.* For x and y in $k_{\mathfrak{p}}$ we write

$$\int_{\mathfrak{p}} x\,dy = S(\operatorname{res} x\,dy). \tag{1}$$

If $x \in k_{\mathfrak{p}}$ and $y \in k_{\mathfrak{p}}^*$ we define

$$\varphi_{\mathfrak{p}}(x,y) = \int_{\mathfrak{p}} x\,\frac{dy}{y} = \int_{\mathfrak{p}} x\,\frac{\frac{dy}{dt}}{y}\,dt.$$

We find easily

$$\varphi_{\mathfrak{p}}(x+x',y) = \varphi_{\mathfrak{p}}(x,y) + \varphi_{\mathfrak{p}}(x',y),$$
$$\varphi_{\mathfrak{p}}(x,yy') = \varphi_{\mathfrak{p}}(x,y) + \varphi_{\mathfrak{p}}(x,y'),$$

and have therefore a pairing of the two groups $k_{\mathfrak{p}}$ and $k_{\mathfrak{p}}^*$ into the additive group $\mathbb{Z}_p$. If x is an integer of $k_{\mathfrak{p}}$ and if y is a unit in $k_{\mathfrak{p}}^*$ then $x\frac{dy}{y}$ has residue 0 and this shows the continuity of this pairing.

LEMMA 5. *Let x be an integer of $k_{\mathfrak{p}}$. Then $\varphi_{\mathfrak{p}}(x,y) = 0$ if and only if either $p \mid \operatorname{ord} y$ or $x = \wp z$ for some $z \in k_{\mathfrak{p}}$.*

PROOF. Let $x = a_0 + a_1 t + \dots$ with $a_i \in k_1$, and $y = t^n \varepsilon$ where ε is a unit of $k_{\mathfrak{p}}^*$. Then $\varphi_{\mathfrak{p}}(x,\varepsilon) = 0$ and consequently

$$\varphi_{\mathfrak{p}}(x,y) = n\int_{\mathfrak{p}} x\,\frac{dt}{t} = nS(a_0).$$

$\varphi_{\mathfrak{p}}(x,y) = 0$ is therefore equivalent with $p \mid n$ or $S(a_0) = 0$ and we have to show that $S(a_0) = 0$ is equivalent with $x = \wp z = z^p - z$.

Suppose first $x = z^p - z$. If the power series for z had poles then these poles would dominate in z^p and x could not be an integer of $k_{\mathfrak{p}}$. If b is the constant term of z then $a_0 = b^p - b$. Raising into the power p is the generating automorphism σ of the finite field $k_1/\mathbb{Z}_p$ so that we have $a_0 = (\sigma - 1)b$, an equation which is equivalent to $S(a_0) = 0$ because the additive Galois cohomology groups of a field are trivial.

If we have conversely $S(a_0) = 0$ then from the last remark $a_0 = (\sigma - 1)b = \wp b$ for a certain $b \in k_1$. Let $x_1 = a_1 t + a_2 t^2 + \dots$ and put $z_1 = -(x_1 + x_1^p + x_1^{p^2} + \dots)$. This series converges in the topology of $k_{\mathfrak{p}}$ and we get easily $x_1 = z_1^p - z_1 = \wp z_1$ and consequently

$$x = \wp(b + z_1). \qquad \square$$

β. *A global pairing.* The trace S in (1) can be broken up into the trace $T_{\mathfrak{p}}$ from the residue class field k_1 of $k_{\mathfrak{p}}$ to the constant field k_0, followed by the trace S' from k_0 down to the prime field $\mathbb{Z}_p$.

Let V be the additive group of valuation vectors of k (see [**2**, XIII, 1]), $\xi \in V$ with local components $\xi_{\mathfrak{p}}$. We select one of the primes and choose t separating in k and local uniformizing parameter for this prime. Put

$$\lambda(\xi) = \sum_{\mathfrak{p}} T_{\mathfrak{p}}(\operatorname{res}_{\mathfrak{p}}(\xi_{\mathfrak{p}}\,dt)).$$

$\lambda(\xi)$ takes values in k_0, is additive and k_0-linear and is not identically 0. If $|\xi_{\mathfrak{p}}|_{\mathfrak{p}}$ is sufficiently small at all primes where t has poles and if $\xi_{\mathfrak{p}}$ is a unit at all other primes, then $\lambda(\xi) = 0$, since each residue is 0. This shows that $\lambda(\xi)$ is continuous

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on V. From [**2**, XVII, 3] it follows that $\lambda(\xi)$ vanishes for all $\xi \in k$, i.e. $\lambda(\xi)$ is a differential of k. Put now

$$\oint \xi\, dt = S'(\lambda(\xi)) = \sum_{\mathfrak{p}} \int_{\mathfrak{p}} \xi_{\mathfrak{p}}\, dt.$$

LEMMA 6. *If ξ is a valuation vector such that $\oint \xi x\, dt = S'(\lambda(\xi x)) = 0$ for all $x \in k$ then ξ is in k.*

PROOF. We may replace x by αx for any $\alpha \in k_0$ and obtain $S'(\alpha \cdot \lambda(\xi x)) = 0$. If $\lambda(\xi x)$ were $\neq 0$ we would have $S'(k_0) = 0$ which is not true since $k_0/\mathbb{Z}_p$ is separable. Therefore $\lambda(\xi x) = 0$ for all $x \in k$. Theorems 5 and 9 of [**2**, XIII, 5] show now that $\xi \in k$. □

t is separating in k. From [**2**, XVII, 4] it follows that for almost all $\mathfrak{p}$ the derivatives $\frac{dt}{dt_{\mathfrak{p}}}$ are local units ($t_{\mathfrak{p}}$ means a local uniformizing parameter at the prime $\mathfrak{p}$).

If $\mathfrak{a} \in J_k$ is an idèle then its local components $\mathfrak{a}_{\mathfrak{p}}$ are in $k_{\mathfrak{p}}^*$. If we form the vector with the components

$$\frac{1}{\mathfrak{a}_{\mathfrak{p}}} \frac{d\mathfrak{a}_{\mathfrak{p}}}{dt} = \frac{1}{\mathfrak{a}_{\mathfrak{p}}} \frac{d\mathfrak{a}_{\mathfrak{p}}}{dt_{\mathfrak{p}}} \left(\frac{dt}{dt_{\mathfrak{p}}}\right)^{-1}$$

then almost all of its components are local integers, it is therefore a valuation vector that we shall denote by $\frac{1}{\mathfrak{a}} \frac{d\mathfrak{a}}{dt}$. We define now the following pairing between elements $x \in k$ and idèles $\mathfrak{a}$, into $\mathbb{Z}_p$:

$$\varphi(x, \mathfrak{a}) = \oint x \frac{1}{\mathfrak{a}} \frac{d\mathfrak{a}}{dt}\, dt = \sum_{\mathfrak{p}} \int_{\mathfrak{p}} x \frac{1}{\mathfrak{a}_{\mathfrak{p}}} \frac{d\mathfrak{a}_{\mathfrak{p}}}{dt}\, dt = \sum_{\mathfrak{p}} \int_{\mathfrak{p}} x \frac{d\mathfrak{a}_{\mathfrak{p}}}{\mathfrak{a}_{\mathfrak{p}}} = \sum_{\mathfrak{p}} \varphi_{\mathfrak{p}}(x, \mathfrak{a}_{\mathfrak{p}}). \tag{2}$$

The last expression shows that we have indeed a pairing. As for continuity, nothing has to be shown for the discrete k; if $\mathfrak{a}_{\mathfrak{p}}$ is very close to 1 then $\frac{d\mathfrak{a}_{\mathfrak{p}}}{dt_{\mathfrak{p}}}$ is very close to 0 and a look at the components of the local integrands shows the required continuity on J_k. The most interesting question, that of the kernels is answered by

LEMMA 7. *The kernel of J_k in our pairing* (2) *is precisely $k^* \cdot J_k^p$, that of k is the additive group $\wp k$, i.e. the elements of the form $z^p - z$ for $z \in k$.* (Note that every element of k has period p and that $J_k/k^* \cdot J_k^p$ is compact; indeed it is isomorphic to C_k/C_k^p. The duality theory may therefore be used.)

PROOF. 1. Suppose $\mathfrak{a}$ is in the kernel. The valuation vector $\xi = \frac{1}{\mathfrak{a}} \frac{d\mathfrak{a}}{dt}$ has then the property that $\oint \xi x\, dt = 0$ for all $x \in k$ and is therefore an element y of k. Taking the $\mathfrak{p}$ component of y at the prime $\mathfrak{p}$, which has the element $t \in k$ as local uniformizing parameter we obtain

$$y = \frac{1}{\mathfrak{a}_{\mathfrak{p}}} \frac{d\mathfrak{a}_{\mathfrak{p}}}{dt}.$$

y is therefore a logarithmic derivative in $k_{\mathfrak{p}}$. The field k is a subfield of $k_{\mathfrak{p}}$ and is stable under the differentiation since t is in k. y is also in this subfield and the corollary of Lemma 4 shows that y is a logarithmic derivative in k:

$$y = \frac{1}{z} \frac{dz}{dt}, \qquad z \in k.$$

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We obtain for our idèle: $\frac{1}{\mathfrak{a}}\frac{d\mathfrak{a}}{dt} = \frac{1}{z}\frac{dz}{dt}$. Put $\mathfrak{c} = z^{-1}\mathfrak{a}$; we find easily $\frac{d\mathfrak{c}}{dt} = -z^{-2}\frac{dz}{dt}\mathfrak{a} + z^{-1}\frac{d\mathfrak{a}}{dt} = 0$. Each component satisfies therefore $\frac{d\mathfrak{c}_\mathfrak{p}}{dt} = 0$. This means that each component is a p-th power whence $\mathfrak{c} \in J_k^p$ and consequently $\mathfrak{a} \in k^* \cdot J_k^p$.

Both J_k^p and k lie trivially in the kernel, k for the reason that λ is a differential.

2. Suppose $x = z^p - z$. By the approximation theorem we can find an element $y \in k$ such that the idèle $y\mathfrak{a}$ is very close to 1 at all primes where x has a pole. At these primes then $\varphi_\mathfrak{p}(x, y\mathfrak{a}_\mathfrak{p}) = 0$. Furtheremore $\varphi(x, \mathfrak{a}) = \varphi(x, y\mathfrak{a})$ since y is in the kernel of J_k. Then

$$\varphi(x, \mathfrak{a}) = \sum_\mathfrak{p} \varphi_\mathfrak{p}(x, y\mathfrak{a}_\mathfrak{p}) = \sum_\mathfrak{q} \varphi_\mathfrak{q}(x, y\mathfrak{a}_\mathfrak{q})$$

where $\mathfrak{q}$ ranges only over those primes where x has no pole and is consequently a local integer. Lemma 5 shows that each $\varphi_\mathfrak{q}(x, y\mathfrak{a}_\mathfrak{q}) = 0$, since $x = z^p - z$. x is therefore in the kernel of k.

3. Suppose that x is in the kernel of k. Then $\varphi(x, \mathfrak{a}) = 0$ for all idèles, especially the local idèles $\mathfrak{a}_\mathfrak{p}$ which have only one component $\neq 1$. One obtains therefore that $\varphi_\mathfrak{p}(x, \mathfrak{a}_\mathfrak{p}) = 0$ for all $\mathfrak{a}_\mathfrak{p} \in k_\mathfrak{p}$. Let $\mathfrak{p}$ be a prime where x has no pole. At such a prime Lemma 5 applies and tells that $x = z_\mathfrak{p}^p - z_\mathfrak{p}$ with $z_\mathfrak{p} \in k_\mathfrak{p}$. Such a prime splits therefore completely in the global extension $k\left(\frac{1}{\wp}x\right)$ and Theorem 2 of Chapter V Section 4 tells us that this global extension must collapse, i.e. $x = z^p - z$, $z \in k$. $\square$

γ. *The Galois group of the maximal elementary p-extension.* The p-extension with the Kummer group k shall be called the maximal elementary p-extension. It is the field $M = k\left(\frac{1}{\wp}k\right)$. The group A associated with it by the Kummer theory is $A = \frac{1}{\wp}k$. This group A is paired with the Galois group $\mathfrak{G}$ by $(\alpha, \sigma) = (\sigma - 1)\alpha$. We have mapped A into k by the map $\alpha \to \wp\alpha$. The image of A is in our case the whole field k and we may view this pairing also as a pairing of k and $\mathfrak{G}$ into $\mathbb{Z}_p$ where

$$[x, \sigma] = (\sigma - 1)\left(\frac{1}{\wp}x\right), \qquad x \in k,\ \sigma \in \mathfrak{G}.$$

The kernel of k is now $\wp k$, the same as in the pairing $\varphi(x, \mathfrak{a})$; that of $\mathfrak{G}$ is 1. Under this pairing the group $\mathfrak{G}$ is naturally isomorphic to the character group of $k/\wp k$ (where $k/\wp k$ has the discrete topology). In the pairing $\varphi(x, \mathfrak{a})$ the group $J_k/k^* J_k^p$ is naturally isomorphic to the same character group. It follows that one has a natural isomorphism between $\mathfrak{G}$ and $J_k/k^* J_k^p$. If σ is the image of the idèle $\mathfrak{a}$ under this map then we should have $\varphi(x, \mathfrak{a}) = [x, \sigma] = (\sigma - 1)\left(\frac{1}{\wp}x\right)$. We denote this map by $\sigma(\mathfrak{a})$ and the previous equation describes for us just the effect of $\sigma(\mathfrak{a})$ on the generators of M, a description which determines $\sigma(\mathfrak{a})$ completely:

$$\sigma(\mathfrak{a})\colon \frac{1}{\wp}x \to \frac{1}{\wp}x + \varphi(x, \mathfrak{a}). \tag{3}$$

The mapping (3) is actually a mapping of the idèle classes into $\mathfrak{G}$ with kernel C_k^p. We have therefore the theorem

THEOREM 6. *The mapping* (3) *is a bicontinuous isomorphism between the group $J_k/k \cdot J_k^p$ (or C_k/C_k^p) and the Galois group of the maximal abelian extension M of exponent p.*

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c. The proof of the second inequality. Let K/k be a cyclic extension of degree p. As subfield of M it is determined by an open subgroup $\mathfrak{h}$ of $\mathfrak{G}$ of index p.

The norm group $k \cdot N_{K/k}(J_K)$ is open (all idèles in a sufficiently small neighborhood of 1 are norms) and of finite index in J_k. According to the first inequality the index is at least p. The image of this group under the isomorphism of Theorem 6 is a certain open subgroup $\mathfrak{h}'$ of $\mathfrak{G}$. The index of $\mathfrak{h}'$ in $\mathfrak{G}$ is finite and at least p. $\mathfrak{h}'$ determines a certain subfield E/k of M/k which is the compositum of cyclic fields of prime degree and each of these fields is left fixed by $\mathfrak{h}'$. To each cyclic subfield K'/k of degree p which is *different* from K we shall exhibit an element of $\mathfrak{h}'$ which does not leave every element of K' fixed. This will show that $\mathfrak{h}'$ must be $\mathfrak{h}$ and prove thereby the second inequality since the index of $\mathfrak{h}$ is p.

According to Chapter V, § 4, Theorem 4 there is a prime $\mathfrak{p}$ of k which splits completely in K and remains prime in K'. If $K' = k(\frac{1}{\wp}x)$ we can select $\mathfrak{p}$ in such a way that x does not have a pole at $\mathfrak{p}$. Let $\mathfrak{a}_\mathfrak{p}$ be an idèle of k which has at $\mathfrak{p}$ a component of *ordinal* 1, and has components 1 at all other primes. It is a norm from K since $\mathfrak{p}$ splits in K. $\sigma(\mathfrak{a}_\mathfrak{p})$ is therefore in $\mathfrak{h}'$. Now we compute $\varphi(x, \mathfrak{a}_\mathfrak{p}) = \varphi_\mathfrak{p}(x, \mathfrak{a}_\mathfrak{p})$. $\mathfrak{p}$ does not split in K', x is therefore not in $\wp k_\mathfrak{p}$. $\mathfrak{a}_\mathfrak{p}$ has ordinal 1 (not divisible by p). According to Lemma 5 we have $\varphi(x, \mathfrak{a}_\mathfrak{p}) \neq 0$. This means according to (3) that $\sigma(\mathfrak{a}_\mathfrak{p})$ moves $\frac{1}{\wp}x$ and is therefore not identity on K'.

5. Infinite Divisibility of the Universal Norms

It is a remarkable feature of the proofs of the second inequality that we obtain as a by-product the Existence Theorem, or rather the infinite divisibility of the universal norms of idèle classes, which turn out to be the kernel of the norm residue symbol.

We shall assume here the *norm index is equal to the degree in all abelian extensions*. This fact will, be proved in the next chapter. (Up to now, it has been proved only for cyclic extensions.)

Under this assumption we shall prove

THEOREM 7. *Let k be a global field. Let $a \in C_k$ be an idèle class which is a universal norm, i.e. a norm from every finite extension. Then for every integer n, a is an n-th power of a universal norm, i.e. there exists a universal norm $b \in C_k$ such that $a = b^n$.*

PROOF. It was shown as a consequence of the compactness of C^0 that the axiomatic conditions of Ch. 14 for the existence theorem are fulfilled. It suffices therefore to prove that for every prime n, a is the n-th power of an idèle class. Furthermore, since we may prove this only in sufficiently large layers, we may assume that all necessary roots of unity lie in k.

Let n be a prime. If $n = p$, the proff of the second inequality *c. above) shows that, under the isomorphism of Theorem 6, a universal norm corresponds to an element of the Galois group which leaves fixed every cyclic extension of degree p, and is therefore a p-th power. We may therefore assume that $n \neq p$. This puts us in the situation of Section 3.

We go back to Lemma 2. We let S_2 be the empty set, $S = S_1$. Then $D_1 = \prod_{\mathfrak{p} \in S} k_\mathfrak{p}^* \times \prod_{\mathfrak{p} \notin S} U_\mathfrak{p} = J_S$. Hence $\Delta_1 = k_S$ and $K_1 = k(k_S^{1/n})$. We also have $k^* D_1 = k^* J_S = J$. Since $k^* D_1 = k^* N_2 J_2$ it follows from the first inequality that $K_2 = k$. The Kummer group of K_2 is $\Delta_2 = D_2 \cap k$ and hence $\Delta_2 = k_S^n$.

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Now from Lemma 2 we obtain

$$(J : kD_2) = [K_1 : k].$$

Supposing the first inequality to be true in all abelian extensions, we must have $k^*D_2 = k^*N_1J_1$. (We have therefore explicitly constructed the class field to k^*D_2. It is then no wonder that the existence theorem comes out as a by-product.)

Let $a \in C_k$ be universal norm. It is in particular norm from K_1. It has therefore a representative idèle $\mathfrak{a} \in D_2$. Any two such representatives differ by an element $\delta \in \Delta_2 = k_S^n$. $\mathfrak{a}$ is n-th power at all primes in S. We shall prove that $\mathfrak{a}$ is also an n-th power at all primes outside S.

If $S' \supset S$ then a has a representative $\mathfrak{b} \in D_2' \subset D_2$. We can write $\mathfrak{b} = \mathfrak{c}^n\mathfrak{b}'$, where $\mathfrak{b}'$ is unit outside S' and is 1 in S', and $\mathfrak{c}$ is 1 outside S'. Let $\beta \in k$ be such that $\mathfrak{c} = \beta\vartheta$ where $\vartheta \in J_S$. Then $\mathfrak{c}^n = \beta^n\vartheta^n$ and $\mathfrak{b} = \beta^n\vartheta^n\mathfrak{b}'$. This shows that $\mathfrak{b}$ is n-th power at all primes in S'. $\mathfrak{b}$ differs from $\mathfrak{a}$ by an n-th power and therefore $\mathfrak{a}$ is also n-th power in S'. The set S' can be made to include any prime, and this concludes the proof of the theorem. □

6. Sketch of the Analytic Proof of the Second Inequality

The Kummer theory proof in Section 3 is basically Chevalley's (cf. [**6**, Sect. 9]). Before 1940, the only known proof of the second inequality was analytic. The analytic proof is quite short, once given the behavior of the zeta functions and L-functions near $s = 1$. It works for arbitrary finite extensions K/k, not necessarily even Galois, but gives only the inequality, not the divisibility. The analytic part of the argument, after the properties of the ζ and L functions are established, is the same as the one Dirichlet used to prove his theorem on primes in arithmetic progressions. He showed that for a prime to m, the set of primes congruent to $a \pmod m$ has what we now call "Dirichlet density" $1/\phi(m)$. A set T of prime ideals of k has Dirichlet density δ, if the function

$$\sum_{\mathfrak{p}\in T} \frac{1}{N\mathfrak{p}^s} - \delta \log(\frac{1}{s-1})$$

is bounded on $1 < s < 1 + \epsilon$. The set of all primes has density 1. This follows from the fact that the zeta-function $\zeta_k(s) = \prod_{\mathfrak{p}} \left(1 - N\mathfrak{p}^{-s}\right)^{-1}$ has a simple pole at $s = 1$, so that $\log \zeta_k(s) - \log\left(\frac{1}{s-1}\right)$ is bounded near $s = 1$, and on the other hand, $\log \zeta_k(s)$ differs from $\sum_{\mathfrak{p}} \frac{1}{N\mathfrak{p}^s}$ by a bounded amount there. A finite set, and any set of primes of degree > 1, have density 0. The analytic proof that $h = |C_k/NC_K|$ is less than $n = [K : k]$ is by showing that the set of primes $\mathfrak{p}$ of k which split completely in K has density $\frac{1}{n}$, and the set of primes whose class is a norm from K has density $\frac{1}{h}$. Since the former set is contained in the latter this implies $h \leq n$.

A sketch of how this goes is as follows. Let B be an open subgroup of finite index in C_k. As explained in (VIII, 2 and 4), we can view C_k/B as a group $\mathcal{C}$ of classes of fractional ideals prime to a "conductor" $\mathfrak{m}$, each class being a coset of a group of "generalized arithmetic progressions" $\pmod{\mathfrak{m}}$. Let X denote the character group of $\mathcal{C}$, the analog of a finite group of Dirichlet characters. The associated L-functions, defined for $\mathrm{Re}(s) > 1$ by $L(s, \chi) = \prod_{p \nmid \mathfrak{m}} \left(1 - \frac{\chi(\mathfrak{p})}{N\mathfrak{p}^s}\right)^{-1}$ have similar properties to those of Dirichlet L-functions. They are holomorphic at $s = 1$, except for $\chi = 1$. The zeta function, $\zeta_k(s) = L(s, 1)$, has a pole of order 1 there.

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One concludes, following Dirichlet, that

$$\sum_{\mathfrak{p}\nmid\mathfrak{m}} \frac{\chi(\mathfrak{p})}{N\mathfrak{p}^s} = -a_\chi \log\left(\frac{1}{s-1}\right) + O(1),$$

for s near 1, where a_χ is the order of $L(s,\chi)$ at $s=1$. Averaging over $\chi \in X$ one finds

$$\sum_{\chi(\mathfrak{p})=1} \frac{1}{N\mathfrak{p}^s} = \frac{1}{h}\Big(1 - \sum_{\chi\neq 1} a_\chi\Big) \log\left(\frac{1}{s-1}\right) + O(1),$$

where $h = |X| = |\mathcal{C}| = |C_k/B|$.

Now suppose K/k is a finite extension, and $B = N_{K/k}C_K$, so that h is the norm index. Let T be the set of primes $\mathfrak{p}$ of k which split completely in K. Each $\mathfrak{p} \in T$ lies below $n = [K:k]$ primes $\mathfrak{P}$ of K, each with $N\mathfrak{P} = N\mathfrak{p}$. Hence, since the set of all primes $\mathfrak{P}$ of K has density 1,

$$\sum_{\mathfrak{p}\in T} \frac{1}{N\mathfrak{p}^s} = \frac{1}{n}\sum_{\mathfrak{P}|T} \frac{1}{N\mathfrak{P}^s} \leq \frac{1}{n}\sum_{all\,\mathfrak{P}} \frac{1}{N\mathfrak{P}^s} = \frac{1}{n}\log\left(\frac{1}{s-1}\right) + O(1).$$

The primes $\mathfrak{p} \in T$ are norms from K, so their classes are norms, i.e. they are among the primes $\mathfrak{p}$ with $\chi(\mathfrak{p}) = 1$. Comparing the last two equations we find therefore

$$\frac{1}{n} \leq \frac{1}{h}\Big(1 - \sum_{\chi\neq 1} a_\chi\Big).$$

Conclusion: $a_\chi = 0$ for $\chi \neq 1$, i.e. $L(s,\chi)$ does not vanish at $s=1$, and $h \leq n$.

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CHAPTER VII

Reciprocity Law

1. Introduction

k is a global field, Ω the separable part of its algebraic closure, and $\mathfrak{G}$ the Galois group of Ω/k. C_Ω are the idèle classes of Ω, $C_\Omega = \bigcup_K C_K$ where $k \subset K \subset \Omega$, K/k finite and normal.

We have seen in the preceding chapter that $(\mathfrak{G}, C_\Omega)$ is a field formation, i.e. $\mathcal{H}^1(\mathfrak{h}, C_\Omega) = 1$ for all open subgroups $\mathfrak{h}$ of $\mathfrak{G}$. It is possible to assign invariants to the element of $\mathcal{H}^2(\mathfrak{G}, C_\Omega)$ in such a way that $(\mathfrak{G}, C_\Omega)$ becomes a class formation in the sense of Chapter XIV. This assignment will be carried out in this chapter.

In function fields, the situation is very much like that in local fields: every idèle class can be given an ordinal, and the constant field extensions in the large can be used in the same way as the unramified extensions were used in local class field theory.

In number fields, it turns out that the cyclotomic extensions can be made to play a role similar to the constant field extensions, in spite of the fact that they ramify and that they are not all cyclic. (As a matter of fact, the constant field extensions themselves are cyclotomic, i.e. they are obtained by adjoining roots of unity.)

It will therefore be necessary to prove first the reciprocity law for cyclotomic extensions of the rational numbers. This is done in 2. The proofs proceed in a completely elementary fashion, except at the very last argument where the second inequality is used. From then on the reciprocity law can be proved for normal extensions K/k of global fields without essentially distinguishing the two cases. We first assign invariants to idèle cocycles by taking the sum of their local invariants. We then prove that number cocycles have invariant 0. This is done by moving a number cocycle from a normal extension to a cyclic cyclotomic extension, where this fact has already been proved. We select the auxiliary cyclic extension so that it splits the cocycle locally everywhere, and then use the triviality of $\mathcal{H}^1(C_K)$ to move the cocycle.

By this procedure, we get invariants for idèle class cocycles, whenever these have a representative idèle cocycle. When they do not, the triviality of $\mathcal{H}^1(C_K)$ allows us to move a cocycle to a cyclic extension K' on which it splits. In K' the cocycle has a representative idèle cocycle, and hence may be given an invariant there. It is easy to show that the invariants thus obtained are independent of the auxiliary constructions performed, and that they satisfy the axioms of a class formation.

Knowing that $(\mathfrak{G}, C_\Omega)$ is a class formation, we can apply the axiomatic development of Chapter XIV. In particular, we get the triviality of the third cohomology group $\mathcal{H}^3(G, C_K)$ in finite layers of the formation.

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We also get the existence of a homomorphism ω of C_k into a dense subgroup of $\mathfrak{G}/\mathfrak{G}'$ given by the norm residue symbol. In function fields, the situation is again completely analogous to that of local class field theory. ω is an isomorphism, but is only *into* $\mathfrak{G}/\mathfrak{G}'$. It is possible to complete C_k to a group $\overline{C}_k$ in such a way that ω extends to an isomorphism of $\overline{C}_k$ onto $\mathfrak{G}/\mathfrak{G}'$, by a procedure similar to the one carried out in the local class field theory.

In number fields, the situation is different: The norm residue symbol ω maps C_k *onto* $\mathfrak{G}/\mathfrak{G}'$, but there exists a non-trivial kernel, the elements which are infinitely divisible. This kernel turns out to be the connected component of C_k, whose structure is discussed in detail in a later chapter.

2. Reciprocity Law over the Rationals

Let $\mathbb{Q}$ be the rationals, $\mathbb{Q}_p$ the completion of $\mathbb{Q}$ at p and p_∞ the archimedean prime of $\mathbb{Q}$. Then $\mathbb{Q}_{p_\infty}$ is the reals and contains the multiplicative group $\mathbb{R}_+$ of the positive reals. $\overline{U}$ denotes as before the product of all U_p for $p \neq p_\infty$, considered as subgroup of the idèles $J_\mathbb{Q}$.

$J_\mathbb{Q}$ contains naturally the direct product $\mathbb{R}_+ \times \overline{U}$ as an open subgroup with the product topology. It consists of those idèles whose infinite component is positive and whose finite components are units. If $\alpha \in \mathbb{Q}$ is a given rational number then the coset $\alpha(\mathbb{R}_+ \times \overline{U})$ consists of the following idèles: their infinite component has the same sign as α and each finite component has the same ordinal as α. Since the ordinals and the sign of a rational number can be arbitrarily prescribed we see that

$$J_\mathbb{Q} = \mathbb{Q}^* \times \mathbb{R}_+ \times \overline{U}$$

with the product topology ($\mathbb{Q}^*$ discrete) since $\mathbb{R}_+ \times \overline{U}$ is open.

Let k be any field. The extension obtained by adjunction of *all* roots of unity shall be called the maximal cyclotomic extension of k and any intermediate field a cyclotomic extension of k. An automorphism of the maximal cyclotomic extension is described by its action on the roots of unity. A root of unity is sent into a power of itself and one derives easily from this that the Galois group is abelian.

Denote by Γ the maximal cyclotomic extension of $\mathbb{Q}$ and by Γ_p the maximal cyclotomic extension of $\mathbb{Q}_p$. Γ_p contains a maximal cyclotomic extension $\overline{\Gamma}$ of $\mathbb{Q}$ as subfield and we have $\Gamma_p = \overline{\Gamma}\mathbb{Q}_p$. $\overline{\Gamma}$ is isomorphic to Γ but there are many isomorphisms of $\overline{\Gamma}$ onto Γ. If t is one of them then λt is the most general one, where λ is any element of the Galois group of $\Gamma/\mathbb{Q}$. Let $\mathfrak{G}$ be the Galois group of $\Gamma/\mathbb{Q}$ and $\overline{\mathfrak{G}}$ the one of $\overline{\Gamma}/\mathbb{Q}$. The map t induces a natural map $\bar{\sigma} = t^{-1}\sigma t$ of $\mathfrak{G}$ onto $\overline{\mathfrak{G}}$. Since $\mathfrak{G}$ is *abelian* one sees that this map is independent of the choice of the map t. This allows a simplified description. We shall choose an identification of the field $\overline{\Gamma}$ with the field Γ, so that we may say $\Gamma_p = \Gamma\mathbb{Q}_p$. This identification results in an identification of $\overline{\mathfrak{G}}$ and $\mathfrak{G}$ and this latter identification is independent of the way in which the identification of $\overline{\Gamma}$ and Γ is done.

The Galois group $\mathfrak{G}_p$ of $\Gamma_p = \Gamma\mathbb{Q}_p$ over $\mathbb{Q}_p$ is now mapped in the well known way canonically into $\mathfrak{G}$, by looking at the effect that an element of $\mathfrak{G}_p$ has on Γ. The group $\mathfrak{G}_p$ becomes in this way a closed subgroup of $\mathfrak{G}$.

For $p = p_\infty$ the field $\Gamma\mathbb{Q}_p$ is the field of complex numbers and $\mathfrak{G}_p$ a group of order 2.

If p is finite, $\Gamma\mathbb{Q}_p/\mathbb{Q}$ contains the *maximal unramified* extension of $\mathbb{Q}_p$, obtained by adjoining to it all m_0-th roots of unity ζ_0 with an m_0 prime to p. The group of

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this unramified extension contains the Frobenius substitution which sends each ζ_0 onto ζ_0^p. We shall denote by φ_p an element of $\mathfrak{G}_p$ which has this effect on the ζ_0.

Our first aim is to prove that $\mathfrak{G}$ is isomorphic to $\overline{U}$ in a natural way.

An element of $\mathfrak{G}$ is determined by its effect on the roots of unity. If ζ is a primitive m-th root of unity then the automorphisms of $\mathfrak{G}$ induce of course the Galois group of $\mathbb{Q}(\zeta)/\mathbb{Q}$ on this subfield. The following lemma which is nothing else than the irreducibility of the cyclotomic equation gives the structure of this group:

LEMMA 1. *Let ζ be a primitive m-th root of unity. An automorphism of $\mathbb{Q}(\zeta)/\mathbb{Q}$ sends ζ into a power ζ^n where n is prime to m. Conversely, to any given n prime to m there is an automorphism σ such that $\zeta^\sigma = \zeta^n$. In short, $G_{\mathbb{Q}(\zeta)/\mathbb{Q}} \approx (\mathbb{Z}/m\mathbb{Z})^*$.*

PROOF. The first part of the lemma is trivial. As to the second part, it suffices to prove the statement if n is a prime p that does not divide m. ζ satisfies the equation $x^m - 1 = f(x) = 0$, and $f'(\zeta) = m\zeta^{m-1}$ is prime to p. The local field $\mathbb{Q}_p(\zeta)/\mathbb{Q}_p$ is therefore unramified. Its Frobenius substitution sends ζ into an m-th root of unity that is congruent to ζ^p. Since $f'(\zeta) = \prod_\mu(\zeta - \zeta^\mu)$ is prime to p it follows, that no two m-th roots of unity are in the same residue class; ζ^p is therefore the image of ζ under the Frobenius substitution. This automorphism of the local field induces an automorphism of the global field $\mathbb{Q}(\zeta)/\mathbb{Q}$ and this proves the lemma. $\square$

For the description of the automorphisms of the infinite field Γ the exponentiation with integers is not convenient and shall be replaced by an exponentiation with elements $u \in \overline{U}$. Let m_p be the p-contributions to m of the prime divisors p of m and u_p the p-components of u. We can find an integer n that satisfies the simultaneous congruence $n \equiv u_p \pmod{m_p}$. We write shortly $n \equiv u \pmod{m}$. This n will be prime to m and its residue class mod m is uniquely determined. If we put $\zeta^u = \zeta^n$ then ζ^u is well defined. If conversely an integer n prime to p is given, there exists a $u \in \overline{U}$ such that $u \equiv n \pmod{m}$. Indeed, it suffices to select $u_p = n$ for all $p \mid m$ and $u_p = 1$ for all other p. One verifies easily that $(\zeta^u)^v = \zeta^{uv}$ for u and v in $\overline{U}$.

According to our lemma we may now say that an automorphism of $\mathbb{Q}(\zeta)/\mathbb{Q}$ sends ζ into some power ζ^u and that each $u \in \overline{U}$ gives rise to an automorphism of this field which maps ζ onto ζ^u. If m divides m' and if ζ' is an m'-th root of unity then $\mathbb{Q}(\zeta) \subset \mathbb{Q}(\zeta')$. Let σ be the automorphism that sends ζ' into $(\zeta')^u$. To describe it we have to find an $n' \equiv u \pmod{m'}$. Then $(\zeta')^\sigma = (\zeta')^{n'}$. ζ is a power of ζ' hence $\zeta^\sigma = \zeta^{n'}$. Since this n' is also congruent to $u \pmod{m}$, $\zeta^\sigma = \zeta^u$. This means that the automorphism that u defines on $\mathbb{Q}(\zeta')$ will induce on the subfield $\mathbb{Q}(\zeta)$ again the automorphism corresponding to u.

Let now σ_u be the following map of Γ into Γ. If $\alpha \in \Gamma$ then α lies in some field $\mathbb{Q}(\zeta)$. The given u defines on $\mathbb{Q}(\zeta)$ a certain automorphism and we let $\sigma_u(\alpha)$ be the image of α under this automorphism of $\mathbb{Q}(\zeta)$. If α lies also in the field $\mathbb{Q}(\zeta')$, we can find a root of unity ζ_1 such that ζ and ζ' are powers of ζ_1. The automorphism of $\mathbb{Q}(\zeta_1)$ corresponding to u agrees with those of $\mathbb{Q}(\zeta)$ and $\mathbb{Q}(\zeta')$ and this shows that our map is well defined. σ_u is an automorphism of Γ since it is an automorphism on every subfield $\mathbb{Q}(\zeta)$. To make its description now very short: σ_u sends *every* root of unity ζ into ζ^u. Clearly $\sigma_u\sigma_v = \sigma_{uv}$.

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The map $u \to \sigma_u$ is an algebraic homomorphism of $\overline{U}$ into $\mathfrak{G}$. Its kernel consists of those u which are $\equiv 1$ modulo every integer m, i.e. $u = 1$. The map is therefore an isomorphism into $\mathfrak{G}$. It is continuous. Indeed a neighborhood of 1 in $\mathfrak{G}$ can be taken to be the group $\mathfrak{h}_m$ that leaves a given $\mathbb{Q}(\zeta)$ fixed and the inverse image of $\mathfrak{h}_m$ is the set of all $u \equiv 1 \pmod{m}$; this set is open in $\overline{U}$. The image of $\overline{U}$ is everywhere dense in $\mathfrak{G}$ since every automorphism of the finite subfield $\mathbb{Q}(\zeta)$ is induced by some σ_u. As continuous image of the compact set $\overline{U}$ it is compact and therefore all of $\mathfrak{G}$. Thus we have proved:

PROPOSITION. *The map $u \to \sigma_u$ is a topological isomorphism of $\overline{U}$ onto $\mathfrak{G}$.*

By means of this description of the group $\mathfrak{G}$ we define now a map $\mathfrak{a} \to \sigma(\mathfrak{a})$ of the idèle group $J_{\mathbb{Q}}$ onto $\mathfrak{G}$:

DEFINITION 1. According to $J_{\mathbb{Q}} = \mathbb{Q} \times \mathbb{R}_+ \times \overline{U}$ we write a given idèle $\mathfrak{a}$ uniquely $\mathfrak{a} = \alpha \cdot \rho \cdot u$ where $\alpha \in \mathbb{Q}$, $\rho \in \mathbb{R}_+$ and $u \in \overline{U}$. Put $\sigma(\mathfrak{a}) = \sigma_{u^{-1}} = \sigma_u^{-1}$. The action on the roots of unity is therefore

$$\zeta^{\sigma(\mathfrak{a})} = \zeta^{u^{-1}}.$$

Since the map is a projection followed by an inversion followed by an exponentiation: $\mathfrak{a} \to u \to u^{-1} \to \zeta^{u^{-1}}$, it is a continuous homomorphism of $J_{\mathbb{Q}}$ onto $\mathfrak{G}$ with kernel $\mathbb{Q} \times \mathbb{R}_+$. Since $\mathbb{Q}$ is in the kernel it is actually a continuous homomorphism of the idèle classes $C_{\mathbb{Q}}$ onto $\mathfrak{G}$ with kernel $D_{\mathbb{Q}} = \mathbb{R}_+$.

(We note parenthetically that $D_{\mathbb{Q}}$ is the connected component of $C_{\mathbb{Q}}$. Namely, $\mathbb{R}_+$ is obviously the connected component of $J_{\mathbb{Q}}$. The continuous image $D_{\mathbb{Q}} = \mathbb{R}_+$ in the idèle classes is therefore connected and $C_{\mathbb{Q}}/\mathbb{R}_+$ is totally disconnected.)

If $\mathfrak{a}_p$ denotes as usual the idèle that has at p the same component as $\mathfrak{a}$ and 1 at all other primes we may write for any finite set of primes

$$\mathfrak{a} = \prod_{\mathfrak{p} \in S} \mathfrak{a}_p \cdot \mathfrak{c}_{\widetilde{S}}.$$

$\widetilde{S}$ is the complement of S in the set of all primes. As S becomes bigger and bigger $\mathfrak{c}_{\widetilde{S}}$ approaches 1. Since $\sigma(\mathfrak{a})$ is continuous, $\lim \sigma(\mathfrak{c}_{\widetilde{S}}) = 1$. A more highbrow and more formal argument is the following: The totality of all sets $\widetilde{S}$ forms a filter in the set of all primes. In this sense $\lim_{\widetilde{S}} \mathfrak{c}_{\widetilde{S}} = 1$ and the continuity implies $\lim \sigma(\mathfrak{c}_{\widetilde{S}}) = 1$. From this we deduce the convergence of the infinite produce $\prod_p \sigma(\mathfrak{a}_p)$ in the sense of this filter and the formula:

$$\sigma(\mathfrak{a}) = \prod_p \sigma(\mathfrak{a}_p).$$

We compute now each single $\sigma(\mathfrak{a}_p)$:

1. $p = p_\infty$. Let the infinite component be $\varepsilon\alpha$ where $\varepsilon = \pm 1$ and α positive real. According to our definition we have to split up $\mathfrak{a}_{p_\infty}$ into three factors: $(\varepsilon, \varepsilon, \dots) \in \mathbb{Q}$, $(\alpha, 1, 1, \dots) \in \mathbb{R}_+$, $u = u^{-1} = (1, \varepsilon, \varepsilon, \dots) \in \overline{U}$. In order to find now $\zeta^{u^{-1}}$ we have to find an integer $n \equiv u^{-1} \pmod{m}$ and notice that $n = \varepsilon$ will always do it. Therefore $\zeta^{\sigma(\mathfrak{a}_{p_\infty})} = \zeta^{\varepsilon}$ or:

$$\zeta^{\sigma(\mathfrak{a}_{p_\infty})} = \zeta^{\operatorname{sign}(\mathfrak{a}_{p_\infty})}.$$

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2. p finite. We shall write idèles in this way: first the component of p_∞ and then that of p and then those of the other primes. Let

$$\begin{aligned}\mathfrak{a}_p &= (1, p^\mu u_p, 1, 1, 1, \dots). \quad \text{The three factors are:}\\ &(p^\mu, p^\mu, p^\mu, p^\mu, \dots) \in \mathbb{Q}\\ &(p^{-\mu}, 1, 1, 1, \dots) \in \mathbb{R}_+\\ u &= (1, u_p, p^{-\mu}, p^{-\mu}, \dots) \in \overline{U}.\end{aligned}$$

Since any root of unity ζ is the product of an m_0-th root of unity ζ_0 with $p \nmid m_0$ and a p^r-th root of unity ζ_{p^r} it suffices to give the action of $\sigma(\mathfrak{a}_p)$ on ζ_0 and ζ_{p_r} separately:

We have to find an integer $\equiv u^{-1} \pmod{m_0}$ and see that p^μ does it for all m_0 if $\mu \geqslant 0$. Hence $\zeta_0^{\sigma(\mathfrak{a}_p)} = \zeta_0^{p^\mu}$. The Frobenius substitution φ_p maps ζ_0 onto ζ_0^p and we may also write:

$$\zeta_0^{\sigma(\mathfrak{a}_p)} = \zeta_0^{\varphi_p^\mu},$$

a description which remains valid if $\mu < 0$ because of the group property.

For the action on ζ_{p^r} we have to find an integer $n \equiv u_p^{-1} \pmod{p^r}$ i.e. simply a sufficiently good approximation of the p-adic number u_p^{-1} and we shall leave it at that. Then

$$\zeta_{p^r}^{\sigma(\mathfrak{a}_p)} = \zeta_{p^r}^{u_p^{-1}} = \zeta_{p^r}^n.$$

All this suggests the introduction of a local symbol defined by:

Definition 2. For every prime let $\sigma_p(\alpha_p)$ denote a map $\alpha_p \mapsto \sigma_p(\alpha_p)$ of $\mathbb{Q}_p^*$ into $\mathfrak{G}$ defined as follows:

1. If $p = p_\infty$, let $\zeta^{\sigma_\infty(\alpha_\infty)} = \zeta^{\operatorname{sign}(\alpha_\infty)}$.
2. If p is finite and $\alpha_p = p^{\operatorname{ord}\alpha_p} u_p$, let the action on ζ_0 resp. ζ_{p^r} be:

$$\zeta_0^{\sigma_p(\alpha_p)} = \zeta_0^{\varphi_p^{\operatorname{ord}\alpha_p}}, \quad \zeta_{p^r}^{\sigma_p(\alpha_p)} = \zeta_p^{u_p^{-1}}. \qquad (\varphi_p \text{ Frobenius subst.})$$

The map $\alpha_p \to \sigma_p(\alpha_p)$ is continuous. The kernel is $\mathbb{R}_+$ at p_∞ and 1 at finite primes. Indeed, to have $\sigma_p = 1$ we must have $\zeta_{p^r}^{n_p^{-1}} = 1$ for all ζ_{p^r} and this implies $u_p^{-1} \equiv 1 \pmod{p^r}$ for all r, whence $u_p = 1$. It also implies $\varphi_p^{\operatorname{ord}(\alpha_p)} = 1$ and this means $\operatorname{ord}(\alpha_p) = 0$.

The symbol $\sigma_p(\alpha_p)$ will eventually turn out to take its values in $\mathfrak{G}_p$ and to he the local norm residue symbol of $\mathbb{Q}_p$. We have obviously:

Theorem 1. *If $\mathfrak{a}$ has components $\mathfrak{a}_p$ then $\sigma(\mathfrak{a}) = \prod_p \sigma_p(\mathfrak{a}_p)$. We also write $\sigma_p(\mathfrak{a})$ for $\sigma_p(\mathfrak{a}_p)$.*

We come now to the keystone of this investigation:

Theorem 2. *Let $K/\mathbb{Q}$ be a finite cyclotomic extension: $\mathbb{Q} \subset K \subset \mathbb{Q}(\zeta)$ where $\zeta^m = 1$. Let $\mathfrak{h}$ be the subgroup of $\mathfrak{G}$ that determines K (is identity on K). If $\mathfrak{a}$ is an idèle $\in \mathbb{Q}^* N_{K/\mathbb{Q}} J_K$ then $\sigma(\mathfrak{a}) \in \mathfrak{h}$.*

Proof. 1. For each p the map $\alpha_p \to \sigma_p(\alpha_p)$ is continuous and $\mathfrak{h}$ is open. Therefore $\sigma_p(\alpha_p) \in \mathfrak{h}$ for all α_p near enough to 1.

2. Let S be the set of all archimedean primes of K and of all primes that divide m. Assume $\mathfrak{a} = \alpha N_{K/\mathbb{Q}}(\mathfrak{A})$ with $\alpha \in \mathbb{Q}$. According to the approximation theorem we can find an element A of K such that $\mathfrak{A} = A\mathfrak{B}$ where $\mathfrak{B}$ is as close to 1 as we like

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at all primes of S. Then $\mathfrak{A} = \alpha N_{K/\mathbb{Q}}(A) N_{K/\mathbb{Q}}(\mathfrak{B}) = \beta N_{K/\mathbb{Q}}(\mathfrak{B})$. Since $\sigma(\beta) = 1$ it suffices to show $\sigma(N_{K/\mathbb{Q}}(\mathfrak{B})) \in \mathfrak{B}$. We decompose according to Theorem 1 this automorphism into its local parts $\sigma_p(N_{K/\mathbb{Q}}(\mathfrak{B})) = \sigma_p$. We prove the contention by showing that each σ_p is in $\mathfrak{h}$ ($\mathfrak{h}$ is closed). The norm mapping is continuous. If we have brought the components of $\mathfrak{B}$ for the primes of S near enough to 1 then $N_{K/\mathbb{Q}}(\mathfrak{B})$ will have components near 1 for p_∞ and for all $p \mid m$. According to what we have seen, σ_p will be in $\mathfrak{B}$ for these primes.

There remains the case when p is finite and does not divide m. The field $\mathbb{Q}_p(\zeta)$ is then unramified and so is the completion $K_\mathfrak{p}$ of K as subfield of $\mathbb{Q}_p(\zeta)$. If n_p is the degree of $K_\mathfrak{p}/\mathbb{Q}_p$ and φ_p the Frobenius substitution then $\varphi_p^{n_p}$ leaves $K_\mathfrak{p}$ and consequently $K \subset K_\mathfrak{p}$ fixed.

The p-component of $N_{K/\mathbb{Q}}(\mathfrak{B})$ is a certain local norm $N_{K_\mathfrak{p}/\mathbb{Q}_p}(\alpha)$ and all we have to know about it is that its ordinal is a certain multiple rn_p of the local degree of $K_\mathfrak{p}$. Since p does not divide m, the action of σ_p on ζ is the same as that of $\varphi_p^{rn_p}$. The automorphism $\sigma_p \varphi_p^{-rn_p}$ leaves therefore $\mathbb{Q}(\zeta)$ and consequently the subfield K fixed. Since $\varphi_p^{rn_p}$ leaves it also fixed it follows, that K is fixed under σ_p in other words $\sigma_p \in \mathfrak{h}$. □

We remark that Theorem 2 contains an independent proof of the first inequality in cyclotomic extensions of the rationals.

The local counterpart of Theorem 2, together with the proof that the map $\alpha_p \to \sigma_p(\alpha_p)$ is a true local map is contained in:

THEOREM 3. *The map $\alpha_p \to \sigma_p(\alpha_p)$ is a continuous homomorphism of $\mathbb{Q}_p^*$ onto an* everywhere dense *subgroup of $\mathfrak{G}_p$. Let $K_\mathfrak{p}$ be a finite cyclotomic extension of $\mathbb{Q}_p$ and $\overline{\mathfrak{h}}$ the (closed) subgroup of $\mathfrak{G}_p$ that determines $K_\mathfrak{p}$. If $\alpha_p \in \mathbb{Q}_p^*$ is a norm from $K_\mathfrak{p}$ then $\sigma_p(\alpha_p) \in \overline{\mathfrak{h}}$.*

PROOF. 1. We begin with a proof of the last part of the theorem. Assume that α_p is a norm from $K_\mathfrak{p}$ and let $\mathfrak{a}_p$ be the idèle with component α_p at p and components 1 at all other primes. Let $\mathfrak{h}$ be any open subgroup of the global group $\mathfrak{G}$ (it has finite index in $\mathfrak{G}$). The group $\mathfrak{h}\overline{\mathfrak{h}}$ is also open and determines a certain finite extension E of $\mathbb{Q}$. The completion of E at a prime dividing p is determined by the subgroup $\mathfrak{h}\overline{\mathfrak{h}} \cap \mathfrak{G}_p$ of $\mathfrak{G}_p$. Since $\overline{\mathfrak{h}}$ is in $\mathfrak{G}_p$ we have $\overline{\mathfrak{h}} \subset \mathfrak{h}\overline{\mathfrak{h}} \cap \mathfrak{G}_p$. This means that the completion of E is a subfield of $K_\mathfrak{p}$ so that α_p is also a norm from the completion of E. The idèle $\mathfrak{a}_p$ is consequently a norm from J_E and Theorem 2 shows now $\sigma(\mathfrak{a}_\mathfrak{p}) \in \mathfrak{h}\overline{\mathfrak{h}}$. $\sigma(\mathfrak{a}_p)$ is therefore in every open neighborhood of the compact, hence closed subgroup $\overline{\mathfrak{G}}$ and consequently in $\overline{\mathfrak{h}}$. Hence $\sigma_p(\mathfrak{a}_p) \in \overline{\mathfrak{h}}$.

2. If we select $K_\mathfrak{p} = \mathbb{Q}_p$, then $\overline{\mathfrak{h}} = \mathfrak{G}_p$ and every α_p is norm from $K_\mathfrak{p}$. Therefore $\sigma_p(\alpha_p) \in \mathfrak{G}_p$. Our map is therefore into $\mathfrak{G}_p$.

3. The image of $\mathbb{Q}_p^*$ will be everywhere dense in $\mathfrak{G}_p$ if every automorphism of a finite extension $\mathbb{Q}_p(\zeta)/\mathbb{Q}_p$ is induced by some $\sigma_p(\alpha_p)$. If we write $\zeta = \zeta_0 \zeta_{p^r}$, where ζ_0 is an m_0-th root of unity (with $p \nmid m_0$) and ζ_{p^r} a p^r-th root of unity, then $\mathbb{Q}_p(\zeta_0)$ is unramified. The action of an automorphism on ζ_0 is therefore the same as that of φ_p^μ where φ_p is the Frobenius substitution. ζ_{p^r} is sent into say $\zeta_{p^r}^n$ where $(n, p) = 1$. According to Definition 2 the element $\sigma_p(n^{-1}p^\mu)$ has the same action. □

It would be quite nice if the next theorem could be proved without appeal to the second inequality, but this seems to be hopeless at present.

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THEOREM 4. 1. *Let $K/\mathbb{Q}$ be a finite cyclotomic extension determined by the subgroup $\mathfrak{h}$ of $\mathfrak{G}$. Then $\sigma(\mathfrak{a}) \in \mathfrak{h}$ if and only if $\mathfrak{a} \in \mathbb{Q}^* N_{K/\mathbb{Q}}(J_K)$*

2. *Let similarily $K_\mathfrak{p}/\mathbb{Q}_p$ be a finite cyclotomic extension of $\mathbb{Q}_p$ determined by the subgroup $\overline{\mathfrak{h}}$ of $\mathfrak{G}_p$. Again $\sigma_p(\alpha_p) \in \overline{\mathfrak{h}}$ if and only if $\alpha_p \in N_{K_\mathfrak{p}/\mathbb{Q}_p}(K_p^*)$.*

PROOF. Both "if" statements have been proved before. If M denotes therefore the idèles of $\mathbb{Q}$ (resp. elements of $\mathbb{Q}_p^*$) that map into $\mathfrak{h}$ (resp. $\overline{\mathfrak{h}}$) then $M \supset \mathbb{Q}^* N_{K/\mathbb{Q}}(J_K)$ (resp. $M \supset N_{K_\mathfrak{p}/\mathbb{Q}_p^*}(K_p^*)$). If we follow our maps by the canonical map of $\mathfrak{G}$ resp. $\mathfrak{G}_p$ onto the factor groups $\mathfrak{G}/\mathfrak{h}$ resp. $\mathfrak{G}_p/\overline{\mathfrak{h}}$ then M is simply the kernel. These last maps are onto the factor groups because the map $\sigma(\mathfrak{a})$ is onto, and $\sigma_p(\alpha_p)$ at least onto an everywhere dense subgroup. This means that the index of M in $J_\mathbb{Q}$ resp. $\mathbb{Q}_p$ is the degree of $K/\mathbb{Q}$ resp. of $K_\mathfrak{p}/\mathbb{Q}_p$. By the global (resp. local) second inequality, the index of $\mathbb{Q} \cdot N_{K/\mathbb{Q}}(J_k)$ (resp. $N_{K_\mathfrak{p}/\mathbb{Q}_p}(K_p^*)$) is at most equal to these degrees. The theorem follows immediately. □

The next and last theorem of this section is a purely local one. We have made several attempts to prove it by purely local means and have not succeeded.[1]

THEOREM 5. *The symbol $\sigma_p(\alpha_p)$ is equal to the local norm residue symbol $(\alpha_p, \mathbb{Q}_p)$ in $\mathbb{Q}_p$. In view of the explicit character of the map $\sigma_p(\alpha_p)$* (Definition 2) *we have therefore before us an explicit determination of the norm residue symbol in $\mathbb{Q}_p$.*

PROOF. For the infinite prime the statement follows from Definition 2. Assume therefore that p is finite. We select in $\mathbb{Q}_p$ any element π of ordinal 1 (not necessarily p). We put $\sigma_p(\pi) = \sigma$ and $(\pi, \mathbb{Q}_p) = \tau$. We want to show that $\sigma = \tau$. Let $\mathfrak{h}$ be an open subgroup of $\mathfrak{G}$, and $K_\mathfrak{p}$ the subfield determined by it. Both automorphisms σ and τ have the property that π is a norm from $K_\mathfrak{p}$ if and only if σ resp. τ are in $\mathfrak{h}$. Denoting by $\mathfrak{h}_\sigma$ resp. $\mathfrak{h}_\tau$ the *closed* subgroups of $\mathfrak{G}$ generated by these elements we see that $\mathfrak{h}$ contains $\mathfrak{h}_\sigma$ if and only if it contains $\mathfrak{h}_\tau$. The two subgroups have therefore the same open neighborhoods and that implies $\mathfrak{h}_\sigma = \mathfrak{h}_\tau$. $\mathfrak{h}_\sigma$ consists of all powers σ^r where $r \in \overline{\mathbb{Z}}$, the completion[2] of the integers $\mathbb{Z}$. It follows that $\tau = \sigma^r$. To prove that $r = 1$ we look at the action of σ and τ on an unramified field of degree m. Both σ and τ act like φ_p (Frobenius substitution) since π has ordinal 1, and $\tau = \sigma^r$ shows that $\varphi_p = \varphi_p^r$ on this field. Since φ_p has period m on this field we get $r \equiv 1 \pmod{m}$ for all m. But this implies $r = 1$.

A unit u_p is the quotient of $\pi = pu_p$ and p (both of ordinal 1). We have therefore also $\sigma_p(u_p) = (u_p, \mathbb{Q}_p)$ and the general statement is now an immediate consequence. □

3. Reciprocity Law

Let k be a global field. Let K/k be a normal extension with group G, degree n. Let $\mathfrak{p}$ be a prime of k. The groups $K_\mathfrak{P}$ are isomorphic for all $\mathfrak{P} \mid \mathfrak{p}$ and so are the decomposition groups $G_\mathfrak{P}$. For each $\mathfrak{p}$ select one of the primes $\mathfrak{P} \mid \mathfrak{p}$

[1] Since these notes first appeared there have been two local approaches found. The first was by Dwork [**7**], see also [**21**, Ch. XIII, Sect. 5]. The second was by Lubin and Tate [**17**], see also [**12**] and [**5**, Ch. VI, Sect. 3].

[2] Completion with respect to the subgroups of finite index: $\overline{\mathbb{Z}} = \varprojlim_m \mathbb{Z}/m\mathbb{Z}$.

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and put $K_{\mathfrak{P}} = K_{\mathfrak{p}}$, $G_{\mathfrak{P}} = G_{\mathfrak{p}}$. $\mathcal{H}^2(G, J_K)$ is a direct sum of semilocal components $\mathcal{H}^2\big(G, \prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}\big)$ and each one of these is isomorphic to any of its local constituents:

$$\mathcal{H}^2\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} K_{\mathfrak{P}}\Big) \simeq \mathcal{H}^2(G_{\mathfrak{p}}, K_{\mathfrak{p}}).$$

Hence $\mathcal{H}^2(G, J_K)$ may be viewed as a direct sum of the local cohomology groups $\mathcal{H}^2(G_{\mathfrak{p}}, K_{\mathfrak{p}})$, by assigning to each cocycle class $c \in \mathcal{H}^2(G, J_K)$ local components $c_{\mathfrak{p}} \in \mathcal{H}^2(G_{\mathfrak{p}}, K_{\mathfrak{p}})$ which determine c completely. We have $c_{\mathfrak{p}} = 1$ for almost all $\mathfrak{p}$.

We recall how these local components behave under inflation, restriction, and verlagerung.[3]

Inflation. Let $L \supset K \supset k$ be normal over k. Let $\mathfrak{P}' \mid \mathfrak{P}$ be a prime of L. A local component of $\operatorname{Inf}_L c$ is the local inflation of a local component $c_{\mathfrak{p}}$ to $L_{\mathfrak{P}'}$:

$$(\operatorname{Inf}_L c)_{\mathfrak{p}} = \operatorname{Inf}_{L_{\mathfrak{P}'}}(c_{\mathfrak{p}}). \tag{1}$$

To discuss the restriction and verlagerung, we suppose that H is a subgroup of G and that E is its fixed field. Let $\mathfrak{q} \mid \mathfrak{p}$ in E.

Restriction. Let $c \in \mathcal{H}^2(G, J_K)$. The local $\mathfrak{q}$ component of $\operatorname{Res}_E c$ is obtained by the local restriction of $c_{\mathfrak{p}}$ to $E_{\mathfrak{q}}$:

$$(\operatorname{Res}_E c)_{\mathfrak{q}} = \operatorname{Res}_{E_{\mathfrak{q}}} c_{\mathfrak{p}}. \tag{2}$$

Verlagerung. Let $c \in \mathcal{H}^2(H, J_K)$. A local $\mathfrak{p}$ component of $V_{E/k}c$ is the product of the local verlagerungs of the components $c_{\mathfrak{q}}$:

$$(V_{E/k}c)_{\mathfrak{p}} = \prod_{\mathfrak{q}|\mathfrak{p}} (V_{E_{\mathfrak{q}}/k_{\mathfrak{p}}} c_{\mathfrak{q}}). \tag{3}$$

In the local class field theory we have attached to each element $c_{\mathfrak{p}} \in \mathcal{H}^2(G_{\mathfrak{p}}, K_{\mathfrak{p}})$ a local invariant, $\operatorname{inv}_{\mathfrak{p}} c_{\mathfrak{p}}$. These invariants are rational mod 1. We know that $\operatorname{inv}_{\mathfrak{p}} c_{\mathfrak{p}}$ completely characterizes $c_{\mathfrak{p}}$. In particular, $c_{\mathfrak{p}} = 1$ if and only if $\operatorname{inv}_{\mathfrak{p}} c_{\mathfrak{p}} = 0$.

We shall now assign invariants to global elements $c \in \mathcal{H}^2(G, J_K)$. Namely we define $\operatorname{inv}_{\mathfrak{p}} c = \operatorname{inv}_{\mathfrak{p}} c_{\mathfrak{p}}$ and let

$$\operatorname{inv} c = \sum_{\mathfrak{p}} \operatorname{inv}_{\mathfrak{p}} c.$$

This sum is finite because $c_{\mathfrak{p}} = 1$ at almost all $\mathfrak{p}$, and hence $\operatorname{inv}_{\mathfrak{p}} c = 0$ at almost all $\mathfrak{p}$.

We have trivially

$$\operatorname{inv} c_1 c_2 = \operatorname{inv} c_1 + \operatorname{inv} c_2.$$

The map

$$c \to \operatorname{inv} c$$

is a homomorphism of $\mathcal{H}^2(G, J_K)$ into the rationals mod 1. We note that $\operatorname{inv} c$ does not characterize c and that $\operatorname{inv} c = 0$ does not imply that $c = 1$. (in contrast with the local set up).

It is convenient to discuss immediately the formalism of these invariants, i.e. how they behave under inflation, restriction, and verlagerung. We note that the invariants we have assigned are functions of K and k. When necessary we shall write $\operatorname{inv}_{K/k}$ to emphasize their dependence on particular fields.

[3] *Verlagerung* is the term used here for the map which today is called corestriction.

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The next three lemmas are immediate consequences of the local class field theory.

LEMMA 1. *Let $L \supset K \supset k$ be normal over k and let $c \in \mathcal{H}^2(G, J_K)$. Then the invariant does not change under inflation to L, i.e.* $\operatorname{inv}_{\mathfrak{p}} \operatorname{Inf}_L c = \operatorname{inv}_{\mathfrak{p}} c$, *and*

$$\operatorname{inv}_{L/k} \operatorname{Inf}_L c = \operatorname{inv}_{K/k} c.$$

PROOF. From the local class field theory we know that the local invariant of $c_{\mathfrak{p}}$ does not change under local inflation to $L_{\mathfrak{P}'}$. The lemma follows immediately from (1). □

REMARK. Lemma 1 shows that our assignment of invariants is consistent with the isomorphic imbedding of $\mathcal{H}^2(G, J_K)$ in $\mathcal{H}^2(\mathfrak{G}, J_\Omega)$. In other words, let K_1/k and K_2/k be two normal extensions with groups G_1 and G_2. Let $c_1 \in \mathcal{H}^2(G_1, J_{K_1})$ and $c_2 \in \mathcal{H}^2(G_2, J_{K_2})$. Let $L = K_1K_2$. If

$$\operatorname{Inf}_L c_1 = \operatorname{Inf}_L c_2$$

then $\operatorname{inv}_{K_1/k} c_1 = \operatorname{inv}_{K_2/k} c_2$. Hence $\operatorname{inv}_{K/k}$ is not a function of K but only of k. We may therefore write legitimately inv_k instead of $\operatorname{inv}_{K/k}$. In particular, given $c \in \mathcal{H}^2(\mathfrak{G}, J_\Omega)$. If $\operatorname{Res}_K c = 1$ we know that c can be deflated to a cocycle class of $\mathcal{H}^2(G, J_K)$ (or equivalently that c is the inflation of a cocycle class of $\mathcal{H}^2(G, J_K)$). To determine the invariant of a cocycle class c, it suffices therefore to determine it in any one of the layers on which it splits.

For the next two lemmas, we let H be a subgroup of G and E the fixed field. Let $\mathfrak{q} \mid \mathfrak{p}$ in E and let $n_{\mathfrak{q}} = [E_{\mathfrak{q}} : k_{\mathfrak{p}}]$ be the local degree of E.

LEMMA 2. *Let $c \in \mathcal{H}^2(\mathfrak{G}, J_K)$. Then* $\operatorname{inv}_{\mathfrak{q}} \operatorname{Res}_E c = n_{\mathfrak{q}} \operatorname{inv}_{\mathfrak{p}} c$, *and*

$$\operatorname{inv}_E \operatorname{Res}_E c = [E : k] \operatorname{inv}_k c.$$

PROOF. From the local class field theory and (2), the statement concerning the local restriction is clear. We recall the elementary fact that $\sum_{\mathfrak{q}|\mathfrak{p}} n_{\mathfrak{q}} = [E : k]$ and therefore

$$\sum_{\mathfrak{q}|\mathfrak{p}} (n_{\mathfrak{q}} \operatorname{inv}_{\mathfrak{p}} c) = \Big(\sum_{\mathfrak{q}|\mathfrak{p}} n_{\mathfrak{q}}\Big) \operatorname{inv}_{\mathfrak{p}} c = [E : k] \operatorname{inv}_{\mathfrak{p}} c.$$

Summing over $\mathfrak{p}$ gives the desired result. □

LEMMA 3. *Let $c \in \mathcal{H}^2(H, J_K)$. Then the verlagerung preserves invariants, i.e.*

$$\operatorname{inv}_k V_{E/k} c = \operatorname{inv}_E c.$$

PROOF. From the local class field theory, we know that the local verlagerung preserves the local invariant, i.e.

$$\operatorname{inv}_{\mathfrak{p}} V_{\mathfrak{q}} c_{\mathfrak{q}} = \operatorname{inv}_{\mathfrak{q}} c_{\mathfrak{q}}.$$

From (3) we obtain

$$\operatorname{inv}_{\mathfrak{p}} (V_{E/k} c)_{\mathfrak{p}} = \sum_{\mathfrak{q}|\mathfrak{p}} \operatorname{inv}_{\mathfrak{p}} (V_{\mathfrak{q}} c_{\mathfrak{q}}) = \sum_{\mathfrak{q}|\mathfrak{p}} \operatorname{inv}_{\mathfrak{q}} c_{\mathfrak{q}}.$$

Taking the sum over $\mathfrak{p}$ proves the lemma. □

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Our purpose is now to determine the kernel of the homomorphism $c \to \operatorname{inv} c$, i.e. to characterize the elements $c \in \mathcal{H}^2(G, J_K)$ which have invariant 0.

To simplify the notation we omit G in writing cohomology groups.

We recall the exact sequence

$$\mathcal{H}^1(C_K) \to \mathcal{H}^2(K^*) \to \mathcal{H}^2(J_K) \to \mathcal{H}^2(C_K).$$

It was a consequence of the first two inequalities (VI, 1, Theorem 1) that $\mathcal{H}^1(C_K)$ is trivial. It follows that $\mathcal{H}^2(K^*)$ is imbedded isomorphically into $\mathcal{H}^2(J_K)$ by inclusion, and that a 2-cocycle in K^* splits globally if and only if it splits locally everywhere.

In view of this isomorphic imbedding we may view elements of $\mathcal{H}^2(K^*)$ as idèle cocycle classes, and may therefore assign local and global invariants to them. In other words, for $c \in \mathcal{H}^2(K^*)$ we can define naturally $\operatorname{inv}_\mathfrak{p} c$ and $\operatorname{inv} c$.

We know that the inclusion mapping i commutes with the inflation, restriction, and verlagerung. Hence the properties of Lemmas 1, 2, and 3 are naturally valid for elements of $\mathcal{H}^2(K^*)$, and the remark made following Lemma 1 is equally valid for elements of $\mathcal{H}^2(\mathfrak{q}, \Omega^*)$. In order to find the invariant of $c \in \mathcal{H}^2(\mathfrak{q}, \Omega^*)$, it suffices to find its invariant in any one of the layers in which it splits.

In the rest of this section we shall identify $\mathcal{H}^2(K^*)$ with the image (under inclusion) in $\mathcal{H}^2(J_K)$.

With this identification, the fundamental result (proved below in Theorem 8) may be expressed as follows: Let $c \in \mathcal{H}^2(J_K)$. Then $\operatorname{inv} c = 0$ if and only if $c \in \mathcal{H}^2(K^*)$.

We begin by examining the invariants more closely in cyclic extensions.

Let K/k be cyclic of degree n, with group G. Let $\widehat{G}$ be the character group of G, generated by χ. The characters take on their values in the rationals mod 1. According to the cyclic theory, any element $c \in \mathcal{H}^2(J_K)$ is of the form $c = \varkappa(\mathfrak{a}) \cup \delta\chi$ where $\mathfrak{a} \in J_K$.

If $\mathfrak{a}$ has $\mathfrak{p}$-component $\mathfrak{a}_\mathfrak{p}$ the local component of c is

$$c_\mathfrak{p} = \operatorname{Pr}_{K_\mathfrak{p}} \operatorname{Res}_{G_\mathfrak{p}} \varkappa(\mathfrak{a}) \cup \delta\chi = \varkappa(\mathfrak{a}_\mathfrak{p}) \cup \delta\chi_\mathfrak{p}$$

where $\chi_\mathfrak{p} = \operatorname{Res}_{G_\mathfrak{p}} \chi$. If $(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p})$ is the local norm residue symbol, then

$$\begin{aligned} \operatorname{inv}_\mathfrak{p} \varkappa(\mathfrak{a}) \cup \delta\chi &= \operatorname{inv}_\mathfrak{p} \varkappa(\mathfrak{a}_\mathfrak{p}) \cup \delta\chi_\mathfrak{p} \\ &= \chi_\mathfrak{p}((\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p})) \\ &= \chi((\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p})), \end{aligned}$$

if we interpret $(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p})$ also as an element of G.

To simplify the notation, we shall omit one parenthesis and write for instance $\chi_\mathfrak{p}(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p})$ instead of $\chi_\mathfrak{p}((\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}))$.

We note that $(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = 1$ at all $\mathfrak{p}$ where $\mathfrak{a}_\mathfrak{p}$ is a unit and $\mathfrak{p}$ is unramified, because $\mathfrak{a}_\mathfrak{p}$ is then a local norm.

Define

$$(\mathfrak{a}, K/k) = \prod_\mathfrak{p} (\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}).$$

This product is finite because $(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = 1$ at almost all $\mathfrak{p}$ and it is defined because G is abelian so that the order of the factors does not matter.

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We note parenthetically that if $k = \mathbb{Q}$ is the rationals and $K/\mathbb{Q}$ is cyclotomic, we have shown in the preceding section that $(\mathfrak{a}, K/k) = \sigma(\mathfrak{a})$. In particular, $(\alpha, K/\mathbb{Q}) = 1$ in this case.

We have trivially

$$(\mathfrak{a}\mathfrak{b}, K/k) = (\mathfrak{a}, K/k)(\mathfrak{b}, K/k).$$

The following identity is obvious:

$$\sum_{\mathfrak{p}} \operatorname{inv}_{\mathfrak{p}} \varkappa(\mathfrak{a}) \cup \delta\chi = \sum_{\mathfrak{p}} \chi(\mathfrak{a}_{\mathfrak{p}}, K_{\mathfrak{p}}/k_{\mathfrak{p}}) = \chi\Big(\prod_{\mathfrak{p}} (\mathfrak{a}_{\mathfrak{p}}, K_{\mathfrak{p}}/k_{\mathfrak{p}})\Big),$$

and consequently

$$\operatorname{inv} \varkappa(\mathfrak{a}) \cup \delta\chi = \chi(\mathfrak{a}, K/k).$$

THEOREM 6. *Let K/k be a normal extension with group G and $c \in \mathcal{H}^2(G, K^*)$. Then* $\operatorname{inv} c = 0$.

PROOF. The following argument shows that in number fields, it suffices to prove the theorem in case $k = \mathbb{Q}$ is the rationals.

Let $L \supset K$ be normal over $\mathbb{Q}$. Then Inf_L c is an element of $\mathcal{H}^2(L/k)$ and by Lemma 1,

$$\operatorname{inv}_k \operatorname{Inf}_L c = \operatorname{inv}_k c.$$

According to Lemma 3, taking the verlagerung does not change the invariant. The cocycle class $V_{k/\mathbb{Q}} \operatorname{Inf}_L c$ is an element of $\mathcal{H}^2(L/\mathbb{Q})$ and we have

$$\operatorname{inv}_k c = \operatorname{inv}_{\mathbb{Q}} V_{k/\mathbb{Q}} \operatorname{Inf}_L c.$$

If the theorem is proved for normal extensions of the rationals, it will follow that $\operatorname{inv}_{\mathbb{Q}} V_{k/\mathbb{Q}} \operatorname{Inf}_L c = 0$, and hence $\operatorname{inv}_k c = 0$. □

We shall now prove the theorem in special cases.

Case 1. $k = \mathbb{Q}$ is the rationals, and $K/\mathbb{Q}$ is a cyclic, cyclotomic extension.

Any cocycle class $c \in \mathcal{H}^2(K)$ is of type $\varkappa(\alpha) \cup \delta\chi$ where $\alpha \in \mathbb{Q}$. We know that

$$\operatorname{inv} \varkappa(\alpha) \cup \delta\chi = \chi(\alpha, K/\mathbb{Q})$$

and we have already remarked that $(\alpha, K/\mathbb{Q}) = 1$. Hence $\operatorname{inv} \chi(\alpha) \cup \delta\chi = 0$, as was to be shown.

Case 2. k is a function field and K is a constant field extension (hence cyclic cyclotomic).

K/k is cyclic and unramified. Let $\mathfrak{a} \in k_{\mathfrak{p}}$. Then

$$(\mathfrak{a}_{\mathfrak{p}}, K_{\mathfrak{p}}/k_{\mathfrak{p}}) = \varphi_{\mathfrak{p}}^{\nu_{\mathfrak{p}}}, \qquad \nu_{\mathfrak{p}} = \operatorname{ord}_{\mathfrak{p}} \mathfrak{a},$$

where $\varphi_{\mathfrak{p}}$ is the Frobenius Substitution and $(\mathfrak{a}_{\mathfrak{p}}, K_{\mathfrak{p}}/k_{\mathfrak{p}})$ acts on the residue class field locally. If q is the number of elements of k_0 then the residue class field of $\mathfrak{p}$ has $q^{f_{\mathfrak{p}}}$ elements. The effect of $\varphi_{\mathfrak{p}}^{\nu_{\mathfrak{p}}}$ on the residue class field is therefore

$$\varphi_{\mathfrak{p}}^{\nu_{\mathfrak{p}}} : x \to \left(x^{q^{f_{\mathfrak{p}}}}\right)^{\nu_{\mathfrak{p}}} = x^{q^{f_{\mathfrak{p}}\nu_{\mathfrak{p}}}},$$

and it induces the same effect on the constant field of K.

Any cocycle class $c \in \mathcal{H}^2(K^*)$ is of type $\varkappa(\alpha) \cup \delta\chi$, $\alpha \in k$. We look at α as an idèle, $\alpha = (\dots, \alpha, \alpha, \alpha, \dots)$, and let $\nu_{\mathfrak{p}} = \operatorname{ord}_{\mathfrak{p}} \alpha$. The product formula gives

$$\sum_{\mathfrak{p}} f_{\mathfrak{p}} \nu_{\mathfrak{p}} = 0.$$

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Since

$$(\alpha, K/k) = \prod_{\mathfrak{p}} (\alpha, K_{\mathfrak{p}}/k_{\mathfrak{p}})$$

we see that the effect of $(\alpha, K/k)$ is identity, namely

$$(\alpha, K/k)\colon x \to x^{q^{\sum f_{\mathfrak{p}} \nu_{\mathfrak{p}}}} = x.$$

Hence $\chi(\alpha, K/k) = 0$ and $\operatorname{inv}(\chi(\alpha) \cup \delta\chi) = 0$, as was to be shown.

We now treat the global cases together again, and suppose that k is either a *function field* or the *rational numbers*. K/k is an arbitrary normal extension. Let $c \in \mathcal{H}^2(K)$. We use K' to denote cyclic cyclotomic extensions of k, and let L be the compositum, $L = KK'$.

In order to prove our theorem, it suffices to prove the existence of an extension K' such that

$$\operatorname{Res}_{K'} \operatorname{Inf}_L c = 1.$$

Indeed, K' is a splitting field for $\operatorname{Inf}_L c$. Using the remark following Lemma 1 we see that $\operatorname{Inf}_L c$ is the inflation to L of a cocycle class $c' \in \mathcal{H}^2(K'^*)$; coupling this with the results of Cases 1 and 2, we see that $\operatorname{inv}_k c = 0$.

The existence of K' will be a consequence of the following elementary existence statement.

Proposition. *Given a finite set of primes $\mathfrak{p}$, and integers $r_{\mathfrak{p}}$ (with the obvious restriction at $\mathfrak{p}_\infty$ that $r_{\mathfrak{p}_\infty} = 1$ or 2 if $k_{\mathfrak{p}_\infty}$ is real, and 1 otherwise), there exists a cyclic cyclotomic extension K'/k such that $[K'_{\mathfrak{q}} : k_{\mathfrak{p}}]$ is divisible by $r_{\mathfrak{p}}$ for each prime $\mathfrak{q} \mid \mathfrak{p}$ in K'.*

Before proving the proposition, we indicate how our theorem results from it.

$\operatorname{inv}_{\mathfrak{p}} c$ will be $\neq 0$ only at a finite number of primes $\mathfrak{p}$, and will have denominators $r_{\mathfrak{p}}$ at these primes. Let K' be a cyclic cyclotomic extension such that $[K\mathfrak{q}' : k_{\mathfrak{p}}]$ is divisible by $r_{\mathfrak{p}}$ for each prime $\mathfrak{q} \mid \mathfrak{p}$. Actually, $[K'_{\mathfrak{q}} : k_{\mathfrak{p}}]$ is the same for all $\mathfrak{q} \mid \mathfrak{p}$. Denote this number by $m_{\mathfrak{p}}$. Abbreviate $\operatorname{Inf}_L c$ by c_L. By Lemma 2,

$$\operatorname{inv}_{\mathfrak{q}} \operatorname{Res}_{K'} c_L = m_{\mathfrak{p}} \operatorname{inv}_{\mathfrak{p}} c_L.$$

By Lemma 1, $\operatorname{inv}_{\mathfrak{p}} c_L = \operatorname{inv}_{\mathfrak{p}} c$ and hence

$$\operatorname{inv}_{\mathfrak{q}} \operatorname{Res}_{K'} c_L = 0$$

for all primes $\mathfrak{q}$ of K'. This means that $\operatorname{Res}_{K'} c_L$ splits locally at all primes $\mathfrak{q}$ of K'. By VI, Section 1, Theorem 2 we conclude that $\operatorname{Res}_{K'} c_L = 1$. This is precisely what we wanted to achieve.

Proof of Proposition. In function fields we may select for K' a suitably large constant field extension, with highly divisible local degrees at the finite set of primes.

In number fields, let ℓ be a prime, and let ℓ^ν be the l.c.m. of all powers of ℓ dividing the numbers $r_{\mathfrak{p}}$. (We need consider only the finite set of primes ℓ for which $\ell^\nu \neq 1$.) It suffices to prove that there exists an extension K'_ℓ of ℓ power degree such that its local degrees at the finite set of primes $\mathfrak{p}$ is divisible by ℓ^ν. Indeed, the compositum K' of the fields K'_ℓ will be cyclic (because the degrees of its components are relatively prime), and its local degrees will be the product of the local degrees of K'_ℓ. Hence K' will satisfy the requirements of our Proposition.

We let ζ_{ℓ^n} denote a primitive ℓ^n-th root of unity.

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Suppose ℓ is odd. The field $\mathbb{Q}(\zeta_{\ell^n})/\mathbb{Q}$ is a compositum of the field $\mathbb{Q}(\zeta_\ell)$ and a cyclic field K'_n of degree ℓ^{n-1}, and $\mathbb{Q}(\zeta_{\ell^n})$ has degree $\leqslant (\ell-1)$ over K'_n. At a finite prime p this implies that $\mathbb{Q}_p(\zeta_{\ell^n})$ has degree $\leqslant (\ell-1)$ over the completion $K'_n\mathbb{Q}_p$ of K'_n. By a local result (left to reader) it follows that the degree $[K'_n\mathbb{Q}_p : \mathbb{Q}_p] \to \infty$ as $n \to \infty$. Since this degree is a power of ℓ, we can find an n such that this degree is divisible by ℓ^ν at a given finite set of finite primes, as desired.

Suppose $\ell = 2$. Let $\zeta = \zeta_{2^n}$ with $n \geqslant 3$. Let $\xi = \zeta - \zeta^{-1}$. Let $K'_n = \mathbb{Q}(\xi)$. We contend that $K'_n/\mathbb{Q}$ is cyclic. The automorphisms of $\mathbb{Q}(\zeta)/\mathbb{Q}$ are given by

$$\sigma_\mu \colon \zeta \to \zeta^\mu \qquad \mu \text{ odd.}$$

We have

$$\sigma_\mu \xi = \zeta^\mu - \zeta^{-\mu}.$$

We note that $\zeta^{2^{n-1}} = -1$ and one verifies directly that

$$\sigma_{-\mu+2^{n-1}}\xi = \sigma_\mu \xi.$$

One of the numbers μ or $-\mu + 2^{n-1}$ is $\equiv 1 \pmod 4$. This implies that the automorphisms of $\mathbb{Q}(\xi)/\mathbb{Q}$ are all induced by automorphisms σ_μ where $\mu \equiv 1 \pmod 4$. This group is cyclic and the Galois group of $\mathbb{Q}(\xi)/\mathbb{Q}$ is a factor group of it, whence cyclic.

Furthermore, $\sigma_{-1}\xi = -\xi$. Hence $\mathbb{Q}(\xi)$ is not real, and its local degree at a real infinite prime is 2, as desired.

Let p be a finite prime. We know that $[\mathbb{Q}_p(\zeta) : \mathbb{Q}_p]$ is a power of 2. Since $[\mathbb{Q}(\zeta) : \mathbb{Q}(\xi)] = 2$ it follows that $[\mathbb{Q}_p(\zeta) : \mathbb{Q}_p(\xi)] \leqslant 2$. Since the degree of $\mathbb{Q}_p(\zeta)/\mathbb{Q}_p$ increases as n increases, it follows that the degree of $\mathbb{Q}_p(\xi)/\mathbb{Q}_p$ must also increase. For large n, it can therefore be made divisible by any given power of 2.

This concludes the proof of the Proposition. □

As an immediate consequence of Theorem 6 we obtain a refinement of VI, Section 1, Theorem 2.

Corollary. *Let K/k be normal, and let $c \in \mathcal{H}^2(K^*)$. If $c_\mathfrak{p} = 1$ at all primes but one, then $c_\mathfrak{p} = 1$ at that prime also, and hence $c = 1$.*

Proof. Obvious, because $\sum \operatorname{inv}_\mathfrak{p} c = 0$. □

This corollary is a prototype of statements which follow from product (or sum) formulas. We shall meet another one of the same kind later on in the theory.

We are now approaching the end of our journey, and assign invariants to cocycle classes of idèle classes.

We recall the map

$$j \colon \mathcal{H}^2(J_K) \to \mathcal{H}^2(C_K).$$

We have defined invariants for elements $c \in \mathcal{H}^2(J_K)$, and the natural way of proceeding would be to assign jc the same invariant as c. However, we encounter the following difficulty: The map j is not always onto. In order to assign invariants to elements of $\mathcal{H}^2(C_K)$ we shall therefore proceed as follows.

We begin by making three auxiliary remarks. The first one is essentially a corollary to Theorem 6.

Remark 1. Let L/k be normal with group G_L. Let $\bar{c} \in \mathcal{H}^2(G_L, C_L)$, and suppose $\bar{c} = jc = jd$ with c, $d \in \mathcal{H}^2(G_L, J_L)$. Then $\operatorname{inv}_k c = \operatorname{inv}_k d$. *Proof*: We

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have $(j(cd^{-1})) = 1$ and hence $cd^{-1} \in \mathcal{H}^2(G_L, L)$. By Theorem 6, it follows that $\operatorname{inv}_k(cd^{-1}) = 0$, whence $\operatorname{inv}_k c = \operatorname{inv}_k d$.

REMARK 2. Let $\bar{c} \in j\mathcal{H}^2(J_K)$, say $\bar{c} = jc$. Let $L \supset K$ be normal over k. Since inflation commutes with j, we have

$$\operatorname{Inf}_L \bar{c} = \operatorname{Inf}_L(jc) = j \operatorname{Inf}_L c.$$

We see therefore that if $\bar{c}$ is in the image of j in any layer, then its inflation to any bigger layer is also in the image of j.

REMARK 3. Let $\bar{c} \in \mathcal{H}^2(C_K)$. Let $L_1 \supset K$ and $L_2 \supset K$ be normal with groups G_1 and G_2 over k. Suppose that

$$\operatorname{Inf}_{L_1} \bar{c} = jc_1 \quad \text{and} \quad \operatorname{Inf}_{L_2} \bar{c} = jc_2$$

with $c_1 \in \mathcal{H}^2(G_1, J_{L_1})$ and $c_2 \in \mathcal{H}^2(G_2, J_{L_2})$. Let $L = L_1L_2$ be the compositum. By the transitivity of inflation and by Remark 2 we get

$$\operatorname{Inf}_L \bar{c} = j \operatorname{Inf}_L c_1 = j \operatorname{Inf}_L c_2.$$

It follows from Remark 1 that $\operatorname{inv}_k \operatorname{Inf}_L c_1 = \operatorname{inv}_k \operatorname{Inf}_L c_2$, and hence by Lemma 1, $\operatorname{inv}_k c_1 = \operatorname{inv}_k c_2$

Let K/k be a given normal extension. An element $\bar{c} \in \mathcal{H}^2(C_K)$ will be called *regular* if it has the following property: There exists a normal extension $L \supset K$ such that

$$\operatorname{Inf}_L \bar{c} = jc$$

for some cocycle class $c \in \mathcal{H}^2(G_L, J_L)$. We denote the subset of regular elements by $\bar{\mathcal{H}}^2$.

We first note that $\bar{\mathcal{H}}^2$ is a group. It suffices to prove that it is closed under multiplication. Let $\bar{c}, \bar{d} \in \bar{\mathcal{H}}^2$. There exists fields L_1 and $L_2 \supset K$ such that $\operatorname{Inf}_{L_1} \bar{c}$ is in the image of j, and $\operatorname{Inf}_{L_2} \bar{d}$ is in the image of j. Let $L = L_1L_2$. By Remark 2, it follows that $\operatorname{Inf}_L \bar{c}$ and $\operatorname{Inf}_L \bar{d}$ is in the image of j, and since

$$\operatorname{Inf}_L(\bar{c}\bar{d}) = \operatorname{Inf}_L \bar{c} \operatorname{Inf}_L \bar{d}$$

this proves that $\bar{c}\bar{d}$ is regular. It proves also that given two regular elements, there exists a common extension L in which their inflation is in the image of j.

We shall now define invariants for regular elements. It will be proved at the very end that all elements are regular.

Let $\bar{c} \in \bar{\mathcal{H}}^2(C_K)$ be regular, and let $L \supset K$ be a normal field such that

$$\operatorname{Inf}_L \bar{c} = jc$$

with $c \in \bar{\mathcal{H}}^2(G_L, J_L)$. We define $\operatorname{inv}_k \bar{c}$ to be $\operatorname{inv}_k c$.

From Remark 1, it follows that our definition is independent of the choice of representative c in the field L. From Remark 3, it follows that our definition is independent of the choice of auxiliary field L. The invariant depends only on the field k, and is well defined.

Given two regular elements $\bar{c}$ and $\bar{d}$, we can find a field L such that $\operatorname{Inf}_L \bar{c} = jc$ and $\operatorname{Inf}_L \bar{d} = jd$. We then have $\operatorname{Inf}_L(\bar{c}\bar{d}) = j(cd)$ and $\operatorname{inv}_k \bar{c}\bar{d} = \operatorname{inv}_k \bar{c} + \operatorname{inv}_k \bar{d}$. The map

$$\bar{c} \to \operatorname{inv}_k \bar{c}$$

is therefore a homomorphism of the regular elements $\bar{\mathcal{H}}^2(C_K)$ into the rationals mod 1. In particular, if $\operatorname{inv}_k \bar{c} \neq 0$ then $\bar{c} \neq 1$.

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LEMMA 4. *Let H be a subgroup of G and let E be the fixed field. Let $\bar{c}$ be a regular element of $\bar{\mathcal{H}}^2(C_K)$. Then $\operatorname{Res}_E \bar{c}$ is regular, and*

$$\operatorname{inv}_E \operatorname{Res}_E \bar{c} = [E : k] \operatorname{inv}_k \bar{c}.$$

PROOF. Let $L \supset K$ and suppose $\operatorname{Inf}_L \bar{c} = jc$, with $c \in \mathcal{H}^2(G_L, J_L)$. By the commutativity of Inf, Res, and j we have

$$\operatorname{Inf}_L \operatorname{Res}_E \bar{c} = \operatorname{Res}_E \operatorname{Inf}_L \bar{c} = \operatorname{Res}_E jc = j \operatorname{Res}_E c.$$

This means that $\operatorname{Res}_E \bar{c}$ can be assigned an invariant. By Lemma 2, we have

$$\operatorname{inv}_E \operatorname{Res}_E c = [E : k] \operatorname{inv}_k c$$

and this proves our lemma, because we have already shown that the assignment of invariants is independent of auxiliary fields and selected representatives. □

Our final task is now to prove that every cocycle class $\bar{c} \in \mathcal{H}^2(C_K)$ is regular. At the same time we shall prove that the map

$$\bar{c} \to \operatorname{inv} \bar{c}$$

is an isomorphism: $C_{K/k} \xrightarrow{\sim} \frac{1}{n}\mathbb{Z}/\mathbb{Z}$, where $n = [K : k]$. We shall first do this in cyclic fields, and need a lemma.

LEMMA 5. *Let K/k be cyclic of degree n. There exists a regular cocycle class $\bar{c} \in \bar{\mathcal{H}}^2(C_K)$ having invariant $1/n$.*

PROOF. It suffices to show that if $\ell \mid n$ is a prime and $\ell^r \mid n$ exactly, then there exists a cocycle class $c \in \mathcal{H}^2(J_K)$ having invariant $1/\ell^r$. Indeed, we let $\bar{c} = jc$. Then $\operatorname{inv}_k \bar{c} = 1/\ell^r$ by definition, and taking a suitable power product of such $\bar{c}$ for all primes ℓ gives a cocycle class of invariant $1/n$.

K/k contains a subfield K_0/k of degree ℓ^r. According to the theorem on the existence of primes, V, Section 4, Theorem 3, there exists a prime $\mathfrak{p}$ in k which remains prime in K_0: the local degree at $\mathfrak{p}$ is therefore ℓ^r. According to the local class field theory, there exists a local cocycle class $c_{\mathfrak{p}}$ having invariant $1/\ell^r$. We form the global cocycle class c having $\mathfrak{p}$-component $c_{\mathfrak{p}}$, and 1 at all other primes. Then $\operatorname{inv}_k c = 1/\ell^r$, and the invariant of its inflation to K remains the same. This proves our lemma. □

LEMMA 6. *Let K/k be* cyclic *of degree n. The map*

$$j \colon \mathcal{H}^2(J_K) \to \mathcal{H}^2(C_K)$$

is onto. In other words, every cocycle class of C_K is regular. Let $\bar{c} \in \mathcal{H}^2(C_K)$. Then $\operatorname{inv}_k \bar{c} = 0$ if and only if $\bar{c} = 1$. Hence $\mathcal{H}^2(C_K)$ is cyclic of degree n.

PROOF. According to Lemma 5 there exists a regular cocycle class $\bar{c} \in \bar{\mathcal{H}}^2(C_K)$ such that $\operatorname{inv}_k \bar{c} = 1/n$. The powers of $\bar{c}$ form a cyclic group of order at least n. Because of the second inequality and the fact that in the cyclic case, $\mathcal{H}^2(C_K) \simeq C_k/N_{K/k}C_K$, the group generated by $\bar{c}$ has order exactly n, and consists of all elements of $\mathcal{H}^2(C_K)$. If $\operatorname{inv}_k \bar{c}^\nu = 0$ it follows that $\bar{c}^\nu = 1$. This proves our lemma. □

We are now ready to generalize the above results to arbitrary normal extensions.

Let K/k be normal of degree n. We shall prove that all elements of $\mathcal{H}^2(C_K)$ are regular, and that $\mathcal{H}^2(C_K)$ is cyclic.

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Let K' be any cyclic field of degree n over k. (The existence of K' follows from the proposition concerning the existence of cyclic cyclotomic fields.) Let $L = KK'$ be the compositum. Let $\bar{c}' \in \mathcal{H}^2(C_{K'})$ have invariant $1/n$. Such a $\bar{c}'$ exists by Lemma 5. Then $\operatorname{Inf}_L \bar{c}'$ has also invariant $1/n$. It follows from Lemma 4 that $\operatorname{Res}_K \operatorname{Inf}_L \bar{c}'$ has invariant 0. But L/K is cyclic. By Lemma 6 this implies that $\operatorname{Res}_K \operatorname{Inf}_L \bar{c}' = 1$. This means that $\operatorname{Inf}_L \bar{c}'$ can be deflated to K, or in other words, that there exists $\bar{c} \in \mathcal{H}^2(C_K)$ such that

$$\operatorname{Inf}_L \bar{c}' = \operatorname{Inf}_L \bar{c}.$$

Since $\operatorname{Inf}_L \bar{c}$ is a cocycle class of invariant $1/n$, it follows that $\bar{c}$ is regular and has invariant $1/n$. The group generated by $\bar{c}$ is cyclic of order at least n. The second inequality shows that the order is exactly n, and that $\bar{\mathcal{H}}^2(C_K) = \mathcal{H}^2(C_K)$ is cyclic.

We have therefore shown that the map

$$\bar{c} \to \operatorname{inv}_k \bar{c}$$

is an isomorphism of $\mathcal{H}^2(C_K)$ into the rationals mod 1. If $\operatorname{inv}_k \bar{c} = 0$ then $\bar{c} = 1$.

We now see that invariants can be assigned to every element $\bar{c} \in \mathcal{H}^2(C_K)$. The invariant is independent of the field L to which c may be inflated, and this allows us to assign invariants to the elements of $\mathcal{H}^2(\mathfrak{G}, C_\Omega)$. Consequently the map

$$\bar{c} \to \operatorname{inv} \bar{c}$$

for $\bar{c} \in \mathcal{H}^2(\mathfrak{G}, C_\Omega)$ is an isomorphism onto the rationals mod 1.

It is onto, because of Lemma 5, and the existence of cyclic fields of any given degree n.

The lemma on restriction (Lemma 4) holds in general, because $\bar{\mathcal{H}}^2 = \mathcal{H}^2$. Hence we have proved the fundamental theorem:

THEOREM 7. *Let k be a global field, and Ω its separable algebraic closure, with group $\mathfrak{G}$. Then $(\mathfrak{G}, C_\Omega)$ is a class formation. The elements $\bar{c} \in \mathcal{H}^2(\mathfrak{G}, C_\Omega)$ can be assigned invariants which satisfy Axiom* 2 *of Ch.* XIV, 3 *according to the procedure described above, and the map $\bar{c} \to \operatorname{inv} \bar{c}$ is an isomorphism onto the rationals* mod 1.

THEOREM 8. *Let k be a global field. Let K/k be a normal extension of degree n. $\mathcal{H}^2(K^*)$ is characterized as the subgroup of elements $c \in \mathcal{H}^2(J_K)$ such that* $\operatorname{inv} c = 0$. *Furthermore, given local invariants $\lambda_\mathfrak{p}$ satisfying the conditions*

1. $n_\mathfrak{p}\lambda_\mathfrak{p} = 0$ ($n_\mathfrak{p}$ = *local degree*)
2. *almost all* $\lambda_\mathfrak{p} = 0$
3. $\sum_\mathfrak{p} \lambda_\mathfrak{p} = 0$,

there exists a cocycle class $c \in \mathcal{H}^2(K)$ such that $\operatorname{inv}_\mathfrak{p} c = \lambda_\mathfrak{p}$.

PROOF. Let $c \in \mathcal{H}^2(J_K)$, $\operatorname{inv} c = 0$. Let $\bar{c} = jc$ be the image of c in $\mathcal{H}^2(C_K)$. Then $\operatorname{inv} \bar{c} = 0$ and consequently $\bar{c} = 1$. Hence c is in the kernel of j, which is precisely $\mathcal{H}^2(K^*)$.

Let now $c_\mathfrak{p} \in \mathcal{H}^2(K_\mathfrak{p}^*)$ be local cocycle classes having the prescribed invariants $\lambda_\mathfrak{p}$. Let $c \in \mathcal{H}^2(J_K)$ have local components $c_\mathfrak{p}$. Then $\operatorname{inv} c = 0$, and by the preceding argument, $c \in \mathcal{H}^2(K^*)$, thereby completing the proof of our theorem. □

According to Theorem 7 we may now do class field theory in $(\mathfrak{G}, C_\Omega)$ and obtain a homomorphism

$$a \to (a, k)$$

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of C_k into a dense subgroup of $\mathfrak{G}/\mathfrak{G}'$. If $\mathfrak{a}$ is any idèle representing a we define $(\mathfrak{a}, k) = (a, k)$.

Locally we have also a class formation and we may form the symbol $(\mathfrak{a}_\mathfrak{p}, k_\mathfrak{p})$, which is an element of $\mathfrak{G}_\mathfrak{p}/\mathfrak{G}'_\mathfrak{p}$ where $\mathfrak{G}_\mathfrak{p}$ is the local group of $\Omega k_\mathfrak{p}/k_\mathfrak{p}$.

We investigate how the global symbol is related to the local one.

THEOREM 9. *Let K/k be a finite abelian extension. Let $a \in C_k$, and let $\mathfrak{a}$ be any representing idèle of a. Then*

$$(a, K/k) = \prod_\mathfrak{p} (\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}).$$

PROOF. The product is finite, because if $\mathfrak{p}$ is unramified and $\mathfrak{a}_\mathfrak{p}$ is a unit, $(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = 1$. To show the equality it suffices to prove that for every character χ of G (the Galois group of K/k) we have

$$\chi(a, K/k) = \chi\Big(\prod_\mathfrak{p} (\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p})\Big).$$

From the properties of the dual mapping we know that

$$\begin{aligned}(a, K/k) &= \operatorname{inv}(\varkappa(a) \cup \delta\chi)\\ &= \sum_\mathfrak{p} \operatorname{inv}_\mathfrak{p}(\varkappa(\mathfrak{a}) \cup \delta\chi)\\ &= \sum_\mathfrak{p} \operatorname{inv}_\mathfrak{p}(\varkappa(\mathfrak{a}_\mathfrak{p}) \cup \delta\chi_\mathfrak{p})\end{aligned}$$

where $\chi_\mathfrak{p} = \operatorname{Res}_{G_\mathfrak{p}} \chi$. From the definition of the local dual mapping it follows that

$$\operatorname{inv}_\mathfrak{p}(\varkappa(\mathfrak{a}_\mathfrak{p}) \cup \delta\chi_\mathfrak{p}) = \chi_\mathfrak{p}(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = \chi(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}).$$

Hence

$$\operatorname{inv}(\varkappa(a) \cup \delta\chi) = \sum_\mathfrak{p} \chi(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = \chi\Big(\prod_\mathfrak{p} (\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p})\Big)$$

as was to be shown. □

COROLLARY 1. *Let K/k be finite abelian. If an element $\alpha \in k$ is a local norm at all primes but one, then it is also a local norm at this prime.*

PROOF. This is an immediate consequence of the product formula for the norm residue symbol,

$$1 = (\alpha, K/k) = \prod_\mathfrak{p} (\alpha, K_\mathfrak{p}/k_\mathfrak{p}),$$

and of the properties of the local symbol. □

COROLLARY 2. *Let A_k/k be the maximal abelian extension, and let $\mathfrak{G}$ be its Galois group. Then*

$$(a, k) = \prod_\mathfrak{p} (\mathfrak{a}_\mathfrak{p}, k_\mathfrak{p}).$$

PROOF. The product is finite on every finite subfield of A_k, and hence converges in $\mathfrak{G}$. Since

$$(a, k) = \prod_\mathfrak{p} (\mathfrak{a}_\mathfrak{p}, k_\mathfrak{p})$$

on each finite subfield, this is also true on A_k, as contended. □

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COROLLARY 3. *The norm residue symbol gives a continuous map from C_k into $\mathfrak{G}$.*

PROOF. Let $\mathfrak{h}$ be a neighborhood of 1 in $\mathfrak{G}$, i.e. an open subgroup of finite index. Let K be its fixed field. Let $\mathfrak{a} \in J_k$. If $\mathfrak{p}$ is unramified in K, and $\mathfrak{a}_\mathfrak{p}$ is a unit then $(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = 1$. If $\mathfrak{p}$ is unramified, we have therefore $(U_\mathfrak{p}, k) \subset \mathfrak{h}$. If $\mathfrak{p}$ is ramified, it follows from the local theory that there exists a neighborhood $N_\mathfrak{p}$ of 1 such that $(N_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = 1$. Hence $(N_\mathfrak{p}, k) \subset \mathfrak{h}$. If we put $N = \prod_{\mathfrak{p}\ \mathrm{unram.}} U_\mathfrak{p} \times \prod_{\mathfrak{p}\ \mathrm{ram.}} N_\mathfrak{p}$ then N is a neighborhood of 1 in J_k and $(N, k) \subset \mathfrak{h}$. Since C_k has the factor topology, and $(k, k) = 1$, the corollary is proved. □

Let K/k be an abelian extension. Let $\mathfrak{a} \in J_k$. Then $(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = \sigma_\mathfrak{p}$ is an element of the local group $G_\mathfrak{p}$. If the idèle $\mathfrak{a}$ is a field element $\alpha \in k$, then $\prod_\mathfrak{p} \sigma_\mathfrak{p} = \prod_\mathfrak{p} (\alpha, K_\mathfrak{p}/k_\mathfrak{p}) = 1$. Conversely, given a set of local automorphisms $\sigma_\mathfrak{p} \in G_\mathfrak{p}$, we wish to determine when there exists $\alpha \in k$ such that $(\alpha, K_\mathfrak{p}/k_\mathfrak{p}) = \sigma_\mathfrak{p}$. It turns out that the obvious necessary conditions to be placed on the $\sigma_\mathfrak{p}$ are also sufficient.

THEOREM 10. *Let K/k be a finite abelian extension with group G. Given a set of automorphisms $\sigma_\mathfrak{p} \in G_\mathfrak{p}$ for each $\mathfrak{p}$ such that*

1. *almost all $\sigma_\mathfrak{p} = 1$*
2. $\prod_\mathfrak{p} \sigma_\mathfrak{p} = 1$

there exists $\alpha \in k$ such that $(\alpha, K_\mathfrak{p}/k_\mathfrak{p}) = \sigma_\mathfrak{p}$.

PROOF. By the local class field theory we can obviously find an idèle $\mathfrak{a} \in J_k$ such that $(\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = \sigma_\mathfrak{p}$. But $(\mathfrak{a}, K/k) = \prod_\mathfrak{p} \sigma_\mathfrak{p} = 1$ implies that $\mathfrak{a}$ lies in the kernel of the norm residue symbol, i.e. $\mathfrak{a} \in k^* N J_K$. We can therefore write $\mathfrak{a} = \alpha N \mathfrak{a}$ for some $\alpha \in k^*$, $\mathfrak{a} \in J_K$. Locally, $(N\mathfrak{a}_\mathfrak{p}, K_\mathfrak{p}/k_\mathfrak{p}) = 1$ and consequently $(\alpha, K_\mathfrak{p}/k_\mathfrak{p}) = \sigma_\mathfrak{p}$, as was to be shown. □

The elements α of k^* for which $(\alpha, K_\mathfrak{p}/k_\mathfrak{p}) = 1$ at all primes $\mathfrak{p}$ are $k^* \cap N J_K$, the elements which are local norms everywhere. The conditions under which these elements are also global norms will be discussed later in this chapter.

4. Higher Cohomology Groups in Global Fields

Let k be a global field, K/k a normal extension of degree n.

We consider the exact sequence

$$\mathcal{H}^2(J_K) \xrightarrow{j} \mathcal{H}^2(C_K) \xrightarrow{\delta} \mathcal{H}^3(K) \xrightarrow{i} \mathcal{H}^3(J_K).$$

The third cohomology group in layers of a class formation is trivial. From the local class field theory we know therefore that $\mathcal{H}^3(G_\mathfrak{p}, K_\mathfrak{p})$ is trivial for all primes $\mathfrak{p}$. Hence $\mathcal{H}^3(J_K)$ is also trivial. This means that

$$\delta \colon \mathcal{H}^2(C_K) \to \mathcal{H}^3(K^*)$$

is onto. $\mathcal{H}^2(C_K)$ is cyclic of order n, generated by the cocycle class $\bar{c}$ having invariant $1/n$. We see therefore that $\mathcal{H}^3(K^*)$ is cyclic, generated by $\delta\bar{c} = t$, the so-called Teichmüller Cocycle. It is easy to determine the order of $\mathcal{H}^3(K^*)$. The kernel, of δ consists precisely of the images of j, i.e. those idèle class cocycles which have an idèle representative.

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THEOREM 11. *Let m be the l.c.m. of all local degrees $n_{\mathfrak{p}} = [K_{\mathfrak{p}} : k_{\mathfrak{p}}]$. The image $j\mathcal{H}^2(J_K)$ in $\mathcal{H}^2(C_K)$ consists precisely of those elements $\bar{c}$ having invariant r/m, where r is an integer.*

PROOF. By selecting suitable components, it is clearly possible to find a cocycle class $c \in \mathcal{H}^2(J_K)$ having invariant $1/m$. Hence we get all possible invariants r/m as images of idèle cocycle classes. Conversely, it is obvious that these are the only invariants we can get from idèle cocycle classes. □

THEOREM 12. *Let K/k be a normal extension of degree n. Let $m =$ l.c.m. of all local degrees $n_{\mathfrak{p}}$. Then $\mathcal{H}^3(K)$ is cyclic of order n/m, generated by $\delta\bar{c}_{K/k}$ where $\bar{c}_{K/k}$ is the fundamental cocycle class of $\mathcal{H}^2(C_K)$.*

We now derive consequences concerning the inflation and restriction of 3-cocycles in global fields.

THEOREM 13. *Let K/k be normal with group G. Let H be a subgroup of G. Then every element of $\mathcal{H}^3(H, K^*)$ is the restriction of an element of $\mathcal{H}^3(G, K^*)$.*

PROOF. We have commutativity in the following diagram:

$$\begin{tikzcd}
\mathcal{H}^2(H, C_K) \arrow[r, "\delta"] & \mathcal{H}^3(H, K^*) \\
\mathcal{H}^2(G, C_K) \arrow[u, "\mathrm{Res}"] \arrow[r, "\delta"] & \mathcal{H}^3(G, K^*) \arrow[u, "\mathrm{Res}"']
\end{tikzcd}$$

Hence $t_{K/E} = \delta\bar{c}_{K/E} = \mathrm{Res}_E\, \delta\bar{c}_{K/k} = \mathrm{Res}_E\, t_{K/k}$. □

THEOREM 14. *Let K/k be normal, $t_{K/k}$ the Teichmüller cocycle. There exists a normal extension L/k, $L \supset K \supset k$, such that $\mathrm{Inf}_L\, t_{K/k}$ splits.*

PROOF. We have commutativity in the following diagram.

$$\begin{tikzcd}
\mathcal{H}^2(C_L) \arrow[r, "\delta"] & \mathcal{H}^3(L^*) \\
\mathcal{H}^2(C_K) \arrow[u, "\mathrm{Inf}"] \arrow[r, "\delta"] & \mathcal{H}^3(K^*) \arrow[u, "\mathrm{Inf}"']
\end{tikzcd}$$

$c_{K/k}$ has invariant $1/n$. All we need to do is select a field L having local degrees at least n. Then $\mathrm{Inf}_L\, c_{K/k}$ has also invariant $1/n$, and is now in the image of j. It is consequently in the kernel of δ, as was to be shown. □

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CHAPTER VIII

The Existence Theorem

1. Existence and Ramification Theorem

Let k be a global field. Let $C_k = C$, $C_k^0 = C_0$. Let A_k be the maximal abelian extension of k, and let $\mathfrak{G}$ be its Galois group over k. Let $\omega\colon C \to \mathfrak{G}$ be the mapping of the norm residue symbol.

We want to prove the Existence Theorem:

THEOREM 1. *Given an open subgroup of finite index B of C, there exists an abelian extension K/k such that $B = NC_K$. K is the fixed field of $\omega(B)$.*

It will be best to prove Theorem 1 in number fields and function fields separately, even though the methods used are quite similar. This will be done in Sections 2 and 3. We shall here simply make further comments on the characterization of abelian extensions of k by their norm groups.

First we have an important corollary.

COROLLARY. *There is a 1–1 correspondence between the open subgroups of finite, index B of C and the finite abelian extensions K/k, given by $B = NC_K$, and K fixed field of $\omega(B)$. If B_1 and B_2 correspond to K_1 and K_2 respectively, then B_1B_2 corresponds to $K_1 \cap K_2$, and $B_1 \cap B_2$ corresponds to K_1K_2. Finally, $K_1 \subset K_2$ if and only if $B_1 \supset B_2$.*

PROOF. The first part of the corollary is a repetition of the theorem, and of the fact that the norm index is equal to the degree of the extension. The second part is an obvious consequence of these facts. □

Let k be a global field. One may say that a finite extension E/k is a *class field* if $[E : k] = (C_k : N_{E/k}C_E)$. We have proved that all abelian extensions are class fields, and that all class fields are abelian. An abelian extension is said to be the *class field* of its norm group.

The reciprocity law allows us to determine the ramification of primes in abelian extensions by investigating the idèle class group NC_K. We imbed each local field $k_\mathfrak{p}$ into C_k via the composition of maps $k_\mathfrak{p} \hookrightarrow J_k \to C_K$. The next theorem gives the connection between the global and local situations.

THEOREM 2. *Let K/k be an abelian extension, $\mathfrak{P} \mid \mathfrak{p}$. Then*

$$NC_K \cap k_\mathfrak{p}^* = N_\mathfrak{P} K_\mathfrak{P}^*.$$

PROOF. In terms of idèles, we have to prove

$$NJ_K \cdot k^* \cap k_\mathfrak{p}^* \cdot k^* = N_\mathfrak{P} K_\mathfrak{P}^* \cdot k.$$

The inclusion $\supset$ is trivial.

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Conversely, let $N\mathfrak{A}\alpha = \mathfrak{a}_\mathfrak{p}\beta$. Then α/β is a local norm at all primes but $\mathfrak{p}$. By the product formula for the norm residue symbol, we know that α/β is a norm at $\mathfrak{p}$ also. This implies that $\mathfrak{a}_\mathfrak{p}$ is a local norm, and concludes the proof of the theorem. □

The Ramification Theorem now shows how the splitting of a prime $\mathfrak{p}$ in a class field is reflected in its idèle class group.

THEOREM 3. *Let K/k be an abelian extension belonging to the group $B = NC_K$. A prime $\mathfrak{p}$ is unramified in K if and only if $U_\mathfrak{p} \subset B$. $\mathfrak{p}$ splits completely in K if and only if $k_\mathfrak{p}^* \subset B$. (If $\mathfrak{p}$ is archimedean, the two notions coincide, and $U_\mathfrak{p} = k_\mathfrak{p}$.)*

PROOF. If $\mathfrak{p}$ is unramified, then all local units are local norms. Hence $U_\mathfrak{p} \subset B$. Conversely, if $U_\mathfrak{p} \subset B$, then by the previous theorem $U_\mathfrak{p} \subset (B \cap k_\mathfrak{p}) = N_\mathfrak{P} K_\mathfrak{P}$. All local units are therefore norms, and by the local class field theory, $\mathfrak{p}$ must be unramified.

The part of the theorem relating to the complete splitting is proved by replacing "$U_\mathfrak{p}$" by "$k_\mathfrak{p}$" and "unramified" by "split completely" in the preceding argument. □

The global problem of determining how a prime ramifies can therefore be solved by investigating the norm group, and is completely reduced to a local problem by the preceding theorems. The higher ramification for finite primes will be studied in detail in a subsequent chapter. Note that Theorems 2 and 3 are valid for $\mathfrak{p}$ finite or not.

2. Number Fields

Let k be a number field. We shall prove the Existence Theorem, and investigate the structure of the open subgroups of finite index in C.

We have first:

THEOREM 4. *The norm residue symbol $\omega\colon a \to (a, k)$ maps C* onto $\mathfrak{G}$.

PROOF. We can write $C = \mathbb{R}_+ \times C_0$. $\mathbb{R}_+$ is infinitely divisible and belongs to the kernel of ω. C_0 is compact and $\omega(C_0) = \omega(C)$ is everywhere dense in $\mathfrak{G}$. It is compact, closed, and hence all of $\mathfrak{G}$. □

The kernel of ω is contained in every open subgroup of finite index. Otherwise some infinitely divisible element would not be in B, and $(C : B)$ would not be finite.

Let B be an open subgroup of finite index. Then B is closed in C. Let $B_0 = B \cap C_0$. B_0 is closed in C_0, hence compact, and $(C_0 : B_0) = (C : B)$. $\omega(B_0) = \mathfrak{h}$ is a closed subgroup of $\mathfrak{G}$, and B_0 is the full inverse image of $\mathfrak{h}$ in C_0 because B contains the kernel of ω. Hence $(\mathfrak{G} : \mathfrak{h}) = (C_0 : B_0) = (C : B)$. Let K be the fixed field of $\mathfrak{h}$. Then NC_K is contained in the inverse image B of $\mathfrak{h}$ under ω. Since $(C : NC_K) = [K : k] = (\mathfrak{G} : \mathfrak{h}) = (C : B)$ it follows that $B = NC_K$. This concludes the proof of the existence theorem in number fields. □

We now turn to a more detailed study of the open subgroups of finite index.

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Let $\mathfrak{p}$ be a prime, $\nu_\mathfrak{p} \geqslant 0$. If $\mathfrak{p}$ is archimedean, $\nu_\mathfrak{p} = 0$ or 1. Let $\mathfrak{a}_\mathfrak{p} \in k_\mathfrak{p}$. We define $\mathfrak{a}_\mathfrak{p} \equiv 1 \pmod{\mathfrak{p}^{\nu_\mathfrak{p}}}$ to mean

the usual congruence,	if $\mathfrak{p}$ finite, $\nu_\mathfrak{p} \geqslant 1$.
$\mathfrak{a}_\mathfrak{p} \in U_\mathfrak{p}$,	if $\mathfrak{p}$ finite, $\nu_\mathfrak{p} = 0$.
$\mathfrak{a}_\mathfrak{p} > 0$,	if $\mathfrak{p}$ real, $\nu_\mathfrak{p} = 1$.

If $\mathfrak{p}$ is complex, or if $\mathfrak{p}$ is real but $\nu_\mathfrak{p} = 0$, then we put no restriction on $\mathfrak{a}_\mathfrak{p}$.

By a *module* $\mathfrak{m}$ we shall mean a formal product $\mathfrak{m} = \prod_\mathfrak{p} \mathfrak{p}^{\nu_\mathfrak{p}}$, where $\nu_\mathfrak{p} \geqslant 0$, and almost all $\nu_\mathfrak{p} = 0$. We can think of a module as a divisor in number fields, with archimedean primes entering in it.

Let $\mathfrak{a}$ be an idèle. We define $\mathfrak{a} \equiv 1 \pmod{\mathfrak{m}}$ to mean $\mathfrak{a}_\mathfrak{p} \equiv 1 \pmod{\mathfrak{p}^{\nu_\mathfrak{p}}}$. Such idèles form a group which we denote by $\mathfrak{S}_\mathfrak{m}$. We let $C_\mathfrak{m} = \mathfrak{S}_\mathfrak{m} k^*/k^*$.

The neighborhoods of 1 in C are obtained from neighborhoods of 1 in J. A fundamental system in J consists of neighborhoods which are constructed as follows:

$$W = \prod_{\mathfrak{p} \nmid \mathfrak{m}} U_\mathfrak{p} \times \prod_{\mathfrak{p} \mid \mathfrak{m}} W_\mathfrak{p}$$

where $W_\mathfrak{p}$ is the group of all $\alpha_\mathfrak{p} \in k_\mathfrak{p}$ such that $\alpha_\mathfrak{p} \equiv 1 \pmod{\mathfrak{p}^{\nu_\mathfrak{p}}}$ if $\mathfrak{p}$ is finite, and if $\mathfrak{p}$ is archimedean, then $W_\mathfrak{p}$ is simply a neighborhood of 1 in $k_\mathfrak{p}$. The sets Wk constitute therefore a fundamental system in C. The group generated by the elements of W is $\mathfrak{S}_\mathfrak{m}$ if $\nu_\mathfrak{p} = 1$ at each real $\mathfrak{p}$. Let $N_\mathfrak{p}$ be the group generated by $W_\mathfrak{p}$. Then $N_\mathfrak{p} = W_\mathfrak{p}$ if $\mathfrak{p}$ is finite, and $N_\mathfrak{p}$ is either the complex, the reals, or the positive reals if $\mathfrak{p}$ is archimedean.

If B is an open subgroup of C then B contains a neighborhood Wk of the type just described. Since B is a group, B contains the corresponding group $C_\mathfrak{m}$.

If $\mathfrak{m} = 1$, then $\mathfrak{S}_1 = J_S$ where S consists precisely of the archimedean primes. We let $k\mathfrak{S}_1 = C_1$. Then C/C_1 is isomorphic to the ideal class group and $(C : C_1) = h$ is finite.

We may write

$$(C : C_\mathfrak{m}) = (C : C_1)(C_1 : C_\mathfrak{m})$$

and $(C_1 : C_\mathfrak{m})$ is obviously finite because $(U_\mathfrak{p} : N_\mathfrak{p})$ is finite for all primes $\mathfrak{p}$. We have proved

THEOREM 5. *Let k be a number field. Let B be an open subgroup of C. Then B is of finite index in C, and B contains one of the groups $C_\mathfrak{m}$.*

Let B be an open subgroup of C. Let $\mathfrak{f}_\mathfrak{p}$ be the least power of $\mathfrak{p}$ such that $\mathfrak{a}_\mathfrak{p} \equiv 1 \pmod{\mathfrak{f}_\mathfrak{p}}$ implies $\mathfrak{a}_\mathfrak{p} \in B$. Then $\mathfrak{f} = \prod_\mathfrak{p} \mathfrak{f}_\mathfrak{p}$ is obviously a module, and $C_\mathfrak{f} \subset B$. In fact it is clear that $C_\mathfrak{m} \subset B$ if and only if $\mathfrak{f} \mid \mathfrak{m}$. f is called the conductor of B. If K is the class field of B, then $\mathfrak{f}$ is the *conductor* of K. It is an immediate consequence of Theorem 2 that $\mathfrak{f}_\mathfrak{p}$ is the local conductor of K, in the sense of the local class field theory.

We specialize the preceding discussion to the rational numbers. A module is given by $\mathfrak{m} = m \cdot p_\infty^\nu$ where m is a positive rational number, p_∞ is the infinite prime, and $\nu = 0$ or 1. Suppose $\nu = 1$, so $\mathfrak{m} = mp_\infty$. Every open subgroup B of C contains a group C_{mp_∞}. The class field to the group C_{mp_∞} is simply the cyclotomic field $\mathbb{Q}(\zeta_m)$, according to what we have proved in Ch. 7, §2. Recalling the Corollary to Theorem 1, this yields Kronecker's Theorem:

THEOREM 6. *Every finite abelian extension of the rational numbers is a cyclotomic field, i.e. is a subfield of $\mathbb{Q}(\zeta_m)$ for some m.*

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Returning to an arbitrary number field, we consider the special module $\mathfrak{m} = 1$. The group C_1 is an open subgroup of C and we note that it has been canonically defined.

By the Existence Theorem, C_1 has a class field K/k. By the Ramification Theorem and the definition of C_1, K/k is unramified at all primes. (Recall that for an archimedian prime, "unramified" means "split completely".) Conversely, if K'/k is an unramified abelian extension, it follows from the ramification theorem that $C_1 \subset N_{K'/k}C_{K'}$.

The compositum of unramified extensions is also unramified. We get

THEOREM 7. *Let k be a number field. The maximal unramified abelian extension K of k is finite over k, and $[K : k] = h$ is the class number. K is class field to C_1, where $C_1 = kJ_S$, S being the set of archimedean primes.*

K is called the *Hilbert Class Field* of k. We abbreviate by HCF. If $h = 1$, then every abelian extension of k ramified, and the HCF is k itself.

One may construct a tower of HCF over a number field k.

THEOREM 8. *Let k he a number field. Let K be its HCF, with group G. Let K_1 be the HCF of K. Then K_1/k is normal with group G_1, and K is the maximal abelian subfield of K_1. In other words, $G = G_1/G_1'$ is the factor commutator group of G_1.*

PROOF. Let σ be an isomorphism of K_1/k. Then σ maps K into itself, so $K^\sigma = K$. Since $K_1^\sigma/K^\sigma = K_1^\sigma/K$ is unramified, it follows that $K_1^\sigma \subset K_1$ whence K_1 is normal over k. □

Let K' be maximal abelian in K_1. Then $K' \supset K$. But K' is unramified. Hence $K' \subset K$ as was to be shown.

The Principal Ideal Theorem applies to K and K_1 of the preceding theorem. It was in fact the way it arose historically.

It is an unsolved problem[1] to determine whether there exists towers of HCF with infinitely many steps. More generally it is an unsolved problem to determine whether there exists infinite non-abelian unramified extensions over number fields.

The HCF has some interesting consequences concerning the class number.

THEOREM 9. *Let K/k be the HCF. Let E/k be a finite extension. If $E \cap K = k$ then $h_k \mid h_E$.*

PROOF. KE/E is abelian and unramified, hence is a subfield of the HCF of E. We have $h_k = [K : k]$. By assumption $h_k = [KE : E]$. It follows that $h_k \mid h_E$. □

EXAMPLE. Let p be a prime, ζ_{p^ν} a primitive p^ν-th root of unity. Then $\mathbb{Q}(\zeta_{p^\nu})/\mathbb{Q}$ is completely ramified at p. Let $F \subset \mathbb{Q}(\zeta_{p^\nu})$. Let K be the HCF of F. Then $\mathbb{Q}(\zeta_{p^\nu}) \cap K = F$. If $F \subset E \subset \mathbb{Q}(\zeta_{p^\nu})$, then $E \cap K = F$ and by the preceding theorem, $h_F \mid h_E$. In particular, the class number of the real subfield divides the class number of $\mathbb{Q}(\zeta_{p^\nu})$.

We finish our discussion of the HCF by a remark concerning unramified extensions in general: Namely, there exists many other unramified extensions besides the HCF. These will of course be non-abelian. One can in fact prove the following result:

[1]This problem was solved by E. S. Golod and I. R. Shafarevich, see their paper [**9**].

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Let we be given any normal extension K/k of a number field with group G. There exist infinitely many finite extensions E/k such that $E \cap K = k$, i.e. KE/E has also group G, and such that KE/E is unramified.

$$\begin{tikzcd}[column sep=small] & KE \arrow[dl, dash] \arrow[dr, dash, "G"] & \\ K \arrow[dr, dash, "G"'] & & E \arrow[dl, dash] \\ & k & \end{tikzcd}$$

3. Function Fields

Let k be a global function field. Let k_0 be the constant field, with q elements. Let K_0 be the algebraic closure of k_0, and let $K = K_0 k$ be the compositum. $\overline{\mathbb{Z}}$ may be interpreted as the group $\mathfrak{G}_0$ of K_0/k_0, or of $K_0 k/k$. Let $\varphi\colon \alpha \to \alpha^q$ be the canonical generator of $\mathfrak{G}_0$, which then consists of powers φ^ν, $\nu \in \overline{\mathbb{Z}}$. (We recall that $\overline{\mathbb{Z}}$ is the completion of $\mathbb{Z}$ under the ideal topology.)

$$\begin{tikzcd}[column sep=small] & K_0 k \arrow[dl, dash] \arrow[dr, dash] & \\ k \arrow[dr, dash] & & K_0 \arrow[dl, dash] \\ & k_0 & \end{tikzcd}$$

Let A_k be the maximal abelian extension of k, and let $\mathfrak{G}$ be its Galois group. Let $\sigma \in \mathfrak{G}$. If $\sigma = \varphi^\nu$ on K_0, we define $\operatorname{ord} \sigma = \nu$.

We have previously assigned ordinals to idèle classes. Namely, if $a \in C$, we can write $a = a_1^\nu \cdot a_0$ where $|a_0| = 1$, $|a_1| = q$, $\nu \in \mathbb{Z}$. We let $\operatorname{ord}_a = \nu$.

Let $\omega\colon a \to (a, k)$ be the map of the norm residue symbol.

THEOREM 10. *Let $a \in C$. Then $\operatorname{ord}(a, k) = \operatorname{ord} a$. In particular, $\operatorname{ord}(a, k)$ is in $\mathbb{Z}$ and not in $\overline{\mathbb{Z}}$.*

PROOF. Let $\mathfrak{a}$ be a representative idèle of a. Then from the Reciprocity Law we know that $(a, k) = \prod_{\mathfrak{p}} (\mathfrak{a}_{\mathfrak{p}}, k_{\mathfrak{p}})$ and we view $(\mathfrak{a}_{\mathfrak{p}}, k_{\mathfrak{p}})$ as an element of $\mathfrak{G}$.

The global and local ordinals are related by the formula

$$\operatorname{ord}(\mathfrak{a}_{\mathfrak{p}}, k_{\mathfrak{p}}) = f_{\mathfrak{p}} \operatorname{ord}_{\mathfrak{p}}(\mathfrak{a}_{\mathfrak{p}}, k_{\mathfrak{p}})$$

because the constant field of $k_{\mathfrak{p}}$ has $q^{f_{\mathfrak{p}}}$ elements. We have from the local class field theory

$$\operatorname{ord}_{\mathfrak{p}}(\mathfrak{a}_{\mathfrak{p}}, k_{\mathfrak{p}}) = \operatorname{ord}_{\mathfrak{p}} \mathfrak{a}_{\mathfrak{p}}.$$

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It follows that $\operatorname{ord}(\mathfrak{a}_\mathfrak{p}, k_\mathfrak{p}) = 0$ at almost all primes, and therefore

$$\begin{aligned}\operatorname{ord}(a,k) &= \sum_\mathfrak{p} \operatorname{ord}(\mathfrak{a}_\mathfrak{p}, k_\mathfrak{p}) \\ &= \sum_\mathfrak{p} f_\mathfrak{p} \operatorname{ord}_\mathfrak{p}(\mathfrak{a}_\mathfrak{p}, k_\mathfrak{p}) \\ &= \sum_\mathfrak{p} f_\mathfrak{p} \operatorname{ord}_\mathfrak{p} \mathfrak{a}_\mathfrak{p} \\ &= \operatorname{ord} a\end{aligned}$$

thereby proving the theorem. □

COROLLARY. *Let $\mathfrak{h}_0$ be the group of A_k/K_0k. Then ω is a topological isomorphism of C_0 onto $\mathfrak{h}_0$. $\mathfrak{G}$ is isomorphic to the direct product of $\mathfrak{h}_0$ and $\mathfrak{G}_0$.*

PROOF. $\omega(C_0)$ is contained in $\mathfrak{h}_0$. Conversely, if $\omega(a) \in \mathfrak{h}_0$ then $\operatorname{ord} a = 0$ and $a \in C_0$. Hence ω is an isomorphism of C_0 onto $\mathfrak{h}_0$. It is continuous, and since C_0 is compact it is a homeomorphism onto.

Let $\sigma \in \mathfrak{G}$ be a lifting of a topological generator φ of $\mathfrak{G}_0$. Then the closure $\overline{\langle\sigma\rangle}$ of the subgroup generated by σ maps isomorphically to $\mathfrak{h}_0$, and $\mathfrak{G} = \overline{\langle\sigma\rangle} \times \mathfrak{G}_0$. □

Let $\operatorname{ord} a_1 = 1$. Write C as a direct product $\{a_1\} \times C_0$. Any $a \in C$ can be written $a = a_1^\nu \times a_0$. We see that C is isomorphic to $\mathbb{Z} \times C_0$ (but not canonically).

We define a new topology on C: The neighborhoods of 1 are to be the open subgroups of finite index in the ordinary topology. This topology will be called the *class topology.*

LEMMA. *The class topology coincides with the old one on C_0. It induces the ideal topology on $\mathbb{Z}$ and the product topology on $\mathbb{Z} \times C_0$.*

PROOF. Let B be an open subgroup of finite index in C. Let $b \in B$ have least positive ordinal. Let $B_0 = B \cap C_0$. Then $B = \bigcup_{\nu\in\mathbb{Z}} b^\nu B_0$. B_0 is open because C_0 is open. We shall now discuss those open subgroups, and prove they are of finite index in C_0.

As in our discussion of number fields, we let $\mathfrak{m} = \prod_\mathfrak{p} \mathfrak{p}^{\nu_\mathfrak{p}}$ ($\nu_\mathfrak{p} \geqslant 0$ and $= 0$ for almost all primes) be a module. A fundamental system of neighborhoods of 1 in J is given by the groups

$$\mathfrak{S}_\mathfrak{m} = \prod_{\mathfrak{p}\nmid\mathfrak{m}} U_\mathfrak{p} \times \prod_{\mathfrak{p}\mid\mathfrak{m}} N_\mathfrak{p}$$

where $N_\mathfrak{p}$ are the groups of $\alpha_\mathfrak{p} \in k_\mathfrak{p}$ such that $\alpha_\mathfrak{p} \equiv 1 \pmod{\mathfrak{p}^{\nu_\mathfrak{p}}}$, for $\mathfrak{p} \mid \mathfrak{m}$. The groups $C_\mathfrak{m} = k\mathfrak{S}_\mathfrak{m}$ are a fundamental system in C, and in fact $C_\mathfrak{m} \subset C_0$. We have

$$(C_0 : C_\mathfrak{m}) = (C_0 : Uk)(Uk : C_\mathfrak{m}),$$

where $U = \prod_{\text{all } \mathfrak{p}} U_\mathfrak{p}$. By the finiteness of class number, $(C_0 : Uk^*) = h$ is finite, and obviously $(Uk : C_\mathfrak{m})$ is finite, because $(U_\mathfrak{p} : N_\mathfrak{p})$ is finite.

The open subgroup B_0 must contain some open group $C_\mathfrak{m}$ and hence $(C_0 : B_0)$ is finite.

It is clear that the induced topology on $\mathbb{Z}$ is the ideal topology, and this proves our lemma. In addition, it gives insight into the structure of open subgroups similar to that obtained in number fields. □

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We form the group $\overline{C} = a_1^{\mathbb{Z}} \times C_0 \simeq \overline{\mathbb{Z}} \times C_0$ by taking the formal Cartesian product $\{a_1^\nu\} \times C_0$ where the exponents ν of a_1 now range over $\overline{\mathbb{Z}}$. Both $\overline{\mathbb{Z}}$ and C_0 are compact, so $\overline{C}$ is compact.

If we look at the effect of the norm residue map on C, we see that ω is uniformly continuous, in the new topology. We may therefore extend ω by continuity to $\bar{C}$. Since $\overline{C}$ is compact and $\omega(\overline{C})$ is dense in $\mathfrak{G}$, it follows that $\omega(\overline{C})$ is onto $\mathfrak{G}$. ω is an isomorphism between $\overline{C}$ and $\mathfrak{G}$, and consequently ω is a homeomorphism.

We can now prove the existence theorem in function fields using the same procedure that was used in number fields.

Let B be an open subgroup of finite index in C. $B_0 = B \cap C_0$, $B = \bigcup_{\nu \in \mathbb{Z}} b^\nu B_0$. If we write $b = a_1^d$ then $(\overline{\mathbb{Z}} : d\overline{\mathbb{Z}}) = d$. Let $\overline{B} = d\overline{\mathbb{Z}} \times B_0$. Then

$$(C : B) = d(C_0 : B_0) = (\overline{C} : \overline{B}).$$

Let $\mathfrak{h} = \omega(\overline{B})$. $\mathfrak{h}$ is closed in $\mathfrak{G}$ and $(C : B) = (\mathfrak{G} : \mathfrak{h})$. Let K be the fixed field of $\omega(\overline{B})$. Then $[K : k] = (C : B)$. We know that $\omega(NC_K) \subset \mathfrak{h}$, and hence $NC_K \subset B$. Since $(C : NC_K) = [K : k]$ we conclude that $NC_K = B$, thereby proving the existence theorem.

As in the local class field theory, we remark that the open subgroups of finite index in $\overline{C}$ are in 1–1 correspondence with those of C. Namely, let B' be an open subgroup of finite index in $\overline{C}$. Let $B_0 = B' \cap C_0$, and let $I' = B' \cap \overline{\mathbb{Z}}$. Then I' is a closed subgroup of finite index of $\overline{\mathbb{Z}}$, and is consequently of type $d\overline{\mathbb{Z}}$ where d is an ordinary integer. We see therefore that B' is simply the closure of the group $d\mathbb{Z} \times B_0$.

The Hilbert Class Field has a certain analogue in function fields. The maximal unramified abelian extension of k is infinite because of the possible constant field extensions. We may ask however for unramified abelian extensions which have the same constant field. Let $C_1 = Uk^*/k^*$. Let $b \in C$, $\operatorname{ord} b = 1$. Then the group $B = \{b\} \times C_1$ is open in C, and $(C : B) = h$ is the class number. The class field of B may be viewed as a HCF. It is a maximal unramified abelian extension having the same constant field as k. We contend that there are h such class fields, and that their Galois groups are isomorphic to C/B.

Let $u_1, \ldots, u_h$ be representatives of C_0/C_1. Let $b_i = bu_i$. The groups $B_i = \{b_i\} \times C_1$ are all distinct and $(C : B_i) = h$. Furthermore the factor groups C/B_i are all isomorphic. The groups B_i give rise to h class fields of the above mentioned type, of degree h over k, with isomorphic Galois groups.

Let $\operatorname{ord} a = 1$. Then $\operatorname{ord} ab^{-1} = 0$ and hence a is in some coset $bu_i \bmod C_1$. This proves that the HCF we have described account for all of them. We denote these HCF by K_i, $i = 1, \ldots, h$.

The class field of $\{b^h\} \times C_0$ is the constant field extension of degree h. Denote it by L. We contend that $LK_i = LK_j = LK_i \ldots K_h$ for any i, j. Indeed,

$$\{b^h\} \times C_0 \cap \{b_i\} \times C_1 = \{b_i^h\} \times C_1.$$

But b_i^h and b_j^h are in the same coset mod C_1. This proves our contention.

The field $K = LK_1 \ldots K_h$ is of degree h^2 over k. It is invariantly defined and is may be viewed as the correct generalization of the Hilbert Class Field. It is class field to $\{b^h\} \times C_1$, where b is any idèle class of ordinal 1.

On the other hand, noting that in number fields the HCF is class field to the group $k^* J_{S_\infty}$, where S_∞ is the set of archimedean primes, one could say that an

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analogue in function fields is the class field to k^*J_S, where S is any non-empty set of primes which we designate as the "infinite" ones. Then the Galois group of the Dedekind ring O_S of functions with no poles outside S, in strict analogy with the number field case, where the Galois group of the HCF is isomorphic to the ideal class group of the ring of integers. The constant field extension in this type of HCF is of degree equal the qcd of the degrees of the primes in S.

4. Decomposition Laws and Arithmetic Progressions

We consider only number fields k, and leave to the reader the task of formulating the analogous results in function fields.

Let $\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}$ be a module in the sense of 2, and $G_{\mathfrak{p}} = k^*\mathfrak{S}_{\mathfrak{m}}$. Let K be the class field of $C_{\mathfrak{m}}$.

Let $J_{\mathfrak{m}}$ be the group of idèles having component 1 at all $\mathfrak{p} \mid \mathfrak{m}$ and arbitrary otherwise. We contend that every idèle class mod $k^*\mathfrak{S}_{\mathfrak{m}}$ has a representative in $J_{\mathfrak{m}}$. Indeed, let $\mathfrak{a}$ be an idèle. By the approximation theorem, there exists $\alpha \in k$ such that $\mathfrak{a}\alpha \equiv 1 \pmod{\mathfrak{p}^{\nu_{\mathfrak{p}}}}$ for $\mathfrak{p} \mid \mathfrak{m}$, $\nu_{\mathfrak{p}} \geqslant 1$. This implies that $\mathfrak{a}\alpha \in J_{\mathfrak{m}}\mathfrak{S}_{\mathfrak{m}}$, or in other words, $J = k^*J_{\mathfrak{m}}\mathfrak{S}_{\mathfrak{m}}$ as contended.

By an elementary isomorphism theorem, we have

$$J/k\mathfrak{S}_{\mathfrak{m}} \simeq J_{\mathfrak{m}}/J_{\mathfrak{m}} \cap k^*\mathfrak{S}_{\mathfrak{m}}.$$

Let

$$\Phi_{\mathfrak{m}} \colon J \to \bar{J}$$

map idèles onto ideals by putting $\Phi_{\mathfrak{m}}(\mathfrak{a}) = \bar{\mathfrak{a}} = \prod_{\substack{\mathfrak{p}\nmid\mathfrak{m} \\ \mathfrak{p}\text{ finite}}} \mathfrak{p}^{\operatorname{ord}_{\mathfrak{p}}\mathfrak{a}}$. We get a map

$$\Phi_{\mathfrak{m}} \colon J_{\mathfrak{m}} \to \bar{J}_{\mathfrak{m}}$$

of $J_{\mathfrak{m}}$ onto the ideals prime to $\mathfrak{m}$. Let $\bar{k}^*_{\mathfrak{m}}$ be the principal ideals represented by elements $\alpha \in k^*$, $\alpha \equiv 1 \pmod{\mathfrak{m}}$. Then $\Phi_{\mathfrak{m}}^{-1}(k^*_{\mathfrak{m}}) = J_{\mathfrak{m}} \cap k^*\mathfrak{S}_{\mathfrak{m}}$. Hence we get

$$J/k^*\mathfrak{S}_{\mathfrak{m}} \simeq J_{\mathfrak{m}}/J_{\mathfrak{m}} \cap k^*\mathfrak{S}_{\mathfrak{m}} \simeq \bar{J}_{\mathfrak{m}}/\bar{k}^*_{\mathfrak{m}}.$$

Let G be the group of K/k. Let

$$\omega \colon J_{\mathfrak{m}} \to G$$

be the norm residue map restricted to $J_{\mathfrak{m}}$. Then ω depends only on cosets of $J_{\mathfrak{m}} \cap k^*\mathfrak{S}_{\mathfrak{m}}$. For every ideal $\bar{\mathfrak{a}}$ in $\bar{J}_{\mathfrak{m}}$ we select an idèle $\mathfrak{a} \in J_{\mathfrak{m}}$ such that $\Phi_{\mathfrak{m}}(\mathfrak{a}) = \bar{\mathfrak{a}}$, and define $\left(\frac{K}{\bar{\mathfrak{a}}}\right)$ to be $\omega(\mathfrak{a})$. We know that ω depends only on cosets of $J_{\mathfrak{m}} \cap k^*\mathfrak{S}_{\mathfrak{m}}$, and consequently $\left(\frac{K}{\bar{\mathfrak{a}}}\right)$ depends only on cosets of $\bar{k}^*_{\mathfrak{m}}$. The map

$$\bar{\mathfrak{a}} \to \left(\frac{K}{\bar{\mathfrak{a}}}\right)$$

is a homomorphism of $\bar{J}_{\mathfrak{m}}$ onto G, with kernel $\bar{k}_{\mathfrak{m}}$. We interpret $\bar{\mathfrak{a}}\bar{k}^*_{\mathfrak{m}}$ as a generalized arithmetic progression mod$\mathfrak{m}$.

Let $\mathfrak{p}$ be a finite prime. It follows from the Ramification Theorem that if $\mathfrak{p} \nmid \mathfrak{m}$ then $\mathfrak{p}$ is unramified. Let $\pi \in k_{\mathfrak{p}}$, $\operatorname{ord}_{\mathfrak{p}} \pi = 1$. The idèle $\pi = (\dots, 1, \pi, 1, \dots)$ having π at $\mathfrak{p}$ and 1 at all other primes is such that $\Phi_{\mathfrak{m}}(\pi) = \mathfrak{p}$, and hence $\omega(\pi) = \left(\frac{K}{\mathfrak{p}}\right)$ is the Frobenius Substitution. This shows how the norm residue symbol can be described just in terms of the unramified primes. If $\mathfrak{q}$ is another prime we have $\left(\frac{K}{\mathfrak{p}}\right) = \left(\frac{K}{\mathfrak{q}}\right)$ if and only if $\mathfrak{p}$ and $\mathfrak{q}$ lie in the same arithmetic progression.

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If $k = Q$ is the rationals, then a module $\mathfrak{m}$ is of type m or mp_∞ where m is a positive rational. Considering the latter type, $\mathfrak{m} = mp_\infty$, we see that $\Phi_\mathfrak{m}$ maps J_{mp_∞} onto the ideals prime to m mod m. A rational number α is $\equiv 1 \pmod{\mathfrak{m}}$ if and only if it is positive and $\equiv 1 \pmod{m}$ in the ordinary sense. This shows that two integral ideals are in the same generalized progression mod mp_∞ if and only if their unique *positive* generators are in the same arithmetic progression mod m. Thus our "generalized progressions" do generalize the classical ones, which justifies our terminology.

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CHAPTER IX

Connected Component of Idèle Classes

1. Structure of the Connected Component

Let k be a global field. Let $\mathfrak{G}$ be the Galois group of its maximal abelian extension, and let $\omega\colon C_k \to \mathfrak{G}$ be the map given by the norm residue symbol. In function fields we have seen that the kernel of ω is trivial, i.e. consists of 1 alone. In number fields, the kernel is the subgroup of C_k of all elements which are infinitely divisible according to Prop. 10 of Ch. 14. We shall prove that this subgroup is the connected component of identity of C_k, which we shall denote by D_k. We shall also determine the structure of D_k by finding a set of representative idèles whose images generate D_k.

We need an auxiliary result (which will reappear in sharper form in the next chapter). We prove a little more than is immediately needed for our purpose. We denote by ζ_m a primitive m-th root of unity.

THEOREM 1. *Let k be a global field, $m = 2^t m'$ (m' odd) an integer and S a finite set of primes. Let $\alpha \in k$ and assume $\alpha \in k_{\mathfrak{p}}^{*m}$ for all $\mathfrak{p} \notin S$.*

A. *If k is a function field, or if k is a number field and the field $k(\zeta_{2^t})/k$ is cyclic (this condition is certainly satisfied if $t \leqslant 2$) then $\alpha \in k^{*m}$.*

B. *Otherwise at least $\alpha \in k^{*m/2}$.*

PROOF. 1. Suppose m and n are relatively prime, $\alpha = \beta^m$ and $\alpha = \gamma^n$ with β, γ in k. Let $rm + sn = 1$ then

$$\alpha = \alpha^{rm}\alpha^{sn} = \gamma^{rmn}\beta^{smn} \in k^{mn}.$$

This shows that it suffices to prove our theorem in case $m = p^r$ a power of a prime and we shall assume this for the rest of the proof.

2. Let p be the characteristic of k. Consider the inseparable extension $K = k(\sqrt[m]{\alpha})$. Since $K^m \subset k$, any valuation $\mathfrak{p}$ of k extends uniquely to K which means that only one prime $\mathfrak{P}$ of K divides $\mathfrak{p}$. From the elementary theory of global fields we know $\sum_{\mathfrak{P}|\mathfrak{p}} e_{\mathfrak{P}} f_{\mathfrak{P}} = \deg(K/k)$ and hence $\deg(K/k) = e_{\mathfrak{P}} f_{\mathfrak{P}} = \deg(K_{\mathfrak{P}}/k_{\mathfrak{P}})$. If we now assume that $\alpha \in k_{\mathfrak{p}}^m$ then $\deg(K_{\mathfrak{P}}/k_{\mathfrak{P}}) = 1$, hence deg $(K/k) = 1$ or $\alpha \in k^m$.

3. Assume $\zeta_m \in k$. The field $k(\sqrt[m]{\alpha}) = K$ is then a well defined Kummer extension of k and our assumption about α means that $K_{\mathfrak{P}_m} = k_{\mathfrak{p}}$ for every $\mathfrak{p} \notin S$. Theorem 2 of Chapter V shows $K = k$ hence $\alpha \in k^m$.

4. If $\zeta_m \notin k$ we form the new ground field $k' = k(\zeta_m)$. The preceding part of the proof shows $\alpha = \beta^m$ with $\beta \in k'$.

The polynomial $x^m - \alpha$ splits in k' into linear factors $x - \beta\zeta_m^{\mu}$. Let $x^m - \alpha = \prod_i f_i(x)$ be its decomposition into irreducible factors in k; each $f_i(x)$ has a root of the form $\beta_i = \beta\zeta_m^{\mu}$, and $k(\beta_i)/k$ is abelian as subfield of k'/k. If $\alpha = \gamma_{\mathfrak{p}}^m$, $\gamma_{\mathfrak{p}} \in k_{\mathfrak{p}}$,

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then $\prod_i f_i(\gamma_{\mathfrak{p}}) = 0$ so there exists one $f_i(x)$ such that $f_i(\gamma_{\mathfrak{p}}) = 0$. This means that our $\mathfrak{p}$ splits completely in one of the subfields $k(\beta_i)$ of k'.

Assume now that k'/k is cyclic of prime power degree. The subfields of k' are then linearly ordered by inclusion and we may assume that $k(\beta_1)$ is the smallest of the subfields $k(\beta_i)$. If $\mathfrak{p} \notin S$ then $\mathfrak{p}$ splits in one of the $k(\beta_i)$ and consequently in $k(\beta_1)$. Theorem 2 of Chapter V shows $k(\beta_1) = k$, $\beta_1 \in k$, $\alpha = \beta_1^m$.

5. Let p be odd and put $k_1 = k(\zeta_p)$. k'/k_1 is cyclic of prime power degree. Considering k_1 as the new ground field we see that we can write $\alpha = \beta^m$ with $\beta \in k_1$. Taking norms from k_1 to k on both sides we see that $\alpha^d \in k^m$ where d is the degree of k_1/k and therefore prime to $m = p^r$ as divisor of $p-1$. Since $\alpha^m \in k^m$ we conclude $\alpha \in k^m$.

6. Only the case $m = 2^t$ remains. The degree of k'/k is then a power of 2. If k'/k is cyclic our problem is settled by part 4 of our proof. This eliminates the case of a function field. We may therefore assume that k is a number field, and that k'/k is not cyclic (especially that $t > 2$).

Put $k_1 = k(\sqrt{-1})$. Since it is well known that k'/k_1 is cyclic we have $k_1 \neq k$ and we can write $\alpha = \beta_1^m$ with $\beta_1 \in k_1$. Taking norms from k_1 to k we obtain $\alpha^2 = \beta^m$ with $\beta \in k$. This means $\alpha = \pm\beta^{2^{t-1}}$ and we must show that the $-$ sign is impossible. Assume that it occurs. Then $-1 = \frac{\alpha}{\beta^{2^{t-1}}}$. The assumption about α together with $t > 2$ allows us to conclude that -1 is a square at all primes $\mathfrak{p} \notin S$. According to Theorem 2 of Chapter V we would have $k_1 = k$ which is not the case. $\square$

Remark. If the m-th roots of unity are not in k, the symbol $k(\sqrt[m]{\alpha})$ has no well defined meaning and a careless use of it may lead to mistakes. The crucial steps of our proof are those of part 4 and a reader that wishes to simplify them is asked to be especially careful at this place. In the next chapter we shall see by the way, that the conjecture that $\alpha \in k^m$ is *not* always true.

We let k be a number field for the rest of this chapter. $J = J_k$, $C = C_k$.

The connected component of J is the product of the multiplicative groups of the complex local fields and the positive real lines at real primes. One knows from the elementary theory of topological groups that the connected component of C is the closure of the image of the connected component of J. However, we shall obtain a more explicit model for it, and will not make use of the abstract topological situation, which would in any case not be sufficient for a discussion of the cohomology of the connected component.

The group U of unit idèles splits into a direct product $\widetilde{U} \times \overline{U}$ where $\widetilde{U} = \prod_{\mathfrak{p}_\infty} k_{\mathfrak{p}_\infty}$ and $\overline{U} = \prod_{\mathfrak{p} \neq \mathfrak{p}_\infty} U_{\mathfrak{p}}$. Accordingly each unit idèle $\mathfrak{a}$ can be written as a product

$$\mathfrak{a} = \tilde{\mathfrak{a}} \cdot \bar{\mathfrak{a}}$$

where $\tilde{\mathfrak{a}}$ has components 1 at all finite primes and $\bar{\mathfrak{a}}$ has components 1 at all archimedean primes. The units ε of the number field k, considered as idèles, can therefore be written accordingly $\varepsilon = \tilde{\varepsilon}\bar{\varepsilon}$. The two factors on the right side are of course only idèles and not any longer principal idèles.

The group $\overline{U}$ is compact and has a fundamental system of neighborhoods of 1 that are subgroups of finite index. It follows that the exponentiation with ordinary integers can be extended to an exponentiation with elements of $\overline{\mathbb{Z}}$ where $\overline{\mathbb{Z}}$ is the completion of $\mathbb{Z}$ under the topology of the ideals of $\mathbb{Z}$. The symbol $\bar{\mathfrak{a}}^x$ is therefore

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defined for every unit idèle and every $x \in \overline{\mathbb{Z}}$ and especially $\bar{\varepsilon}^x$ is defined for units ε of k and $x \in \overline{\mathbb{Z}}$. It is of course again only a unit idèle of $\overline{U}$.

As usual we denote by r_1 the number of real archimedean primes, by r_2 the number of complex primes and put $r = r_1 + r_2 - 1$.

THEOREM 2. *Let $\varepsilon_1, \ldots, \varepsilon_r$ be independent units of k. The corresponding idèles $\bar{\varepsilon}_1, \ldots, \bar{\varepsilon}_r$ are independent relative to exponentiation with $\overline{\mathbb{Z}}$. In other words, the relation*

$$\bar{\varepsilon}_1^{x_1} \bar{\varepsilon}_2^{x_2} \ldots \bar{\varepsilon}_r^{x_r} = 1 \qquad x_i \in \overline{\mathbb{Z}} \tag{1}$$

implies that all $x_i = 0$.

PROOF. The group generated (with ordinary integers as exponents) by the ε_i is of finite index d in the group of all units of k. Let m be any integer. We find to each x_i in (1) an approximating ordinary integer ν_i: $\nu_i \equiv x_i \pmod{2dm}$, $\nu_i \in \mathbb{Z}$. Then $\varepsilon = \varepsilon_1^{\nu_1} \ldots \varepsilon_r^{\nu_r}$ is an element of k and we can write

$$\varepsilon = \tilde{\varepsilon}_1^{\nu_1} \tilde{\varepsilon}_2^{\nu_2} \ldots \tilde{\varepsilon}_r^{\nu_r} \bar{\varepsilon}_1^{\nu_1} \bar{\varepsilon}_2^{\nu_2} \ldots \bar{\varepsilon}_r^{\nu_r}.$$

Dividing the right side by the left side of (1) we obtain

$$\varepsilon = \tilde{\varepsilon}_1^{\nu_1} \tilde{\varepsilon}_2^{\nu_2} \ldots \tilde{\varepsilon}_r^{\nu_r} \bar{\varepsilon}_1^{\nu_1 - x_1} \bar{\varepsilon}_2^{\nu_2 - x_2} \ldots \bar{\varepsilon}_r^{\nu_r - x_r}.$$

At a finite prime each $\tilde{\varepsilon}_i$ has component 1; the remaining factors have exponents that are divisible by $2dm$. ε is therefore a $2dm$-th power at every finite prime. Theorem 1 shows that it is a dm-th power in k: $\varepsilon = \eta^{dm}$, $\eta \in k$. This η must be a unit of k and consequently its d-th power in the subgroup generated by the ε_i : $\eta^d = \varepsilon_1^{\mu_1} \ldots \varepsilon_r^{\mu_r}$, $\mu_i \in \mathbb{Z}$. Consequently $\varepsilon = \varepsilon_1^{\nu_1} \varepsilon_2^{\nu_2} \ldots \varepsilon_r^{\nu_r} = \varepsilon_1^{\mu_1 m} \varepsilon_2^{\mu_2 m} \ldots \varepsilon_r^{\mu_r m}$. Since ε_i are independent, we must have $\nu_i = \mu_i m$. This shows that every x_i is divisible by m. Since m was an arbitrary integer it follows that each $x_i = 0$ as was to be shown. □

Let V be the integral valuation vectors of the rational numbers $\mathbb{Q}$. We may view V as a direct sum: $V = \overline{\mathbb{Z}} + \mathbb{R}$ ($\mathbb{R}$ the reals) with the product topology. Any element $\lambda \in V$ can therefore be written as pair: $\lambda = (x, s)$ where $x \in \overline{\mathbb{Z}}$ and $s \in \mathbb{R}$. The integers $m \in \mathbb{Z}$ are canonically imbedded in V by the map $m \to (m, m)$.

If ε is a totally positive unit of k (that means $\varepsilon_{\mathfrak{p}_\infty} > 0$ for every real $\mathfrak{p}_\infty$) we define (not quite canonically) an exponentiation of ε by an element $\lambda = (x, s)$ of V by putting

$$\varepsilon^\lambda = \tilde{\varepsilon}^s \bar{\varepsilon}^x.$$

The idèle $\tilde{\varepsilon}^s$ is defined as follows: it has component 1 all finite primes and component $e^{s \log \varepsilon_{\mathfrak{p}_\infty}}$ at $\mathfrak{p}_\infty$; the log has any fixed value for a complex prime but is the real log if $\mathfrak{p}_\infty$ is real (this last restriction is necessary in order to obtain for the component an element of $k_{\mathfrak{p}_\infty}$). This ε^λ is again an idèle of U. It is the ordinary power if λ is an integer and has always volume $= 1$. Indeed $|\varepsilon^\lambda| = |\tilde{\varepsilon}^s| = |\tilde{\varepsilon}|^s = 1$.

The map $\lambda \to \varepsilon^\lambda$ (for a fixed ε) is a continuous map of V into U.

If $\mathfrak{p}_1, \ldots, \mathfrak{p}_{r_2}$ are the complex primes of k, we have unit circles in each of the corresponding local fields. We describe the μ-th unit circle by the idèle $\phi_\mu(t_\mu)$ which has at $\mathfrak{p}_\mu$ the component $e^{2\pi i t_\mu}$ ($t_\mu \in \mathbb{R}$) and 1 at all other components.

Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_r$ be a system of independent totally positive units of k (if ε_i is not totally positive its square will be).

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We note first that the idèles of the form

$$\varepsilon_1^{\lambda_1}\varepsilon_2^{\lambda_2}\dots\varepsilon_r^{\lambda_r}\phi_1(t_1)\dots\phi_{r_2}(t_{r_2}) \qquad \lambda_1 \in V,\ t_i \in \mathbb{R} \tag{2}$$

form a group and have volume 1. We want to know when an idèle of the form (2) is principal.

LEMMA 1. *Let $\lambda_i = (x_i, s_i)$. An idèle of the form (2)is equal to an element α of k if and only if each λ_i and each t_i is an integer. For the λ_i this means that x_i is an integer (in the natural imbedding in $\overline{\mathbb{Z}}$) and that s_i is the same integer.*

PROOF. The sufficiency of the condition is obvious. Suppose conversely that the idèle of (2) is $= \alpha \in k$. Looking only at the finite primes we must have

$$\bar{\varepsilon}_1^{x_1}\dots\bar{\varepsilon}_r^{x_r} = \overline{\alpha}. \tag{3}$$

The element α must be a unit. α^d lies therefore in the subgroup generated by the ε_i, say $\alpha^d = \varepsilon_1^{\mu_1}\dots\varepsilon_r^{\mu_r}$ with $\mu_i \in \mathbb{Z}$. Raising (3) into the d-th power we obtain:

$$\bar{\varepsilon}_1^{dx_1-\mu_1}\dots\bar{\varepsilon}_r^{dx_r-\mu_r} = 1.$$

Theorem 2 shows that $dx_i - \mu_i = 0$. This means (if one considers the ordinals of x_i at the primes p of $\overline{\mathbb{Z}}$) that the integer μ_i is divisible by the integer d in the ordinary sense, and since d is not a divisor of 0 in $\overline{\mathbb{Z}}$, this means that x_i is an ordinary integer. This makes now

$$\varepsilon = \varepsilon_1^{x_1}\dots\varepsilon_r^{x_r}$$

an element of k. This ε is equal to α at all finite primes. Even if we had this only at one prime we could conclude $\alpha = \varepsilon$. The infinite components of α are therefore $\tilde{\varepsilon}_1^{x_1}\tilde{\varepsilon}_2^{x_2}\dots\tilde{\varepsilon}_r^{x_r}$ on one hand, whereas (2) gives $\tilde{\varepsilon}_1^{s_1}\tilde{\varepsilon}_2^{s_2}\dots\tilde{\varepsilon}_r^{s_r}\phi_1(t_1)\dots\phi_{r_2}(t_{r_2})$.

The relation is:

$$\tilde{\varepsilon}_1^{s_1-x_1}\tilde{\varepsilon}_2^{s_2-x_2}\dots\tilde{\varepsilon}_r^{s_r-x_r}\phi_1(t_1)\dots\phi_{r_2}(t_{r_2}) = 1. \tag{4}$$

If we take components at $\mathfrak{p}_\infty$ and then the absolute value we obtain

$$|\varepsilon_1|_{\mathfrak{p}_\infty}^{s_1-x_1}\dots|\varepsilon_r|_{\mathfrak{p}_\infty}^{s_r-x_r} = 1.$$

From the independence of our units we see immediately that $s_i = x_i$. If we substitute this result in (4) it follows now trivially that all the t_i are also integers. This proves the lemma. □

The reader is invited to investigate how it comes that the roots of unity of k did not play a role in our proof.

Consider the map

$$(\lambda_1, \dots, \lambda_r, t_1, \dots, t_{r_2}) \to \varepsilon_1^{\lambda_1}\dots\varepsilon_r^{\lambda_r}\phi_1(t_1)\dots\phi_{r_2}(t_{r_2}).$$

We shall use the vector notation, and abbreviate this map by

$$(\lambda, t) \to \varepsilon^\lambda\phi(t).$$

It is a homomorphism of $V^r \times \mathbb{R}^{r_2}$ into the idèles and it is continuous.

We may follow this map by the canonical homomorphism into the idèle classes. According to the preceding lemma, the kernel consists precisely of $\mathbb{Z}^r \times \mathbb{Z}^{r_2}$. Each $\mathbb{Z}$ is closed in V or $\mathbb{R}$, and we obtain a continuous isomorphism of $(V/\mathbb{Z})^r \times (\mathbb{R}/\mathbb{Z})^{r_2}$ into C_0. It will be shown below that it is a topological isomorphism *onto* the connected component of C_0.

The following lemma concerning the topology and the group structure of $V/\mathbb{Z}$ will be useful.

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LEMMA 2. *$V/\mathbb{Z}$ is compact. The reals $\mathbb{R}$, naturally imbedded in $V/\mathbb{Z}$, are everywhere dense in it, and $V/\mathbb{Z}$ is consequently connected. $V/\mathbb{Z}$ is infinitely and uniquely divisible.*

PROOF. 1. One sees easily that every element of $V/\mathbb{Z}$ has a representative $\lambda = (x, s)$ with $0 \leqslant s \leqslant 1$. If we denote by N the set of all $\lambda = (0, s)$ with $0 \leqslant s \leqslant 1$ then our representative is in $\overline{\mathbb{Z}} + N$ and this means that $V/\mathbb{Z}$ is the image of the compact set $\overline{\mathbb{Z}} + N$ under the canonical map of V onto $V/\mathbb{Z}$. $V/\mathbb{Z}$ is therefore compact.

2. Let $\lambda = (x, s)$ be any element of V. If m is a given integer (describing a neighborhood of $\overline{\mathbb{Z}}$), let $\mu \equiv x \pmod{m}$, $\mu \in \mathbb{Z}$ and subtract the image of μ in V, so the pair (μ, μ) from λ. This leads to the pair $(x - \mu, s - \mu)$ which represents the same element as λ in $V/\mathbb{Z}$. This pair belongs to the set $(m\overline{\mathbb{Z}}, 0) + (0, s - \mu)$ which is mapped into a neighborhood of the real $s - \mu$ in $V/\mathbb{Z}$. Therefore $\mathbb{R}$ is dense in $V/\mathbb{Z}$. Since the closure of a connected set is connected, it follows that $V/\mathbb{Z}$ is connected.

3. The same computation allows us to write $\lambda \equiv m\left(\frac{x-\mu}{m}, \frac{s-\mu}{m}\right) \pmod{\mathbb{Z}}$ and shows divisibility by m. If $m\lambda \equiv 0 \pmod{\mathbb{Z}}$ then there would be an integer $\mu \in \mathbb{Z}$ such that $mx = \mu$ in $\overline{\mathbb{Z}}$ and $ms = \mu$ in $\mathbb{R}$. The first equation implies (as in the previous lemma) that x is an integer and the second shows now that s is the same integer. But this means $\lambda \equiv 0 \pmod{\mathbb{Z}}$, and proves the unique divisibility. $\square$

The group $V/\mathbb{Z}$ is called the solenoid. It can be shown that it is simply the compact dual of the discrete additive group $\mathbb{Q}$ of rationals.

We return to our continuous isomorphism of $(V/\mathbb{Z})^r \times (\mathbb{R}/\mathbb{Z})^{r_2}$ into C_0. Each circle $\mathbb{R}/\mathbb{Z}$ is compact and infinitely (but not uniquely) divisible. It is obviously connected. The group $(V/\mathbb{Z})^r \times (\mathbb{R}/\mathbb{Z})^{r_2}$ is therefore compact, connected and infinitely divisible. Its image D_0 in C_0 is therefore compact, connected and infinitely divisible and our map an isomorphism, both algebraically and topologically.

We contend now that every infinitely divisible idèle class of volume 1 lies in D_0. Let a be such a class. Write $a = b^{2hm}$ where h is the class number of k and m a highly divisible integer. The class b^h can be represented by a unit idèle, the class b^{2h} by an idèle $\mathfrak{c}$ of U that is totally positive. a is therefore represented by the idèle $\mathfrak{c}^m = \tilde{\mathfrak{c}}^m \bar{\mathfrak{c}}^m$. The idèle $\bar{\mathfrak{c}}^m$ is in as small a neighborhood of 1 as we like for a suitably highly divisible m. We shall prove that the idèle class c of $\tilde{\mathfrak{c}}^m$ belongs to D_0. From this it follows that a is in the closure of D_0. But D_0 is a closed set and consequently we will have proved that a is in D_0.

Let $\tilde{\mathfrak{c}}^m = \tilde{\mathfrak{a}}$. Then $\tilde{\mathfrak{a}}$ is totally positive and $|\tilde{\mathfrak{a}}| = 1$.

Let $\mathfrak{p}_\nu$ $(\nu = 1, \ldots, r+1)$ range over the archimedean primes.

From the independence of the units we can solve for $s_1, \ldots, s_r$ the system of equations

$$|\tilde{\mathfrak{a}}|_{\mathfrak{p}_\nu} = |\tilde{\varepsilon}_1|_{\mathfrak{p}_\nu}^{s_1} \ldots |\tilde{\varepsilon}_r|_{\mathfrak{p}_\nu}^{s_r} \qquad \nu = 1, \ldots, r.$$

Since $|\tilde{\mathfrak{a}}| = 1$ we also have

$$|\tilde{\mathfrak{a}}|_{\mathfrak{p}_{r+1}} = |\tilde{\varepsilon}_1|_{\mathfrak{p}_{r+1}}^{s_1} \ldots |\tilde{\varepsilon}_r|_{\mathfrak{p}_{r+1}}^{s_r}.$$

Let $\lambda_i = (0, s_i) \in V$. Since $\tilde{\mathfrak{a}}$ is totally positive, we have

$$\tilde{\mathfrak{a}}_{\mathfrak{p}_\infty} = (\tilde{\varepsilon}_1^{s_1} \ldots \tilde{\varepsilon}_r^{s_r})_{\mathfrak{p}_\infty}$$

for any real $\mathfrak{p}_\infty$. At a complex prime they differ by an element of value 1. Hence we can write

$$\tilde{\mathfrak{a}} = \tilde{\varepsilon}_1^{s_1} \ldots \tilde{\varepsilon}_r^{s_r} \phi_1(t_1) \ldots \phi_{r_2}(t_{r_2})$$

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thereby proving our contention, that D_0 contains the infinitely divisible idèle classes of volume 1. Since D_0 itself is infinitely divisible, it is therefore equal to the infinitely divisible idèle classes of volume 1.

This shows that D_0 is the intersection of all open subgroups of finite index in C_0. It contains therefore the connected component of identity in C_0. Since D_0 is connected, it now follows that it is the connected component of 1 in C_0.

In order to get the full connected component of C, we select at some archimedean prime $\mathfrak{p}_\infty$ a positive real line $\mathbb{R}_+ \subset k_{\mathfrak{p}_\infty}$. Then it is clear that every element from the connected component is uniquely represented by an idèle of the group

$$\mathbb{R}_+ \cdot \varepsilon^\lambda \cdot \phi(t).$$

Thus we have proved

THEOREM 3. *The mapping* $(\lambda, t) \to \varepsilon^\lambda \phi(t)$ *of* $(V/\mathbb{Z})^r \times (\mathbb{R}/\mathbb{Z})^{r_2}$ *into* C_0 *is a topological isomorphism* onto *the connected component of* C_0. *The structure of the connected component of* C *is that of a direct product of one real line* $\mathbb{R}_+$, $r = r_1 + r_2 - 1$ *solenoids, and* r_2 *circles and is the closure in* C *of the image of the connected component of* J.

2. Cohomology of the Connected Component

Let K/k be normal with group G. As we shall deal only with the idèle classes of K, we let

$$C = C_K, \quad D = D_K, \quad D_0 = D_K \cap C_K^0.$$

Then, as is easily seen, it is possible to choose a splitting such that $D = D_0 \times \mathbb{R}_+$ and G has trivial action on $\mathbb{R}_+$

Let $\{\mathfrak{P}\}$ range over the *complex* primes of K. The circles of the preceding section are now denoted by $\phi_{\mathfrak{P}}(t_{\mathfrak{P}})$. Such a circle is the idèle having all components 1 except the $\mathfrak{P}$-component, which is $e^{2\pi i t_{\mathfrak{P}}}$. We can write

$$\prod_{\mathfrak{P}} \phi_{\mathfrak{P}} = \prod_{\mathfrak{p}} \Big(\prod_{\mathfrak{P}|\mathfrak{p}} \phi_{\mathfrak{P}} \Big).$$

Indeed, each $\mathfrak{P}$ of K extends exactly one prime $\mathfrak{p}$ of k, which may be real or complex. It is clear from the definition of the action of G on idèles that each semilocal component $\prod_{\mathfrak{P}|\mathfrak{p}} \phi_{\mathfrak{P}}$ is invariant under G, and hence the full product is invariant under G. (This comes from the fact that each local module $\phi_{\mathfrak{P}}$ consists precisely of those elements of the local complex field having absolute value equal to 1.)

Let D' be the subgroup of D given by the canonical image of the circles in C, i.e.

$$D' = \prod_{\mathfrak{P}} \phi_{\mathfrak{P}}(t_{\mathfrak{P}}) K.$$

Then D' is G-isomorphic to the product $\prod_{\mathfrak{P}} \phi_{\mathfrak{P}}$, and is a subgroup of D, invariant under G. From the preceding section, we see that D/D' is G-isomorphic to $\mathbb{R}_+ \times (\prod V/\mathbb{Z})$. Since both $\mathbb{R}_+$ and $V/\mathbb{Z}$ are uniquely divisible, it follows from IV, that $\mathcal{H}^r(D/D') = 1$ for all r. From the exact sequence

$$\mathcal{H}^{r-1}(D/D') \to \mathcal{H}^r(D') \to \mathcal{H}^r(D) \to \mathcal{H}^r(D/D')$$

we get

THEOREM 4. *For all* r, $\mathcal{H}^r(D') \simeq \mathcal{H}^r(D)$.

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The cohomology of D' is now easily determined. For each semilocal pair $(G, \prod_{\mathfrak{P}|\mathfrak{p}} \phi_{\mathfrak{P}})$ let $(G_{\mathfrak{p}}, \phi_{\mathfrak{p}})$ be a local component. From the semilocal theory, we know that

$$\mathcal{H}^r\Big(G, \prod_{\mathfrak{P}|\mathfrak{p}} \phi_{\mathfrak{P}}\Big) \simeq \mathcal{H}^r(G_{\mathfrak{p}}, \phi_{\mathfrak{p}}),$$

and since the cohomology group of a product is the product of the cohomology groups, we have

$$\mathcal{H}^r(D') \simeq \prod_{\mathfrak{p}} \mathcal{H}^r(G_{\mathfrak{p}}, \phi_{\mathfrak{p}}).$$

If $\mathfrak{p}$ is unramified, i.e. splits, then $G_{\mathfrak{p}}$ is trivial, and so is the local cohomology group. If $\mathfrak{p}$ is real, $G_{\mathfrak{p}}$ is cyclic of order 2, generated by the complex conjugation. From the cyclic theory, we know that the cohomology groups are periodic of period 2, and hence our computation is reduced to the dimensions -1 and 0.

In dimension -1, we know that the cohomology group is isomorphic to the elements with norm 1, modulo those which have obviously norm 1 (i.e. those of type $a^{1-\sigma}$). One verifies immediately that for the circles, this factor group is trivial.

In dimension 0, we know that

$$\mathcal{H}^0(G_{\mathfrak{p}}, \phi_{\mathfrak{p}}) \simeq (\phi_{\mathfrak{p}}^{G_{\mathfrak{p}}} : N_{\mathfrak{p}}\phi_{\mathfrak{p}}),$$

i.e. the fixed elements modulo the norms. But the fixed elements are simply ± 1, and the norms are trivial, i.e. equal to 1. Hence the factor group is cyclic of order 2. Thus we have proved:

THEOREM 5. *Let K/k be normal with group G. Then $\mathcal{H}^{2r}(D')$ is of type $(2, 2, \ldots, 2)$ and order 2^{μ} where μ is the number of ramified archimedean primes. $\mathcal{H}^{2r+1}(D') = 1$ for all r.*

In the exact sequence

$$1 = \mathcal{H}^1(C) \to \mathcal{H}^1(C/D) \to \mathcal{H}^2(D) \to \mathcal{H}^2(C) \to \mathcal{H}^2(C/D) \to 1$$

the map i maps each local factor of $\mathcal{H}^2(D)$ onto the kernel of multiplication by 2 in the cyclic group $\mathcal{H}^2(C)$. Thus, if $\mu > 0$, then $\mathcal{H}^1(C/D)$ is of type $(2, 2, \ldots, 2)$ and order $2^{\mu-1}$, and $cH^2(C/D)$ is cyclic of order $\frac{1}{2}[K : k]$.

We note that $\mathcal{H}^1(C/D)$ is not necessarily trivial, and that consequently the inclusion $i\colon \mathcal{H}^2(D) \to \mathcal{H}^2(C)$ is not always an isomorphism.

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CHAPTER X

The Grunwald–Wang Theorem[1]

1. Interconnection between Local and Global m-th Powers

Let k be a global field, m any integer and S a finite (possibly empty) set of primes. Denote by $P(m, S)$ the group of those elements $\alpha \in k^*$ which are in $k_{\mathfrak{p}}^{*m}$ for all $\mathfrak{p} \notin S$.

k^{*m} is a subgroup of $P(m, S)$ and we ask now for the precise structure of the factor group. Theorem 1 of the preceding chapter provides already an answer in certain cases and we must now investigate the remaining possibilities. If k is a function field, then $P(m, S) = k^{*m}$.

Assume therefore from now on that k is a numberfield and let $m = 2^t m'$ (m' odd). The answer is again $P(m, S) = k^m$ if $k(\zeta_{2^t})/k$ is cyclic (see IX, Section 1, Theorem 1).

We are therefore led to investigate the conditions under which $k(\zeta_{2^t})$ is a non-cyclic field.

Changing the notation slightly denote by ζ_r ($r = 1, 2, \dots$) a primitive 2^r-th root of unity such that

$$\zeta_{r+1}^2 = \zeta_r. \tag{1}$$

Put:

$$\eta_r = \zeta_r + \zeta_r^{-1} \tag{2}$$

and observe that $\eta_r = 0$ if and only if $r = 2$. We also obtain:

$$\eta_{r+1}^2 = 2 + \eta_r \tag{3}$$

$$\zeta_{r+1}\eta_{r+1} = \zeta_r + 1. \tag{4}$$

(3) shows that a field containing η_r also contains all η_μ with $\mu \leqslant r$. (4) shows that a field containing η_{r+1} and ζ_r will contain ζ_{r+1} if $r \geqslant 2$. If therefore $\zeta_2 = i$ and $\eta_r (r > 2)$ are in a field, then ζ_r is also in this field. $k(\zeta_r)$ is therefore the compositum of the fields $k(i)$ and $k(\eta_r)$ for $r > 2$.

[1] In 1932 W. Grunwald proved a theorem like Theorem 5 of this chapter in his dissertation (supervised by H. Hasse, cf. [**10**]). However he did not notice the need for an extra condition in a certain special case. Ten years later, G. Whaples published a new proof of Grunwald's theorem [**30**] also overlooking the special case. This new proof was presented by Bill Mills in Artin's seminar in the spring of 1948. A few days later, one of the listeners, Shianghaw Wang, came to Artin's office with a counterexample to a key lemma of that proof, and later the same day produced a counterexample to the Theorem itself, by proving that there does not exist a cyclic extension of $\mathbb{Q}$ of degree 8 in which 2 stays prime (see the 'consequence' before Theorem 2 below). In his Ph.D. thesis, [**26**] Wang published a corrected version of Grunwald's statement, which is now known as the Grunwald–Wang Theorem. This chapter is Artin's own reworking of this theorem, its proof and related questions, in the light of Wang's discovery.

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The extensions $k(\eta_r)/k$ are cyclic. To see this it is sufficient to show that $\mathbb{Q}(\eta_r)/\mathbb{Q}$ is cyclic since the group of $k(\eta_r)/k$ will be a subgroup. Every automorphism of $\mathbb{Q}(\zeta_r)$ is induced by an automorphism $\sigma_\mu(\zeta_r) = \zeta_r^\mu$ of $\mathbb{Q}(\eta_r)$ and (2) shows that σ_μ and $\sigma_{-\mu}$ will induce the same automorphism of $\mathbb{Q}(\eta_r)$. We may therefore assume that $\mu \equiv 1 \pmod 4$. These automorphisms σ_μ of $\mathbb{Q}(\zeta_r)$ form a cyclic group generated by σ_5 and this shows that $\mathbb{Q}(\eta_r)$ is cyclic.

If k would contain all η_r then $k(i)$ would contain all ζ_r which is impossible.

There is therefore an integer $s \geqslant 2$ such that $\eta_s \in k$ but $\eta_{s+1} \notin k$.

Because of (3) $k(\eta_{s+1})/k$ is quadratic.

Suppose now that all $k(\zeta_r)/k$ are cyclic. $k(\eta_{s+1})$ is then the only quadratic subfield they can have. $k(i)$ is either k or quadratic and consequently $i \in k(\eta_{s+1})$. If conversely $i \in k(\eta_{s+1}) \subset k(\eta_r)$ (for $r > s+1$) then $k(\zeta_r) = k(\eta_r)$ and is therefore cyclic.

The non-cyclic case is therefore characterized by the fact that $F = k(i, \eta_{s+1}) = k(\zeta_{s+1})$ is a four group field over k. In this case, F contains the three quadratic subfields $k(i)$, $k(\eta_{s+1})$ and $k(i\eta_{s+1})$ and (3) shows that this fact can be expressed within k by the condition that the three elements -1, $2+\eta_s$, $-(2+\eta_s)$ are non-squares in k.

Since $\eta_s \in k$, $k(i) = k(\zeta_s)$ and we see that $k(\zeta_t)$ is cyclic for $t \leqslant s$. But $F = k(\zeta_{s+1})$ and all $k(\zeta_t)$ with $t > s$ are non-cyclic. Let σ be the automorphism $\neq 1$ of $k(i)/k$. $\zeta_s + \zeta_s^{-1} = \eta_s \in k$ and $\zeta_s\zeta_s^{-1} = 1 \in k$ show that ζ_s and ζ_s^{-1} are conjugate: $\zeta_s^\sigma = \zeta_s^{-1}$.

Suppose now that there exists an $\alpha \in P(m, S)$ which is not in k^{*m}. Then $\alpha \notin P(2^t, S)$ but $\alpha \in k^{*2^t}$ (since $\alpha \in k^{*m'}$). We see immediately that we must have the non-cyclic case and that $t > s$.

With $k(i)$ as ground field we have the cyclic case so we can write $\alpha = A^{2^t}$. Raising to the power $1-\sigma$ we obtain:

$$(A^{1-\sigma})^{2^t} = 1.$$

$A^{1-\sigma}$ is a 2^t-th root of unity in $k(i) = k(\zeta_s)$ and $\zeta_{s+1} \notin k(i)$ (it generates F). Therefore $A^{1-\sigma} = \zeta_s^\mu$. In $\alpha = A^{2^t}$ we can replace A by $A_1 = A\zeta_s^\lambda$. Then $A_1^{1-\sigma} = \zeta_s^{\mu+2\lambda}$. If μ were even we could achieve $A^{1-\sigma} = 1$ or $A_1 \in k$ contradicting $\alpha \notin k^{2^t}$. So μ is odd and now we can achieve that μ is any given odd number especially m'.

$$A^{1-\sigma} = \zeta_s^{m'} = \left(\frac{1+\zeta_s}{1+\zeta_s^{-1}}\right)^{m'} = (1+\zeta_s)^{(1-\sigma)m'}.$$

It follows that $\beta = A(1+\zeta_s)^{-m'}$ satisfies $\beta^{1-\sigma} = 1$ or $\beta \in k$. From $A = \beta(1+\zeta_s)^{m'}$ we get $\alpha = \alpha_0\beta^{2^t}$ where

$$\begin{aligned} \alpha_0 &= (1+\zeta_s)^m = \eta_{s+1}^m = (i\eta_{s+1})^m = (2+\eta_s)^{m/2} \\ &= (-2-\eta_s)^{m/2} = \left((2+\eta_s)^{2^{t-1}}\right)^{m'}. \end{aligned} \tag{5}$$

These various expressions follow from (3), (4) using $t > s \geqslant 2$.

This α_0 is in $k^{*m'}$; $\alpha\alpha_0^{-1} \in k^{*m'}$ together with $\alpha\alpha_0^{-1} = \beta^{2^t}$ shows $\alpha\alpha_0^{-1} \in k^{*m}$ or $\alpha \in \alpha_0 k^{*m}$.

Thus far we have seen that $P(m, S) \subset k^m \cup \alpha_0 k^m$. We have still to investigate whether actually $\alpha_0 \in P(m, S)$ and whether the two cosets are different.

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Let S_0 be the (possibly empty) set of all primes $\mathfrak{p}$ where $F_\mathfrak{p}/k_\mathfrak{p}$ is of degree 4 (and consequently a four group field). Since unramified fields are cyclic, and $k(\zeta_{s+1})$ is only ramified for primes dividing 2, and since archimedean extensions are of degree 2, it follows that S_0 consists only of certain divisors of 2.

Assume now that $\alpha_0 \in k^m$ or that it is in some $k_\mathfrak{p}^m$ with $\mathfrak{p} \in S_0$. Then it would be in k^{2^t} resp. $k_\mathfrak{p}^{2^t}$. Since m' is odd, $(2+\eta_s)^{2^{t-1}}$ would be in k^{2^t} resp. $k_\mathfrak{p}^{2^t}$. Extracting a 2^{t-1}-th root $(2+\eta_s)\zeta$ would be in k^2 resp. $k_\mathfrak{p}^2$ where ζ is a 2^{t-1}-th root of unity of k resp. $k_\mathfrak{p}$. But $k(i)$ resp. $k_\mathfrak{p}(i)$ are genuine extensions so ζ can only be ± 1. The fact that F/k resp. $F_\mathfrak{p}/k_\mathfrak{p}$ is a four group field shows that $\pm(2+\eta_s)$ is not a square and we have a contradiction.

This shows that $\alpha_0 k^m \neq k^m$ but it shows also that α_0 can be in $P(m,S)$ only if $S_0 \subset S$.

Assume now $S_0 \subset S$ and let $\mathfrak{p} \notin S$. $F_\mathfrak{p}/k_\mathfrak{p}$ collapses so one of the three quadratic fields of F must collapse. Either ζ_s or η_{s+1} or $i\eta_{s+1}$ is in $k_\mathfrak{p}$ and a suitable expression from (5) shows $\alpha_0 \in k_\mathfrak{p}^m$, $\alpha_0 \in P(m,S)$. We have now proved the theorem:

THEOREM 1. *Let k be a global field, m an integer, S a finite set of primes and $P(m,S)$ the group of all $\alpha \in k^*$ such that $\alpha \in k_\mathfrak{p}^m$ for all $\mathfrak{p} \notin S$. Then $P(m,S) = k^{*m}$ except under the following conditions which will be referred to as the* special case:

1. *k is a number field.*
2. *-1, $2+\eta_s$ and $-(2+\eta_s)$ are non-squares in k, where $s \geqslant 2$ is an integer such that $\eta_s \in k$, $\eta_{s+1} \notin k$.*
3. *$m = 2^t m'$ where m' is odd, and $t > s$.*
4. *$S_0 \subset S$, where S_0 is the set of those primes $\mathfrak{p} \mid 2$ where -1, $2+\eta_s$, and $-(2+\eta_s)$ are non-squares in $k_\mathfrak{p}$.*

In this special case $P(m,S) = k^{*m} \cup \alpha_0 k^{*m}$ where α_0 is given by (5). Obviously $\alpha_0 \in k^{*m/2}$, but $\alpha_0 \notin k^{*m}$.

As an example let $k = \mathbb{Q}$. Then $s = 2$, $F = \mathbb{Q}(\zeta_3) = \mathbb{Q}(i, \sqrt{2})$. S_0 contains the single prime 2. The special case occurs if $8 \mid m$ and if S contains 2. $\eta_3 = \sqrt{2}$ therefore

$$\alpha_0 = (1+i)^m = (\sqrt{2})^m = (\sqrt{-2})^m = 2^{\frac{1}{2}m}.$$

Taking especially $m = 8$ (the smallest integer where the exceptional case occurs) we have

$$\alpha_0 = 16 = (1+i)^8 = (\sqrt{2})^8 = (\sqrt{-2})^8$$

and 16 is an 8-th power at ∞ and at all odd primes. Indeed, if p is odd, the extension $\mathbb{Q}_p(i, \sqrt{2})$ is unramified and must therefore be cyclic of degree 2. This means that either i or $\sqrt{2}$ or $\sqrt{-2}$ lies in $\mathbb{Q}_p$ and the formulas show that 16 is an 8-th power. On the other hand 16 is not an 8-th power in $\mathbb{Q}$ or in the 2-adic field.

CONSEQUENCE: Let $K/\mathbb{Q}$ be any cyclic field of degree 8. Then 16 is a norm at all p except possibly 2, and hence at 2 also by the product formula for the norm residue symbol. It follows that $K_2/\mathbb{Q}_2$ cannot be unramified of degree 8 (the ordinal of 16 is not divisible by 8). This shows that one cannot prescribe local conditions at certain primes and find a global cyclic field of given degree satisfying these conditions. The circumstances when this can be done are discussed in the next section.

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As a corollary to Theorem 1 we prove the following theorem which gives us some insight into the structure of the idèle class group:

THEOREM 2. *Let k be a global field, m an integer. Denote by J_m resp. C_m the group of idèles resp. idèle classes whose period divides m and let $\bar{J}_m$ be the canonical image of J_m in the idèle classes. We contend $C_m = \bar{J}_m$ unless we are in the special case with the additional restriction that $S_0 = \varnothing$ (empty). Then there exists an idèle class $a_0 \in \bar{J}_m$ such that $C_m = \bar{J}_m \cup a_0\bar{J}_m$. In function fields of characteristic $p : C_p = 1$.*

PROOF. Let $\mathfrak{a}$ represent an $a \in C_m$. Then $\mathfrak{a}^m = \alpha$ and consequently $\alpha \in P(m, \varnothing)$. Therefore $\alpha = \beta^m$ unless we are in the special case with $S_0 = \varnothing$ when α may be $\alpha_0\beta^m$; this α_0 is an m-th power at all $\mathfrak{p}$ so there exists an idèle $\mathfrak{a}_0$ such that $\alpha_0 = \mathfrak{a}_0^m$. The idèle class a_0 of $\mathfrak{a}_0$ is not in $\bar{J}_m$ or else $\mathfrak{a}_0 = \beta_0\mathfrak{c}$ with $\mathfrak{c}^m = 1$; this would yield $\alpha_0 = \beta_0^m$ which is not the case. We obtain now $\mathfrak{a}_m = \beta^m$ resp. $\mathfrak{a}_m = (\mathfrak{a}_0\beta)^m$ so that $\frac{\mathfrak{a}}{\beta}$ resp. $\frac{\mathfrak{a}}{\beta\mathfrak{a}_0}$ is in J_m. This shows that a resp. $\frac{a}{a_0}$ is in $\bar{J}_m$. $\square$

If k is a function field then trivially $J_p = 1$ and consequently $C_p = \bar{J}_p = 1$.

REMARK 1. The structure of J_m is pretty obvious: each component of and idèle of J_m must be a local m-th root of unity. Since the kernel of the canonical map of J_m onto $\bar{J}_m$ is very simple namely the m-th roots of unity of the global field k (as far as they exist), the structure of $\bar{J}_m$ is also understood.

Since S_0 is not empty in case $k = \mathbb{Q}$, this is not an example for the special case of Theorem 2.

Consider $k = \mathbb{Q}(\sqrt{7})$. Over $\mathbb{Q}_2$ we have $\sqrt{7} = i\sqrt{-7}$ where $\sqrt{-7} \in \mathbb{Q}_2$ since $-7 \equiv 1 \pmod 8$. $k(\zeta_3) = k(\sqrt{2}, i) = F$ is a four group field which collapses over $k_2 = \mathbb{Q}_2\sqrt{7}$ since $i \in k_2$. Therefore S_0 is empty, and we have the special case of Theorem 2 if we take $m = 8$.

REMARK 2. In the special case (if only $t > s$), a_0 is independent of m. Formula (5) shows indeed that $(\mathfrak{a}_0)_\mathfrak{p} = 1 + \zeta_s$ or η_{s+1} or $i\eta_{s+1}$ depending on which of the three quadratic fields collapses at $\mathfrak{p}$.

2. Abelian Fields with Given Local Behavior

Consider the idèle class group C of a field k. The corresponding local fields $k_\mathfrak{p}$ have been imbedded naturally in C. This imbedding is done in such a way that the local norm residue symbol $(\alpha, k_\mathfrak{p})$ coincides with the global one (α, k).

An abelian extension K/k is described by its norm group $N = NC_K$ which is an open subgroup of finite index in C, and every open subgroup of finite index is conversely the norm group of exactly one abelian extension K/k.

The norm groups $N_\mathfrak{P}K_\mathfrak{P}^*$ of the local completions are given by:[2]

$$N_\mathfrak{P}K_\mathfrak{P}^* = N \cap k_\mathfrak{p}^*.$$

Let S be a finite set of primes. Via the injection $P \to J_k$ we see that C contains the direct product $P = \prod_{\mathfrak{p}\in S} k_\mathfrak{p}^*$. If S contains more than one prime the topology that C induces on P is not any more the product topology. P will even not be closed in C. The intersection

$$P_0 = N \cap P$$

[2]This follows from (VII, Sect. 3, Corollary 1 of Theorem 9).

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defines a subgroup P_0 of P of finite index which is open in P. It describes the local extensions $K_{\mathfrak{P}}/k_{\mathfrak{p}}$ for $\mathfrak{p} \in S$ and the interconnections between them. Indeed, $N_{\mathfrak{P}} K_{\mathfrak{P}}^* = P_0 \cap k_{\mathfrak{p}}^*$ and more generally, P_0 is the kernel of the norm residue map for elements $a \in P$:

$$(a, K/k) = \prod_{\mathfrak{p} \in S} (a_{\mathfrak{p}}, K_{\mathfrak{P}}/k_{\mathfrak{p}})$$

Two questions arise now:

1. Let $\widetilde{P} = \prod_{\mathfrak{p} \in S} k_{\mathfrak{p}}^*$ but with the product topology. The map of $\widetilde{P}$ onto P is an algebraic isomorphism. We may view $\widetilde{P}$ as a subgroup of J and the map as the canonical map into C. It is therefore continuous. Do the open subgroups of finite index of $\widetilde{P}$ and P correspond to each other? The answer is affirmative.

2. Let P_0 be a given open subgroup of finite index in P. Does there exist in open subgroup $N \subset C$ of finite index such that $P_0 = N \cap P$? This would give the existence of an abelian extension K/k whose local behavior at the primes of S is the one described by P_0. The answer is also affirmative.

LEMMA 1. *Let A and B be subsets of a topological group, A compact and B closed. Then AB is closed.*

PROOF. We have to show that the complement of AB is open. We remark that $c \notin AB$ is equivalent to $A^{-1}c \cap B = \varnothing$. We have therefore to show: If $A^{-1}c \cap B = \varnothing$, then there exists a neighborhood V of 1 such that $A^{-1}cV \cap B = \varnothing$. We denote by W's open neighborhoods of identity. Since $A^{-1}c$ is compact a change of notation transforms our question into the following:

Let A be compact and B closed and assume $A \cap B = \varnothing$. Find a V such that $AV \cap B = \varnothing$. Since B is closed we can find to each $a \in A$ a W'_a such that $aW'_a \cap B = \varnothing$ and a W_a such that $W_a W_a \subset W'_a$. Since A is compact we can cover A by a finite number $a_i W_i$ of the sets aW_a. Now we determine a $V \subset \bigcap_i W_i$. If $a \in A$ is in $a_i W_i$ then $aV \subset a_i W_i V \subset a_i W_i W_i \subset a_i W'_{a_i}$ and does not intersect B. □

LEMMA 2. *$P \cap C^m = P^m$ unless we are in the special case with a non-empty set S_0. In this special case let c_m be the idèle class of P that has components α_0 at the primes of S_0 and 1 at the other primes of S. Then $P \cap C^m = P^m \cup c_m P^m$.*

PROOF. An element $a \in P \cap C^m$ can be represented by an idéle $\mathfrak{a}$ with components 1 outside S and we must have an equation of the form $\mathfrak{a} = \alpha \mathfrak{c}^m$. This shows that $\alpha \in P(m, S)$ and therefore $\alpha = \beta^m$ unless we are in the special case when $\alpha = \alpha_0 \beta^m$ is also possible. This α_0 is an m-th power outside S_0 and we may write $\alpha_0 = \mathfrak{c}_m \vartheta^m$ where $\mathfrak{c}_m$ has components α_0 in S_0 and 1 everywhere else. This shows that $\mathfrak{a} = \mathfrak{n}^m$ or, in the special case if S_0 is not empty possibly $= \mathfrak{c}_m \mathfrak{n}^m$. Comparing components outside S we see that $\mathfrak{n}$ can be taken in $\prod_{\mathfrak{p} \in S} k_{\mathfrak{p}}$. Going back to P we obtain our lemma. □

LEMMA 3. *P^m is closed in P and the factor groups P/P^m and $\widetilde{P}/\widetilde{P}^m$ are compact.*

PROOF. C^m is closed as we know. Therefore $P \cap C^m$ is closed in P. This shows that P^m is closed in P unless we are in the special case. In the special case k is a number field and the index of P^m in P is finite. Since $P^{2m} \subset (P \cap C^{2m}) \subset P^m$ and $P \cap C^{2m}$ is closed, we see that P^m is closed in this case also.

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The map of $\widetilde{P}/\widetilde{P}^m$ onto P/P^m is continuous. It suffices therefore to show that $\widetilde{P}/\widetilde{P}^m$ is compact. This amounts to show that $k_\mathfrak{p}/k_\mathfrak{p}^m$ is compact. For archimedean primes the factor group is finite of order 1 or 2. For non-archimedean primes $k_\mathfrak{p} \simeq \mathbb{Z} \times U_\mathfrak{p}$, and therefore $k_\mathfrak{p}/k_\mathfrak{p}^m \simeq \mathbb{Z}/m\mathbb{Z} \times U_\mathfrak{p}/U_\mathfrak{p}^m$, a product of a finite group and a continuous image $U_\mathfrak{p}/U_\mathfrak{p}^m$ of the compact group $U_\mathfrak{p}$. □

THEOREM 3. *The open subgroups of finite index of $\widetilde{P}$ and of P correspond to each other under the mapping.*

PROOF. Since the map of $\widetilde{P}$ onto P is continuous, an open subgroup of P comes from an open subgroup of $\widetilde{P}$. Let conversely $\widetilde{P}_0$ be an open subgroup of finite index of $\widetilde{P}$ and select m in such a way that $\widetilde{P}^m \subset \widetilde{P}_0$. $\widetilde{P}_0/\widetilde{P}^m$ is compact so the image P_0/P^m is compact and therefore closed. It is of finite index in P/P^m and therefore open. Consequently P_0 is open. □

LEMMA 4. *If P_0 is an open subgroup of finite index in P then P_0C^n is closed in C.*

PROOF. $\widetilde{P}_0$ is open in $\widetilde{P}$. Let $\widetilde{W} = \prod_{\mathfrak{p}\in S} \widetilde{W}_\mathfrak{p}$ be a neighborhood of 1 in $\widetilde{P}$ that is contained in $\widetilde{P}_0$, where the $\widetilde{W}_\mathfrak{p}$ are compact neighborhoods of 1 in $k_\mathfrak{p}$. $\widetilde{W}$ is compact. Determine an m divisible by n such that $\widetilde{P}^m \subset \widetilde{P}_0$. Since $\widetilde{P}_0/\widetilde{P}^m$ is compact one can find a finite covering $\widetilde{P}_0 = \bigcup_i p_i\widetilde{P}^m\widetilde{W}$. Going over to P one obtains $P_0 = \bigcup_i p_iP^mW$. The point is that W is compact as continuous image of a compact set (although not necessarily a neighborhood of 1 in P). Multiplying by C^n and observing that P^m is absorbed by C^n one has $P_0C^n = \bigcup_i p_iWC^n$. W is compact and C^n is closed. Lemma 1 shows the contention. □

LEMMA 5. *C/C^n is compact.*

PROOF. In number fields C has the structure $\mathbb{R}_+ \times C_0$ so that $C/C^n \simeq C_0/C_0^n$ an image of the compact group C_0. In function fields the structures are $\mathbb{Z} \times C_0$ and $\mathbb{Z}/n\mathbb{Z} \times C_0/C_0^n$ and the conclusion is the same. □

LEMMA 6. *C has a fundamental system of neighborhoods V such that the sets C^nV are open subgroups of finite index in C.*

PROOF. Let $W = \prod_\mathfrak{p} W_\mathfrak{p}$ be a fundamental system of neighborhoods of J, selected in such a way that the $W_\mathfrak{p}$ are compact subgroups of $k_\mathfrak{p}^*$ if $\mathfrak{p}$ is non-archimedean. At the archimedean primes let them be small neighborhoods of 1. J^nW is a group since for the archimedean primes ${}^{*n}_{\mathfrak{p}}W_\mathfrak{p} = k_\mathfrak{p}^{*n}$. The images V of the W in C are a fundamental system of neighborhoods in C and the images of J^nW are C^nV.

Read modulo C^n the C^nV are neighborhoods of 1 in C/C^n. Since C/C^n is compact, a finite number of translations cover the whole group and this proves that the index is finite. □

LEMMA 7. *Let P_0 be an open subgroup of finite index of P. There exists (if n is a given integer) an open subgroup N of finite index of C such that $P \cap N = P_0(P \cap C^n)$. N contains C^n.*

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PROOF. [3] Both groups P_0C^n and PC^n are closed in C. Since P_0 is of finite index in P, P_0C^n is of finite index in PC^n. The complement of P_0C^n in PC^n consists of a finite number of (closed) cosets of P_0C^n and is therefore closed in the topology of PC^n, P_0C^n is therefore open in the group PC^n. This means that there exists a neighborhood V of 1 in C such that $PC^n \cap V \subset P_0C^n$. Put $N = P_0C^nV$. According to Lemma 6, N is an open subgroup of finite index of C. $PC^n \cap P_0C^nV = P_0C^n(PC^n \cap V) = P_0C^n$. Therefore $P \cap N = P \cap PC^n \cap P_0C^nV = P \cap P_0C^n = P_0(P \cap C^n)$.

$$\begin{tikzcd}
 & & C \arrow[d, dash] & & \\
 & & PN \arrow[dl, dash] \arrow[dr, dash] & & \\
 & PC^n \arrow[dl, dash] \arrow[dr, dash] & & N = P_0C^nV \arrow[dl, dash] & \\
P \arrow[dr, dash] & & P_0C^n \arrow[dl, dash] \arrow[dr, dash] & & \\
P_0(P \cap C^n) = P \cap P_0C^n & & & PC^n \cap V &
\end{tikzcd}$$

□

THEOREM 4. *Let P_0 be an open subgroup of finite index of P. There exists an open subgroup N of finite index of C such that $P \cap N = P_0$. The smallest integer n that one can achieve as exponent of the factor group C/N is the smallest n for which $P \cap C^n \subset P_0$. Trivially $PN/N \simeq P/P_0$. Let m be the exponent of P/P_0. Then $n = m$ if $P \cap C^m \subset P_0$; otherwise, $n = 2m$.*

PROOF. 1. Suppose an N is found such that $P \cap N = P_0$ and such that the factor group C/N is of exponent n. Then $C^n \subset N$ whence $P \cap C^n \subset P \cap N = P_0$.

2. Suppose $P \cap C^n \subset P_0$. Using Lemma 7 with this n we find an N such that $P \cap N = P_0$.

3. Since P_0 is of finite index there is an n such that $P^n \subset P_0$. Under any circumstances $P \cap C^{2n} \subset P^n \subset P_0$. □

This result suggests that one should impose further global conditions. We study first the special case when P/P_0 is a cyclic group of order m. We may view in this case P_0 as the kernel of a continuous character χ_S of P of period m. The restrictions of χ_S to $k_\mathfrak{p}^*$ (if $\mathfrak{p} \in S$) define local characters $\chi_\mathfrak{p}$ on $k_\mathfrak{p}^*$ whose kernels are $P_0 \cap k_\mathfrak{p}$ (i.e. the local norm groups of the global field K belonging to N). If $n_\mathfrak{p}$ are the periods of the $\chi_\mathfrak{p}$, then m is the least common multiple of the $n_\mathfrak{p}$. If the $\chi_\mathfrak{p}$ are given then $\chi_S = \prod_{\mathfrak{p} \in S} \chi_\mathfrak{p}$.

The question arises whether χ_S can be extended to a continuous character χ of C and what the minimal period of such an extension would be. The kernel N will be an open subgroup of finite index of C and the exponent of the factor group will be the period n of χ. Theorem 4 shows that the minimal value for n that one can hope to achieve is $n = m$ unless we have $P \cap C^m \not\subset P_0$ when $n = 2m$ will be needed. Let n be defined in this way.

[3] We make use of the following lemma: If B is a group, A a subset of B and C any other set then $B \cap AC = A \cdot (B \cap C)$.

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According to Theorem 4 we can find an open subgroup N_1 of finite index of C such that $P \cap N_1 = P_0$, $PN_1/N_1 \simeq P/P_0$ and such that C/N_1 has exponent n. We conclude first that χ_S can be extended naturally to a character of the group PN_1 with kernel N_1. If we view this extension as a character of the finite group PN_1/N_1, we can extend it to a character of the finite group C/N_1. If we view this extension as a character on C it will be an extension of χ_S to C, will be continuous since its kernel N contains N_1 and will satisfy $\chi^n = 1$ since C/N_1 has exponent n.

Finally we look at the condition that determines the value of n. Lemma 2 shows (since $P^m \subset P_0$) that we have $n = 2m$ if and only if we are in the special case for $P(m, S)$ with a non-empty set S_0 and if the idèle class c_m of Lemma 2 is not in P_0. In other words if $\chi_S(c_m) = \prod_{\mathfrak{p} \in S_0} \chi_{\mathfrak{p}}(\alpha_0) = 1$. We have proved:

THEOREM 5. *Let S be a finite set of primes, $\chi_{\mathfrak{p}}$ local characters of periods $n_{\mathfrak{p}}$ for each $\mathfrak{p} \in S$ and m the least common multiple of the $n_{\mathfrak{p}}$. There exists a global character χ on C whose local restrictions are the given $\chi_{\mathfrak{p}}$. Its period can be made $= m$ provided that in the special case the condition*:

$$\prod_{\mathfrak{p} \in S_0} \chi_{\mathfrak{p}}(\alpha_0) = 1$$

is satisfied. If the condition is not satisfied one can only achieve the period $2m$.

We describe very briefly the corresponding question if P/P_0 is abelian. Let $\widetilde{\chi}_1, \widetilde{\chi}_2, \ldots, \widetilde{\chi}_r$ be a basis for the group of characters of P/P_0. Let e_i be the period of $\widetilde{\chi}_i$. The question is whether one can find an N such that $C/N \simeq P/P_0$ and $P \cap N = P_0$. This turns out to be equivalent to the condition that each of the $\widetilde{\chi}_i$ can be extended to a character χ_i of C with period e_i. If this is possible, one takes for N the common kernel of the χ_i.

We return now to the cyclic case.

REMARK. Assume that we are in the special case, $\mathfrak{p} \in S_0$ and $K_{\mathfrak{p}}/k_{\mathfrak{p}}$ the local extension described by the given character $\chi_{\mathfrak{p}}$ of period $n_{\mathfrak{p}}$. If $n_{\mathfrak{p}}$ is even, let $K'_{\mathfrak{p}}/k_{\mathfrak{p}}$ be its quadratic subfield. Then $\chi_{\mathfrak{p}}(\alpha_0) = \pm 1$ where the $-$ sign occurs if and only if $\frac{m}{n_{\mathfrak{p}}}$ is odd and $-(2 + \eta_s)$ is not a norm from $K'_{\mathfrak{p}}/k_{\mathfrak{p}}$.

PROOF. According to (5) $\alpha_0 = (-(2 + \eta_s))^{m/2}$. If $\frac{m}{n_{\mathfrak{p}}}$ is even, then $\alpha_0 \in k_{\mathfrak{p}}^{n_{\mathfrak{p}}}$ and consequently $\chi_{\mathfrak{p}}(\alpha_0) = 1$. If $\frac{m}{n_{\mathfrak{p}}}$ is odd we write:

$$\chi_{\mathfrak{p}}(\alpha_0) = \left(\chi_{\mathfrak{p}}^{\frac{n_{\mathfrak{p}}}{2}}(-(2 + \eta_s))\right)^{\frac{m}{n_{\mathfrak{p}}}}.$$

$\chi^{\frac{n_{\mathfrak{p}}}{2}}$ is the character of period 2 that describes the field $K'_{\mathfrak{p}}/k_{\mathfrak{p}}$ so that we can drop the exponent $\frac{m}{n_{\mathfrak{p}}}$. This proves the remark. □

LEMMA 8. *Assume that we are in the special case, that n is even and $\mathfrak{p} \in S_0$. There exists a cyclic local extension of degree $n_{\mathfrak{p}}$, such that $-(2 + \eta_s)$ is a norm from its quadratic subfield.*

PROOF. It suffices to achieve $\eta_{s+1} \in K_{\mathfrak{p}}$. Indeed, $K'_{\mathfrak{p}}$ would be $= k_{\mathfrak{p}}(\eta_{s+1})$ and $N_{K_{\mathfrak{p}}/k_{\mathfrak{p}}}(\eta_{s+1}) = -\eta_{s+1}^2 = -(2 + \eta_s)$. To find such a $K_{\mathfrak{p}}$, observe that the fields $k_{\mathfrak{p}}(\eta_\mu)$, $\mu > s$ are all cyclic and have as degree a power of 2. Among them there are fields of arbitrarily high degree since a given field will contain only a finite number of the η_μ. All of them contain η_{s+1}. It suffices to take as K the compositum of such a field with an unramified field of suitable odd degree. □

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We can now prove a theorem that has applications in the theory of algebras over global fields:

THEOREM 6. *Let k be a global field, S a finite set of primes and $n_{\mathfrak{p}}$ positive integers associated with each $\mathfrak{p} \in S$. If $\mathfrak{p}$ is archimedean $n_{\mathfrak{p}}$ should be a possible degree for an extension of $k_{\mathfrak{p}}$. Then there exists a cyclic extension K/k whose degree n is the least common multiple of the $n_{\mathfrak{p}}$, and such that the completions $K_{\mathfrak{P}}/k_{\mathfrak{p}}$ have degree $n_{\mathfrak{p}}$ for all $\mathfrak{p} \in S$.*

PROOF. To each $\mathfrak{p} \in S$ select a cyclic field $K_{\mathfrak{p}}/k_{\mathfrak{p}}$ in the following way:

1. If $\mathfrak{p}$ is archimedean select the extension in the only way that is anyhow possible.
2. If $\mathfrak{p}$ is non-archimedean but not in S_0 select the unramified extension of degree $n_{\mathfrak{p}}$. For primes in S_0 select the extension as in the previous lemma.

If we describe each of these local extensions by characters, we can find a global character of period n whose restrictions to $k_{\mathfrak{p}}$ are the given local ones since we have taken care that in the special case $\chi_{\mathfrak{p}}(\alpha_0) = 1$. □

The following corollary is directly adapted to the intended application to algebras:

COROLLARY. *Let k be a global field and $c \in \mathcal{H}^2(\mathfrak{G}, \Omega)$ a 2-cocycle of period n. Then c has a cyclic splitting field of degree n over k.*

PROOF. We know that $c_{\mathfrak{p}} = 1$ at almost all primes. The period of c is the least common multiple of the periods $n_{\mathfrak{p}}$ of the $c_{\mathfrak{p}}$. Each $c_{\mathfrak{p}}$ is split by any local field of degree $n_{\mathfrak{p}}$. Furthermore c splits globally if and only if it splits locally everywhere. By Theorem 5 we can find a global cyclic field K of degree n which has local degree $n_{\mathfrak{p}}$ at all primes where $c_{\mathfrak{p}} \neq 1$. This field will split all $c_{\mathfrak{p}}$ and consequently c. □

3. Cyclic Extensions

It is possible to ask a somewhat different question. Namely, given a cyclic extension K/k of degree n, and a prime p. We wish to determine when there exists a field $L \supset K$ which is cyclic of degree p^r over K, and also cyclic over k.

If $p \nmid n$, then the question reduces to the one already treated. We suppose that $p \mid n$.

To simplify the notation, let C denote the idèle classes modulo the connected component in the case of number fields, and the compactified idèle classes in the case of function fields. We have therefore $C = \overline{C}$ in the notation used previously. Denote by $\widehat{C}$ the character group of C.

Let N be the norm group of K, to which K is class field, and $N_{\mathfrak{p}} = N \cap k_{\mathfrak{p}}$. Let χ be a character with kernel N. A field L of the prescribed type will exist if and only if there exists a character $\psi \in \widehat{C}$ such that $\chi = \psi^{p^r}$, or in other words, $\chi \in \widehat{C}^{p^r}$. Indeed, the kernel of such a character ψ would be an open subgroup $M \subset N$ of finite index $p^r n$ in C, and its class field L would satisfy our requirements.

The subgroup of C orthogonal to $\widehat{C}^{p^r}$ is precisely $1^{1/p^r}$ (the elements of order dividing p^r in C) because $a \in (C^{p^r})^* \Leftrightarrow \psi^{p^r}(a) = 1$ for all $\psi \in \widehat{C} \Leftrightarrow \psi(a^{p^r}) = 1$ for all $\psi \in \widehat{C} \Leftrightarrow a^{p^r} = 1$.

In function fields we note that the elements $1^{1/p^r}$ have ordinal 0 and hence are genuine idèle classes.

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The following theorem is essentially trivial if we take into account Theorem 2 and the relation between the local and global groups $N_{\mathfrak{p}}$ and N described at the beginning of Section 2.

THEOREM 7. *Let the notation be that described in the preceding paragraphs. The following statements are equivalent*:

1. *There exists an extension L of K such that L/K is cyclic of degree p^r and L/k is also cyclic.*
2. $\chi \in \widehat{C}^{p^r}$.
3. *At each prime $\mathfrak{p}$, $\chi_{\mathfrak{p}}(\zeta) = 1$ for every p^r-th root of unity ζ in $k_{\mathfrak{p}}$, and in the special case, $\chi(c_{p^r}) = 1$ (where c_{p^r} is the idèle class of Lemma 2).*
4. *At each prime $\mathfrak{p}$, $\zeta \in N_{\mathfrak{p}}$ for every p^r-th root of unity in $k_{\mathfrak{p}}$, and in the special case, $c_{p^r} \in N_{\mathfrak{p}}$.*

COROLLARY 1. *If k is a function field and p is the characteristic then the construction of Theorem 7 is always possible.*

PROOF. There are no idèle classes of period p, and there is no special case. □

COROLLARY 2. *If k is a global field and ζ is a primitive p^r-th root of unity in k, then the four statements of the theorem are equivalent to the condition that ζ is a norm from K, i.e. $\zeta = N\alpha$, $\alpha \in K$.*

PROOF. Since ζ lies in k, there is no special case. If ζ is a global norm it is a local norm everywhere. Conversely, if ζ is a local norm everywhere, it is a global norm because K/k is cyclic. This proves the equivalence of our statement with 4 and hence with the others. □

The conditions formulated in Corollary 2 and in Theorem 7, 1, are completely algebraic, i.e. do not involve arithmetic objects. It can be shown that they are equivalent in any abstract field k. Corollary 1 is likewise true over any field k of characteristic p. See for instance N. Bourbaki, *Elements*, Ch. V, Algebre, 11, Exercises.

We see that the algebraic theorems and criteria are special cases of the arithmetic theorems for global fields, but that the latter hold whether the roots of unity are in k or not.

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CHAPTER XI

Higher Ramification Theory

1. Higher Ramification Groups

We recall some elementary facts of local valuation theory, and begin by fixing the notation.

Let k be complete under a discrete valuation, with perfect residue class field $\bar{k}$. The characteristic of $\bar{k}$ is $p \geqslant 0$.

K/k denotes a normal extension The prime ideals are $\mathfrak{P}$ and $\mathfrak{p}$ respectively, the rings of integers $\mathfrak{O}$ and $\mathfrak{o}$. Π has order 1 in K, and π has order 1 in k. We write $\alpha \sim \beta$ if α and β have the same value.

E/k denotes a finite separable extension. $\vartheta_{E/k}$ is the different. E/k is unramified if and only if $\vartheta_{E/k} = 1$. If $K \supset E \supset k$ then $\vartheta_{K/k} = \vartheta_{K/E} \cdot \vartheta_{E/k}$. If the powers of α are a minimal basis for E/k, and $f(x)$ is the equation for α over k, then $\vartheta_{E/k} = f'(\alpha)\mathfrak{O}$.[1]

In a tower $K \supset E \supset k$, we distinguish objects of K/E by the sign $\tilde{}$, and those of E/k by the sign $\bar{}$ (which will not be confused with the residue class field). We would write

$$\vartheta = \tilde{\vartheta}\bar{\vartheta}.$$

Let K/k be normal with group G. We define a descending sequence of groups $G = V_{-1} \supset V_0 \supset V_1 \supset V_2 \supset \dots$ as follows. V_i is the group of all $\sigma \in G$ such that

$$\sigma\alpha \equiv \alpha \pmod{\mathfrak{P}^{i+1}}$$

for all $\alpha \in \mathfrak{O}$.

We also write this condition symbolically

$$(\sigma - 1)\mathfrak{O} \subset \mathfrak{P}^{i+1}.$$

V_i is called the *i-th ramification group* of K/k. V_i and V_{i+1} are not necessarily distinct. V_0 is the inertia group and its fixed field T is the maximal unramified subfield of K. We note that the groups V_i are normal in G. Indeed, let $\sigma \in V_i$, $\tau \in G$. Then

$$\tau\sigma\tau^{-1}\alpha - \alpha = \tau(\sigma\tau^{-1}\alpha - \tau^{-1}\alpha) \in \tau\mathfrak{P}^{i+1} = \mathfrak{P}^{i+1}$$

and hence $\tau\sigma\tau^{-1} \in V_i$, thereby showing that V_i is normal.

Suppose that the powers of α are a minimal basis: $\mathfrak{O} = \sum \mathfrak{o}\alpha^\nu$. Then obviously $\sigma \in V_i$ if and only if $(\sigma - 1)\alpha \in \mathfrak{P}^{i+1}$. This shows that $V_i = 1$ for large i. If K/k is completely ramified, then $\{\Pi^\nu\}$ is a minimal basis and therefore $\sigma \in V_i$ if and only if $\sigma\Pi \equiv \Pi \pmod{\mathfrak{P}^{i+1}}$.

The different is given in terms of the ramification groups as follows:

[1]Such elements α exist. If β is an element of $\mathfrak{O}$ whose residue class generates the residue class field extension and Π is a prime element of $\mathfrak{O}$, then either β or $\beta + \Pi$ will be such an α. See [**16**, III, § 1, Prop. 3].

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THEOREM 1. *Let K/k be a normal extension. Then*

$$\operatorname{ord}_K \vartheta = \sum_{i=0}^{\infty} (\#V_i - 1).$$

PROOF. If α is a minimal basis, and $f(x)$ the irreducible equation for α over k, then $\vartheta = f'(\alpha) = \prod_{\sigma \neq 1}(\sigma\alpha - \alpha)$. For $\sigma \in V_i$ but $\sigma \notin V_{i+1}$, we have by definition $\operatorname{ord}_K(\sigma\alpha - \alpha) = i+1$. It follows that $\operatorname{ord}_K \vartheta = \sum_{i=0}^{\infty}(i+1)(\#V_i - \#V_{i+1})$ and this formula is easily seen to be equal to the one of the theorem, for instance as follows. Suppose $\#V_t = 1$. Then our sum is equal to

$$\begin{aligned}
&(\#V_0 - \#V_1) + 2(\#V_1 - \#V_2) + 3(\#V_2 - \#V_3) + \cdots + t(\#V_{t-1} - \#V_t) \\
&\quad = \#V_0 + \#V_1 + \cdots + \#V_{t-1} - t\#V_t = \\
&\quad = (\#V_0 - 1) + (\#V_1 - 1) + (\#V_2 - 1) + \cdots + (\#V_{t-1} - 1) = \\
&\quad = \sum_{i=0}^{\infty}(\#V_i - 1).
\end{aligned}$$

□

We shall now derive some useful structure theorems concerning the ramification groups.

We begin by remarking that we may assume K/k completely ramified without loss of generality. Indeed, we have

THEOREM 2. *Let K/k be normal with group G. Let E be an intermediate field fixed under H. Let $\widetilde{V}_i$ be the ramification groups of K/E. Then $\widetilde{V}_i = V_i \cap H$.*

PROOF. Clear. □

If E/k is normal it is much more difficult to determine the relationship between the ramification groups $\bar{V}_i$ of E/k, and those of K/k. The next section will be devoted to this enterprise. For the moment, we stay in one field.

Denote by k_i the multiplicative group of units of k which are $\equiv 1 \pmod{\mathfrak{p}^i}$ for $i > 0$. Complete this definition by letting k_0 be the group of units of k, and $k_{-1} = k^*$ itself. We have

$$k_{-1} \supset k_0 \supset k_1 \supset k_2 \supset \ldots.$$

Let K/k be normal with group G.

The factor group K_0/K_1 is naturally isomorphic with $\overline{K}^*$, the multiplicative group of the residue class field. If K/k is completely ramified, then $\overline{K} = \bar{k}$ and G operates with trivial action.

Let K/k be completely ramified, $V_0 = G$. We have

$$\begin{aligned}
\sigma \in V_i &\Leftrightarrow \sigma\Pi \equiv \Pi \pmod{\mathfrak{P}^{i+1}} \\
&\Leftrightarrow \Pi^{\sigma-1} \equiv 1 \pmod{\mathfrak{P}^i} \\
&\Leftrightarrow \Pi^{\sigma-1} \in K_i.
\end{aligned}$$

In fact, we see that $\Pi^{\sigma-1} \in Z^1(V_0, K_0)$ is a unit cocycle.

THEOREM 3. *Let K/k be completely ramified. V_0/V_1 is isomorphic to a finite subgroup of $\overline{K}^*$ (multiplicative), and is cyclic. Its order e_0 is prime to p.*

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PROOF. The map $\sigma \to \Pi^{\sigma-1} \pmod{\mathfrak{P}}$ is a homomorphism of V_0 into the multiplicative group of the residue class field, because $\Pi^{\sigma-1}$ is a 1-cocycle with trivial action in $K_0/K_1 = \overline{K}^*$. The kernel is precisely V_1, and this proves that V_0/V_1 is isomorphic to a subgroup of $\overline{K} = \bar{k}$. A finite subgroup of the multiplicative group of a field is cyclic, so V_0/V_1 is cyclic. If e_0 is its order, then we must have e_0 prime to the characteristic of $\bar{k}$, as was to be shown. □

THEOREM 4. *Let K/k be completely ramified, $i \geqslant 1$. If $p = 0$, then $V_1 = 1$. If $p \neq 0$, then V_i/V_{i+1} is isomorphic to an additive subgroup of $\overline{K}$, and is of type $(p, p, \ldots, p)$. V_1 is consequently a p-group.*

PROOF. Again the map $\sigma \to \Pi^{\sigma-1} \pmod{K_{i+1}}$ for $\sigma \in V_i$ is a 1-cocycle in K_i/K_{i+1}. Since V_i has trivial action on K_i, it is a homomorphism, with kernel V_{i+1}. It follows that V_i/V_{i+1} is isomorphic to a subgroup of $\overline{K} = \bar{k}$ (additive) because the map $\chi \mapsto 1 + \Pi^i\chi$ induces an isomorphism of the additive group of the residue field and the multiplicative group K_i/K_{i+1}. If $p = 0$ then the group is trivial. If $p > 0$ then every element has period p, and since V_i/V_{i+1} is a finite group, it is of type $(p, p, \ldots, p)$, as was to be shown. □

THEOREM 5. *Let $\tau \in V_0$ and let $\sigma \in V_i$ $(i \geqslant 1)$. Then $\tau\sigma\tau^{-1}\sigma^{-1} \in V_{i+1}$ if and only if either $\sigma \in V_{i+1}$, or $\tau^i \in V_1$.*

PROOF. We may clearly assume that K/k is completely ramified.

Let $\tau\Pi = \varepsilon\Pi$, where $\varepsilon \in K_0$. Let

$$\sigma\Pi \equiv \Pi + \beta\Pi^{i+1} \pmod{\mathfrak{P}^{i+2}},$$

where $\beta \in \mathfrak{O}$. Since $\sigma \in V_i$, we have $\sigma\varepsilon \equiv \varepsilon \pmod{\mathfrak{P}^{i+1}}$. Hence

$$\begin{aligned}\sigma\tau\Pi = \sigma(\varepsilon\Pi) &\equiv \varepsilon\Pi^\sigma \pmod{\mathfrak{P}^{i+2}}\\ &\equiv \varepsilon\Pi + \varepsilon\beta\Pi^{i+1} \pmod{\mathfrak{P}^{i+2}}\\ &\equiv \varepsilon\Pi + \big(\varepsilon\Pi\big)^{i+1}\beta\varepsilon^{-i} \pmod{\mathfrak{P}^{i+2}}.\end{aligned}$$

On the other hand, $\tau\beta \equiv \beta \pmod{\mathfrak{P}}$ because K is completely ramified. Hence

$$\begin{aligned}\tau\sigma\Pi &\equiv \tau\Pi + \beta^\tau(\tau\Pi)^{i+1} \pmod{\mathfrak{P}^{i+2}}\\ &\equiv \varepsilon\Pi + \beta(\varepsilon\Pi)^{i+1} \pmod{\mathfrak{P}^{i+2}}.\end{aligned}$$

Combining the two congruences we get

$$(\sigma\tau - \tau\sigma)\Pi \equiv (\varepsilon\Pi)^{i+1}\beta(\varepsilon^{-i} - 1) \pmod{\mathfrak{P}^{i+2}}.$$

Replacing Π by $\big(\tau^{-1}\sigma^{-1}\Pi\big)$ we get

$$\Pi - \tau\sigma\tau^{-1}\sigma^{-1}\Pi \equiv (\varepsilon\Pi)^{i+1}\beta(\varepsilon^{-i} - 1) \pmod{\mathfrak{P}^{i+2}}.$$

We see that

$$\begin{aligned}\tau\sigma\tau^{-1}\sigma^{-1} \in V_{i+1} &\Leftrightarrow \beta(\varepsilon^{-i} - 1) \equiv 0 \pmod{\mathfrak{P}}\\ &\Leftrightarrow \beta \equiv 0 \pmod{\mathfrak{P}} \quad \text{or} \quad \varepsilon^i \equiv 1 \pmod{\mathfrak{P}}\end{aligned}$$

But from the definition of σ, we clearly have $\sigma \in V_{i+1}$ if and only if $\beta \equiv 0 \pmod{\mathfrak{P}}$.

Since K is completely ramified, $\tau\varepsilon \equiv \varepsilon \pmod{\mathfrak{P}}$ and therefore

$$\tau^i\Pi \equiv \varepsilon^i\Pi \pmod{\mathfrak{P}^2}.$$

This proves that $\tau^i \in V_1$ if and only if $\varepsilon^i \equiv 1 \pmod{\mathfrak{P}}$ and concludes the proof of our theorem. □

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COROLLARY 1. *The group V_i/V_{i+1} is contained in the center of V_1/V_{i+1}.*

PROOF. If we take τ in V_1 then we must have $\tau\sigma\tau^{-1}\sigma^{-1} \in V_{i+1}$. This is the corollary. □

COROLLARY 2. *If K/k is abelian, and $V_i \neq V_{i+1}$, then e_0 divides i.*

PROOF. The commutator is 1. Let τ be a generator of V_0/V_1. The period of τ is e_0. By hypothesis we can select $\sigma \notin V_{i+1}$. This implies that $e_0 \mid i$, as contended. □

If K/k is not abelian, we can generalize Corollary 2 as follows. Consider elements $\sigma \in V_i$ but $\sigma \notin V_{i+1}$. Let τ generate V_0/V_1. Note that $\tau^\nu\sigma\tau^{-\nu} \notin V_{i+1}$ for any integer ν, because V_{i+1} is normal. Separate $V_i - V_{i+1}$ into equivalence classes by letting $\sigma \sim \sigma'$ if $\sigma' = \tau^\nu\sigma\tau^{-\nu}$ for some integer ν. Let the equivalence class of an element σ have r_σ elements. Then r_σ is obviously the least integer ν such that $\sigma = \tau^\nu\sigma\tau^{-\nu}$. We have $\tau^{r_\sigma}\sigma\tau^{-r_\sigma}\sigma^{-1} = 1$, and since $\sigma \notin V_{i+1}$ we conclude from the theorem that $\tau^{r_\sigma i} \in V_1$. Hence $e_0 \mid ir_\sigma$ for each σ.

The above facts will be used in our discussion of the conductor later, to prove that the conductor is an integral ideal.

2. Ramification Groups of a Subfield

Let k be complete under a discrete valuation with perfect residue class field $\bar{k}$, of characteristic p.

Let K/k be a normal extension with group G. Let H be a subgroup, and E the fixed field. If E/k is normal, it will turn out that the ramification groups of E/k are given by V_iH/H, but the numbering does not fit, i.e. $\bar{V}_i$ is not necessarily V_iH/H. We shall find the function of i which gives the precise correspondence between the ramification groups.

LEMMA 1. *Let $\mathfrak{O}_K = \mathfrak{o}[\alpha]$, and $\mathfrak{O}_E = \mathfrak{o}[\beta]$. Then for any $\sigma \in G$,*

$$\sigma\beta - \beta \sim \prod_{\tau\in\sigma H}(\tau\alpha - \alpha) = \prod_{\gamma\in H}(\sigma\gamma\alpha - \alpha).$$

PROOF. Let $f(x) = \mathrm{Irr}(\alpha, E) = \prod_{\gamma\in H}(x - \gamma\alpha)$. The coefficients of f are in $\mathfrak{O}_E$. We have $f^\sigma(x) = \mathrm{Irr}(\sigma\alpha, \sigma E) = \prod_{\gamma\in H}(x - \sigma\gamma\alpha)$. But $f(x) - f^\sigma(x)$ has coefficients divisible by $\sigma\beta - \beta$ and hence $(\sigma\beta - \beta) \mid f^\sigma(\alpha) - f(\alpha) = f^\sigma(\alpha)$.

Conversely, let $\beta = g(\alpha)$, $g(x) \in \mathfrak{o}[x]$. Then $g^\sigma = g$. Since $g(x) - \beta$ has root α and coefficients in E, it follows that $g(x) - \beta = f(x)h(x)$ with $h(x) \in \mathfrak{O}_E[x]$. Hence $g(x) - \beta^\sigma = f^\sigma(x)h^\sigma(x)$ and $g(\alpha) - \beta^\sigma = f^\sigma(\alpha)h^\sigma(\alpha) = \beta - \beta^\sigma$. This proves our lemma. □

Let S be a subset of G and let $\delta(\sigma, S) = \begin{cases} 1 & \sigma \in S \\ 0 & \sigma \notin S \end{cases}$. Then $\#S = \sum_{\sigma\in G}\delta(\sigma, S)$.

Let $i(\sigma) + 1 = \mathrm{ord}_K(\sigma\alpha - \alpha)$ for any number α giving a minimal basis. We put $i(1) = \infty$. Equivalently, we might have defined $i(\sigma)$ by $\sigma \in V_{i(\sigma)}$ but $\sigma \notin V_{i(\sigma)+1}$.

We obtain

$$i(\sigma) + 1 = \sum_{\nu=0}^{\infty}\delta(\sigma, V_\nu). \tag{1}$$

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Let E/k be normal, fixed under H. Let $G = \bigcup H\bar{\sigma}$. We have functions $\bar{i}(\bar{\sigma})$ for E/k. The following lemma gives $\bar{i}(\bar{\sigma})$ in terms of the indices in K/E.

LEMMA 2. *If $\sigma H \cap V_i$ is not empty for some i, then*

$$\bar{i}(\bar{\sigma}) + 1 = \sum_{\nu=0}^{\ell(\sigma)} \frac{1}{(\widetilde{V}_0 : \widetilde{V}_\nu)}$$

where $\ell(\sigma)$ is the largest integer ν such that $\sigma H \cap V_\nu$ is not empty.

We may also describe $\ell(\sigma)$ as the integer such that $\sigma \in V_{\ell(\sigma)}H$ but $\sigma \notin V_{\ell(\sigma)+1}H$.

PROOF. If we take $\mathrm{ord}_K(\sigma\beta - \beta)$ in Lemma 1, we obtain

$$\begin{aligned}
\tilde{e}(\bar{i}(\bar{\sigma}) + 1) &= \sum_{\gamma \in H} (i(\sigma\gamma) + 1) \\
&= \sum_{\gamma \in H} \sum_{\nu=0}^{\infty} \delta(\sigma\gamma, V_\nu) \\
&= \sum_{\nu=0}^{\infty} \sum_{\gamma \in H} \delta(\sigma\gamma, V_\nu) \\
&= \sum_{\nu=0}^{\ell(\sigma)} \#(\sigma H \cap V_\nu).
\end{aligned}$$

Suppose $\sigma H \cap V_\nu$ is not empty, say $\sigma\gamma_0 \in V_\nu$. Then

$$\begin{aligned}
\sigma\gamma \in V_\nu &\Leftrightarrow \sigma\gamma_0\gamma_0^{-1}\gamma \in V_\nu \\
&\Leftrightarrow \gamma_0^{-1}\gamma \in (\sigma\gamma_0)^{-1}V_\nu \\
&\Leftrightarrow \gamma_0^{-1}\gamma \in V_\nu \cap H \\
&\Leftrightarrow \gamma \in \gamma_0(V_\nu \cap H).
\end{aligned}$$

This means that there is a 1–1 correspondence between elements of $\gamma_0(V_\nu \cap H)$, $(V_\nu \cap H)$, and $(\sigma H \cap V_\nu)$. Hence

$$\#(\sigma H \cap V_\nu) = \#(H \cap V_\nu) = \widetilde{V}_\nu.$$

Returning to our sum we substitute the number obtained and get

$$\tilde{e}(\bar{i}(\bar{\sigma}) + 1) = \sum_{0}^{\ell(\sigma)} \#\widetilde{V}_\nu.$$

We note that $\tilde{e} = \#\widetilde{V}_0$ by definition. Dividing by $\tilde{e}$ yields

$$\bar{i}(\bar{\sigma}) + 1 = \sum_{0}^{\ell(\sigma)} \frac{1}{(\widetilde{V}_0 : \widetilde{V}_\nu)}$$

as desired. $\square$

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Let K/k be normal. Let x be a real variable. If x is an integer $\geqslant 0$, we define the real function $\varphi_{K/k}(x)$ by putting

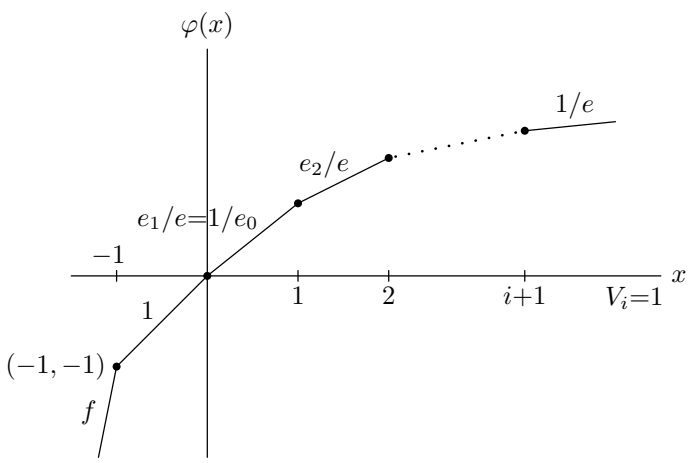
$$\varphi_{K/k}(x) + 1 = \sum_{\nu=0}^{x} \frac{1}{(V_0 : V_\nu)}.$$

We often omit the subscript of $\varphi_{K/k}$ and write briefly φ. We have

$$\varphi(x) = \sum_{\nu=1}^{x} \frac{1}{(V_0 : V_\nu)}.$$

We note that $\varphi(0) = 0$.

We extend this function by making it linear between integers. Its graph looks roughly as follows:



Let $\{x\}$ = least integer $\geqslant x$. Define $V_x = V_{\{x\}}$. Then it is easily seen that

$$\varphi(x) = \int_0^x \frac{1}{(V_0 : V_t)}\, dt \qquad x \geqslant 0.$$

We extend φ to the left by putting $(V_0 : V_x) = (V_x : V_0)^{-1}$ for $x \leqslant 0$. Then $\varphi(-1) = -1$ and the slope of φ to the left of -1 is f.

Properties of $\varphi(x)$.

1. $\varphi(x)$ is continuous, strictly monotone increasing, and $\varphi(0) = 0$.

2. $\varphi(x)$ has right and left derivatives everywhere, φ'_r and φ'_ℓ. If x is not an integer, or if $V_x = V_{x+1}$ then

$$\varphi'_r(x) = \varphi'_\ell(x) = \frac{1}{(V_0 : V_x)}.$$

At an integer i where $V_i \neq V_{i+1}$ we have

$$\varphi'_\ell(i) = \frac{1}{(V_0 : V_i)} \quad \text{and} \quad \varphi'_r(i) = \frac{1}{(V_0 : V_{i+1})}.$$

3. $\varphi'(-\infty) = f$ and $\varphi'(+\infty) = 1/e$.

4. $\varphi(x)$ is convex.

5. $\varphi(x)$ has an inverse function $\psi(x) = \varphi^{-1}(x)$.

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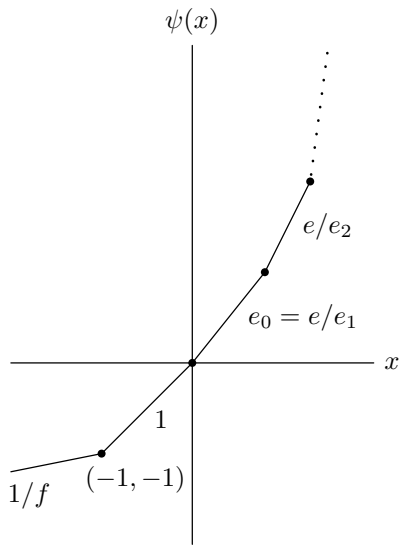
Properties of $\psi(x)$.

1. $\psi(x)$ is continuous, strictly monotone increasing, and $\psi(0) = 0$.
2. $\psi(x)$ has right and left derivatives everywhere, ψ'_r and ψ'_ℓ. By elementary calculus, $\psi'(x) = 1/\varphi'(\psi(x))$, and $\psi'_r(x)$ or $\psi'_\ell(x)$ is always an integer for $x \geqslant -1$, except for $\psi_\ell(-1)$.
3. $\psi'(-\infty) = 1/f$ and $\psi'(+\infty) = e$.
4. $\psi(x)$ is convex.
5. If i is an integer, then $\psi(i)$ is also an integer, for $i \geqslant -1$.

The properties of $\varphi(x)$ and $\psi(x)$ are obvious from their definitions, except possibly 5 which follows from 2.



We may formulate Lemma 2 with the aid of the φ function as follows:

$$\bar{i}(\bar{\sigma}) + 1 = \widetilde{\varphi}(\ell(\sigma)) + 1. \tag{2}$$

We use the notation $\overline{\varphi} = \varphi_{E/k}$ and $\widetilde{\varphi} = \varphi_{K/E}$.

THEOREM 6. *Let $K \supset E \supset k$, with K/k and E/k both normal, E fixed under H. The ramification groups of E/k are given by*

$$\bar{V}_{\widetilde{\varphi}(x)} = V_x H/H, \quad \textit{or} \quad V_x = V_{\widetilde{\psi}(x)} H/H.$$

PROOF. We have

$$\begin{aligned}
\widetilde{\sigma} \in \bar{V}_{\widetilde{\varphi}(x)} &\Leftrightarrow \bar{i}(\bar{\sigma}) \geqslant \widetilde{\varphi}(x) \\
&\Leftrightarrow \widetilde{\varphi}(\ell(\sigma)) \geqslant \widetilde{\varphi}(x) \quad \text{(by (2))} \\
&\Leftrightarrow \ell(\sigma) \geqslant x \\
&\Leftrightarrow \ell(\sigma) \geqslant \{x\} \\
&\Leftrightarrow \sigma \in V_{\{x\}} H \\
&\Leftrightarrow \sigma \in V_x H
\end{aligned}$$

thereby proving the theorem. □

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THEOREM 7. *Let $K \supset E \supset k$ be as above. Then*

$$\varphi(x) = \overline{\varphi}(\widetilde{\varphi}(x))$$
$$\psi(x) = \widetilde{\psi}(\overline{\psi}(x)).$$

PROOF. Both functions φ and $\overline{\varphi}(\widetilde{\varphi})$ are continuous, have value 0 at 0 and are differentiable except at a finite set of values. By the chain rule for differentiation, at the points where derivatives exist, we have

$$\begin{aligned}\frac{d}{dx}(\overline{\varphi}(\widetilde{\varphi}(x))) &= \overline{\varphi}'(\widetilde{\varphi}(x))\widetilde{\varphi}'(x)\\ &= \frac{1}{(\bar{V}_0 : \bar{V}_{\widetilde{\varphi}(x)})}\frac{1}{(\widetilde{V}_0 : \widetilde{V}_x)}\\ &= \frac{1}{(V_0H : V_xH)}\frac{1}{(V_0 \cap H : V_x \cap H)}\\ &= \frac{1}{(V_0 : V_x)}\\ &= \varphi'(x).\end{aligned}$$

Hence at the points where the functions are differentiable, their derivatives are equal. By elementary calculus, it follows that $\varphi = \overline{\varphi}(\widetilde{\varphi})$, as contended.

In order to avoid mentioning explicitly the function φ, we may define

$$V^x = V_{\psi(x)} \quad \text{or} \quad V^{\varphi(x)} = V_x.$$

Then the result of Theorem 6 may be expressed as follows:

$$\bar{V}^x = V^xH/H.$$

The new numbering of the ramification groups is now set up in such a way that we may define ramification groups in infinite extensions. Indeed, let Ω/k be the algebraic closure of k (any other infinite Galois extension would also do). Let $\mathfrak{G}$ be its Galois group, and let $\mathfrak{h}$ range over normal open subgroups of finite index.

For real x let $\mathfrak{Y}^x_\mathfrak{h}$ be the subgroup of $\mathfrak{G}$ containing $\mathfrak{h}$ such that $\mathfrak{Y}^x_\mathfrak{h}/\mathfrak{h} = V^x$ for the normal extension $\Omega^\mathfrak{h}/k$. Let $\mathfrak{h}_0 \supset \mathfrak{h}$. It follows from Theorem 6 that

$$\mathfrak{Y}^x_{\mathfrak{h}_0} = \mathfrak{Y}^x_\mathfrak{h} \cdot \mathfrak{h}_0.$$

Define $\mathfrak{Y}^x = \bigcap_\mathfrak{h} \mathfrak{Y}^x_\mathfrak{h}$.

The groups for a finite normal extension are now easily recovered from $\mathfrak{Y}^x$. Indeed, we contend that

$$\mathfrak{Y}^x_{\mathfrak{h}_0} = \mathfrak{Y}^x \cdot \mathfrak{h}_0.$$

The inclusion $\supset$ is trivial. Conversely, let $\tau \in \mathfrak{Y}^x_{\mathfrak{h}_0}$. We can write $\tau = \tau_\mathfrak{h}\gamma_\mathfrak{h}$ where $\tau_\mathfrak{h} \in \mathfrak{Y}^x_\mathfrak{h}$ and $\gamma_\mathfrak{h} \in \mathfrak{h}_0$. Hence $\tau\gamma_\mathfrak{h}^{-1} = \tau_\mathfrak{h} \in \mathfrak{Y}^x_\mathfrak{h}$. The sets $\mathfrak{Y}^x_\mathfrak{h} \cap \tau_\mathfrak{h}\mathfrak{h}_0$ are not empty and are closed. Since the group is compact, the intersection $\bigcap_{\mathfrak{h}\subset\mathfrak{h}_0}(\mathfrak{Y}^x_\mathfrak{h} \cap \tau_\mathfrak{h}\mathfrak{h}_0) \subset \mathfrak{Y}^x$ is not empty. Let $\sigma \in \mathfrak{Y}^x$ be in it. Then $\sigma\mathfrak{h}_0 = \tau_\mathfrak{h}\mathfrak{h}_0$ for all $\mathfrak{h}$. This shows that $\tau = \sigma\gamma$, $\gamma \in \mathfrak{h}_0$, and proves our contention. □

3. The General Residue Class Field

The fields studied in this brief section will be used as the residue class fields of a complete field in the next section. It turns out that local class field theory and the ramification theory can be carried out in these more general complete fields.

We suppose therefore that k is a field satisfying *Moriya's Axiom*:

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k is perfect, and for each positive integer n there exists exactly one extension of degree n of k.

We note that any finite extension of k also satisfies the axiom.

LEMMA 3. *All finite extensions of k are cyclic.*

PROOF. It is obviously sufficient to prove that an arbitrary finite normal extension K/k is cyclic. Let G be its Galois group and let $[K:k]=n$. For each divisor d of n let $\psi(d)$ be the number of elements $\sigma \in G$ with the exact period d. Any two automorphisms with the same period must have the same fixed subfield, and must therefore generate the same group: they are powers of each other. Therefore $\psi(d) \leqslant \varphi(d)$. We have

$$\sum_{d/n} \psi(d) = \text{order of } G = n = \sum_{d/n} \varphi(d).$$

It follows that $\psi(d)=\varphi(d)$. Taking $d=n$ shows that G is cyclic. □

The following algebraic proposition will be useful in Lemma 4.

PROPOSITION. *Let F be any field of characteristic $p \geqslant 0$.*

Let ℓ be a prime $\neq p$. Let $a \in F$. The polynomial $x^\ell - a$ is either irreducible in F, or a is an ℓ-th power in F.

Let $p>0$. Let a, b, $c \in F$ and $abc \neq 0$. Let $f(x)=ax^p-bx$. The polynomial $f(x)-c$ is either irreducible in F or has a root in F.

PROOF. Suppose $x^\ell - a$ is reducible in F. The constant term of one of the factors with coefficients in F is of type $\zeta\sqrt[\ell]{a^r}$ where $r<\ell$, and ζ is an ℓ-th root of unity. From $\zeta\sqrt[\ell]{a^r} \in F$ we deduce $a^r \in F^\ell$. Trivially $a^\ell \in F^\ell$ and hence $a^{(r,\ell)} = a \in F^\ell$, as was to be shown.

We turn to the additive case. Obviously, $f(x+y)=f(x)+f(y)$. Let α be one root of $f(x)=c$. It is trivially verified that all the roots are given by $\alpha+\nu\beta$ where $\nu=0,1,\ldots,p-1$ and $\beta = \sqrt[p-1]{b/a}$. Suppose $f(x)-c$ is reducible. The coefficient of x^{p-1} in one of the factors in F is a sum of roots, and hence of type $r\alpha+\mu\beta$ where $0<r<p$. It lies in F, and hence $\frac{1}{r}(r\alpha+\mu\beta)=\alpha+(\mu/r)\beta$ is in F and is a root of $f(x)=c$, as contended. □

LEMMA 4. *Every element of k is norm from any finite extension.*

PROOF. Using the transitivity of the norm and the fact that every finite extension of k also satisfies the axiom, we see that it suffices to prove the lemma in case of cyclic extensions of prime degree ℓ. Note that all ℓ-th powers in k are norms. We distinguish three cases.

$\ell = p$. Then $k^\ell = k$ because k is perfect. Everything is a norm.

$\ell \neq p$ and ℓ is odd. Let $a \in k$. Consider the equation $x^\ell - a$. If it is irreducible, then a is clearly a norm. If it is reducible, then a is ℓ-th power by the proposition, and hence a norm.

$\ell \neq p$ and $\ell = 2$. If x^2-a is reducible then a is a square and hence a norm. Otherwise, its root generates the extension of degree 2, and $-a$ is a norm. We shall now show that -1 is also a norm. Let K be the extension of degree 2 over k. We know that k has exactly one extension of degree 4, say $K(\sqrt{\alpha})$ with $\alpha \in K$. Let σ be the non-trivial automorphism of K/k. Then $K(\sqrt{\alpha^\sigma})$ has also degree 4 over k

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and hence $K(\sqrt{\alpha}) = K(\sqrt{\alpha^\sigma})$. By Kummer theory

$$\alpha^{1-\sigma} = \beta^2 \qquad \beta \in K.$$

Taking norms to k yields

$$1 = (N\beta)^2.$$

Therefore $N\beta = \pm 1$. Suppose $N\beta = 1$. Then $\beta = \gamma^{\sigma-1}$ by Hilbert's Theorem, and substituting in the preceding formula gives $(\alpha\gamma^2)^{1-\sigma} = 1$ whence $\alpha\gamma^2 = a \in k$. We get $K(\sqrt{\alpha}) = K(\sqrt{a})$. But $[k(\sqrt{a}) : k] \leqslant 2$, and since there is only one extension of degree 2 over k, we must have $\sqrt{a} \in K$. This is a contradiction, and $N\beta = -1$, as was to be shown. □

LEMMA 5. *If $\ell \neq p$ then $(k : k^\ell) \leqslant \ell$.*

PROOF. Suppose $(k : k^\ell) \neq 1$. There exists an element $a \notin k^\ell$. The polynomial $x^\ell - a$ is irreducible by the proposition. By Lemma 3, one root of this polynomial generates a cyclic field K of degree ℓ which is therefore splitting field of the equation. This implies that a primitive ℓ-th root of unity ζ lies in K. But ζ satisfies an equation of degree $< \ell$ over k. Hence ζ must lie in k. We are therefore in the situation of Kummer theory, and since there is only one extension of degree ℓ, $(k : k^\ell) = \ell$, as was to be shown. □

LEMMA 6. *Let $a, b \in k$, $ab \neq 0$. Let $f(x) = ax^p - bx$. Then the additive index $(k : f(k))$ is $\leqslant p$.*

PROOF. Suppose $(k : f(k)) \neq 1$. There exists an element $c \in k$, $c \notin f(k)$ and the polynomial $f(x) - c$ is irreducible by the proposition. By Lemma 3, one root generates a cyclic splitting field K of degree p. This implies that $\beta = \sqrt[p-1]{b/a}$ lies in K, and hence in k, because β satisfies an equation of degree $< p$ over k. It is easily verified that

$$\wp(\beta x) = a^{-1}\beta f(x)$$

and we have trivially $\wp(\beta k) = \wp(k)$. By the additive Kummer theory, and since there is only one extension of degree p, we obtain

$$(k : f(k)) = (k : a^{-1}\beta f(k)) = (k : \wp(k)) \leqslant p$$

as was to be shown. □

4. General Local Class Field Theory

Let k be complete under a discrete valuation with residue class field $\bar{k}$ satisfying Moriya's Axiom as discussed in the preceding section. Such a field will be called a *general local field*.

We note that all finite extensions of k are solvable. Indeed, we need only consider normal extensions. Such an extension is a tower whose first step is unramified, and hence cyclic by Lemma 3. The second step is tamely ramified and can be obtained by adjoining an n-th root. The final step is strongly ramified, and its group is a p-group which is consequently solvable.

LEMMA 7. *Let k be a general local field and let K/k be an unramified extension. Then every unit of k is norm from K.*

PROOF. The proof is based on the following result, valid in complete fields. We use N for the norm and S for the trace.

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PROPOSITION. *Let k be complete under any discrete valuation and let K/k be a normal unramified extension. Let u be a unit in k, $u \equiv 1 \pmod{\mathfrak{p}}$. Then $u = N\varepsilon$, where ε is a unit of K.*

PROOF. Since K/k is unramified, the prime $\mathfrak{p}$ remains prime in K. An element π of order 1 in k also has order 1 in K. We show that given a unit $u_m \equiv 1 \pmod{\mathfrak{p}^m}$, there exists a unit ε_m of K such that $\varepsilon_m \equiv 1 \pmod{\mathfrak{p}^m}$ and $u_m \equiv N\varepsilon_m \pmod{\mathfrak{p}^{m+1}}$.

Put $\varepsilon = 1 + \pi^m y_m$ where y_m is some integer of K to be determined. Write $u_m = 1 + \pi^m \alpha_m$ where α_m is an integer of k. We have

$$N(1 + \pi^m y_m) = \prod_\sigma (1 + \pi^m y_m^\sigma) \equiv 1 + \pi^m S(y_m) \pmod{\mathfrak{p}^{m+1}}.$$

We may interpret the trace in the residue class field. K corresponds to a normal extension K_0 of the residue class field k_0, and not all elements of K_0 have trace 0. Since the trace is homogeneous with respect to the elements of k_0 we can obviously find an integer y_m such that $S(y_m) \equiv \alpha_m \pmod{\mathfrak{p}}$. For this value of y_m we get

$$N\varepsilon_m \equiv 1 + \pi^m S(y_m) \equiv 1 + \pi^m \alpha_m \pmod{\mathfrak{p}^{m+1}}$$

as contended.

We have $u \equiv N\varepsilon_1 \pmod{\mathfrak{p}^2}$. Put $u/N\varepsilon_1 = u_2$. We can find a unit $\varepsilon_2 \equiv 1 \pmod{\mathfrak{p}^2}$ such that $u_2 \equiv N\varepsilon_2 \pmod{\mathfrak{p}^3}$. Put $u_2/N\varepsilon_2 = u_3$. Proceed inductively in this manner. The infinite product $\varepsilon_1\varepsilon_2\varepsilon_3\ldots$ converges to a unit of K, and clearly $u = N\varepsilon$, as was to be shown. □

Returning to our general local fields, we let u be any unit of k. According to Lemma 4, there exists a unit ε_0 of K such that $u \equiv N\varepsilon_0 \pmod{\mathfrak{p}}$. By the proposition, $u/N\varepsilon_0 = N\varepsilon_1$ for some unit ε_1 in K. It follows that $u = N\varepsilon$ where $\varepsilon = \varepsilon_0\varepsilon_1$, thereby proving Lemma 7. □

The next lemmas are preparatory to the proof of the second inequality, and to the determination of the conductor in cyclic extensions of prime degree.

If K/k is a normal extension, Π denotes a prime of K, π a prime of k, and Π_s an element of order $\geqslant s$ in K. We use N for the norm, and S for the trace.

LEMMA 8. *Let K/k be cyclic of prime degree ℓ. Let x be an integer of k. Then*

$$N(1 + x\Pi_s) \equiv 1 + xS(\Pi_s) + x^\ell N\Pi_s \pmod{S(\mathfrak{P}^{2s})}.$$

PROOF. Let σ generate the group of K/k. We have

$$N(1 + x\Pi_s) = (1 + x\Pi_s)(1 + x\Pi_s^\sigma)\big(1 + x\Pi_s^{\sigma^2}\big)\ldots\big(1 + x\Pi_s^{\sigma^{\ell-1}}\big).$$

Beside the three terms in the expansion given above, we have terms of the type $x^\nu \Pi_s^{\varphi(\sigma)}$ where $\varphi(\sigma)$ is a polynomial in σ of degree $\leqslant \ell - 1$, with coefficients either 0 or 1 and involving at least two monomials. Furthermore all such polynomials will actually occur as exponents of Π_s.

If $\varphi(\sigma) = \sigma\varphi(\sigma)$, then $\varphi(\sigma) = \sigma^2\varphi(\sigma) = \sigma^3\varphi(\sigma) = \ldots$ and hence $\varphi(\sigma) = 1 + \sigma + \sigma^2 + \cdots + \sigma^{\ell-1}$. This $\varphi(\sigma)$ would give the norm, already taken care of. We have therefore $\varphi(\sigma) \neq \sigma\varphi(\sigma)$. We lump together the terms involving $\varphi(\sigma)$, $\sigma\varphi(\sigma), \ldots, \sigma^{\ell-1}\varphi(\sigma)$, and by taking these together we obtain terms of $x^\nu S(\Pi_s^{\varphi(\sigma)})$. Since $\varphi(\sigma)$ has at least two monomials, this proves our lemma. □

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LEMMA 9. *Let K/k be normal of degree ℓ, and completely ramified*: $\mathfrak{p} = \mathfrak{P}^\ell$. *Let $\vartheta = \mathfrak{P}^m$ be the different. Then $S(\mathfrak{P}^s) = \mathfrak{p}^r$ where $r = [(m+s)/\ell]$.*

PROOF. Let $S(\mathfrak{P}^s) = \mathfrak{p}^r$. We must determine r. We have

$$\mathfrak{p}^{-r} S(\mathfrak{P}^s) = S(\mathfrak{p}^{-r}\mathfrak{P}^s) = S(\mathfrak{P}^{s-\ell r}) \subset \mathfrak{O}.$$

By the definition of the different, we see that

$$\mathfrak{P}^{s-\ell r} \subset \vartheta^{-1} \quad \text{and} \quad \vartheta \subset \mathfrak{P}^{\ell r - s}.$$

This means that $m \geqslant \ell r - s$, or equivalently, $r \leqslant (m+s)/\ell$.

Conversely, we have

$$\begin{aligned} S(\mathfrak{P}^s) \not\subset \mathfrak{p}^{r+1} &\Rightarrow \mathfrak{p}^{-(r+1)} S(\mathfrak{P}^s) \not\subset \mathfrak{O} \\ &\Rightarrow S(\mathfrak{P}^{s-\ell(r+1)}) \not\subset \mathfrak{O} \\ &\Rightarrow \mathfrak{P}^{s-\ell(r+1)} \not\subset \vartheta^{-1} \\ &\Rightarrow \vartheta \not\subset \mathfrak{P}^{\ell(r+1)-s} \\ &\Rightarrow (r+1)\ell - s > m \\ &\Rightarrow r+1 > (m+s)/\ell. \end{aligned}$$

This yields $r = [(m+s)/\ell]$ as desired. □

We are now in a position to investigate the cyclic extensions of prime degree.

Let K/k be cyclic of prime degree ℓ, with group G. Since G has no subgroups except itself and 1, the pattern of the ramification groups of K/k will be:

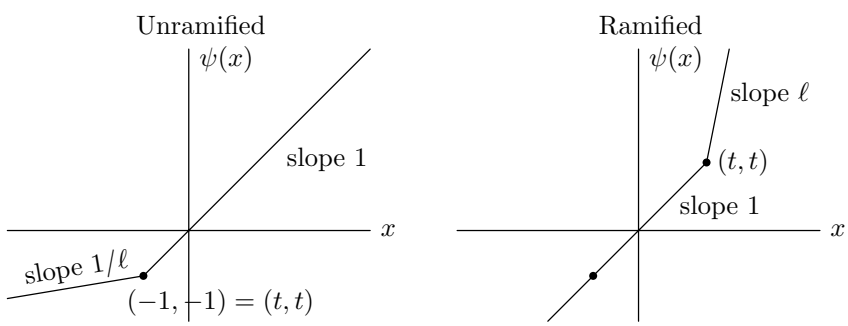
$$\begin{gathered} G = V_{-1} = V_0 = \cdots = V_t \\ V_{t+1} = V_{t+2} = \cdots = 1, \end{gathered}$$

where t is an integer $\geqslant -1$. The order of the different is

$$\operatorname{ord}_K \vartheta = \sum_{i=0}^{\infty} (\#V_i - 1) = (t+1)(\ell - 1),$$

according to Theorem 1.

K/k is unramified if and only if $t = -1$. The graphs of $\psi(x)$ in the ramified and unramified cases are as follows:



$$\psi(x) = x \text{ for } x \geqslant -1 \qquad \psi(x) = \begin{cases} x & -1 \leqslant x \leqslant t \\ t + \ell(x-t) & x \geqslant t \geqslant 0 \end{cases}$$

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THEOREM 8. *Let K/k be cyclic of prime degree ℓ. For an integer $i \geqslant -1$ we have*

1. $NK_{\psi(i)} \subset k_i$ *and* $NK_{\psi(i)+1} \subset k_{i+1}$.
2. $(k_i : k_{i+1}NK_{\psi(i)}) \leqslant \begin{cases} 1 & \text{if } i \neq t \\ \ell & \text{if } i = t \end{cases}$

where t is the integer such that $V_t \neq V_{t+1}$.

PROOF. We distinguish several cases.

$i = -1$. Then $\psi(-1) = -1$. The inclusion statements are obvious, and so is the index which $= 1$ if K/k is ramified ($t \neq -1$) and $= \ell$ if K/k is unramified ($t = -1$).

$i = 0$. Then $\psi(0) = 0$. Obviously, $NK_0 \subset k_0$. We also have $NK_1 \subset k_1$ and we see that the inclusion statements are obvious.

If $t = -1$ (K/k unramified) then every unit is a norm by Lemma 7, and $(k_0 : k_1NK_0) = 1$.

Let $^-$ denote residue classes $\pmod{\mathfrak{P}}$. We have

$$\begin{aligned}(k_0 : k_1NK_0) &= (\bar{k}_0 : \overline{k_1NK_0})(k_1 : k_1 \cap k_1NK_0) \\ &= (\bar{k} : \overline{k_1NK_0}) \\ &\leqslant (\bar{k} : \bar{k}^\ell).\end{aligned}$$

If $t = 0$ then K/k is tamely ramified and $\ell \neq p$. By Lemma 5 the index is $\leqslant \ell$. If $t > 0$ then K/k is strongly ramified and $\ell = p$. Since $\bar{k}$ is perfect, the index is 1. This settles the case $i = 0$.

We shall now treat the cases where $i \geqslant 1$. We distinguish several values of t, and use Lemma 9 constantly, $m = (t + l)(\ell - 1)$.

$t = -1$. We have $\psi(i) = i$. Therefore $S\mathfrak{P}^{\psi(i)} = \mathfrak{p}^i$. Also, we see that $S\mathfrak{P}^{2\psi(i)}$ and $S\mathfrak{P}^{\psi(i)+1}$ are contained in $\mathfrak{p}^{i+1}$ We have trivially $N\mathfrak{P}^{\psi(i)} = \mathfrak{p}^{\ell i} \subset \mathfrak{p}^{i+1}$. Hence by Lemma 8 we have

$$N(1 + \mathfrak{P}^{\psi(i)}) \equiv 1 + \mathfrak{p}^i \pmod{\mathfrak{p}^{i+1}}$$

which gives $NK_{\psi(i)} \cdot k_{i+1} = k_i$, as was to be shown.

$t \geqslant 0$ and $i \geqslant t$. We have $\psi(i) = t + \ell(i - t) = -(\ell - 1)t + \ell i$. Therefore $\psi(i) + (t + 1)(\ell - 1) = \ell i + (\ell - 1)$. From this and Lemma 9 it follows that

$$S\mathfrak{P}^{\psi(i)} = \mathfrak{p}^i \quad \text{and} \quad S\mathfrak{P}^{\psi(i)+1} = \mathfrak{p}^{i+1}.$$

Furthermore, $S\mathfrak{P}^{2\psi(i)} \subset \mathfrak{p}^{i+1}$. Since $\psi(i) \geqslant i$, we have

$$N\mathfrak{P}^{\psi(i)} \subset \mathfrak{p}^i \quad \text{and} \quad N\mathfrak{P}^{\psi(i)+1} \subset \mathfrak{p}^{i+1}.$$

The inclusion statements follows from these remarks and Lemma 8.

To get the index we consider two cases.

$i > t$. Then $\psi(i) > i$ and $N\mathfrak{P}^{\psi(i)} \subset \mathfrak{p}^i$ Hence

$$N(1 + \mathfrak{P}^{\psi(i)}) \equiv 1 + \mathfrak{p}^i \pmod{\mathfrak{p}^{i+1}}$$

and $k_{i+1}NK_{\psi(i)} = k_i$.

$i = t$. We have already considered $i = 0$, so that we may assume $i = t > 0$. We are in strong ramification. We have $\psi(i) = i = t$. By the inclusion statements, there exists an element $\alpha \in \mathfrak{P}^t$, $\alpha \notin \mathfrak{P}^{t+1}$ such that $S\alpha \notin \mathfrak{p}^{t+1}$ and $S\alpha = b\pi^t$,

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$b \notin \mathfrak{p}$. Then $N\alpha = \pi^t a$, where $a \not\equiv 0 \pmod{\mathfrak{p}}$. For any integer x of k we have by Lemma 8:

$$\begin{aligned} N(1+x\alpha) &\equiv 1 + xS\alpha + x^\ell N\alpha \pmod{\mathfrak{p}^{t+1}} \\ &\equiv 1 + \pi^t(ax^\ell + bx) \pmod{\mathfrak{p}^{t+1}} \end{aligned}$$

where $ab \not\equiv 0 \pmod{\mathfrak{p}}$. We have considered the polynomial $f(x) = ax^\ell + bx$ in Lemma 6, and found the index $(\bar{k} : f(\bar{k})) \leqslant p$. Multiplicatively, this gives $(k_t : k_{t+1}NK_t) \leqslant p$, as was to be shown.

$t \geqslant 0$ and $i < t$.

We have $\psi(i) = i < t$. Therefore $\psi(i) + (t+1)(\ell-1) = i + (t+1)(\ell-1) > i + (i+1)(\ell-1) = \ell i + \ell - 1$. This proves that $\psi(i) + (t+1)(\ell-1) \geqslant 1(i+l)$. Hence by Lemma 9, $S\mathfrak{P}^{\psi(i)} \subset \mathfrak{p}^{i+1}$, and a fortiori, $S\mathfrak{P}^{\psi(i)+1}$ and $S\mathfrak{P}^{2\psi(i)} \subset \mathfrak{p}^{i+1}$. For the norm, we are in the ramified case, and therefore $N\mathfrak{P}^{\psi(i)} = \mathfrak{p}^i$, and $N\mathfrak{P}^{\psi(i)+1} = \mathfrak{p}^{i+1}$. Using Lemma 8 proves the inclusion statements. Furthermore,

$$N(1 + \mathfrak{P}^{\psi(i)}) = 1 + \mathfrak{p}^i \pmod{\mathfrak{p}^{i+1}}$$

and this means multiplicatively $k_{i+1}NK_{\psi(i)} = k_i$, thereby completing the proof of our theorem. □

Having settled the case of a cyclic extension of prime degree, we can treat the general normal extension by showing that the statements we wish to make are transitive.

We recall the notation: $\psi'_r(x)$ and $\psi'_\ell(x)$ are the right and left derivatives of $\psi(x)$. We denote by $\psi'_{r/\ell}(x)$ the quotient $\psi'_r(x)/\psi'_\ell(x)$.

THEOREM 9. *Let k be a general local field, and K/k a normal extension. For an integer $i \geqslant -1$ we have*:

1. $NK_{\psi(i)} \subset k_i$, *and* $NK_{\psi(i)+1} \subset k_{i+1}$.
2. $(k_i : k_{i+1}NK_{\psi(i)}) \leqslant \psi'_{r/\ell}(i)$.

PROOF. The two statements have been proved in Theorem 8 in a cyclic extension of prime degree. We know that K/k is solvable, and it suffices therefore to prove that the two statements are transitive. This transitivity is essentially a trivial consequence of the transitivity of the norm, and of the ψ function (proved in Theorem 7).

Let $K \supset E \supset k$ be two normal extensions and assume the theorem for K/E and E/k. We have

$$N = \bar{N}\widetilde{N} \quad \text{and} \quad \psi = \widetilde{\psi}\bar{\psi}.$$

By assumption,

$$\begin{aligned} \bar{N}E_{\bar{\psi}(i)} \subset k_i \quad &\text{and} \quad \widetilde{N}K_{\widetilde{\psi}(\bar{\psi}(i))} \subset E_{\bar{\psi}(i)}. \\ \bar{N}E_{\bar{\psi}(i)+1} \subset k_{i+1} \quad &\text{and} \quad \widetilde{N}K_{\widetilde{\psi}(\bar{\psi}(i))+1} \subset E_{\bar{\psi}(i)+1}. \end{aligned}$$

The first statement is now trivial, because

$$NK_{\psi(i)} = \bar{N}\widetilde{N}K_{\psi(i)} \subset \bar{N}E_{\bar{\psi}(i)}.$$

To prove the second statement, we write

$$(k_i : k_{i+1}NK_{\psi(i)}) = (k_i : k_{i+1}\bar{N}E_{\bar{\psi}(i)})\big(k_{i+1}\bar{N}E_{\bar{\psi}(i)} : k_{i+1}\bar{N}\widetilde{N}K_{\widetilde{\psi}(\bar{\psi}(i))}\big).$$

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We can insert the group $E_{\bar{\psi}(i)+1}$ in the last index because of the inclusion $NE_{\bar{\psi}(i)+1} \subset k_{i+1}$ which we have just proved:

$$k_{i+1}\bar{N}\left[E_{\bar{\psi}(i)+1}\widetilde{N}K_{\widetilde{\psi}(\bar{\psi}(i))}\right] = k_{i+1}\bar{N}\widetilde{N}K_{\widetilde{\psi}(\bar{\psi}(i))}.$$

Hence our index becomes by induction

$$\begin{aligned}
&\leqslant \bar{\psi}'_{r/\ell}(i)\big(E_{\bar{\psi}(i)} : E_{\bar{\psi}(i)+1}\widetilde{N}K_{\widetilde{\psi}(\bar{\psi}(i))}\big)\\
&\leqslant \bar{\psi}'_{r/\ell}(i)\widetilde{\psi}'_{r/\ell}(\bar{\psi}(i))\\
&= \psi'_{r/\ell}(i).
\end{aligned}$$

This final step follows from the chain rule for right and left differentiation which holds because our functions are strictly monotone. □

If we take i very large in the preceding theorem, we get $\psi'_{r/\ell}(i) = 1$. This means that for all sufficiently large i,

$$(k_i : k_{i+1}NK_{\psi(i)}) = 1$$

and from this index we see that every unit of k_i is congruent to a norm from $K_{\psi(i)}$ mod $\mathfrak{p}_{i+1}$. Such a unit can therefore be refined to a norm by an obvious argument.

Corollary. *Let k be a general local field. Let K/k be a normal extension. Then $k_i \subset NK$ for some integer i.*

The conductor of K/k is the least power $\mathfrak{p}^\nu$ of $\mathfrak{p}$ such that $k_\nu \subset NK$, and we see that the conductor exists. We denote it by $f_{K/k}$, or briefly by f if the reference to the field is clear.

We may now write the norm index as a product:

$$(k : NK) = (k : k_0NK)(k_0NK : k_1NK)\dots(k_{s-1}NK : k_sNK).$$

If s is big enough, $k_s \subset NK$. Hence

$$(k : NK) = \prod_{-1}^{\infty}(k_iNK : k_{i+1}NK). \tag{1}$$

Furthermore, by an elementary isomorphism theorem, we have

$$\begin{aligned}
(k_iNK : k_{i+1}NK)(k_i \cap NK : k_{i+1}NK_{\psi(i)} \cap NK) &= (k_i : k_{i+1}NK_{\psi(i)})\\
&\leqslant \psi'_{r/\ell}(i) \quad \text{(by Theorem 9)}.
\end{aligned} \tag{2}$$

From (1) and (2) we obtain

$$(k : NK) \leqslant \prod_{-1}^{\infty}\psi'_{r/\ell}(i) \leqslant n \tag{3}$$

where $n = [K : k]$. Indeed $\psi'_r(i) \leqslant \psi'_\ell(i+1)$ whence $\frac{\psi'_r(i)}{\psi'_\ell(i+1)} \leqslant 1$. Furthermore $\psi'_\ell(-1) = 1/f$, and $\psi'_r(\infty) = e$. Hence the product is $\leqslant ef = n$.

The index inequality of (3) is the second inequality. The class field theory in general local fields may now be developed in exactly the same way as the theory in the classical case. As a consequence of the second inequality, we can prove that $h_2(K/k) \leqslant [K : k]$, and then use the same method as that of Chapter XIV. Instead of the Frobenius Substitution, we select in the Galois group of the algebraic closure of $\bar{k}$ an element $\bar{\sigma}$ which operates non trivially on every finite extension of

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$\bar{k}$. The effect of $\bar{\sigma}$ on each finite extension is to generate the Galois group, which is cyclic. The automorphism $\bar{\sigma}$ has a corresponding automorphism σ in the Galois group of the maximal unramified extension of k, which can play the same role as the Frobenius Substitution. It has not been canonically selected, but this does not make any difference to the proofs, as long as we develop only a local theory. It is only in the global theory, when the local theories were pieced together, that it became important to choose the proper automorphism in the local fields.

Having done local class field theory, we obtain all the results of Chapter XIV. In particular:

THEOREM 10. *Let k be a general local field. Let Ω be its algebraic closure, and $\mathfrak{G}$ the Galois group. Then $(\mathfrak{G}, \Omega^*)$ is a class formation.*

We may now return to the indices computed to prove the second inequality. We know that in abelian extensions the norm index is equal to the degree. This means that the doubtful indices of Theorem 8, 9 and (2), (3) above for abelian fields are no more doubtful, but are actually equal.

THEOREM 11. *Let k be a general local field, and let K/k be a finite abelian extension.*

1. $k_i \cap NK = k_{i+1}NK_{\psi(i)} \cap NK \subset k_{i+1}NK_{\psi(i)}$.
2. $(k_i : k_{i+1}NK_{\psi(i)}) = (k_iNK : k_{i+1}NK) = \psi'_{r/\ell}(i)$.
3. *All breaks in $\psi(x)$ occur at integral arguments. In other words, $\psi'_{r/\ell}(x) = 1$ if x is not an integer.*
4. $k_x \subset NK$ *if and only if* $V^x (= V_{\psi(x)}) = 1$.

PROOF. Statements 1 and 2 arise from the equality in (2) and (3) above. To prove 3 we note that $\psi(x)$ has a finite number of breaks. If we take the product $\Pi\psi'_{r/\ell}(x)$ over the numbers x for which there is a break, we get $ef = n$. But each $\psi'_{r/\ell}(x)$ is > 1. Since the product over integers already yields n, this means that the integers give all the breaks.

Using the equality of the indices in 2 we have:

$$\begin{aligned} k_x \subset NK &\Leftrightarrow (k_j : k_{j+1}NK_{\psi(j)}) = 1 \quad \text{for all } j \geqslant x \\ &\Leftrightarrow \psi'_{r/\ell}(j) = 1 \quad \text{for all } j \geqslant x \\ &\Leftrightarrow \psi'_r(x) = \psi'_\ell(x) = \psi'(\infty) = e \\ &\Leftrightarrow (\#V_0 : \#V_{\psi(x)}) = (\#V_0 : \#V_\infty) \\ &\Leftrightarrow V_{\psi(x)} = V_\infty = 1 \\ &\Leftrightarrow V^x = 1 \end{aligned}$$

thereby proving 4. □

THEOREM 12. *Let k be a general local field, and ω the norm residue mapping, into the Galois group $\mathfrak{G}$ of the maximal abelian extension of k. Then $\omega(k_x)$ is everywhere dense in $\mathfrak{Y}^x$.*

PROOF. Let K/k be a finite abelian extension. K is left fixed by

$$\begin{aligned} \mathfrak{Y}^x &\Leftrightarrow V^x_{K/k} = 1 \\ &\Leftrightarrow k_x \subset NK \\ &\Leftrightarrow \omega(k_x) \text{ leaves } K \text{ fixed} \end{aligned}$$

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and this proves the theorem. □

COROLLARY. *Let K/k be a finite abelian extension of a general local field. Then $\omega(k_i) = V^i$.*

5. The Conductor

Let k be complete under a discrete valuation with perfect residue class field $\bar{k}$. We shall investigate the conductor of a normal extension of k attached to a character of its Galois group, and defined formally in terms of the ramification groups. Most of our discussion will be valid in the field k, and it will be explicitly stated when k is to be specialized to a general local field.

Let K/k be normal with group G. We assume known the representation theory of G over the complex numbers.

Let $f(\sigma)$ be a function of G into the complex. Let H be a subgroup of G. Then f is also a function on H by restriction. We let

$$\int_H f(\sigma)\, d\sigma = \frac{\sum_H f(\sigma)}{\#H}.$$

The integral is an average over H.

Let χ be a character of G. Its restriction to H is a character of H. On H, we can express χ as a sum of irreducible characters:

$$\chi = \sum \mu_i \chi_i.$$

We call μ_i the multiplicity of χ_i in χ. It follows from representation theory that

$$\sum_H \chi(\sigma) = \mu \# H$$

where μ is the multiplicity of 1 in χ. (We refer to the irreducible character $\chi(\sigma) = 1$ for all σ as 1.)

We have

$$\int_H \chi(\sigma)\, d\sigma = \mu$$

and the integral is therefore an integer $\geqslant 0$.

A character χ is linear on the group ring. We have

$$\int_H \chi(1-\sigma)\, d\sigma = \dim \chi - \mu$$

where μ is again the multiplicity of 1 in χ. From this we see that $\int_H \chi(1-\sigma)\, d\sigma$ is an integer $\geqslant 0$.

Given a character χ on G, we define a real number $\nu(\chi)$ as follows:

$$\nu(\chi) = \int_{-1}^{\infty} \left(\int_{V^t} \chi(1-\sigma)\, d\sigma \right) dt.$$

Then $V^t = 1$ for large t, and $\chi(1-1) = 0$. Hence $\int_{V^t} \chi(1-\sigma) d\sigma = 0$ for large t. The inner integral is always $\geqslant 0$ according to the preceding remarks, and therefore $\nu(\chi)$ is a well defined non-negative real number. We define formally the *conductor* belonging to this character to be $\mathfrak{p}^{\nu(\chi)} = \mathfrak{f}_\chi$.

The function $f(t) = \int_{V^t} \chi(1-\sigma)\, d\sigma$ is easily seen to be a step function. Suppose that G is abelian and that χ is an ordinary character of G (i.e. a homomorphism into the unit circle), $\chi \neq 1$. It follows from representation theory that $\sum_{V^t} \chi(\sigma) = 0$ if

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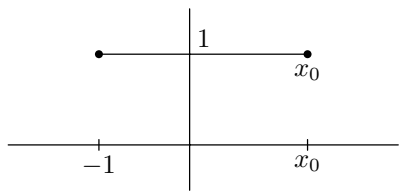
and only if χ is nontrivial on V^t. Under these circumstances, $f(t) = 1$. Hence the graph of $f(t)$ is as follows:



Let $x_0 = \inf\{x\colon f(x) = 0\}$. Then we see that $\nu(\chi) = x_0 + 1$. Furthermore, χ is not trivial on V^{x_0} but is trivial on $V^{x_0+\delta}$ for $\delta > 0$.

Suppose in particular that k is a general local field. By definition, $\psi(x_0)$ must have a break. According to Theorem 11 of section 3 the breaks in $\psi(x)$ occur only at integers in the abelian extensions of general local fields. This proves

THEOREM 13. *Let k be a general local field and K/k an abelian extension. Let χ be a character of G. Then $\nu(\chi)$ is an integer. (If $\chi = 1$ then $\nu(\chi) = 0$ trivially.)*

Let ω be the mapping of the norm residue symbol. Then $\chi \cdot \omega$ is a homomorphism of k into the unit circle. From the existence of a conductor for K/k (Theorem 9, Cor.) we see that $\chi \cdot \omega$ is continuous in the class topology on k.

To get precise information we use the Corollary to Theorem 12. We know that $\omega(k_i) = V^i$. Combining this with the preceding discussion we obtain

THEOREM 14. *Let k be a general local field and K/k an abelian extension. Then $\nu(\chi)$ is the least integer i such that $\chi \cdot \omega(k_i) = 1$.*

COROLLARY. *Let $\mathfrak{f}$ be the conductor of K/k. Then*

$$\mathfrak{f} = \mathrm{l.\,c.\,m.}\, \mathfrak{f}_\chi.$$

We see that the formal conductors obtained from characters of the Galois group are the same as the conductors arising in the class field theory.

Having shown that the conductors coincide when class field theory holds, we shall return to the abstract case, and investigate the formal properties of $\nu(\chi)$. We assume therefore that the residue class field of k is perfect.

THEOREM 15. *Let K/k be normal and let χ_1, χ_2 be two characters of G. Then*

$$\nu(\chi_1 + \chi_2) = \nu(\chi_1) + \nu(\chi_2).$$

PROOF. Immediate from the definition of $\nu(\chi)$, because characters are additive. $\square$

THEOREM 16. *Let $K \supset E \supset k$ be two normal extensions. Let $\bar{\chi}$ be a character of $G/H = \bar{G}$, so that $\bar{\chi}$ may be viewed as a character χ of G also. Then*

$$\bar{\nu}(\bar{\chi}) = \nu(\chi).$$

PROOF. Let $V^t = \bigcup \bar{\sigma}(V^t \cap H)$. We know that $\bar{V}^t = V^t H/H$, and the $\bar{\sigma}$ are naturally elements of $\bar{V}^t$, and $\bar{\chi}(\bar{\sigma}) = \chi(\bar{\sigma}\gamma)$ for all $\gamma \in H$.

It suffices therefore to show that

$$\int_{V^t} \chi(1-\sigma)\, d\sigma = \int_{\bar{V}^t} \bar{\chi}(1-\bar{\sigma})\, d\bar{\sigma}.$$

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Note that $\#V^t = \#\bar{V}^t \cdot \#(V^t \cap H)$, and hence

$$\int_{V^t} \chi(1-\sigma)\,d\sigma = \frac{\sum_{V^t} \chi(1-\sigma)}{\#V^t} = \frac{\#(V^t \cap H)}{\#V^t} \sum_{\bar{V}^t} \bar{\chi}(1-\bar{\sigma}) = \int_{\bar{V}^t} \bar{\chi}(1-\bar{\sigma})\,d\bar{\sigma}. \quad \square$$

THEOREM 17. $\nu(\chi) = \frac{1}{e}\sum_{i=0}^{\infty}\sum_{V_i} \chi(1-\sigma)$.

PROOF. Since $V^t = V_{\psi(t)}$, we have

$$\nu(\chi) = \int_{-1}^{\infty} \left(\int_{V_{\psi(t)}} \chi(1-\sigma) d\sigma \right) dt.$$

The function $f(t) = \int_{V_{\psi(t)}} \chi(1-\sigma)d\sigma$ has only a finite number of discontinuities, and we may take the sum of the integrals in the intervals where no discontinuity occurs. Let $s = \psi(t)$ and $t = \varphi(s)$. Then $dt = \varphi'(s)ds$ and by the chain rule:

$$\begin{aligned} &= \int_{-1}^{\infty} \left(\int_{V_s} \chi(1-\sigma) d\sigma \right) \frac{\#V_s}{\#V_0}\, ds \\ &= \frac{1}{e} \int_{-1}^{\infty} \left(\sum_{V_s} \chi(1-\sigma) \right) ds \\ &= \frac{1}{e} \sum_{i=0}^{\infty} \sum_{V_i} \chi(1-\sigma) \end{aligned}$$

because $V_{i+\delta} = V_{i+1}$ for $0 < \delta < 1$, and

$$\int_i^{i+1} \left(\sum_{V_s} \chi(1-\sigma) \right) ds = \sum_{V_{i+1}} \chi(1-\sigma). \quad \square$$

The formula giving $\nu(\chi)$ in the preceding theorem might have been taken as a definition. We have selected the integral because the important Theorems 13, 14, 15, and 16 were immediate consequences of our definition.

THEOREM 18. *Let K/k be normal with group G. Let H be a subgroup, and E the fixed field. Let ψ be a character of H, and χ the induced character of G. Then*

$$\mathfrak{f}_\chi = D_{E/k}^{\psi(1)} N_{E/k} \mathfrak{f}_\psi. \quad (D = \text{discriminant.})$$

PROOF. Let $G = \bigcup H\bar{c}$. The induced character χ has the value

$$\chi(\tau) = \sum_c \psi(\bar{c}\tau\bar{c}^{-1})$$

where $\psi(\sigma) = 0$ if $\sigma \notin H$. Starting with the expression derived for $\nu(\chi)$ in Theorem 17 we have therefore:

$$\begin{aligned} \nu(\chi) &= \frac{1}{e} \sum_{i=0}^{\infty} \sum_{V_i} \chi(1-\sigma) \\ &= \frac{1}{e} \sum_{0}^{\infty} \sum_{V_i} \sum_c \psi(1-\bar{c}\sigma\bar{c}^{-1}). \end{aligned}$$

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But $V_i^{\bar{c}^{-1}} = V_i$ because V_i is normal. We can therefore take out the $\bar{c}^{-1}$:

$$= \frac{1}{e}\sum_0^\infty \sum_{V_i}\sum_c \psi(1-\sigma)$$

$$= \frac{1}{e}\sum_0^\infty \sum_c\sum_{V_i} \psi(1-\sigma)$$

$$= \frac{1}{e}\sum_0^\infty \Big(\bar{e}\bar{f}\sum_{V_i} \psi(1-\sigma)\Big).$$

Since $e = \bar{e}\tilde{e}$ and $\psi(\sigma) = 0$ unless $\sigma \in V_i \cap H$,

$$= \frac{\bar{f}}{\tilde{e}}\sum_0^\infty \Big(\#V_i\psi(1) - \#(V_i\cap H)\psi(1) + \sum_{V_i\cap H}\psi(1-\sigma)\Big).$$

But $V_i \cap H = \widetilde{V}_i$. We may add 1 and subtract 1, thereby giving

$$= \frac{\bar{f}}{\tilde{e}}\sum_0^\infty(\#V_i - 1)\psi(1) + \frac{\bar{f}}{\tilde{e}}\sum_0^\infty(\#\widetilde{V}_i - 1)\psi(1) + \bar{f}\frac{1}{\tilde{e}}\sum_0^\infty\sum_{\widetilde{V}_i}\psi(1-\sigma).$$

Using Theorem 1 for the order of the different, and Theorem 17 giving the expression for $\nu(\psi)$:

$$= \frac{\bar{f}}{\tilde{e}}\psi(1)\big[\operatorname{ord}_K\vartheta - \operatorname{ord}_K\tilde{\vartheta}\big] + \bar{f}\nu(\psi).$$

But $\operatorname{ord}_K\vartheta - \operatorname{ord}_K\tilde{\vartheta} = \operatorname{ord}_K\overline{\vartheta}$. Also, $\frac{1}{\tilde{e}}\operatorname{ord}_K = \operatorname{ord}_E$. Furthermore, $\operatorname{ord}_k N_{E/k} = \bar{f}\operatorname{ord}_E$. Combining these remarks, and writing the formula multiplicatively, we have

$$\mathfrak{f}_x = \mathfrak{p}^{\nu(\chi)} = D_{E/k}^{\psi(1)}N_{E/k}\mathfrak{f}_\psi$$

where $D_{E/k} = N_{E/k}\overline{\vartheta}$ is the discriminant. This proves our theorem. □

COROLLARY 1. *Let $\psi = 1$ and let the induced character $\chi = \sum_i \mu_i\chi_i$ where χ_i are the irreducible characters of G. Then*

$$D_{E/k} = \mathfrak{f}_\chi = \prod_i \mathfrak{f}_{\chi_i}^{\mu_i}.$$

PROOF. Immediate from the additivity of the characters. □

COROLLARY 2. *Let K/k be abelian. Then*

$$D_{K/k} = \prod_\chi \mathfrak{f}_\chi$$

where the product is taken over all the ordinary characters of G.

PROOF. Let $E = K$ in the preceding corollary. The character χ induced by 1 is the character of the regular representation. The irreducible characters have dimension 1 so $\mu_i = 1$ in the product. The irreducible characters are simply the ordinary characters of G when G is abelian, and this proves our corollary. □

The preceding theorems giving the formalism of $\nu(\chi)$ are valid in a complete field with any perfect residue class field. We shall now specialize to a general local field to obtain one more statement concerning the conductor:

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THEOREM 19. *Let k be a general local field, and K/k a normal extension. Then $\nu(\chi)$ is an integer.*

PROOF. A character is a linear combination with integer coefficients of irreducible characters. Using the linearity of $\nu(\chi)$ we see that it suffices to prove the theorem for irreducible characters.

Suppose next that χ is 1-dimensional. Such a character will be called more briefly an abelian character. It is a homomorphism of G, and its kernel G_0 is a normal subgroup. Furthermore, $\bar{G} = G/G_0$ is cyclic. Let $\chi = \bar{\chi}$ on $\bar{G}$. Then $\nu(\chi) = \bar{\nu}(\bar{\chi})$ by Theorem 16, and by Theorem 13 we know that $\bar{\nu}(\bar{\chi})$ is an integer. This proves the theorem if χ is abelian.

Let ψ be an abelian character of a subgroup H. Then $\widetilde{\nu}(\psi)$ is an integer by the preceding remarks, and $\mathfrak{f}_\psi$ is integral. Let χ be the induced character. The formula

$$\mathfrak{f}_\chi = D_{E/k}^{\psi(1)} N_{E/k} \mathfrak{f}_\psi$$

of Theorem 18 shows that $\mathfrak{f}_\chi$ is also integral, i.e. that $\nu(\chi)$ is an integer.

Brauer has proved that every character is a linear combination with integer coefficients of characters induced by abelian characters. In view of the linearity of $\nu(\chi)$ it follows that $\nu(\chi)$ is an integer.

We shall give here a proof independent of Brauer's Theorem, by reducing the problem to p-groups. Indeed, for a p-group, it is shown fairly easily that every irreducible character is induced by an irreducible abelian character,[2] and the preceding argument can then be applied directly.

We may of course assume K/k completely ramified. $e = e_0 p^r = \#V_0$. We must show that e divides $\sum_{i=0}^{\infty} \sum_{V_i} \chi(1-\sigma)$.

We begin by treating e_0. Let $\lambda(\chi) = e\nu(\chi)$ be the double sum. Then

$$\begin{aligned}
\lambda(\chi) &= \sum_{i=0}^{\infty} \Big[\chi(1)\#V_i - \sum_{V_i} \chi(\sigma)\Big] \\
&= \sum_{i=0}^{\infty} i\Big[\chi(1)\#V_i - \sum_{V_i} \chi(\sigma)\Big] - \sum_{i=0}^{\infty} (i-1)\Big[\chi(1)\#V_i - \sum_{V_i} \chi(\sigma)\Big] \\
&= \sum_{i=1}^{\infty} i\Big[\chi(1)\#V_i - \sum_{V_i} \chi(\sigma)\Big] - \sum_{i=-1}^{\infty} i\Big[\chi(1)\#V_{i+1} - \sum_{V_{i+1}} \chi(\sigma)\Big] \\
&= \chi(1)\#V_0 - \sum_{V_0} \chi(\sigma) + \sum_{i=1}^{\infty} i\Big[\chi(1)(\#V_i - \#V_{i+1}) - \sum_{V_i - V_{i+1}} \chi(\sigma)\Big].
\end{aligned}$$

But $e_0 \mid \#V_0$ and $\sum_{V_0} \chi(\sigma) = \#V_0 \cdot \mu$ where μ is the multiplicity of 1 in χ. So the first term is divisible by e_0.

To handle the sums, we recall Theorem 5 and the discussion following it. The sum $\sum_{V_i - V_{i+1}} \chi(\sigma)$ breaks up into a sum over equivalence classes and we have

$$\sum_{V_i - V_{i+1}} \chi(\sigma) = \sum_{\sigma} r_\sigma \chi(\sigma)$$

where the second sum is taken over a representing the equivalence classes. (We can do this because a character depends only on the conjugate class of a group element.)

[2]See the corollary of Theorem 3 in the appendix which follows.

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We know that $\sum_H \chi(\sigma)$ is a rational integer for any subgroup H of G. Hence $\sum_{V_i - V_{i+1}} \chi(\sigma) = \sum_{V_i} \chi(\sigma) - \sum_{V_{i+1}} \chi(\sigma)$ is a rational integer, say m_i. We also know that $e_0 \mid ir_\sigma$. Hence

$$m_i = e_0 \sum_\sigma (ir_\sigma/e_0)\chi(\sigma).$$

But $\sum_\sigma (ir_\sigma/e_0)\chi(\sigma) = m_i/e_0$ is an algebraic integer, and a rational number. It is therefore an integer, and this proves that e_0 divides the questionable sum.

The preceding discussion is valid for any character χ. Taking $\chi = 1$ yields the term $i(\#V_i - \#V_{i+1})$ which is therefore also divisible by e_0.

All that remains to be shown is that $\nu(\chi)$ is divisible by p^r.

We have $\lambda(\chi) = \sum_{i=1}^\infty \sum_{V_i} \chi(1-\sigma) + \sum_{V_0} \chi(1-\sigma)$. The sum $\sum_{V_0} \chi(1-\sigma)$ is divisible by $\#V_0 = e$. Let E be the fixed field of V_1. We know that $\widetilde{V}_i = V_i \cap V_1$ and therefore

$$\sum_{i=1}^\infty \sum_{V_i} \chi(1-\sigma) = \sum_{i=1}^\infty \sum_{\widetilde{V}_i} \widetilde{\chi}(1-\sigma)$$

where $\widetilde{\chi}$ is the restriction of χ to V_1. We have $\widetilde{V}_0 = \widetilde{V}_1$. Our sum differs from $\widetilde{\lambda}(\widetilde{\chi})$ in E by

$$\sum_{\widetilde{V}_0} \widetilde{\chi}(1-\sigma) = \sum_{V_1} \chi(1-\sigma)$$

and this sum is divisible by $\#V_1 = p^r$. Hence it suffices to prove that $\widetilde{\lambda}(\widetilde{\chi})$ is divisible by p^r and reduces therefore the theorem to the case of the p-group V_1. We saw already at the beginning of our proof that the theorem follows in full, if we use the fact that every character of a p-group is induced by an abelian character of a subgroup. $\square$

This concludes our discussion of the higher ramification in general local fields.

Appendix: Induced Characters

For the convenience of the reader we shall develop here the theory of induced characters, used in the preceding section.

We begin by recalling basic notation.

Let G be a finite group, F a field, algebraically closed and of characteristic 0. All the spaces we deal with will be finite dimensional F-spaces. This will not be mentioned explicitly any more. We denote F-spaces by $U, V, W, \ldots$.

Let V be a G-space. G acts on V linearly, and is represented by linear transformations of V. If a basis of V is selected, then the transformations may be given by matrices and we have a homomorphism

$$\sigma \to M_\sigma$$

of G into a group of matrices.

The character χ of the representation is the function on G given by

$$\chi(\sigma) = S(M_\sigma)$$

where S is the trace (sum of diagonal terms). The fundamental theorem states that the character is an invariant of the representation, and that in fact it characterizes

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it: Two representations of G are defined to be equivalent if their spaces are G-isomorphic, and two representations are equivalent if and only if they have the same character.

Induced representation. Let H be a subgroup of G. For each coset $c = H\sigma$, let $\bar{c}$ be a representative element, so that $G = \coprod \bar{c}^{-1}H$, disjoint union. Let V be a G-space, W and H-space, and $i : W \to V$ an H-homomorphism. One says that V *is induced by* W *via* i if i is an isomorphism into and $V = \bigoplus \bar{c}^{-1}iW$, direct sum. For each W there does exist such a V, for example, $V = F[G] \otimes_{F[H]} W$, with $i(w) = 1 \otimes w$. For any such V it is easy to check that if V' is another G-space the map $f \mapsto f \circ i$ is an isomorphism $\mathrm{Hom}_G(V, V') \xrightarrow{\sim} \mathrm{Hom}_H(W, V')$. From this it follows that V is uniquely determined by W up to a unique G-isomorphism f such that $f \circ i = i'$. Therefore we usually view i as an inclusion, don't mention it, and write simply $V = V_W$.

THEOREM 1. *Let $G \supset H \supset I$, and let W be an I-space. Let U_W be the G-space induced by W and V_W the H-space induced by W. Then $U_W \simeq U_{V_W}$.*

If W_1 and W_2 are two I-spaces, then $V_{W_1+W_2} \simeq V_{W_1} + V_{W_2}$ where $+$ means the direct sum.

PROOF. The theorem is an immediate consequence of the uniqueness of the induced representation. □

Let $G \supset H$, and let W be an H-space. Let ψ be the character of the representation of H in W. The character χ of V_W is called the *induced character*. The next theorem gives a formula which allows us to compute the values of χ in terms of the values of ψ.

THEOREM 2. *Let $G = \cup H\bar{c}$. Then*

$$\chi(\sigma) = \sum_c \psi(\bar{c}\sigma\bar{c}^{-1})$$

where we let $\psi(\tau) = 0$ unless $\tau \in H$.

PROOF. Let $\xi_1, \ldots, \xi_n$ be a basis for W over F. We know that $V = \sum \bar{c}^{-1}W$. Let σ be an element of G. The elements $\{\overline{c\sigma}^{-1}\xi_i\}_{c,i}$ form a basis for V.

Remark that

$$\sigma(\overline{c\sigma}^{-1}\xi) = \bar{c}^{-1}(\bar{c}\sigma\overline{c\sigma}^{-1})\bar{\xi}.$$

The action of σ on this basis is therefore given by

$$\begin{aligned}\sigma(\overline{c\sigma}^{-1}\xi_i) &= \bar{c}^{-1}\sum_\mu (\bar{c}\sigma\overline{c\sigma}^{-1})_{i\mu}\xi_\mu \\ &= \sum_\mu (\bar{c}\sigma\overline{c\sigma}^{-1})_{i\mu}(\bar{c}^{-1}\xi_\mu).\end{aligned}$$

By definition,

$$\chi(\sigma) = \sum_{c\sigma=c}\sum_i (\bar{c}\sigma\overline{c\sigma}^{-1})_{ii}.$$

But $c\sigma = c$ if and only if $\bar{c}\sigma\bar{c}^{-1} \in H$. Furthermore,

$$\psi(\bar{c}\sigma\bar{c}^{-1}) = \sum_i (\bar{c}\sigma\bar{c}^{-1})_{ii}.$$

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Hence

$$\chi(\sigma) = \sum_c \psi(\bar{c}\sigma\bar{c}^{-1})$$

as was to be shown. □

THEOREM 3. *Let G be a p-group. Let V be an irreducible G-space. If V is not 1-dimensional then there exists a proper subgroup H_1 and an irreducible H_1-subspace W of V such that (G, V) is induced by (H_1, W).*

PROOF. We begin by recalling that an irreducible representation of an abelian group is 1-dimensional. This implies in particular that if V is not 1-dimensional, then G is not abelian.

We shall first give the proof of our theorem under the additional assumption that V gives a faithful representation of G. (This means, if $\sigma\xi = \xi$ for all $\xi \in V$ then $\sigma = 1$.) It will be easy to remove this restriction at the end.

Let H be a normal subgroup of G which is abelian and contains the center properly.

(Proof of existence of H: G has a non-trivial center G_0. Let $\bar{G} = G/G_0$. Let $\bar{\sigma}$ be an element of period p in the non-trivial center of $\bar{G}$ and let $\bar{H}$ be the subgroup of $\bar{G}$ generated by $\bar{\sigma}$. Then $\bar{H}$ is normal in $\bar{G}$. Its inverse image H in the natural map $G \to \bar{G}$ is normal in G, and is generated by an inverse image σ of $\bar{\sigma}$, and by G_0. Furthermore, $\sigma \notin G_0$ so $H \supset G_0$ properly. Finally, σ commutes with its powers and with G_0 (the center of G) so that H is abelian.)

We denote the elements of H by γ.

As an H-space, V is a direct sum of irreducible H-spaces which are 1-dimensional.

Let $\xi \in V$ generate a 1-dimensional H-space. Let ψ be its character, so $\gamma\xi = \psi(\gamma)\xi$, where $\psi(\gamma) \in F$. If η gives an equivalent representation, then $\gamma\eta = \psi(\gamma)\eta$. If $a, b \in F$ then $\gamma(a\xi + b\eta) = \psi(\gamma)(a\xi + b\eta)$. Hence the vectors of V giving rise to the same irreducible representation of H form an H-space W.

We contend that $V \neq W$. Assume $V = W$. Let ξ be any element of $W = V$. Let $\sigma \in G$. Then $\sigma^{-1}\xi$ is a 1-dimensional H-space by assumption, and has character ψ. Hence

$$\gamma(\sigma^{-1}\xi) = \psi(\gamma)\sigma^{-1}\xi$$
$$(\sigma\gamma\sigma^{-1})\xi = \sigma\psi(\gamma)\sigma^{-1}\xi = \psi(\gamma)\xi.$$

This shows that $\gamma^\sigma\xi = \gamma\xi$ for all $\xi \in W = V$. Since H is not contained in the center, $\gamma \neq \gamma^\sigma$ for some $\gamma \in H$, $\sigma \in G$, and we have contradicted the fact that V is faithful.

We may therefore assume that $V = W_1 + \cdots + W_r$ where each W_i is the space generated by all the equivalent 1-dimensional H-spaces having the same character ψ_i. The sum is direct, and $r > 1$.

We shall prove that G permutes the spaces W_i transitively.

For definiteness, we consider the effect of G on one particular W_i, say W_1.

Let σ be an element of G. Then $\sigma W_1 \subset W_i$ for some i.

PROOF. Let $\xi \in W_1$. For any $\gamma \in H$,

$$\gamma(\sigma\xi) = \sigma(\sigma^{-1}\gamma\sigma)\xi = \sigma\psi(\gamma^{\sigma^{-1}})\xi = \psi_\sigma(\gamma)(\sigma\xi)$$

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where the function ψ_σ given by $\psi_\sigma(\gamma) = \psi(\gamma^{\sigma^{-1}})$ is a character on H and depends only on σ. For a fixed σ, we see therefore that $\psi_\sigma = \psi_i$ for some i. Since W_i contains all the spaces with character ψ_i, it follows that $\sigma W_1 \subset W_i$. □

σ is an isomorphism of W_1 into W_i. By symmetry, $\sigma^{-1}W_i$ must be contained in W_1, and since W_1 and W_i are finite dimensional, this implies that their dimensions are equal. Hence $\sigma W_1 = W_i$.

The preceding argument proves that G permutes the spaces W_i. All that remains to be shown is the transitivity. Let $V' = GW_1$. If $V' \neq V$ then $GV' = V'$ and this contradicts the irreducibility of V. Hence $GW_1 = GW_i = V$. This proves that the spaces are permuted transitively.

Let H_1 be the stabilizer of W_1. Then $H_1 \supset H$, and $H_1 \neq G$ because $r > 1$. Furthermore, (H_1, W_1) is a local constituent of (G, V).

We contend that W_1 is irreducible for H_1. This will finish the proof of our theorem (in the case that V is faithful).

Suppose W_1 is not irreducible for H_1. Then there exists a space $U \subset W_1$, $U \neq W_1$ such that $H_1 U \subset U$. Let $G = \cup H_1 \bar{c}$. We know that $V = \sum_c \bar{c}^{-1} W_1$. Let $V' = \sum \bar{c}^{-1} U$. Then $V' \neq V$, and we shall prove that V' is a G-space, thereby contradicting the irreducibility of V.

We have

$$\sigma \overline{c\sigma}^{-1} U = \bar{c}^{-1}(\bar{c}\sigma\overline{c\sigma}^{-1})U \subset \bar{c}^{-1}U$$

because $(\bar{c}\sigma\overline{c\sigma}^{-1}) \in H_1$. But $c\sigma$ runs through all cosets as c does, and this proves that $\sigma V' \subset V'$. Hence V' is a G-space, contradiction.

Suppose now that V is not faithful. Let G_0 be the normal subgroup of G consisting of all σ such that $\sigma\xi = \xi$ for all $\xi \in V$. Then V is an irreducible space of $\bar{G} = G/G_0$ and gives a faithful representation of $\bar{G}$. If V is not 1-dimensional, then $\bar{G}$ is not abelian and there exists a proper subgroup $\bar{H}$ of $\bar{G}$ and an irreducible $\bar{H}$-space W such that $(\bar{H}, W)$ induces $(\bar{G}, V)$. Let H be the inverse image of $\bar{H}$ in the natural map $G \to \bar{G}$. Then $H \supset G_0$, and W is naturally an irreducible H-space. Furthermore we contend that the stabilizer H_1 of W is H. Certainly, $H_1 \supset H$. Suppose $\sigma \in H_1$, $\sigma \notin H$, then $\sigma W \subset W$, $\bar{\sigma} \notin \bar{H}$. Since $\bar{\sigma}$ has the same effect as σ on W, this contradicts the fact that $\bar{H}$ is the stabilizer of W.

This proves that (H, W) induces (G, V) and concludes the proof of our theorem. □

Corollary. *Let G be a p-group. Let χ be an irreducible character. If χ is not 1-dimensional then χ is induced by a 1-dimensional character ψ of a subgroup H of G.*

Proof. Let V be the irreducible space of χ. A subgroup of a p-group is a p-group. We may apply the preceding result step by step using Theorem 1 until we get a subspace W which is 1-dimensional. The character ψ of W will then induce χ. □

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CHAPTER XII

Explicit Reciprocity Laws

1. Formalism of the Power Residue Symbol

The global norm residue symbol was obtained from local ones. The definition of the local symbol was obtained indirectly from a non-constructive proof of the fact that all cocycles have an unramified splitting field. We are therefore faced with the following unsolved problem: To determine explicitly the effect of the norm residue symbol on totally ramified extensions. (On the unramified extensions, it is the Frobenius Substitution.)

We shall treat special cases of Kummer extensions.

We need an auxiliary algebraic statement.

LEMMA 1. *Let F be a field of characteristic $p \geqslant 0$, containing the n-th roots of unity, $p \nmid n$. Let $\alpha \in F$. Then $-\alpha$ and $1-\alpha$ are norms from $F(\alpha^{1/n})$.*

PROOF. We distinguish two cases.

$[F(\alpha^{1/n}) : F] = n$. Then $x^n - \alpha$ is irreducible and its roots generate $F(\alpha^{1/n})$. We have

$$x^n - \alpha = \prod_{\zeta_n} (x - \zeta_n \alpha^{1/n})$$

where ζ_n ranges over all n-th roots of unity, and $\zeta_n \alpha^{1/n}$ are the conjugates of $\alpha^{1/n}$. Putting $x = 0$ shows $-\alpha$ is a norm. Putting $x = 1$ shows $1 - \alpha$ is a norm.

$[F(\alpha^{1/n}) : F] = d < n$. By Kummer theory, $d \mid n$. Let $\alpha^d = \beta^n$ where $d \mid n$ is the period of $\alpha \pmod{F^n}$. We have $F(\alpha^{1/n}) = F(\beta^{1/d})$ and by selecting $\alpha^{1/n}$ and $\beta^{1/d}$ suitably (i.e. by multiplying them by a suitable root of unity) we may assume $\alpha^{1/n} = \beta^{1/d}$. The conjugates of $a^{1/n}$ are therefore $\zeta_d \alpha^{1/n}$ where ζ_d ranges over all d-th roots of unity. The polynomial

$$x^n - \alpha = \prod_{\zeta_n} (x - \zeta_n \alpha^{1/n})$$

is not irreducible, but for each factor $(x - \zeta_n a^{1/n})$ it contains also all its conjugates. Putting $x = 0$ and $x = 1$ as before shows that $-\alpha$ and $1 - \alpha$ are norms. $\square$

We now let k be a local field, p the characteristic of its residue class field if k is not archimedean. $\mathfrak{p}$ is the prime of k. We assume that k contains the n-th roots of unity, $p \nmid n$.

Let $\sigma = (\alpha, k)$. On each abelian field K/k of finite degree, $\sigma = (\alpha, K/k)$. In particular, on $k(\beta^{1/n})$ we have

$$\sigma = (\alpha, k(\beta^{1/n})/k).$$

Define (α, β) to be $(\beta^{1/n})^{\sigma - 1}$. Then (α, β) is an n-th root of unity and does not depend on the n-th root $\beta^{1/n}$ selected. If k is complex then $(\alpha, \beta) = 1$ for all

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α, β. If k is the reals, then $n = 2$. The explicit values of (α, β) will be discussed in section 3.

The symbol (α, β) satisfies the following formal properties which are in most cases immediate consequences of the properties of the local symbol $\sigma = (\alpha, k)$.

Properties[1] **of** (α, β)**.**

1. $(\alpha, \beta) = 1$ if and only if α is a norm from $k(\beta^{1/n})$.
2. $(\alpha_1\alpha_2, \beta) = (\alpha_1, \beta)(\alpha_2, \beta)$.
3. $(\alpha, \beta_1\beta_2) = (\alpha, \beta_1)(\alpha, \beta_2)$

Proof. We have

$$(\beta_1^{1/n}\beta_2^{1/n})^{\sigma-1} = (\beta_1^{1/n})^{\sigma-1}(\beta_2^{1/n})^{\sigma-1}$$

because the norm residue symbol is consistent. □

4. $(-\alpha, \alpha) = (1 - \alpha, \alpha) = 1$.

Proof. Use Property 1 and Lemma 1. □

5. $(\alpha, \beta)(\beta, \alpha) = 1$ (inversion theorem).

Proof. We have by the preceding property:

$$\begin{aligned}(\alpha, \beta)(\beta, \alpha) &= (\alpha, \beta)(-\beta, \beta)(-\alpha, \alpha)(\beta, \alpha)\\ &= (-\alpha\beta, \beta)(-\alpha\beta, \alpha)\\ &= (-\alpha\beta, \alpha\beta) = 1.\end{aligned}$$ □

6. $(\alpha, \beta) = (\alpha, \alpha+\beta)(\alpha+\beta, \beta)(-1, \alpha+\beta)$ (provided, of course, that $\alpha+\beta \neq 0$).

Proof. Let $\gamma = \alpha + \beta$. Then

$$\begin{aligned}1 &= (1 - \alpha\gamma^{-1}, \alpha\gamma^{-1}) = (\beta\gamma^{-1}, \alpha\gamma^{-1})\\ &= (\beta, \alpha)(\beta, \gamma^{-1})(\gamma^{-1}, \alpha)(\gamma^{-1}, \gamma^{-1}).\end{aligned}$$

But $(\gamma^{-1}, \gamma^{-1}) = (-\gamma^{-1}, \gamma^{-1})(-1, \gamma^{-1}) = (-1, \gamma^{-1})$ by Property 4. The property now follows by transposing the suitable terms, and taking the exponent -1 outside the symbol. □

7. The symbol (α, β) is continuous in both arguments.

Proof. It is continuous in the first argument because if α is close to 1 then α is a norm, and Property 1 can be used. It is continuous in the second argument by the inversion theorem. □

[1] If F is a field and A an abelian group, a map $F^* \times F^* \to A$ with properties 2, 3, and 4 is called a Steinberg symbol and can be interperted as a homomorphism $K_2F \to A$.

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2. Local Analysis

Let k be complete under a discrete valuation with perfect residue class field $\bar{k}$, and integers $\mathfrak{o}$. We shall specialize k later to an ordinary $\mathfrak{p}$-adic field.

We recall some elementary facts.

A series $\sum_{\nu=0}^{\infty} a_\nu$ converges in k if and only if $\lim a_\nu = 0$.

A product Πa_ν converges in k if and only if $\lim a_\nu = 1$.

Let $\mathfrak{o}\{t\}$ be the ring of formal power series $f(t) = \sum_{\nu=0}^{\infty} a_\nu t^\nu$ with $a_\nu \in \mathfrak{o}$. The map

$$f(t) \to f(\pi)$$

is a homomorphism of $\mathfrak{o}\{t\}$ onto $\mathfrak{o}$. This is clear for the polynomial ring and extends to the power series by continuity.

Let F/k be a finite, completely ramified extension. Then $\bar{F} = \bar{k}$. Let Π be a prime in F, and $\varphi(t) = \operatorname{Irr}(\Pi, k, t)$. Every $\alpha \in \mathfrak{O}$ can be written $\alpha = \sum a_\nu \Pi^\nu$ where $a_\nu \in \mathfrak{o}$ because $\bar{F} = \bar{k}$. The map

$$f(t) \to f(\Pi)$$

where $f(t) \in \mathfrak{o}\{t\}$ is therefore onto $\mathfrak{O}$. We contend that its kernel is the principal ideal generated by $\varphi(t)$ in $\mathfrak{o}\{t\}$. Indeed, suppose

$$0 = f(\Pi) = a_0 + a_1\Pi + a_2\Pi^2 + \dots$$

with $a_\nu \in \mathfrak{o}$. Then $\Pi \mid a_0$ and hence $\pi \mid a_0$. Since $\varphi(t)$ is an Eisenstein polynomial with last coefficient divisible by π exactly, we have, for suitable d_0

$$f(t) - d_0\varphi(t) = b_1 t + b_2 t^2 + b_3 t^3 + \dots$$

and $\sum_{\nu=1}^{\infty} b_\nu t^\nu$ is in the kernel. Hence by the same argument as before, $\pi \mid b_1$ and we can find $d_1 \in \mathfrak{o}$ such that

$$f(t) - (d_0 + d_1 t)\varphi(t) = c_2 t^2 + c_3 t^3 + \dots.$$

Repeating this argument shows that $f(t) = g(t)\varphi(t)$ where $g(t) \in \mathfrak{o}\{t\}$, as contended.

Let $\alpha \in \mathfrak{O}$, $\alpha = f(\Pi) = \sum_{\nu=0}^{\infty} a_\nu \Pi^\nu$ where $f(t) \in \mathfrak{o}\{t\}$. We may take the derivative,

$$\alpha' = \sum \nu a_0 \Pi^{\nu-1}.$$

Then α' is not well defined in K. However, we know that any other expression for α as a power series is of type $\alpha = f(\Pi) + g(\Pi)\varphi(\Pi)$ where $g(t) \in \mathfrak{o}\{t\}$. Taking the derivative shows that

$$\alpha' = f'(\Pi) + g(\Pi)\varphi'(\Pi).$$

But $\varphi'(\Pi)$ is the different of F/k. Hence we have proved

LEMMA 2. *Let F/k be completely ramified. Let $\alpha \in \mathfrak{O}$ and write $\alpha = f(\Pi)$ where $f(t) \in \mathfrak{o}\{t\}$. Then the derivative $\alpha' = f'(\Pi)$ is well defined modulo the different of F/k.*

We now specialize the field k: Suppose that k is a completion at a finite prime of an algebraic number field. In other words, k is an ordinary $\mathfrak{p}$-adic field. We shall consider special series in k, analogous to the log and exponent. Let $\mathfrak{p} \mid p$. By ord we shall mean ord_p. If $p = \mathfrak{p}^e$ then $\operatorname{ord}_\mathfrak{p} = e \operatorname{ord}_p$.

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THEOREM 1. *The series*

$$\log(1+x) = x - x^2/2 + x^3/3 - \dots$$

converges for all $x \in k$ *such that* $\operatorname{ord} x > 0$. *If* $\operatorname{ord} x > 1/(p-1)$ *then* $\operatorname{ord} x^n/n > \operatorname{ord} x$ *for* $n \geqslant 2$, *and* $\operatorname{ord} x = \operatorname{ord} \log(1+x)$.

PROOF. Let $p^r \leqslant n < p^{r+1}$. We have $\operatorname{ord} x^n/n = n \operatorname{ord} x - \operatorname{ord} n \geqslant n \operatorname{ord} x - r$. An elementary limit computation shows that $\operatorname{ord} x^n/n \to \infty$ as $n \to \infty$, provided that $\operatorname{ord} x > 0$. This proves the convergence statement.

Suppose $\operatorname{ord} x > 1/(p-1)$. We must show that $\operatorname{ord} x^n/n > \operatorname{ord} x$ for $n \geqslant 2$, or equivalently, $\operatorname{ord} x^n/n - \operatorname{ord} x > 0$. But

$$\begin{aligned}\operatorname{ord} x^n/n - \operatorname{ord} x &= (n-1)\operatorname{ord} x - \operatorname{ord} n \\ &> (n-1)/(p-1) - r \quad (\text{because } n \geqslant 2) \\ &= ((n-1) - r(p-1))/(p-1) \\ &\geqslant 0.\end{aligned}$$

To justify this last step, note that it suffices to show $r(p-1) \leqslant n-1$. Since $p^r \leqslant n$ it suffices to show $r(p-1) \leqslant p^r - 1$. If $r = 0$ this is clear. Otherwise, dividing by $p-1$ shows that the inequality is equivalent to $r \leqslant 1 + p + \dots + p^{r-1}$, which is true because there are r terms on the right.

This concludes our proof because of the strict inequality in the second step. □

REMARK. If $\alpha \equiv 1 \pmod{\mathfrak{p}}$ we define $\log \alpha = \log(1 - (1-\alpha))$.

THEOREM 2. *Let* α *and* β *be* $\equiv 1 \pmod{\mathfrak{p}}$. *Then*

$$\log \alpha\beta = \log \alpha + \log \beta.$$

PROOF. The identity in formal series is known, and all the series involved converge by the preceding theorem. □

COROLLARY. *Let* ζ *be a* p^s *root of unity in* k. *Then* $\log \zeta$ *is defined and* $= 0$.

PROOF. We certainly have $1 - \zeta \equiv 0 \pmod{\mathfrak{p}}$ and hence $\log(1-(1-\zeta))$ converges. By the functional equation,

$$0 = \log 1 = \log \zeta^{p^s} = p^s \log \zeta.$$

Hence $\log \zeta = 0$. □

Although we shall not need the exponential function in the sequel, we give it here anyway for completeness.

THEOREM 3. *The series*

$$\exp x = 1 + x + x^2/2! + x^3/3! + \dots$$

converges for all x *such that ord* $x > 1/(p-1)$. *In that case,* $\operatorname{ord} x^n/n! > \operatorname{ord} x$ *for* $n \geqslant 2$, *and* $\operatorname{ord} x = \operatorname{ord}(\exp x - 1)$.

PROOF. Write n in the p-adic scale:

$$n = a_0 + a_1 p + \dots + a_r p^r$$

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where a_ν are rational integers, $0 \leqslant a_\nu \leqslant p-1$. Then

$$\begin{aligned}
[n/p] &= a_1 + a_2 p + \cdots + a_r p^{r-1} \\
[n/p^2] &= \qquad a_2 \quad + \cdots + a_r p^{r-2} \\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\
[n/p^r] &= \qquad\qquad\qquad\qquad a_r.
\end{aligned}$$

We clearly have

$$\begin{aligned}
\mu = \operatorname{ord} n! &= [n/p] + [n/p^2] + \cdots + [n/p^r] = \\
&= a_1 + (1+p)a_2 + \cdots + (1+p+\cdots+p^{r-1})a_r
\end{aligned}$$

and therefore

$$\begin{aligned}
(p-1)\mu &= (1-1)a_0 + (p-1)a_1 + (p^2-1)a_2 + \cdots + (p^r-1)a_r \\
&= n - s_n
\end{aligned}$$

where $s_n = a_0 + a_1 + \cdots + a_r$. This gives

$$\mu = (n - s_n)/(p-1).$$

Now

$$\begin{aligned}
\operatorname{ord} x^n/n! &= n \operatorname{ord} x - \mu \\
&= n\left(\operatorname{ord} x - \frac{1}{p-1}\right) + s_n/(p-1).
\end{aligned}$$

Since s_n is positive we see that $\operatorname{ord} x^n/n! \to \infty$ when $\operatorname{ord} x > 1/(p-1)$, and the series converges.

To get the second statement we have to show that $\operatorname{ord} x^n/n! - \operatorname{ord} x > 0$. We have

$$\begin{aligned}
\operatorname{ord} x^n/n! - \operatorname{ord} x &= (n-1)\operatorname{ord} x - (n - s_n)/(p-1) \\
&= (n-1)\Big(\operatorname{ord} x - \frac{1}{p-1}\Big) + (s_n - 1)/(p-1).
\end{aligned}$$

If $n \geqslant 2$ the result is obvious because $s_n \geqslant 1$ always. □

THEOREM 4. *If* $\operatorname{ord} x$ *and* $\operatorname{ord} y > 1/(p-1)$ *then*

$$\exp(x+y) = (\exp x)(\exp y).$$

PROOF. The formal identity is known, and all the series converge by the previous theorem. □

THEOREM 5. *If* $\operatorname{ord} x > 1/(p-1)$ *then*

$$\begin{aligned}
\exp\log(1+x) &= 1 + x \\
\log\exp x &= x.
\end{aligned}$$

PROOF. The formal identities are known, and all the series converge by the previous theorems. □

Let $\delta = [e/(p-1)]+1$. The log and exponent give mutually inverse isomorphisms of $1 + \mathfrak{p}^\delta$ onto $\mathfrak{p}^\delta$.

We leave it as an exercise to the reader to prove the converse of the Corollary to Theorem 2, i.e. the kernel of the map

$$\alpha \to \log \alpha$$

for $\alpha \equiv 1 \pmod{\mathfrak{p}}$ is exactly the p^ν-th roots of unity contained in the field k.

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3. Computation of the Norm Residue Symbol in Certain Local Kummer Fields

We treat the simplest cases of Kummer fields, and begin with the infinite prime.

Let $\mathbb{R}$ be the reals, a, $b \in \mathbb{R}$. We may interpret the results of Section 1 with $n = 2$ over the reals. We obtain

$$(a, b)_{p_\infty} = (-1)^{\frac{\operatorname{sign} a - 1}{2} \frac{\operatorname{sign} b - 1}{2}}.$$

Next we consider finite primes. Let ℓ be a prime and $k = \mathbb{Q}_\ell(\zeta_\ell)$ where $\mathbb{Q}_\ell$ is the ℓ-adic completion of the rationals and ζ_ℓ is a primitive ℓ-th root of unity. We shall determine (α, β) in the field $k(\beta^{1/\ell})$ explicitly.

Since the symbol is continuous and multiplicative in both arguments it will suffice to determine it for a multiplicative basis of k.

We first consider $\ell = 2$, $k = \mathbb{Q}_2$. A multiplicative basis[2] of $\mathbb{Q}_2$ is given by 2, -1, and 5.

THEOREM 6. *Let a, $b \in U_2$ be units of $\mathbb{Q}_2$. Then*

$$(2, 2) = 1$$
$$(2, a) = (-1)^{(a^2-1)/8}$$
$$(a, b) = (b, a) = (-1)^{\varepsilon(a)\varepsilon(b)}$$

where $\varepsilon(a) \equiv (a-1)/2 \pmod 2$.

PROOF. We see that $(a, b) = (b, a)$ from the inversion theorem and the fact that $(a, b) = \pm 1$.

We prove that the exponent $\varepsilon(a)$ is multiplicative. We have

$$a_1 a_2 - a_1 - a_2 - 1 = (a_1 - 1)(a_2 - 1) \equiv 0 \pmod 4$$

and therefore

$$\frac{a_1 a_2 - 1}{2} \equiv \frac{a_1 - 1}{2} + \frac{a_2 - 1}{2} \pmod 2.$$

Thus

$$\varepsilon(a_1 a_2) = \varepsilon(a_1) + \varepsilon(a_2) \pmod 2.$$

For the other exponent we have

$$(a_1 a_2)^2 - a_1^2 - a_1^2 - 1 = (a_1^2 - 1)(a_2^2 - 1) \equiv 0 \pmod{64}$$

and therefore

$$\frac{(a_1 a_2)^2 - 1}{8} \equiv \frac{a_1^2 - 1}{8} + \frac{a_2^2 - 1}{8} \pmod 2$$

as was to be shown.

Both sides of our equations are multiplicative. It suffices to verify the statements for a basis of k.

We note that:

$\mathbb{Q}_2(i)$ is ramified (because $(1+i)^2 = 2i$).

$\mathbb{Q}_2(\sqrt{5})$ is unramified. Indeed, let $\alpha = (\sqrt{5} - 1)/2$.

Then α generates $\mathbb{Q}_2(\sqrt{5})$ and satisfies the equation $\alpha^2 + \alpha - 1 = 0$. Read mod 2, it is the canonical equation for an extension of the residue class field. Hence $\mathbb{Q}_2(\alpha) = \mathbb{Q}_2(\sqrt{5})$ is unramified.

Now:

[2]Here and in the next four pages, "multiplicative basis for" means "a set of generators for a dense subgroup of".

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$(2,2) = (-1,2) = 1$ because $-1 = N(1+\sqrt{2})$ is a norm.
$(2,5) = -1$ because the prime 2 is not a norm from the unramified extension $\mathbb{Q}_2(\sqrt{5})$.
$(-1,5) = 1$ because the unit -1 is norm from $\mathbb{Q}_2(\sqrt{5})$.
$(5,5) = 1$ for the same reason.
$(-1,-1) = -1$ because -1 is not a norm from $\mathbb{Q}_2(\sqrt{-1})$. (Otherwise, $-1 = x^2 + y^2$ which is impossible mod 4.)

One verifies directly that the values we have just found coincide with $(-1)^{\varepsilon(a)\varepsilon(b)}$ in each case, and this proves our theorem. □

We suppose from now on that ℓ is an odd prime and that $k = \mathbb{Q}_\ell(\zeta)$ where ζ is a primitive ℓ-th root of unity.

Then $[k : \mathbb{Q}_\ell] = \ell - 1$ and k is completely ramified over $\mathbb{Q}_\ell$. Let $\lambda = 1 - \zeta$. Then λ is a prime in k, and $\ell \sim \lambda^{\ell-1}$. It is actually easy to determine $\ell/\lambda^{\ell-1} \pmod{\lambda}$. Namely: $x^{\ell-1} + \cdots + 1 = \prod_{\nu=1}^{\ell-1}(x - \zeta^\nu)$ and therefore $\ell = \prod_{\nu=1}^{\ell-1}(1 - \zeta^\nu)$ We get

$$\ell/\lambda^{\ell-1} = \frac{\Pi(1-\zeta^\nu)}{\Pi(1-\zeta)} = \prod_{\nu=1}^{\ell-1}(1 + \zeta + \cdots + \zeta^{\nu-1}).$$

But $\zeta \equiv 1 \pmod{\lambda}$. By Wilson's theorem, we see that

$$\frac{\ell}{\lambda^{\ell-1}} \equiv (\ell-1)! \equiv -1 \pmod{\lambda}.$$

We shall abbreviate the modulus, and write (λ) for $\pmod{\lambda}$.

Let $\eta_i = 1 - \lambda^i$, $i \geqslant 1$.

LEMMA 3. *The η_i form a multiplicative basis of the units of $k \equiv 1(\lambda)$. A full multiplicative basis is given by the η_i, the $(\ell-1)$-th roots of unity, and the powers of a prime.*

PROOF. The residue class field of k is $\mathbb{Z}_\ell = \overline{\mathbb{Q}}_\ell$ because $k/\mathbb{Q}_\ell$ is completely ramified. The $(\ell-1)$ roots of unity lie in $\mathbb{Q}_\ell$ and obviously any unit can be multiplied by such a root of unity to make if $\equiv 1(\lambda)$.

Every unit $\varepsilon \equiv 1(\lambda)$ can be written $\varepsilon = 1 + a_1\lambda + a_2\lambda^2 + \dots$ where a_ν are rational integers. We can obviously solve formally

$$(1 + a_1 t + a_2 t + \dots) = (1-t)^{-a_1}(1-t^2)^{-b_2}(1-t^3)^{-c_3}\dots$$

for integers b_2, c_3, by a recursive process. The power series obtained by putting $t = \lambda$ will converge, and this proves our lemma. □

We are now interested in computing the symbols (η_i, λ). We note that

$$(\eta_i, \lambda^i) = (1 - \lambda^i, \lambda^i) = (1 - \lambda^i, \lambda)^i = 1$$

by Properties 1 and 4 of the symbol. This shows that if $\ell \nmid i$ then $(\eta_i, \lambda) = 1$ because (η_i, λ) is an ℓ-th root of unity.

LEMMA 4. *Let ε be a unit of k, $\varepsilon \equiv x^\ell(\lambda^{\ell+1})$ for some $x \in k$. Then ε is an ℓ-th power in k.*

PROOF. Suppose $\varepsilon \equiv x^\ell(\lambda^\nu)$ with $\nu \geqslant \ell + 1$. We try to refine x. We contend that

$$(x + y\lambda^\nu/\ell)^\ell \equiv x^\ell + y\lambda^\nu x^{\ell-1}(\lambda^{\nu+1}).$$

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Indeed, the remaining terms of the binomial expansion are of type

$$\binom{\ell}{\mu} x^{\ell-\mu} y^{\mu} \left(\frac{\lambda^r}{\ell}\right)^{\mu}.$$

Since $\ell \mid \binom{\ell}{\nu}$ the intermediate terms are divisible by $\ell\left(\frac{\lambda^r}{\ell}\right) \sim \lambda^{2\nu-\ell+1}$ and hence are $\equiv 0(\lambda^{\nu+1})$. Furthermore $\lambda^{\nu\ell}/\ell = \lambda^{\nu}\left(\lambda^{\nu(\ell-1)-\ell(\ell-1)}\right)$. Also, $\nu-\ell \geqslant 1$, and $\ell-1 \geqslant 2$. Hence the last term is also $\equiv 0(\lambda^{\nu+1})$. This proves that we can solve for y such that

$$x^{\ell} + yx^{\ell-1}\lambda^{\nu} \equiv \varepsilon(\lambda^{\nu+1})$$

and by a standard refinement process, we can find an ℓ-th root for ε, as contended. □

THEOREM 7. *For all $i \neq \ell$ we have $(\eta_i, \lambda) = 1$.*

PROOF. If $i < \ell$ then $\ell \nmid i$ and we have already considered this case. If $i \geqslant \ell+1$ then $\eta_i \equiv 1(\lambda^{\ell+1})$ and $\eta_i = \alpha^{\ell}$ by the lemma. Hence $(\eta_i, \lambda) = (\alpha, \lambda)^{\ell} = 1$ thereby proving the theorem. □

It will be much harder to prove the following results.

THEOREM 8. *$(\eta_\ell, \lambda) = \zeta^{-1}$, or equivalently, $(\lambda, \eta_\ell) = \zeta$.*

PROOF. The inversion follows from the inversion property for the symbol. We shall deal with (λ, η_ℓ), because we enjoy the advantage of the following statement:

$k(\eta_\ell^{1/\ell})/k$ is unramified.

PROOF. Let $A = \eta_\ell^{1/\ell}$. Then $A^{\ell} - 1 + \lambda^{\ell} = 0$. Let $B = (A-1)/\lambda$. Then $A = \lambda B + 1$ and therefore B satisfies the equation

$$\begin{aligned} 0 &= (\lambda B + 1)^{\ell} - 1 + \lambda^{\ell} \\ &= \lambda^{\ell} B^{\ell} + \binom{\ell}{1} \lambda^{\ell-1} B^{\ell-1} + \cdots + \ell\lambda B + \lambda^{\ell}. \end{aligned}$$

But $\ell \sim \lambda^{\ell-1}$. Divide the equation by λ^{ℓ}. Since $\ell \mid \binom{\ell}{r}$ we see that B satisfies

$$x^{\ell} + (\ell\lambda/\lambda^{\ell})x + 1 \equiv 0 \quad (\lambda).$$

We have seen at the beginning of our discussion that $(\ell\lambda/\lambda^{\ell}) \equiv -1\ (\lambda)$ and our equation becomes therefore

$$x^{\ell} - x + 1 \equiv 0 \quad (\lambda).$$

Since $k/\mathbb{Q}_\ell$ is totally ramified, the residue class fields $\bar{k}$ and $\overline{\mathbb{Q}}_\ell$ are equal and are simply the prime field $\mathbb{Z}_\ell$. Hence the equation above is irreducible in the residue class field and this proves that $k(A)/k$ is unramified. □

We conclude that $\sigma = (\lambda, k(A)/k) = \varphi$ is the Frobenius Substitution. We must therefore prove that $A^{\varphi-1} = \zeta$.

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We know that $B^\varphi \equiv B^\ell \ (\lambda)$, and therefore $B^\varphi \equiv B - 1 \ (\lambda)$. To get the effect on A we have

$$\begin{aligned} A^{\varphi-1} &= (1+\lambda B)^\varphi/(1+\lambda B) \\ &= (1+\lambda B^\varphi)(1-\lambda B+\lambda^2 B^2 - \dots) \\ &\equiv 1+\lambda(B^\varphi - B) \quad (\lambda^2) \\ &\equiv 1-\lambda \quad (\lambda^2) \\ &\equiv \zeta \quad (\lambda^2). \end{aligned}$$

But if $\zeta^\nu \equiv \zeta^\mu \ (\lambda^2)$ then $\zeta^\nu(1-\zeta^{\mu-\nu}) \equiv 0 \ (\lambda^2)$. Since $(1-\zeta^{\mu-\nu}) \sim \lambda$ unless $\nu = \mu \pmod{\ell}$ we conclude that $A^{\varphi-1} = \zeta$ thereby proving our theorem. □

Our next problem is to determine (η_i, η_j) in terms of (η_i, λ).

We begin by deriving a certain functional equation for (η_i, η_j). We see immediately from the definition that

$$\eta_j + \lambda^j \eta_i = \eta_{i+j}.$$

Note that $(-1)^\ell = -1$ because ℓ is an odd prime, and hence $(-1, \beta) = 1$ for all β. From the addition theorem we get

$$(\eta_j, \lambda^j \eta_i) = (\eta_j, \eta_{i+j})(\eta_{i+j}, \lambda^j \eta_i).$$

We use the multiplicativity of the symbol and

$$(\eta_j, \lambda^j) = (1-\lambda^j, \lambda^j) = 1 \tag{1}$$

to get

$$(\eta_j, \eta_i) = (\eta_j, \eta_{i+j})(\eta_{i+j}, \eta_i)(\eta_{i+j}, \lambda)^j$$

and invert everything:

$$(\eta_i, \eta_j) = (\eta_i, \eta_{i+j})(\eta_{i+j}, \eta_j)(\eta_{i+j}, \lambda)^{-j}. \tag{2}$$

Proposition.

$$(\eta_i, \eta_j) = \prod_{\substack{r,s \geqslant 1 \\ (r,s)=1}} (\eta_{ri+sj}, \lambda)^{-(r_0 i + s_0 j)}$$

where r, s are positive integers, relatively prime, and for each pair (r, s) the pair (r_0, s_0) is one solution of the equation $rs_0 - sr_0 = 1$.

Proof. We must first show that the expression is well defined, i.e. does not depend on the choice of (r_0, s_0). Indeed, it is easy to verify that any other solution (r_1, s_1) is given by $r_1 = r_0 + rt$ and $s_1 = s_0 + st$ where t is an integral parameter. This implies that the exponent changes by $(ri + sj)t$. Let $\mu = ri + sj$. From (1) we know that $(\eta_\mu, \lambda)^\mu = 1$ and therefore our expression is well defined.

Let $[\eta_i, \eta_j]$ be the expression. We prove that $[\eta_i, \eta_j]$ satisfies the functional equation (2) as a function of i, j. We consider different values of r, s.

$r = s$. Then $r = s = 1$ because $(r, s) = 1$. We may choose $r_0 = 0$ and $s_0 = 1$. The term gives the contribution $(\eta_{i+j}, \lambda)^{-j}$.

$r > s$. Let $s = s'$ and $r = r' + s'$. This gives a 1–1 correspondence between (r', s') and (r, s). The product over such terms gives the contribution

$$\prod_{r',s'} (\eta_{r'i+s'(i+j)}, \lambda)^{-((r_0-s_0)i+s_0(i+j))}$$

where $r', s' \geqslant 1$ and $(r', s') = 1$. This is precisely $[\eta_i, \eta_{i+j}]$.

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$r < s$. Let $r = r'$ and $s = r' + s'$. Again we have a 1–1 correspondence between (r', s') and (r, s). The product over such terms gives the contribution

$$\prod_{r',s'} (\eta_{r'(i+j)s'j}, \lambda)^{-(r_0(i+j)-(s_0-r_0))}$$

where $r', s' \geqslant 1$ and $(r', s') = 1$. This is precisely $[\eta_{i+j}, \eta_i]$.

We have therefore shown that $[\eta_i, \eta_j]$ satisfies (2). Taking the quotient we obtain

$$\frac{(\eta_i, \eta_j)}{[\eta_k, \eta_j]} = \frac{(\eta_i, \eta_{i+j})(\eta_{i+j}, \eta_j)}{[\eta_i, \eta_{i+j}][\eta_{i+j}, \eta_j]}. \tag{3}$$

For large ν and μ we know that η_ν and η_μ are ℓ-th powers and therefore

$$(\eta_i, \eta_\nu) = [\eta_i, \eta_\nu] = 1$$
$$(\eta_\mu, \eta_j) = [\eta_\mu, \eta_j] = 1.$$

Using (3) recursively to increase the indices we see that the quotient is 1, thereby proving the proposition. □

We have proved in Theorem 7 that $(\eta_\mu, \lambda) = 1$ unless $\mu = \ell$. Hence in the product only terms with $ri + sj = \ell$ will give a contribution. From Theorem 8 we get

$$(\eta_i, \eta_j) = \prod_{(r,s)} \zeta^{r_0 i + s_0 j}$$

where the product is taken for relatively prime positive integers r, s such that $ri + sj = \ell$. If no such integers exist, $(\eta_i, \eta_j) = 1$.

If $ri + sj = \ell$ then r and $s < \ell$ and

$$r_0 i + s_0 j = \frac{r_0}{r}(ri + sj) + \frac{j}{r} \equiv j/r \pmod{\ell}$$

where $1/r$ is the integer inverse to r mod ℓ. We have proved

THEOREM 9. *$(\eta_i, \eta_j) = \prod_{(r,s)} \zeta^{j/r}$ where the product is taken over all integers $r, s \geqslant 1$ such that $ri + sj = \ell$. j/r is interpreted* $\bmod \ell$, *and $(\eta_i, \eta_j) = 1$ if there exist no (r, s) such that $ri + sj = \ell$.*

Having found (η_i, η_j) we may now get (α, β) for all $\alpha, \beta \in k$. By factoring out powers of λ, we see that it suffices to compute the symbol for units, and λ separately.

REDUCTION 1. *Let α, β be units. Then*

$$\alpha^{\ell-1} \equiv \beta^{\ell-1} \equiv 1 \quad (\lambda)$$

and $(\alpha^{\ell-1}, \beta^{\ell-1}) = (\alpha, \beta)$. This reduces our problem to units $\equiv 1$ (λ).

REDUCTION 2. *If $\alpha \equiv 1$ (λ) then we can write $\alpha \equiv 1 + \mu\lambda$ (λ^2) where μ is a rational integer (because $\bar{k} = \overline{\mathbb{Q}}_\ell = \mathbb{Z}_\ell$). Since $\zeta^\mu = (1 - \lambda)^\mu \equiv 1 - \mu\lambda$ (λ^2) we obtain $\alpha\zeta^\mu \equiv 1$ (λ^2). This reduces the computation of (α, β) to the case where $\alpha \equiv 1$ (λ^2), and to (ζ, β) where $\beta \equiv 1$ (λ).*

REDUCTION 3. *We have $(\zeta, \zeta) = (-\zeta, \zeta)(-1, \zeta) = 1$. This reduces the computation of (ζ, β) to the case where $\beta \equiv 1$ (λ^2).*

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Before computing the symbol for the above cases, we make a definition. Let β be a unit of k, $\beta = \sum_{\nu=0}^{\infty} a_\nu \lambda^\nu$ where a_ν are rational integers. Define $D\log\beta$ to be β'/β where β' is the derivative of the power series, as discussed in section 2. Then $D\log\beta$ is well defined mod $\lambda^{\ell-2}$ according to Lemma 2. Indeed, the different of $k/\mathbb{Q}_\ell$ is $\lambda^{\ell-2}$ because $k/\mathbb{Q}_\ell$ is tamely ramified. We have obviously

$$D\log(\alpha\beta) \equiv D\log\beta + D\log\alpha \quad (\lambda^{\ell-2}).$$

Note that $\log\beta$ may not be defined since β is not necessarily $\equiv 1\ (\lambda)$. Even in this case, $\log\beta$ may not be a unit, so $D\log\beta$ is not necessarily $(\log\beta)'$.

Remark. The good radius of convergence for the log in k is precisely λ^2. Indeed, if $\operatorname{ord}_\lambda x \geqslant 2$, or equivalently, if $\operatorname{ord}_\lambda x > 1$ then $\operatorname{ord} x > 1/(\ell-1)$ which is the number of Theorem 1.

We let S denote the trace from k to $\mathbb{Q}_\ell$.

Theorem 10. *(α,β) is determined according to the following scheme.*
1. *$\alpha \equiv 1\ (\lambda^2)$ and $\beta \equiv 1\ (\lambda)$. Then*

$$(\alpha,\beta) = \zeta^{\frac{1}{\ell}S(\zeta\log\alpha D\log\beta)}.$$

2. *$\alpha \equiv 1\ (\lambda)$. Then*

$$(\zeta,\alpha) = \zeta^{\frac{1}{\ell}S(\log\alpha)}.$$

3. *$\alpha \equiv 1\ (\lambda)$. Then*

$$(\alpha,\lambda) = \zeta^{\frac{1}{\ell}S(\frac{\zeta}{\lambda}\log\alpha)}.$$

The previous statements hold in the sense that the exponents involving $\log\alpha$ and $D\log\beta$ are well defined mod ℓ and hence ζ can be raised to such exponents.

Proof. We begin with the first formula and must prove that the exponent is well defined mod ℓ. From $\alpha \equiv 1\ (\lambda^2)$ we get $\log\alpha \equiv 0\ (\lambda^2)$ by Theorem 1. We know that $D\log\beta$ is well defined mod $\lambda^{\ell-2}$. Hence $\log\alpha D\log\beta$ is well defined mod $\lambda^2\lambda^{\ell-2} = \lambda^\ell = \ell\lambda\varepsilon$ where ε is a unit. We see that

$$\frac{1}{\ell}S(\ell\lambda\gamma) = S(\lambda\gamma) = 0 \pmod{\ell}$$

where γ is an integer. This proves that the exponent is well defined.

From the functional equation of log and $D\log$ we see that the exponent is multiplicative in α and β, and that it is continuous in both arguments. It suffices therefore to verify our theorem for a multiplicative basis, i.e. for (η_i,η_j) according to Lemma 3. In fact, since $\alpha \equiv 1\ (\lambda^2)$ we may assume $i \geqslant 2$.

Considering Theorem 9 we must prove that

$$\sum_{\substack{r,s\geqslant 1\\ ri+sj=\ell}} j/r \equiv \frac{1}{\ell}S(\zeta\log\eta_i D\log\eta_j) \pmod{\ell}$$

We have

$$\log\eta_i = \log(1-\lambda^i) = -\sum_{r=1}^{\infty}\lambda^{ri}/r$$

$$D\log\eta_j = \frac{-j\lambda^{j-1}}{1-\lambda^j} = -\sum_{s=1}^{\infty} j\lambda^{sj-1}.$$

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Multiplying the two series:

$$\frac{1}{\ell}S\Big(\zeta\sum_{r,s\geqslant 1}\frac{j}{r}\lambda^{ri+sj-1}\Big)=\sum_{r,s\geqslant 1}\frac{j}{r}\cdot\frac{1}{\ell}\cdot S(\zeta\lambda^{ri+sj-1}).$$

Let $m=ri+sj$. Let $\ell^\nu\mid r$ but $\ell^{\nu+1}\nmid r$. We shall prove that

$$\frac{1}{\ell^{r+1}}S(\zeta\lambda^{m-1})\equiv\begin{cases}1 & (\ell) \quad \text{if } m=\ell,\ \nu=0\\ 0 & (\ell) \quad \text{otherwise.}\end{cases}$$

This will establish the fact that our two exponents are congruent mod ℓ.

We have $\ell^\nu\leqslant r$ and $2\ell^\nu\leqslant m$ because $i\geqslant 2$. Also, $2r\leqslant m=ri+sj$.

Case 1. $2\leqslant m\leqslant\ell$ and $\nu=0$. We expand λ^{m-1} and take ζ into the sum to get

$$\begin{aligned}\frac{1}{\ell}S(\zeta\lambda^{m-1})&=\frac{1}{\ell}S\left(\sum_{\mu=0}^{m-1}\binom{m-1}{\mu}(-1)^\mu\zeta^{\mu+1}\right)\\&=\frac{1}{\ell}\sum_{\mu=0}^{m-1}\binom{m-1}{\mu}(-1)^\mu S(\zeta^{\mu+1}).\end{aligned}$$

But $m\leqslant\ell$ implies $\mu+1\leqslant\ell$. We have

$$S(\zeta^{\mu+1})=\begin{cases}-1 & \text{if } \mu+1<\ell\\ \ell-1 & \text{if } \mu+1=\ell.\end{cases}$$

Indeed, if $\mu+1<\ell$ then $\zeta^{\mu+1}\neq 1$ is a primitive ℓ-th root of unity satisfying $x^{\ell-1}+x^{\ell-2}+\cdots+1=0$ over $\mathbb{Q}_\ell$. Hence $S(\zeta^{\mu+1})=-(\operatorname{coeff}x^{\ell-2})=-1$. If $\mu+1=\ell$ then $\zeta^\ell=1$ and $S(1)=\ell-1$.

We note that

$$0=(1-1)^{m-1}=\sum_{\mu=0}^{m-1}\binom{m-1}{\mu}(-1)^\mu.$$

Hence the sum is 0 except when $m=\ell$. In that case we obtain

$$\begin{aligned}\frac{1}{\ell}S(\zeta\lambda^{m-1})&=-\frac{1}{\ell}\sum_{\mu=0}^{m-2}\binom{m-1}{\mu}(-1)^\mu+\frac{1}{\ell}\binom{m-1}{m-1}(-1)^{m-1}(\ell-1)\\&=-\frac{1}{\ell}\sum_{\mu=0}^{m-1}\binom{m-1}{\mu}(-1)^\mu+\frac{\ell}{\ell}\\&=1.\end{aligned}$$

This concludes our proof.

Case 2. $m>\ell$.

We show that our expression in (4) is $\equiv 0\ (\ell)$. Taking $\ell^{-(\nu+1)}$ inside the trace gives

$$\frac{1}{\ell^{\nu+1}}S(\zeta\lambda^{m-1})=S\big(\zeta\varepsilon\lambda^{m-1-(\nu+1)(\ell-1)}\big)$$

where ε is a unit. It suffices to prove therefore that $m-1-(\nu+1)(\ell-1)>0$. This is a hypothesis for $\nu=0$. We may therefore assume $\nu\geqslant 1$. We have trivially

$$m-1-(\nu+1)(\ell-1)\geqslant 2\ell^\nu-1-(\nu+1)(\ell-1)=f(\nu)$$

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and we shall prove that $f(\nu) > 0$. We have

$$\begin{aligned} f(\nu+1) - f(\nu) &= 2\ell^{\nu+1} - 1 - (\nu+2)(\ell-1) - (2\ell^{\nu} - 1 - (\nu+1)(\ell-1)) \\ &= 2\ell^{\nu}(\ell-1) - (\ell-1) \\ &= (2\ell^{\nu} - 1) - (\ell - 1) \\ &> 0 \end{aligned}$$

for $\nu > 1$. But $f(1) = 2\ell - 1 - 2(\ell-1) = 1$. Hence $f(\nu) > 0$ for all $\nu \geqslant 1$, as was to be shown.

This concludes the proof for the first formula of the theorem.

We shall obtain the second as an immediate corollary. We first consider the case $\alpha \equiv 1 \ (\lambda^2)$, and must show

$$(\zeta, \alpha) = \zeta^{\frac{1}{\ell} S(\log \alpha)}.$$

Since $\zeta \equiv 1 \ (\lambda)$ we can use the preceding result on (α, ζ). We have $\zeta = 1 - \lambda$, $\zeta' = -1$ and $D \log \zeta = \zeta'/\zeta = -1/\zeta$. This gives

$$\log \alpha D \log \zeta = -\log \alpha$$

and by the preceding result,

$$(\alpha, \zeta) = \zeta^{-\frac{1}{\ell} S(\log \alpha)}.$$

The inversion property now gives

$$(\zeta, \alpha) = (\alpha, \zeta)^{-1} = \zeta^{\frac{1}{\ell} S(\log \alpha)}$$

as desired.

We have remarked in Reduction 2 that if $\alpha \equiv 1 \ (\lambda)$ then $\alpha\zeta^{\mu} = 1 \ (\lambda^2)$ for a suitable power of ζ. We know that $(\zeta, \alpha) = (1 - \lambda, \lambda) = 1$, and we also know that $\log \zeta = 0$ by the Corollary of Theorem 2. Hence

$$(\alpha, \zeta) = (\alpha\zeta^{\mu}, \zeta) \quad \text{and} \quad \log(\alpha\zeta) = \log \alpha.$$

This proves that we may assume $\alpha \equiv 1 \ (\lambda^2)$, and that the formula is valid for $\alpha \equiv 1$ (λ).

We now come to the third formula involving the prime λ. The same argument concerning α that has just been made shows that we may assume $\alpha \equiv 1 \ (\lambda^2)$. We must prove

$$(\alpha, \lambda) = \zeta^{\frac{1}{\ell} S\left(\frac{\zeta}{\lambda} \log \alpha\right)}.$$

The exponent is defined because $\log \alpha \equiv 0 \ (\lambda^2)$ according to Theorem 1 and hence $\ell \mid S\left(\frac{\zeta}{\lambda} \log \alpha\right)$.

It suffices to verify the formula for the multiplicative basis η_i $(i \geqslant 2)$. It has been proved in Theorems 7 and 8 that

$$(\eta_i, \lambda) = \begin{cases} 1 & i \neq \ell \\ \zeta^{-1} & i = \ell. \end{cases} \tag{5}$$

Our exponent for ζ gives

$$\frac{1}{\ell} S\left(\frac{\zeta}{\lambda} \log(1 - \lambda^i)\right) = -\sum_{r=1}^{\infty} \frac{1}{r\ell} S(\zeta\lambda^{ir-1}).$$

We refer to the computation carried out previously, i.e. to the result (4), with $m = ri$. The only term giving a non-zero contribution will be the term with $ri = \ell$

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and $\nu = 0$. Since ℓ is a prime and $i \geqslant 2$, this combination can occur only if $r = 1$ and $i = \ell$. In that case, we get $1 \pmod{\ell}$ from the sum, and the minus sign in front. These values are seen to coincide with those of (5), and this concludes the proof of our theorem. $\square$

4. The Power Reciprocity Law

We make first some remarks on finite fields. Let F be a finite field with q elements. Then $x^{q-1} = 1$ for all $x \in F$, $x \neq 0$. Furthermore, $m \mid (q-1)$ if and only if all m-th roots of unity lie in F. Suppose this is the case. We have

$$x^{q-1} - 1 = \prod_{\zeta} (x^{(q-1)/m} - \zeta)$$

where ζ are the m-th roots of unity. The mapping

$$x \to x^{(q-1)/m}$$

has $(q-1)/m$ elements in its kernel and $x^{(q-1)/m} = 1$ if and only if $x = y^m$ for some y.

We let k be a global field, $\mathfrak{p}$ a finite prime, $q = N\mathfrak{p}$. $\mathfrak{p} \nmid m$, and we suppose that a primitive m-th root of unity lies in k. From this it follows that $m \mid (N\mathfrak{p} - 1)$ and that the preceding remarks on finite fields can be applied to the residue class field of $\mathfrak{p}$.

We recall the elementary fact that $\zeta^\nu \equiv \zeta^\mu \pmod{\mathfrak{p}}$ if and only if $\nu \equiv \mu \pmod{m}$.

Suppose $\mathfrak{p} \nmid \alpha$. We let $\left(\frac{\alpha}{\mathfrak{p}}\right)$ be the m-th root of unity defined by

$$\left(\frac{\alpha}{\mathfrak{p}}\right) \equiv \alpha^{(N_{\mathfrak{p}}-1)/m} \pmod{\mathfrak{p}}.$$

The existence and uniqueness of the root of unity follows from the preceding remarks. It follows immediately that

$$\left(\frac{\alpha_1\alpha_2}{\mathfrak{p}}\right) = \left(\frac{\alpha_1}{\mathfrak{p}}\right)\left(\frac{\alpha_2}{\mathfrak{p}}\right).$$

THEOREM 11. $\left(\frac{\alpha}{\mathfrak{p}}\right) = 1$ *if and only if* $\alpha = \beta^m \pmod{\mathfrak{p}}$ *for some* β.

PROOF. Immediate from the definitions and the remarks on finite fields. $\square$

Let $A = \alpha^{1/m}$ be one of the m-th roots of α. From Kummer theory we know that $k(A)/k$ is unramified at p. Let $\varphi_{\mathfrak{p}}$ be the Frobenius Substitution at $\mathfrak{p}$. Then

$$\begin{aligned} A^{\varphi_{\mathfrak{p}}} &\equiv A^{N_{\mathfrak{p}}} && \pmod{\mathfrak{p}} \\ A^{\varphi_{\mathfrak{p}}-1} &\equiv A^{N_{\mathfrak{p}}-1} && \pmod{\mathfrak{p}} \\ &\equiv \alpha^{(N_{\mathfrak{p}}-1)/m} && \pmod{\mathfrak{p}}. \end{aligned}$$

The congruences are written mod $\mathfrak{p}$ because they hold for each $\mathfrak{P} \mid \mathfrak{p}$.

For β satisfying the conditions

1. β is prime to α,
2. β is prime to m,

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define[3]

$$\left(\frac{\alpha}{\beta}\right) = \prod_{\mathfrak{p}\neq\mathfrak{p}_\infty} \left(\frac{\alpha}{\mathfrak{p}}\right)^{\operatorname{ord}_{\mathfrak{p}}\beta}.$$

From this definition we have immediately

$$\left(\frac{\alpha}{\beta_1\beta_2}\right) = \left(\frac{\alpha}{\beta_1}\right)\left(\frac{\alpha}{\beta_2}\right).$$

We wish to investigate the number $\left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{\alpha}\right)^{-1}$. We let $(\beta,\alpha)_{\mathfrak{p}}$ be the local symbol (β,α) discussed at the beginning of this chapter. (Note that $(\beta,\alpha)_{\mathfrak{p}}$ is defined for $\mathfrak{p}=\mathfrak{p}_\infty$.)

Before proving the theorem which gives the connection between our various symbols, we make a useful remark: Let σ, τ be two automorphisms over k. Then $A^{\sigma-1} = \zeta_\sigma$ and $A^{\tau-1} = \zeta_\tau$ where ζ_σ and ζ_τ are two roots of unity. We have $A^{\sigma\tau} = (\zeta_\tau A)^\sigma = \zeta_\sigma\zeta_\tau A$. This proves that

$$A^{\sigma-1}A^{\tau-1} = A^{\sigma\tau-1}.$$

This remark will be used in the course of our proofs.

THEOREM 12. *If* $\mathfrak{p}\nmid mp_\infty$ *then*[4]

$$\left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{\alpha}\right)^{-1}_{\mathfrak{p}} = (\beta,\alpha)_{\mathfrak{p}}.$$

PROOF. If $\mathfrak{p}\nmid\alpha$ then

$$\left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{\alpha}\right)^{-1}_{\mathfrak{p}} = \left(\frac{\alpha}{\beta}\right)_{\mathfrak{p}} = A^{\operatorname{ord}_{\mathfrak{p}}\beta(\sigma_{\mathfrak{p}}-1)}.$$

But $k_{\mathfrak{p}}(A)/k_{\mathfrak{p}}$ is unramified and therefore $\sigma_{\mathfrak{p}} = (\beta, k_{\mathfrak{p}}(A)/k_{\mathfrak{p}}) = \varphi_{\mathfrak{p}}^{\operatorname{ord}_{\mathfrak{p}}\beta}$. By definition,

$$(\beta,\alpha)_{\mathfrak{p}} = A^{\sigma_{\mathfrak{p}}-1} = A^{\operatorname{ord}_{\mathfrak{p}}\beta(\varphi_{\mathfrak{p}}-1)}$$

by the remark on automorphisms. This proves our theorem, if $\mathfrak{p}\nmid\alpha$.

If $\mathfrak{p}\mid\alpha$ then $\mathfrak{p}\nmid\beta$ by assumption, and

$$\left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{\alpha}\right)^{-1} \equiv \left(\frac{\beta}{\alpha}\right)^{-1}.$$

The rest follows by symmetry. □

THEOREM 13. *Let* α, $\beta\in k$. *Then*

$$\prod_{\mathfrak{p}}(\beta,\alpha)_{\mathfrak{p}} = 1$$

where the product is taken over all primes (*including the infinite primes*!).

---

[3]For a prime $\mathfrak{p}$ which divides α, so that $\left(\frac{\alpha}{\mathfrak{p}}\right)$ is not defined, the exponent $\operatorname{ord}_{\mathfrak{p}}\beta$ is 0, and the corresponding factor in the product is to be interpreted as 1.

[4]A slightly more general statement, true for arbitrary $\alpha,\beta\in k_{\mathfrak{p}}^*$, is

$$(\beta,\alpha)_{\mathfrak{p}} = \left(\frac{(-1)^{ij}\alpha^j\beta^{-i}}{\mathfrak{p}}\right),$$

where $i=\operatorname{ord}_{\mathfrak{p}}\alpha$ and $j=\operatorname{ord}_{\mathfrak{p}}\beta$.

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PROOF. We know that $\sigma_{\mathfrak{p}} = (\beta, k_{\mathfrak{p}}(A)/k_{\mathfrak{p}})$. Our theorem is an immediate consequence of the product formula for the norm residue symbol:

$$\prod_{\mathfrak{p}} \sigma_{\mathfrak{p}} = 1$$

and of the remark on automorphisms. □

COROLLARY. *Let α, β be relatively prime. Then*

$$\left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{\alpha}\right)^{-1} = \prod_{\mathfrak{p} \nmid mp_\infty} (\beta, \alpha)_{\mathfrak{p}} = \prod_{\mathfrak{p} \mid mp_\infty} (\alpha, \beta)_{\mathfrak{p}}.$$

The computation of $\left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{\alpha}\right)^{-1}$ is reduced to a computation at a finite number of primes, namely the primes $\mathfrak{p} \mid mp_\infty$.

We must still consider the special primes dividing m. Let $\lambda \in k$ be divisible only by primes of m. Define

$$\left(\frac{\lambda}{\alpha}\right) = \prod_{\mathfrak{p} \nmid mp_\infty} \left(\frac{\lambda}{\mathfrak{p}}\right)^{\operatorname{ord}_{\mathfrak{p}} \alpha}.$$

Then by the same arguments as before we have

THEOREM 14. *Let λ be divisible only by primes dividing m. Then*

$$\left(\frac{\lambda}{\alpha}\right) = \prod_{\mathfrak{p} \nmid mp_\infty} (\alpha, \lambda)_{\mathfrak{p}} = \prod_{\mathfrak{p} \mid mp_\infty} (\lambda, \alpha)_{\mathfrak{p}}.$$

EXAMPLE 1. *Quadratic Reciprocity Law.*

Let $k = \mathbb{Q}$ be the rationals, $m = 2$. Let a, b be two relatively prime odd integers. Since $(a, b)_p = \pm 1$ we have $(a, b)_p = (b, a)_p$. Using Theorem 13 and its Corollary, and the formula found in Theorem 6, we obtain

$$\begin{aligned}\left(\frac{a}{b}\right)\left(\frac{b}{a}\right) &= (a, b)_2 (a, b)_\infty \\ &= (-1)^{\frac{a-1}{2}\frac{b-1}{2}} (-1)^{\frac{\operatorname{sign} a - 1}{2}\frac{\operatorname{sign} b - 1}{2}}.\end{aligned}$$

The complementary result coming from Theorem 14 and Theorem 6 is, for odd a,

$$\left(\frac{2}{a}\right) = (2, a)_2 = (-1)^{\frac{a^2-1}{8}}.$$

EXAMPLE 2. Let $k = \mathbb{Q}(\zeta)$ where ℓ is an odd prime, and ζ a primitive ℓ-th root of unity. Take $m = \ell \sim \lambda^{\ell-1}$, where $\lambda = 1 - \zeta$.

Only one prime divides ℓ, namely λ. All $\mathfrak{p}_\infty$ are complex so they will give no contribution in the power residue symbol.

Suppose $\alpha \equiv 1 \ (\lambda^2)$ and $\beta \equiv 1 \ (\lambda)$. Then by combining Theorems 10 and 13 we get

$$\left(\frac{\alpha}{\beta}\right)\left(\frac{\beta}{a}\right)^{-1} = (\alpha, \beta)_\lambda = \zeta^{\frac{1}{\ell} S(\zeta \log \alpha D \log \beta)}.$$

Noting that $\left(\frac{\alpha}{\zeta}\right) = 1$ because ζ is a unit, and that

$$\left(\frac{\zeta}{\alpha}\right)\left(\frac{\alpha}{\zeta}\right)^{-1} = (\zeta, \alpha)_\lambda$$

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we also obtain

$$\left(\frac{\zeta}{\alpha}\right) = \zeta^{(N\alpha-1)/\ell} = \zeta^{\frac{1}{\ell}S\left(\frac{\zeta}{\lambda}\log\alpha)\right)}$$

(where N is of course the absolute norm).

The complementary statement for λ is

$$\left(\frac{\lambda}{\alpha}\right) = (\lambda, \alpha)_\lambda = \zeta^{-\frac{1}{\ell}S\left(\frac{\zeta}{\lambda}\log\alpha\right)}.$$

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CHAPTER XIII

Group Extensions

1. Homomorphisms of Group Extensions

This section concerns the elementary facts about group extensions and homomorphisms of them. Since these things are quite well known, the treatment is sketchy throughout, many details being left to the reader. The main results are summarized in Theorems 1 and 2 at the end of the section.

Let G be a group, not necessarily finite, and let A be an abelian group. A *group extension of A by G* is an exact sequence of group homomorphisms

$$\{1\} \to A \xrightarrow{i} U \xrightarrow{j} G \to \{1\}.$$

Then U is a group containing a normal subgroup iA which is isomorphic to A, such that the factor group U/iA is isomorphic to G. Since i is an isomorphism it is no essential loss of generality to assume that i is an inclusion, and we will habitually do so. Then we have $A \subset U$ and $U/A \approx G$.

Associated with such an extension there is a natural operation of G on A which can be described informally as follows: Since A is a normal subgroup of U, the elements of U operate on A by conjugation, and since we have assumed A commutative, the elements of A operate trivially on A; therefore the factor group $U/A \approx G$ operates on A.

We shall write A multiplicatively, and we will denote the effect of an element $\sigma \in G$ on an element $a \in A$ by a^σ. According to the discussion above we have

$$a^\sigma = a^{u_\sigma} = u_\sigma a u_\sigma^{-1},$$

where u_σ is any element of U such that $ju_\sigma = \sigma$.

To analyze the structure of an extension U one selects for each $\sigma \in G$ a definite representative element $u_\sigma \in U$ such that $ju_\sigma = \sigma$, i.e. such that $u_\sigma A$ is the coset of A in U corresponding to the element $\sigma \in G$. Then each a element $u \in U$ can be written uniquely in the form

$$u = au_\sigma, \qquad \sigma \in G,\ a \in A.$$

Indeed, the σ and a are given explicitly in terms of u by $\sigma = ju$, and $a = uu_\sigma^{-1}$. For each pair of elements $\sigma, \tau \in G$, there is an element $a_{\sigma,\tau} \in A$ such that

$$u_\sigma u_\tau = a_{\sigma,\tau} u_{\sigma\tau}.$$

These elements $a_{\sigma,\tau} = u_\sigma u_\tau u_{\sigma\tau}^{-1}$ constitute what is called a *factor set*.

The multiplication of elements of U can be described in terms of the multiplication in A and in G, the operation of G on A, and the factor set $a_{\sigma,\tau}$. Indeed, if $u = au_\sigma$ and $v = bu_\tau$ are any two elements of U we have

$$\begin{aligned} uv &= au_\sigma bu_\tau = au_\sigma bu_\sigma^{-1} u_\sigma u_\tau \\ &= ab^\sigma u_\sigma u_\tau = ab^\sigma a_{\sigma,\tau} u_{\sigma\tau}. \end{aligned}$$

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The associativity of the multiplication of elements in U imposes a condition on the factor set $a_{\sigma,\tau}$. On the one hand we have

$$u_\rho(u_\sigma u_\tau) = u_\rho a_{\sigma,\tau} u_{\sigma\tau} = a^\rho_{\sigma,\tau} u_\rho u_{\sigma\tau} = a^\rho_{\sigma,\tau} a_{\rho,\sigma\tau} u_{\rho\sigma\tau}$$

and on the other hand

$$(u_\rho u_\sigma)u_\tau = a_{\rho,\sigma} u_{\rho\sigma} u_\tau = a_{\rho,\sigma} a_{\rho\sigma,\tau} u_{\rho\sigma\tau}.$$

Comparing the two expressions we find

$$a^\rho_{\sigma,\tau} a_{\rho,\sigma\tau} = a_{\rho,\sigma} a_{\rho\sigma,\tau}$$

or

$$a^\rho_{\sigma,\tau} a^{-1}_{\rho\sigma,\tau} a_{\rho,\sigma\tau} a^{-1}_{\rho,\sigma} = 1.$$

The left hand side of this last equation is the multiplicative form of the coboundary formula for standard 2-cochains (cf. the discussion of $H^2(A)$ in §2 of Ch. II). Therefore, our associativity condition means simply that $a_{\sigma,\tau}$ is a standard 2-cocycle of G with values in the G-module A.

Working backwards, we can easily prove that, given any mode of operation of G on A, and any 2-cocycle $a_{\sigma,\tau}$ of G in A with respect to that operation, there exists a group extension U from which they come. We define U to be the set of all symbols $a * u_\sigma$, $a \in A$, $\sigma \in G$, with the multiplication rule

$$(a * u_\sigma)(b * u_\tau) = (ab^\sigma a_{\sigma,\tau}) * (u_{\sigma\tau}).$$

The associativity of this multiplication is an immediate consequence of the coboundary, relation $(\delta a)_{\rho,\sigma,\tau} = 1$, and of the usual properties of the exponentiation a^σ. For fixed α and $\beta \in U$ the equations

$$\alpha\xi = \beta \qquad \eta\alpha = \beta$$

are easily seen to be solvable in U, and it follows that U is a group, i.e. has a unit element and inverses. They are explicitly given by

$$1 = a^{-1}_{1,1} * u_1$$
$$(a * u_\sigma)^{-1} = a^{-\sigma^{-1}} a^{-1}_{\sigma^{-1},\sigma} a^{-1}_{1,1} * u_{\sigma^{-1}}.$$

(Various identities which arise in this connection, and which are of course direct consequences of the coboundary relation are:

$$a_{\rho,1} = a^\rho_{1,1} \qquad a_{1,\tau} = a_{1,1}$$
$$a_{\sigma,\sigma^{-1}} a_{1,1} = a^\sigma_{1,1} a^\sigma_{\sigma^{-1},\sigma}.)$$

Putting $j(a * u_\sigma) = \sigma$ we obtain a homomorphism of U onto G whose kernel is the set of all elements of the form $a * u_1$. This subgroup is isomorphic to A under the correspondence $a \leftrightarrow (aa^{-1}_{1,1} * u_1)$. We may therefore identify A with this subgroup, i.e. put $a = aa^{-1}_{1,1} * u_1$, and if at the same time we put $u_\sigma = 1 * u_\sigma$, we find

$$\begin{aligned} au_\sigma &= (aa^{-1}_{1,1} * u_1)(1 * u_\sigma) = aa^{-1}_{1,1} a_{1,\sigma} * u_\sigma = a * u_\sigma \\ u_\sigma a &= (1 * u_\sigma)(aa^{-1}_{1,1} * u_1) = a^\sigma a^{-\sigma}_{1,1} a_{\sigma,1} * u_\sigma = a^\sigma * u_\sigma = a^\sigma u_\sigma \\ u_\sigma u_\tau &= (1 * u_\sigma)(1 * u_\tau) = a_{\sigma,\tau} * u_{\sigma\tau} = a_{\sigma,\tau} u_{\sigma\tau}. \end{aligned}$$

Therefore, the operation of G on A, and the 2-cocycle $a_{\sigma,\tau}$, with which we began the construction are those derived from the group extension we have constructed.

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A homomorphism of one group extension into another is a collection of homomorphisms f, F, and φ such that the diagram

$$\begin{tikzcd}
(1) \arrow[r] & A \arrow[r, "i"] \arrow[d, "f"'] & U \arrow[r, "j"] \arrow[d, "F"'] & G \arrow[r] \arrow[d, "\varphi"'] & (1) \\
(1) \arrow[r] & B \arrow[r, "i"] & V \arrow[r, "j"] & H \arrow[r] & (1)
\end{tikzcd}$$

is commutative. Since we are viewing i as an inclusion, this means in particular that f is the restriction of F to A. Now suppose f and φ are given. Under what conditions does F exist, and how many F's are there? To study this question we select representatives $u_\sigma \in U$ and $v_\lambda \in V$. Supposing for the moment that an F is given, we have for arbitrary $u = au_\sigma \in U$

$$Fu = F(au_\sigma) = (Fa)(Fu_\sigma) = (fa)(Fu_\sigma).$$

Hence to describe F completely it is enough to give the elements Fu_σ; and since

$$j(Fu_\sigma) = \varphi j u_\sigma = \varphi\sigma = j(v_{\varphi\sigma}),$$

there are elements $c_\sigma \in B$ such that

$$Fu_\sigma = c_\sigma v_{\varphi\sigma}.$$

Thus, F is described by a function $\sigma \to c_\sigma$ from G to B.

If F is to be a homomorphism it is furthermore necessary that

$$F(u_\sigma a u_\sigma^{-1}) = (Fu_\sigma)(fa)(Fu_\sigma)^{-1}$$

and

$$F(u_\sigma u_\tau) = F(u_\sigma)F(u_\tau).$$

Writing out these conditions we find that they amount to

$$f(a^\sigma) = (fa)^{\varphi\sigma}, \qquad a \in A, \quad \sigma \in G$$

and

$$(*) \qquad fa_{\sigma,\tau} = b_{\varphi\sigma,\varphi\tau}(c_\tau^{\varphi\sigma} c_{\sigma\tau}^{-1} c_\sigma),$$

where $a_{\sigma,\tau}$ and $b_{\lambda,\mu}$ are the 2-cocycles associated with our two extensions and our selected representatives u_σ and v_λ. To interpret these conditions, we must view B as a G-module under the operation $b^{\varphi\sigma}$ which is induced by the homomorphism $\varphi\colon G \to H$. Then the first condition means simply that f is a G-homomorphism of A into B; and the second condition means that the 2-cocylcles $fa_{\sigma,\tau}$ and $b_{\varphi\sigma,\varphi\tau}$ of G in B differ by the coboundary of the 1-cochain c_σ of G in B.

Thus, necessary conditions for the existence of F are that f be a G-homomorphism when G operates on B through ϕ, and that $fa_{\sigma,\tau}$ and $b_{\varphi\sigma,\varphi\tau}$ be cohomologous 2-cocycles of G in B. Conversely, it is easy to check that these conditions are sufficient; if c_σ is a 1-cochain satisfying $(*)$, then a suitable F is obtained by putting $F(au_\sigma) = (fa)c_\sigma v_{\varphi\sigma}$.

In order to investigate the number of different F's for given f and φ, we consider the totality of 1-cochains $c_\sigma = (Fu_\sigma)v_{\varphi\sigma}^{-1}$ which describe them. If F and F' are any two F's, then, as we have seen, the corresponding c and c' have the same coboundary, so their quotient

$$d_\sigma = c'_\sigma c_\sigma^{-1} = (F'u_\sigma)v_{\varphi\sigma}^{-1} v_{\varphi\sigma}(F'u_\sigma)^{-1} = (F'u_\sigma)(Fu_\sigma)^{-1}$$

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is a 1-cocycle of G in B. This 1-cocycle is independent of the choice of representatives u_σ because upon replacing u_σ by au_σ, $a \in A$, we find that d_σ changes into $(fa)d_\sigma(fa)^{-1} = d_\sigma$. Therefore there is a natural operation of 1-cocycles $d = d_\sigma$ on the set of homomorphisms $\{F\}$, defined by $(d \cdot F)(u_\sigma) = d_\sigma(Fu_\sigma)$; the rule $d'(dF) = (d'd)F$ is obvious. From the above considerations it is clear that this operation is transitive, and without fixed points, i.e. if F_0 is one fixed F, then the correspondence $d \leftrightarrow dF_0$ is a (1–1)-correspondence between the d's and the F's.

It is natural to ask what is the significance of the 1-coboundaries, $d_\sigma = (\delta b)_\sigma = b^{\varphi\sigma}b^{-1}$, $b \in B$, in this connection? We have

$$((\delta b) \cdot F)u_\sigma = b^{-1}b^{\varphi\sigma}(Fu_\sigma) = b^{-1}(Fu_\sigma)b.$$

Therefore, $(\delta b) \cdot F$ is obtained by following F by the inner automorphism $v \to b^{-1}vb$ of V. We shall call two F's equivalent if they differ by such an inner automorphism. Then it follows from what we have proved that the 1-dimensional cohomology classes of G in B operate transitively and without fixed points on the equivalence classes of F's.

Putting all these results together and considering special cases we obtain the following theorems:

THEOREM 1. *Let us call two group extensions U and U' of A by G equivalent if there exists an isomorphism $F\colon U \approx U'$ such that the following diagram is commutative*:

$$\begin{tikzcd} A \arrow[r] \arrow[d, "\text{identity}"'] & U \arrow[r] \arrow[d, "F"'] & G \arrow[d, "\text{identity}"'] \\ A \arrow[r] & U \arrow[r] & G. \end{tikzcd}$$

Then the equivalence classes are in natural (1–1)*-correspondence with the* 2*-dimensional cohomology classes of G in A, for the various modes of operation of G on A.*

THEOREM 2. *Let $U/A \approx G$ and $U'/A' \approx G'$ be two group extensions. Then, if homomorphisms $f\colon A \to A'$ and $\varphi\colon G \to G'$ are given, there exists a homomorphism $F\colon U \to U'$ such that the diagram*

$$\begin{tikzcd} A \arrow[r] \arrow[d, "f"'] & U \arrow[r] \arrow[d, "F"'] & G \arrow[d, "\varphi"'] \\ A' \arrow[r] & U' \arrow[r] & G' \end{tikzcd}$$

is commutative, if and only if:

1) *f is a G-homomorphism, G operating on A' through φ, and*

2) *$f_*\alpha = \varphi^*\alpha'$, where $\alpha \in H^2(G, A)$ is the class of the extension U, and $\alpha' \in H^2(G', A')$ is the class of U'. (Here f_* and φ^* stand for the homomorphisms induced by $(1, f)\colon (G, A) \to (G, A')$ and $(\varphi, 1)\colon (G', A') \to (G, A')$, respectively.)*

Furthermore, if we call two such F's equivalent whenever they differ by an inner automorphism of U' by an element $a' \in A'$, then the group $H^1(G, A')$ operates transitively without fixed points on the set of equivalence classes. In particular, if $H^1(G, A') = 0$, then all F's are equivalent.

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2. Commutators and Transfer in Group Extensions

Let G be a finite group, A a (multiplicatively written) G-module, and let U be a group extension of A by G:

$$1 \to A \xrightarrow{i} U \xrightarrow{j} G \to 1.$$

There is no loss of generality if we view A as subgroup of U, i.e. treat i as an inclusion. Then, if $u_\sigma \in U$ are representatives of $\sigma \in G$, the elements of U are of the form $au_\sigma, a \in A$, $\sigma \in G$. They are multiplied according to the rules

$$u_\sigma a = a^\sigma u_\sigma \qquad u_\sigma u_\tau = a_{\sigma,\tau} u_{\sigma\tau},$$

where $a_{\sigma,\tau}$ is a standard 2-cocycle of G in A whose cohomology class $\alpha \in H^2(G, A)$ is the class of the extension U.

Let U^c denote the commutator subgroup of U. We wish to describe the subgroup $U^c \cap A$ of A. According to the corollary of Theorem 16 of Ch. V, §4, a subgroup of an abelian group A can be characterized by the set of all characters ($\mathbb{Z}$-homomorphisms $f\colon A \to \mathbb{Q}/\mathbb{Z}$) which vanish on it, and this is the approach we shall use. It is clear that a character f of A vanishes on $U^c \cap A$ if and only if it can be extended to a character F of U. Indeed, if f vanishes on $U^c \cap A$, then f can be viewed as a character of U^cA/U^c by means of the parallelogram in the accompanying diagram, and can then be extended to U, by Theorem 16 of Ch. V, §4. Conversely, any character of U vanishes on U^c because $\mathbb{Q}/\mathbb{Z}$ is abelian.

$$\begin{tikzcd}[column sep=small]
 & & U \arrow[dl, dash] \\
 & U^cA \arrow[dl, dash] \arrow[dr, dash] & \\
U^c \arrow[dr, dash] & & A \arrow[dl, dash] \\
 & U^c \cap A &
\end{tikzcd}$$

Given a character f of A we must therefore investigate the question, whether it can be extended to a character F of U. This can be viewed as a question about homomorphisms of group extensions as in the following diagram:

$$\begin{tikzcd}
A \arrow[r, "i"] \arrow[d, "f"'] & U \arrow[r, "i"] \arrow[d, "F"'] & G \arrow[d] \\
\mathbb{Q}/\mathbb{Z} \arrow[r, "\mathrm{id.}"] & \mathbb{Q}/\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

Since the lower extension is trivial, we see from Theorem 2 of §1 that necessary and sufficient conditions for the existence of F are:

(1) f must be a G-homomorphism, G operating trivially on $\mathbb{Q}/\mathbb{Z}$, and
(2) $f_*\alpha = 0$ in $H^2(G, \mathbb{Q}/\mathbb{Z})$,

where $\alpha \in H^2(G, A)$ is the class of our extension $U/A \approx G$. The first condition means simply that f must vanish on IA, the submodule of A generated by the elements $a^{\sigma-1}$. To transform the second condition we use the corollary of Theorem 18 of Ch. V, §4, which states that $H^2(G, \mathbb{Q}/\mathbb{Z})$ is the dual of $H^{-3}(G, \mathbb{Z})$. It follows that $f_*\alpha$ is the zero element of $H^2(G, \mathbb{Q}/\mathbb{Z})$, if and only if it yields the zero character of $H^{-3}(G, \mathbb{Z})$, i.e. if and only if $(f_*\alpha)\zeta = 0$ for all $\zeta \in H^{-3}(G, \mathbb{Z})$. We have $(f_*\alpha)\zeta = f_*(\alpha\zeta)$ because $(fa)n = f(an)$ for $a \in A$, $n \in \mathbb{Z}$; hence, our second condition is equivalent to

$$(2') \qquad f_*(\alpha\zeta) = 0 \qquad \text{for all } \zeta \in H^{-3}(G, \mathbb{Z});$$

in other words, to the vanishing of f_* on the subgroup $\alpha \cdot H^{-3}(G, \mathbb{Z})$ of $H^{-1}(G, A)$. The effect of f_* on $H^{-1}(G, A)$ is related to the effect of f on A_N (= the kernel of

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the norm map $a \to \prod_\sigma a^\sigma$) by the following commutative diagram

$$\begin{tikzcd} A_N \arrow[r, "\text{Ж}"] \arrow[d, "f"'] & H^{-1}(G, A) \arrow[d, "f_*"'] \\ (\mathbb{Q}/\mathbb{Z})_N \arrow[r, "\text{Ж}"] & H^{-1}(G, \mathbb{Q}/\mathbb{Z}), \end{tikzcd}$$

in which the lower horizontal arrow is an isomorphism. We see finally that our conditions amount to the vanishing of f on the subgroup $\text{Ж}^{-1}(\alpha \cdot H^{-3}(G, \mathbb{Z}))$ of A. Thus we have shown that the characters of A which vanish on $U^c \cap A$ are exactly those vanishing on $\text{Ж}^{-1}(\alpha \cdot H^{-3}(G, \mathbb{Z}))$, both conditions being equivalent to extendability to U. It follows that the two subgroups are equal, so we have proved:

THEOREM 3. *Let $U/A \approx G$ be a group extension belonging to the class $\alpha \in H^2(G, A)$. Then, U^c denoting the commutator subgroup of U, we have*

$$U^c \cap A = \text{Ж}^{-1}(\alpha \cdot H^{-3}(G, \mathbb{Z})).$$

In other words, if IA is the subgroup of A generated by the elements $a^{\sigma-1}$, and A_N is the kernel of the norm map, $a \to Na = \prod_{\sigma \in G} a^\sigma$, then

$$IA \subset U^c \cap A \subset A_N,$$

and the factor group $(U^c \cap A)/IA$ corresponds to $\alpha \cdot H^{-3}(G, \mathbb{Z})$ under the natural isomorphism $A_N/IA \approx H^{-1}(G, A)$.

It should be remarked that the two inclusions $IA \subset U^c \cap A \subset A_N$ can be proved directly. The first is obvious; for any $a \in A$, $\sigma \in G$ we have $a^{\sigma-1} = u_\sigma a u_\sigma^{-1} a^{-1} \in U^c \cap A$. The second is best understood by means of the group theoretical transfer from U to A which we will now investigate. It is a homomorphism

$$V_{U,A} \colon (U/U^c) \to (A/A^c) = A$$

mapping the factor commutator group of U into that of A, the latter being A itself since A is abelian. It is defined by

$$V_{U,A}(uU^c) = \prod_{\sigma \in G} u_\sigma u u_{\tilde{\sigma}}^{-1}.$$

where, for each given $u \in U$, and each $\sigma \in G$, $\tilde{\sigma} = j(u_\sigma u)$ is the unique element of G such that $u_\sigma u u_{\tilde{\sigma}}^{-1} \in A$. Being multiplicative, V is of course determined by its effect on A, and on the representatives u_τ. For these special cases we obtain

$$V_{U,A}(aU^c) = \prod_{\sigma \in G} u_\sigma a u_\sigma^{-1} = \prod_\sigma a^\sigma = Na,$$

and

$$V_{U,A}(u_\tau U^c) = \prod_{\sigma \in G} u_\sigma u_\tau u_{\sigma\tau}^{-1} = \prod_\sigma a_{\sigma,\tau},$$

where $a_{\sigma,\tau}$ is the 2-cocycle associated with the representatives u_σ. Thus, in particular, the effect of V on A is that of the norm, and this explains the inclusion $U^c \cap A \subset A_N$, because V annihilates U^c. V carries the representative u_τ into the image of τ under the Nakayama map $\tau \to \prod_\tau a_{\sigma,\tau}$ which we have studied in §5 of Ch. V, this being the other interpretation of the Nakayama map to which we alluded there. Since the norm and the Nakayama map both have values in A^G, we

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see that the image of U/U^c under V is contained in A^G. This can also be seen directly, for it is a special consequence of the following fact about transfer: If W is any subgroup of finite index in U, then

$$V_{U,uWu^{-1}}(u') = u(V_{U,W}u')u^{-1}$$

for all $u, u' \in U$.

Recalling that the Nakayama map can be used to compute the cup product $\alpha \cdot \zeta_\tau$ we see that we can abstract from the above considerations the following statement which is independent of the choice of representatives u_σ.

THEOREM 4. *Let $U/A \approx G$ be a group extension belonging to the class $\alpha \in H^2(G, A)$. Then the following diagram is commutative*[1]

$$\begin{tikzcd}
0 \arrow[r] & A/(U^c \cap A) \arrow[r, "\bar{i}"] \arrow[d, "\bar{N}"'] & U/U^c \arrow[r, "\bar{j}"] \arrow[d, "V_{U,A}"'] & G/G^c \approx H^{-2}(G, \mathbb{Z}) \arrow[r] \arrow[d, "\alpha"'] & 0 \\
0 \arrow[r] & NA \arrow[r, "\text{incl}"] & A^G \arrow[r, "\varkappa"] & H^0(G, A) \arrow[r] & 0,
\end{tikzcd}$$

where the right-hand vertical arrow denotes multiplication by α, and $\bar{N}$, $\bar{i}$, and $\bar{j}$ denote the maps induced in the obvious way by $N\colon A \to NA$, and by $A \xrightarrow{i} U \xrightarrow{j} G$.

COROLLARY. *Let $U/A \approx G$ be a group extension belonging to the class $\alpha \in H^2(G, A)$. Consider the following three homomorphisms:*

$$V\colon U/U^c \to A^G$$
$$\alpha_{-3}\colon H^{-3}(G, \mathbb{Z}) \to H^{-1}(G, A) \qquad \alpha_{-2}\colon H^{-2}(G, \mathbb{Z}) \to H^0(G, A).$$

We contend that there exist exact sequences

$$0 \to H^{-1}(G, A)/\operatorname{Im}\alpha_{-3} \to \operatorname{Ker} V \to \operatorname{Ker}\alpha_{-2} \to 0$$
$$0 \to A^G/\operatorname{Im} V \to H^0(G, A)/\operatorname{Im}\alpha_{-2} \to 0.$$

In particular, V is an isomorphism into if and only if α_{-3} is onto and α_{-2} is an isomorphism into; V is onto if and only if α_{-2} is onto.

PROOF. Notice that the horizontal rows of the diagram of Theorem 4 are exact. Let us augment that diagram by putting zeros above and below in the following

[1]The diagram gives another interpretation of the Nakayama map α, as the map induced by the Verlagerung in the group extension corresponding to the class α. In that version it is implicit in the proof of the Principal Ideal Theorem due to Artin and Iyanaga which we give in Section 4 of this chapter. It was published in S. Iyanaga [**14**]. Iyanaga was in Hamburg at the time and states that the greater part of the proof is due to Artin, who generously let him publish it under his own name.

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manner:

$$\begin{tikzcd}
 & \arrow[d] & \arrow[d] & \arrow[d] & \\
0 \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & * \arrow[r] \arrow[d, "\bar{N}"'] & * \arrow[r] \arrow[d, "V"'] & * \arrow[r] \arrow[d, "\alpha_{-2}"'] & 0 \\
0 \arrow[r] & * \arrow[r] \arrow[d] & * \arrow[r] \arrow[d] & * \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \\
 & \arrow[d] & \arrow[d] & \arrow[d] &
\end{tikzcd}$$

Viewing the columns as chain complexes, we obtain an exact homology sequence, and the essential part of this sequence is:

$$0 \to \operatorname{Ker} \bar{N} = A_N/(U^c \cap A) \to \operatorname{Ker} V \to \operatorname{Ker} \alpha_{-2} \to$$
$$\to 0 \to A^G/\operatorname{Im} V \to H^0(G, A)/\operatorname{Im} \alpha_{-2} \to 0.$$

Using the description of $U^c \cap A$ given by Theorem 3, we see $A_N/(U^c \cap A) \approx H^{-1}(G, A)/\operatorname{Im} \alpha_{-3}$. This concludes the proof of the corollary. $\square$

3. The Akizuki–Witt Map $v\colon H^2(G, A) \to H^2(G/H, A^H)$

Let A be a multiplicatively written G-module and let U be a group extension of A by G:

$$1 \to A \xrightarrow{i} U \xrightarrow{j} G \to 1.$$

If $H \subset G$ is any subgroup and we put $W = j^{-1}(H)$, then W is obviously an extension of A by H:

$$1 \to A \xrightarrow{i} W \xrightarrow{j} H \to 1.$$

We might call W a *subextension* of U.

If $\alpha \in H^2(G, A)$ is the 2-dimensional class describing U, then $\operatorname{res}_{G,H} \alpha \in H^2(H, A)$ is that describing W. Indeed, if u_σ, $\sigma \in G$, are representatives of G in U, then the elements u_h, $h \in H$, will serve as representatives of H in W. Hence by restricting the 2-cocycle $a_{\sigma,\tau} = u_\sigma u_\tau u_{\sigma\tau}^{-1}$ to H, we obtain a 2-cocycle describing W.

The cosets of H in G correspond to the cosets of W in U, and if H is normal in G we have $U/W \approx G/H$. Therefore we can form the extension

$$1 \to W \xrightarrow{\text{incl}} U \to G/H \to 1.$$

However to obtain an extension of the type we have been considering in which the subgroup is abelian, we must divide out by the commutator group of W:

$$1 \to W/W^c \to U/W^c \to G/H \to 1.$$

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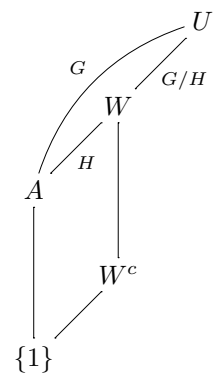
This last extension may be called a *factor extension* of U. Perhaps a lattice diagram will help in visualizing these various groups.



Now let $V = V_{W,A}$ denote the group theoretical transfer from W into A. It maps W/W^c into A^H, and is easily seen to be a (G/H)-homomorphism. Consequently it induces a homomorphism V_* of the $(G/H, W/W^c)$ cohomology into the $(G/H, A^H)$ cohomology. (Needless to say, V_* is *not* the cohomological transfer; it is simply an induced map.) The 2-dimensional class $\beta \in H^2(G/H, W/W^c)$ associated with the factor extension is carried into a certain 2-dimensional class $V_*\beta \in H^2(G/H, A^H)$. Obviously $V_*\beta$ depends only on the equivalence class of the original extension $U/A \approx G$, that is to say, on the element $\alpha \in H^2(G,A)$ which characterizes the extension U. We have therefore constructed a map $\alpha \to V_*\beta$ which we denote by

$$v_{G,(G/H)}\colon H^2(G,A) \to H^2(G/H, A^H).$$

To review the definition of $v_{G,(G/H)}$: Given $\alpha \in H^2(G,A)$ one constructs an extension $U/A \approx G$ belonging to α, goes to the factor extension $(U/W^c)/(W/W^c) \approx G/H$, and then takes the image of its 2-dimensional class β under the homomorphism induced by the (G/H)-homomorphism $V\colon W/W^c \to A^H$; the result is $v_{G,(G/H)}\alpha$. Following this prescription we can easily derive an explicit formula for the computation of $v_{G,(G/H)}$ in terms of standard 2-cocycles. Let $a_{\sigma,\tau}$ be a 2-cocycle in the class of α. Then a suitable group extension U consists of the elements au_σ, $a \in A$, $\sigma \in G$, with the multiplication rules

$$u_\sigma a = a^\sigma u_\sigma \qquad u_\sigma u_\tau = a_{\sigma,\tau} u_{\sigma\tau}.$$

The subgroup W consists of all au_h with $h \in H$. In order to compute a 2-cocycle $b_{r,s} \in W/W^c$ (for $r, s \in G/H$) which describes the factor extension, we first select representatives $\bar r \in G$ of the elements $r \in G/H$. We may then use the elements $u_{\bar r}$, $r \in G/H$, as representatives of the cosets of W in U, and the corresponding 2-cocycle $b_{r,s}$ is determined by the equations

$$u_{\bar r} u_{\bar s} = b_{r,s} u_{\overline{rs}} \pmod{W^c},$$

Introducing the abbreviation $\gamma = \bar r\, \bar s\, \overline{rs}^{-1} \in H$ and comparing the equations

$$u_{\bar r}\, u_{\bar s} = a_{\bar r,\, \bar s} u_{\bar r\, \bar s}$$
$$u_\gamma u_{\overline{rs}} = a_{\gamma, \overline{rs}}\, u_{\bar r \bar s}$$

we find

$$b_{r,s} = a_{\bar r, \bar s}\, a^{-1}_{\gamma, \overline{rs}}\, u_\gamma \pmod{W^c}.$$

Now we must transfer these elements from W to A. Using the rules derived in the discussion preceding Theorem 4 of §2, we find

$$V b_{r,s} = N_H(a_{\bar r, \bar s}\, a^{-1}_{\gamma, \overline{rs}}) \prod_h a_{h,\gamma} = \prod_h a^h_{\bar r, \bar s}\, a^{-h}_{\gamma, \overline{rs}}\, a_{h,\gamma}.$$

THEOREM 5. *Let G be a finite group, H a normal subgroup, and let the elements $\bar r \in G$ be representatives of the cosets $r = \bar r H$ of H in G. Then the cohomology map*

$$v\colon H^2(G,A) \to H^2(G/H, A^H)$$

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is induced by the cocycle map $a_{\sigma,\tau} \to (va)_{r,s}$, *which is defined by*

$$(va)_{r,s} = N_H(a_{\bar{r},\bar{s}}\, a^{-1}_{\gamma,\overline{rs}}) \cdot \prod_h a_{h,\gamma},$$

where $\gamma = \gamma_{r,s} = \bar{r}\,\bar{s}\,\overline{rs}^{-1} \in H$.[2]

Corollary. *The cohomology map* v *is multiplicative.*

Theorem 6. *For* $\alpha \in H^2(G, A)$ *we have*

$$\inf_{(G/H),G} v_{G,(G/H)}\alpha = \alpha^{(H:1)}.$$

Proof. If $\sigma \in G$, we denote by $\bar{\sigma}$ the representative of the coset in which σ lies; i.e. $\bar{\sigma} = \overline{\sigma H}$. Then $\inf v\alpha$ is represented by the 2-cocycle

$$(\inf va)_{\sigma,\tau} = (va)_{\sigma H,\tau H} = \prod_h a^{h}_{\bar{\sigma},\bar{\tau}}\, a^{-h}_{\bar{\sigma}\bar{\tau}\,\overline{\sigma\tau}^{-1},\overline{\sigma\tau}}\, a_{h,\bar{\sigma}\bar{\tau}\,\overline{\sigma\tau}^{-1}}.$$

Using the coboundary relation to pull the h's down from the exponents we obtain a formula in which two pair of terms cancel:

$$\prod_h a_{h\bar{\sigma},\bar{\tau}}\, \cancel{a^{-1}_{h,\bar{\sigma}\bar{\tau}}}\, a_{h,\bar{\sigma}}\, a^{-1}_{h\bar{\sigma}\bar{\tau}\,\overline{\sigma\tau}^{-1},\overline{\sigma\tau}}\, \cancel{a_{h,\bar{\sigma}\bar{\tau}}}\, \cancel{a^{-1}_{h,\bar{\sigma}\bar{\tau}\overline{\sigma\tau}^{-1}}}\, \cancel{a_{h,\bar{\sigma}\bar{\tau}\overline{\sigma\tau}^{-1}}}.$$

Since $\bar{\sigma}$ is in the same coset of H as σ is, and since H is normal, we can replace the product over $h\bar{\sigma}$ in the first factor by the product over σh. Similarly, we can replace $h\bar{\sigma}\bar{\tau}\,\overline{\sigma\tau}^{-1}$ by h in the third uncancelled term, because $\bar{\sigma}\bar{\tau}\,\overline{\sigma\tau}^{-1} \in H$. This yields:

$$(\inf va)_{\sigma,\tau} = \prod_h a_{\sigma h,\bar{\tau}}\, a_{h,\bar{\sigma}}\, a^{-1}_{h,\overline{\sigma\tau}}.$$

Now the 1-cochain $c_\sigma = \prod_h a_{h,\bar{\sigma}}$ has the coboundary

$$(\delta c)_{\sigma,\tau} = \prod_h a^{\sigma}_{h,\bar{\tau}}\, a^{-1}_{h,\overline{\sigma\tau}}\, a_{h,\bar{\sigma}}.$$

Dividing our 2-cocycle by this coboundary we obtain the following equivalent 2-cocycle in the class $\inf va$

$$e_{\sigma,\tau} = \prod_h a_{\sigma h,\bar{\tau}}\, a^{-\sigma}_{h,\bar{\tau}}.$$

It consists of two factors of the coboundary $(\delta a)^{-1}_{\sigma,h,\bar{\tau}}$, and since this coboundary is 1 we can write

$$e_{\sigma,\tau} = \prod_h a_{\sigma,h\bar{\tau}}\, a^{-1}_{\sigma,h} = \prod_h a_{\sigma,\tau h}\, a^{-1}_{\sigma,h}.$$

[2]Multiplying our formula for $(va)_{r,s}$ by $1 = \prod_{h\in H}(\delta a)_{h,\gamma,\overline{rs}}$ one finds another expression: $(va)_{r,s} = N_H a_{\bar{r},\bar{s}} \prod_h \frac{a_{h,\bar{r}\bar{s}}}{a_{h,\overline{rs}}}$. In this form our map v was discovered independently by Witt and Akizuki in 1935 (see [**31**] and [**1**]). Both these authors proved our Theorem 6. If the inflation-restriction sequence is exact in dimension 2, e.g. if $\mathcal{H}^1(H, A) = \{0\}$, then it is clear that there exists a unique map like v satisfying Theorem 6, and Witt's motivation was to find an explicit formula for it in terms of cocycles. Akizuki proved a result relating v to the Nakayama map which we can express as $v\alpha\cdot\zeta_{\sigma H} = (\alpha\cdot\zeta_\sigma)N_{G/H}A^H$. He used this with $H = G'$ to show that Nakayama's theorem about the injectivity of his map for abelian G implies with the same hypothesis that for arbitrary G the kernel is G' (cf. footnote at the end of Section 2 of the Chapter "Preliminaries").

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Using the coboundary relation to put the σ in the first factor back in the exponent we find

$$\begin{aligned} e_{\sigma,\tau} &= \prod_h a_{\tau,h}^{-\sigma} a_{\sigma\tau,h} a_{\sigma,\tau} a_{\sigma,h}^{-1} \\ &= \Big(\prod_h a_{\tau,h}^{\sigma} a_{\sigma\tau,h}^{-1} a_{\sigma,h}\Big)^{-1} \cdot a_{\sigma,\tau}^{(H:1)}. \end{aligned}$$

Since the quantity in the parentheses is the coboundary of the 1-cochain $c'_\sigma = \prod_h a_{\sigma,h}$, we see finally

$$e_{\sigma,\tau} \sim a_{\sigma,\tau}^{(H:1)}.$$

In terms of cohomology classes this means $\inf v\alpha = \alpha^{(H:1)}$, as contended. □

4. Splitting Modules and the Principal Ideal Theorem

Let G be a group and A a G-module. Let $\alpha \in H^r(G, A)$ be a cohomology class of G in A. A *splitting module* for the class α is a G-module B containing A, such that the image of α under the homomorphism induced by the inclusion of A in B is zero. In other words, such that a cocycle in the class of α becomes a coboundary when it is viewed in the bigger module B. Splitting modules exist for cohomology classes of all dimensions, but we will content ourselves with a discussion of a special case, namely a canonical type of splitting module for 2-dimensional classes. This special case is the most important one because of its intimate relationship with group extensions.

Let $a_{\sigma,\tau}$ be a standard 2-cocycle of G in A, and for convenience assume $a_{1,\tau} = a_{\sigma,1} = 0$. (Any 2-dimensional class contains such a 2-cocycle; from the point of view of group extensions, the normalization $a_{1,\tau} = a_{\sigma,1} = 0$ corresponds to choosing the identity element of the extension U as the representative of the identity element of the factor group G.) We wish now to construct a G-module B containing A, in which there is a 1-cochain x_σ such that

$$(\delta x)_{\sigma,\tau} = \sigma x_\tau - x_{\sigma\tau} + x_\sigma = a_{\sigma,\tau}.$$

As the additive group of B we take the direct sum of A and of a $\mathbb{Z}$-free module whose basis consists of symbols x_τ, one for each element $\tau \in G$ except the identity, i.e. we put

$$B = A + \sum_{\tau \neq 1} \mathbb{Z} x_\tau \qquad \text{(direct sum)}.$$

For notational convenience we define x_1 to be the zero element of B. We extend the operation of G from A to B in just such a way that the 2-cocycle $a_{\sigma,\tau}$ becomes the coboundary of the 1-cochain x_τ. This is done by defining

$$(*) \qquad \sigma x_\tau = x_{\sigma\tau} - x_\sigma + a_{\sigma,\tau}$$

for $\sigma, \tau \in G$. In order to justify this definition, we discuss the general question of $\mathbb{Z}$-homomorphisms of B into an arbitrary $\mathbb{Z}$-module C. A $\mathbb{Z}$-homomorphism $f\colon B \to C$ is obviously determined by its restriction f_A to A, and by the values $f(x_\tau)$ for $\tau \neq 1$. Conversely, given any $\mathbb{Z}$-homomorphism $g\colon A \to C$ and given arbitrary elements $c_\tau \in C$ for $\tau \neq 1$, there exists a homomorphism $f\colon B \to C$ such that $f_A = g$ and $f(x_\tau) = c_\tau$. Now taking C to be B itself we can therefore associate with each $\sigma \in G$ the unique homomorphism $\sigma\colon B \to B$ such that σ_A is the originally given effect of σ on A, and such that σx_τ is given by equation $(*)$ for

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$\tau \in 1$. The equation $(*)$ then holds automatically for $\tau = 1$ because $a_{\sigma,1} = 0$. The unit element of G acts as identity on A because

$$1x_\tau = x_\tau - x_1 + a_{1,\tau} = x_\tau.$$

(Recall that $x_1 = 0$ and $a_{1,\tau} = 0$.) The associative law for our operation follows from the coboundary relation for the 2-cocycle $a_{\sigma,\tau}$, namely:

$$\begin{aligned}\rho(\sigma x_\tau) &= \rho(x_{\sigma\tau} - x_\sigma + a_{\sigma,\tau})\\ &= x_{\rho\sigma\tau} - x_\rho + a_{\rho,\sigma\tau} - x_{\rho\sigma} + x_\rho - a_{\rho,\sigma} + \rho a_{\sigma,\tau}\\ &= x_{\rho\sigma\tau} - x_{\rho\sigma} + (\rho a_{\sigma,\tau} + a_{\rho,\sigma\tau} - a_{\rho,\sigma})\\ &= x_{(\rho\sigma)\tau} - x_{\rho\sigma} + a_{\rho\sigma,\tau}\\ &= (\rho\sigma)x_\tau.\end{aligned}$$

We have shown that the operation of G on B has all the required properties; B is a G-module.

The main properties of the splitting module B are stated in the next several lemmas. These lemmas will be used later to prove the principal ideal theorem.

LEMMA 1. *The factor module B/A is G-isomorphic to the augmentation ideal I. More precisely, the following is an exact sequence of G-homomorphisms*

$$0 \to A \xrightarrow{i} B \xrightarrow{j} I \to 0,$$

where i is the inclusion, and j is defined by $jA = 0$, and by $jx_\tau = (\tau - 1)$ for $\tau \neq 1$.

PROOF. Since the elements $\tau - 1$ for $\tau \neq 1$ are a $\mathbb{Z}$-basis for I, and since the elements x_τ are a $\mathbb{Z}$-basis for B/A, we see that j is a $\mathbb{Z}$-homomorphism of B onto I with kernel A, so the sequence is exact. To see that j is a G-homomorphism we note that the formula $jx_\tau = \tau - 1$ holds for all τ, because it holds automatically for $\tau = 1$, and consequently

$$\begin{aligned}j\sigma x_\tau &= j(x_{\sigma\tau} - x_\sigma + a_{\sigma,\tau})\\ &= (\sigma\tau - 1) - (\sigma - 1) = \sigma\tau - \sigma = \sigma(\tau - 1)\\ &= \sigma j x_\tau.\end{aligned}$$

□

LEMMA 2. *Let γ be an element of the group ring of G. Then $\gamma B \subset A$ if and only if γ is a multiple of the trace, i.e. $\gamma = e(\sum_\sigma \sigma) = eS$ for some integer e.*

PROOF. By Lemma 1 we have $B/A \approx I$. Hence $\gamma B \subset A \Leftrightarrow \gamma I = 0 \Leftrightarrow \gamma(\tau - 1) = 0$ for all $\tau \in G$. Writing $\gamma = \sum e_\sigma \sigma$ we see that this means all coefficients e_σ are equal. □

Now let us introduce the group extension $U/A \approx G$ constructed with the 2-cocycle $a_{\sigma,\tau}$. When A occurs in this connection, we will write it multiplicatively, though we still preserve the additive way of writing A when we view it as a submodule of the splitting module B. Thus, U consists of the elements au_σ with the multiplication rule

$$au_\sigma bu_\tau = ab^\sigma a_{\sigma,\tau} u_{\sigma\tau}.$$

In particular, we have $u_1 u_1 = a_{1,1} u_1 = u_1$, hence $u_1 = 1$, because we have assumed $a_{1,1}$ is the identity element of A.

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LEMMA 3. *The abelian group B/IB is isomorphic to the factor commutator group U/U^c of U under the correspondence*

$$au_\sigma U^c \leftrightarrow a + x_\sigma + IB.$$

PROOF. Define a map $\log\colon U \to B/IB$ by $\log au_\sigma \equiv a + x_\sigma \pmod{IB}$. Then the log of the product is the sum of the logs. Indeed

$$\begin{aligned}\log au_\sigma bu_\tau &= \log(ab^\sigma a_{\sigma,\tau} u_{\sigma\tau})\\ &= a + \sigma b + a_{\sigma,\tau} + x_{\sigma\tau},\end{aligned}$$

whereas,

$$\log au_\sigma + \log bu_\tau \equiv a + x_\sigma + b + x_\tau,$$

and subtracting these two expressions we obtain an element of IB, namely

$$\sigma b - b + (x_{\sigma\tau} - x_\sigma + a_{\sigma,\tau}) - x_\tau = (\sigma - 1)b + (\sigma - 1)x_\tau.$$

Since B/IB is abelian, our log homomorphism induces a homomorphism

$$\overline{\log}\colon U/U^c \to B/IB.$$

On the other hand, we can define a homomorphism $\exp\colon B \to U/U^c$ by putting

$$\begin{cases}\exp a = aU^c, & a \in A\\ \exp x_\tau = u_\tau U^c, & \tau \neq 1.\end{cases}$$

As usual, the second of these formulas holds for $\tau = 1$ as well because $x_\tau = 0$ and $u_\tau = 1$. This homomorphism exp vanishes on IB, because

$$\exp(\sigma - 1)a = a^{\sigma-1}U^c = u_\sigma a u_\sigma^{-1} a^{-1} U^c = U^c$$

and

$$\begin{aligned}\exp(\sigma - 1)x_\tau &= \exp(x_{\sigma\tau} - x_\sigma + a_{\sigma,\tau} - x_\tau)\\ &\equiv \frac{a_{\sigma,\tau}u_{\sigma\tau}}{u_\sigma u_\tau} \equiv 1 \pmod{U^c}.\end{aligned}$$

Consequently, exp induces a homomorphism $\overline{\exp}\colon B/IB \to U/U^c$. A glance at the definitions shows that $\overline{\log}$ and $\overline{\exp}$ are mutually inverse maps; hence they are both isomorphisms onto. This concludes the proof. □

LEMMA 4. *The transfer map $V_{U,A}\colon U/U^c \to A$ corresponds, under the isomorphism $U/U^c \approx B/IB$, to the map $\bar{S}\colon B/IB \to A$ which is induced by the trace map $S\colon B \to B$.* (Notice that S carries B into A (cf. Lemma 1) and S vanishes on IB.)

PROOF. For the representatives u_σ we have

$$\begin{aligned}V(u_\tau U^c) &= \prod_\sigma a_{\sigma,\tau} \quad \text{(viewing } A \text{ in } U)\\ &= \sum_\sigma a_{\sigma,\tau} \quad \text{(viewing } A \text{ in } B)\\ &= \sum_\sigma \sigma x_\tau - \not{x}_{\sigma\tau} + \not{x}_\sigma\\ &= \sum_\sigma \sigma x_\tau = Sx_\tau = \bar{S}\,\overline{\log}\, u_\tau,\end{aligned}$$

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and for $a \in A$ we have

$$\begin{aligned} V(aU^c) &= Na \quad (\text{in } U) \\ &= Sa \quad (\text{in } B) \\ &= \bar{s}(a + IB) = \bar{s}\,\overline{\log}\, a. \end{aligned}$$

Since any element of U is of the form au_τ, this concludes the proof. □

THEOREM (Principal Ideal Theorem). *Let U be a group whose commutator subgroup U^c is of finite index and is finitely generated. Then the transfer map $V_{U,U^c}\colon U/U^c \to U^c/(U^c)^c$ is the zero map.*

PROOF. Dividing out by $(U^c)^c$ does not affect the transfer map, so it is no loss of generality to assume $(U^c)^c = 1$, i.e. U^c abelian. Furthermore, replacing U^c by an arbitrary abelian subgroup A containing U^c, we see that it is enough to prove:

THEOREM 7′. *Let U be a group whose commutator subgroup U^c is of finite index and is finitely generated. Suppose that A is an abelian subgroup of U containing U^c. Then, if $e = (A : U^c)$, e times the transfer map $V_{U,A}\colon U/U^c \to A$ is the zero map.*

PROOF. Since $A \supset U^c, A$ is normal in U and the factor group $G = U/A$ is a finite abelian group. Let its order be $n = (U : A)$. Then $(U : U^c) = (U : A)(A : U^c) = ne$. Let B be a splitting module for a 2-cocycle coming from the extension $U/A \approx G$, as described in the paragraphs above. Then B is finitely generated over $\mathbb{Z}$, because $B/A \approx I$ is a free $\mathbb{Z}$-module on $(n-1)$-generators, A/U^c a finite group, and U^c is assumed to be finitely generated. By Lemma 3, the factor group B/IB is isomorphic to U/U^c, and is therefore a finite abelian group of order ne. Let $b_1, b_2, \ldots, b_m$ be elements of B which are representatives, mod IB, of a basis for B/IB. Let e_i be the period of b_i mod IB. Then $ne = e_1 e_2 \ldots e_m$. Let $b_{m+1}, b_{m+2}, \ldots, b_s$ be generators for IB (as subgroup of a finitely generated abelian group, B, IB is finitely generated). Put $e_{m+1} = e_{m+2} = \cdots = e_s = 1$. Then we have achieved the following three things:

1) The elements $b_1, b_2, \ldots, b_s$ generate B.
2) $e_i b_i \in IB$, $i = 1, 2, \ldots, s$.
3) $\prod_{i=1}^s e_i = ne$.

From 1) we have $B = \sum_{j=1}^s \mathbb{Z} b_j$, hence $IB = \sum_{j=1}^s I b_j$. Therefore, by 2) we see that there exist elements $\theta_{ij} \in I$ such that

$$e_i b_i = \sum_{j=1}^s \theta_{ij} b_j, \qquad i = 1, 2, \ldots, s.$$

Putting $\gamma_{ij} = e_i \delta_{ij} - \theta_{ij}$, where δ_{ij} is the Kronecker delta, we obtain an $s \times s$ matrix (γ_{ij}) of elements of the group ring Γ of G such that

$$(*) \qquad \sum_{j=1}^s \gamma_{ij} b_j = 0, \qquad i = 1, 2, \ldots, s.$$

Since G is an abelian group, the group ring Γ is commutative; hence the notion of a determinant of a square matrix with elements in Γ makes sense. If $(\widetilde{\gamma}_{ij})$ is the matrix whose elements are the cofactors of the elements of the matrix (γ_{ij}) we have

$$\sum_i \widetilde{\gamma}_{ik} \gamma_{ij} = \gamma \cdot \delta_{kj},$$

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where $\gamma = \det \gamma_{ij}$ and δ is the Kronecker delta. Multiplying $(*)$ by $\widetilde{\gamma}_{ik}$ and summing over i we obtain

$$\gamma b_j = 0$$

for all j. Since the b_j generate B it follows that γ annihilates B, and a fortiori, $\gamma B \subset A$. Hence, by Lemma 2, γ is a multiple of the trace, say $\gamma = tS$. To determine the value of t, it is enough to compute the image of γ under the ring homomorphism $\varepsilon\colon \Gamma \to \mathbb{Z}$ which is defined by $\varepsilon\sigma = 1$ for all $\sigma \in G$, and whose kernel is I. On the one hand we have

$$\varepsilon\gamma = \varepsilon(tS) = t(\varepsilon S) = tn.$$

On the other hand, going back to the definition of γ, we have

$$\begin{aligned}\varepsilon\gamma &= \varepsilon \det(\gamma_{ij}) = \det(\varepsilon\gamma_{ij}) = \det(e_i\delta_{ij})\\ &= \prod_{i=1}^{s} e_i = en\end{aligned}$$

because $\gamma_i = e_i\delta_{ij} - \theta_{ij} \equiv e_i\delta_{ij} \pmod{I}$. Comparison of the two equations above shows $t = e$, hence $\gamma = eS$. This proves the theorem, because according to Lemma 4, the transfer map $V_{U,A}$ is induced by the trace map S on B and, as we have already remarked, γ annihilates B.

$\square$

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CHAPTER XIV

Abstract Class Field Theory

1. Formations

In this section we will introduce an abstract algebraic object called a *formation*. In order to motivate the definition and terminology we first consider an important special case arising from ordinary galois theory. Let k be any field and let Ω be the separable part of the algebraic closure of k. Let G be the galois group of Ω over k, i.e. the group of all automorphisms of Ω which are identity on k. (Of course G is usually infinite.) Let Σ be the set of all *finite* extensions of k in Ω; we denote them by F, E, K, L etc. For each $F \in \Sigma$, let G_F denote the subgroup of G corresponding to F, i.e. consisting of all automorphisms of Ω which are identity on F. Then G can be made into a topological group by taking the family of subgroups $\{G_F\}_{F\in\Sigma}$ as a fundamental system of neighborhoods of the identity, and it follows then from galois theory that every open subgroup of G is of the form G_F for some $F \in \Sigma$.

Let A denote the multiplicative group of Ω, viewed as G-module. For each finite extension F of k, let A_F denote the multiplicative group of F. Then from galois theory we know that $A_F = A^{G_F}$, i.e. A_F, is the submodule of A consisting of all elements which are left fixed by G_F.

If F and K are two finite extensions of k, we have $F \subset K$ if and only if $G_F \supset G_K$, in which case K/F is a finite extension of degree $[K : F] = (G_F : G_K)$, and $A_F \subset A_K$. The extension K/F is normal if and only if G_K is a normal subgroup of G_F, and then its galois group $G_{K/F}$, is isomorphic to the factor group G_F/G_K. In this situation the finite galois group $G_{K/F}$, operates on the multiplicative group A_K of the normal extension K; hence we have cohomology groups $H^r(G_{K/F}, A_K)$. It is this type of cohomology group in which we are interested.

However, if the fields K and F are finite algebraic number fields, the galois group $G_{K/F}$ operates not only on the multiplicative group of K, but also on the idèle group of K, and on the idèle class group of K; and in global class field theory it is essential to study the cohomology of all three situations. The idèles of the various finite algebraic number fields F can be assembled into one big group, the idèle group of the field of all algebraic numbers, and the same goes for the idèle classes. Thus, we will be able to treat each of these situations as special cases of the following abstraction:

Definition 1. A *formation* $(G, \{G_F\}; A)$ consists of:

1) *A* group G, usually infinite, together with a non-empty indexed family $\{G_F\}_{F\in\Sigma}$ of subgroups of G satisfying the following conditions:

a) Each member of the family $\{G_F\}$ is of finite index in G.
b) Each subgroup of G which contains a member of the family $\{G_F\}$ also belongs to the family.

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c) The intersection of two members of the family $\{G_F\}$ also belongs to the family.
d) Any conjugate of a member of the family $\{G_F\}$ is also a member of the family.
e) The intersection of all members of the family $\{G_F\}$ is the identity:

$$\bigcap_{F\in\Sigma} G_F = 1.$$

2) A G-module A such that $A = \bigcup_{F\in\Sigma} A^{G_F}$, in other words, such that every element of A is left fixed by some member of the family $\{G_F\}$.

Notice that in this definition the symbol F has been reduced to the humble status of a mere index. Logically it would be more reasonable to drop it altogether and use the subgroups G_F themselves as indices. However we retain the F for psychological purposes, and in order that our terminology will fit the applications.

The requirements concerning the family of subgroups $\{G_F\}$ could have been stated more briefly by our saying: G is a Hausdorff topological group in which there is a fundamental system of neighborhoods of the identity consisting of subgroups of finite index, and $\{G_F\}$ is the family of open subgroups of G. However, until much later there will be no need to consider G seriously as a topological group, so we prefer the explicit algebraic description of the properties of the family $\{G_F\}$ to the more succinct topological one. For the time being, the only hint of the topological aspect will be our use of the convenient term *open subgroup* as a synonym for "a member of the family $\{G_F\}$".

The terminology of formations which we now outline can be best understood if one keeps in mind the special example discussed in the opening paragraphs of this section, in which G is an infinite galois group and A is the multiplicative group of the infinite normal extension. If $(G, \{G_F\}, A)$ is a formation, we call G the *galois group of the formation*, and we call A the *formation module*. The indices F are referred to as *fields*. Corresponding to each field F we define a submodule A_F of A by $A_F = A^{G_F}$. We call these submodules A_F the various *levels* of the formation and we say that A_F is the F-level. If F and K are a pair of fields, we write $F \subset K$ whenever $G_F \supset G_K$, and say then that F is a *subfield* of K, or K an extension of F. When this is the case, the F-level is contained in the K-level: $A_F \subset A_K$. A pair of fields such that $F \subset K$ is said to determine a *layer* of the formation, and this layer is referred to by the symbol K/F. A_F is called the *ground level*, and A_K the *top level*, of the layer. The index $(G_F : G_K)$, which is finite by assumption la) of the definition of a formation, is called the *degree* of the layer K/F and is denoted by $[K : F]$. When G_K is a normal subgroup of G_F the layer K/F is called a *normal layer*. Then the finite factor group G_F/G_K operates naturally on the top level of the layer, i.e. on $A_K = A^{G_K}$. This factor group G_F/G_K is called the *galois group of the normal layer* and is denoted by $G_{K/F}$. We have clearly

$$A_F = A^{G_F} = (A^{G_K})^{G_F/G_K} = A_K^{C_{K/F}};$$

in other words, the ground level of a normal layer consists exactly of all elements of the top level which are left fixed by the operation of the galois group of the layer. As in galois theory, a normal layer is called solvable, abelian, or cyclic, if the galois group is solvable, abelian, or cyclic.

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By the *cohomology groups of a normal layer* K/F we mean those associated with the operation of the galois group $G_{K/F}$, on the top level A_K. They are denoted briefly by $H^r(K/F)$. Thus we have by definition

$$H^r(K/F) = H^r(G_{K/F}, A_K) = H^r(G_F/G_K, A^{G_K}).$$

Of course these cohomology groups are those of finite groups $G_{K/F}$, in spite of the fact that the galois group G of the whole formation is usually infinite. Therefore all of the theory which we have developed in the preceding chapters applies to them. In particular, we have at our disposal all the natural homomorphisms. We now list these, giving first the old notation, and following it by a new abbreviated notation which we are hereby introducing:

Transfer[1] **and Restriction:** If $F \subset E \subset K$ and K/F is normal, then we have

$$V_{(G_E/G_K),(G_F/G_K)} \colon H^r\big(G_E/G_K, A^{(G_K)}\big) \to H^r\big(G_F/G_K, A^{(G_K)}\big)$$

or,

$$V_{(K/E),(K/F)} \colon H^r(K/E) \to H^r(K/F);$$

and

$$\mathrm{res}_{(G_F/G_K),(G_E/G_K)} \colon H^r\big(G_F/G_K, A^{(G_K)}\big) \to H^r\big(G_E/G_K, A^{(G_K)}\big)$$

or,

$$\mathrm{res}_{(K/F),(K/E)} \colon H^r(K/F) \to H^r(K/E).$$

Conjugation: If K/F is normal and $\sigma \in G$, then we have

$$\sigma_* \colon H^r\big(G_F/G_K, A^{(G_K)}\big) \to H^r\big(G_F^\sigma/G_K^\sigma, \sigma A^{(G_K)}\big)$$

or,

$$\sigma_* \colon H^r(K/F) \to H^r(K^\sigma/F^\sigma),$$

where K^σ is defined by the equation $G_{K^\sigma} = G_K^\sigma = \sigma G_K \sigma^{-1}$. (This is permissible by assumption 1d) of the definition of a formation, which states that the family of subgroups $\{G_F\}$ is closed under conjugation.)

Inflation: If $F \subset K \subset L$ and L/F and K/F are normal, then we have for $r \geqslant 1$

$$\inf_{(G_F/G_K),(G_F/G_L)} \colon H^r\big(G_F/G_K, A^{(G_K)}\big) \to H^r\big(G_F/G_L, A^{(G_L)}\big)$$

or,

$$\inf_{(K/F),(L/F)} \colon H^r(K/F) \to H^r(L/F).$$

The purpose of a formation is to facilitate the study of the cohomology groups of its normal layers. By putting these layers all together into one big object, the formation, we achieve the fullest possible freedom in studying their interrelationships, so that we can glean information about "difficult" layers from our knowledge of simple layers.

A special notational device which we will employ is that of using K and L to denote the top field in normal layers and using E to denote the top field of non-normal layers. For example, K/F and L/E are automatically understood to be normal, whereas E/F and E_1/E are not necessarily normal.

Most features of ordinary galois theory carry over to abstract formations. Two important examples of this general fact are furnished by the following propositions.

[1]The map "transfer" is nowadays often called "corestriction".

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PROPOSITION 1. *Given any finite set of layers over the same ground field, $E_1/F, E_2/F, \ldots, E_m/F$, there exists a normal layer K/F containing all of them, i.e. such that $F \subset E_i \subset K$ for $i = 1, \ldots, m$.*

PROOF. Each of the subgroups G_{E_i} is of finite index in G_F and has therefore only a finite number of distinct conjugates $G^{\sigma}_{E_i}$ in G_F. These conjugates are open subgroups of G by assumption d) of definition 1. Hence, by assumption c) of that same definition, their intersection,

$$\bigcap_{\substack{\sigma \in G_F \\ 1 \leqslant i \leqslant m}} G^{\sigma}_{E_i}$$

is also open, i.e. is of the form G_K for some field K. Since G_K is obviously normal in G_F, and is contained in G_{E_i} for each i, it follows that K/F is normal, and $K \supset E_i$ for each i, as contended. □

PROPOSITION 2. *Every subgroup of the galois group $G_{K/F}$ of a normal layer K/F is of the form $G_{K/E}$ for some $E, F \subset E \subset K$.*

PROOF. We have $G_{K/F} = G_F/G_K$. Hence any subgroup is of the form H/G_K for some group H such that $G_F \supset H \supset G_K$. Since $H \supset G_K$ there is a field E such that $H = G_E$, by assumption b) of Definition 1. Then $F \subset E \subset K$ and our original subgroup of $G_{K/F}$ is $G_E/G_K = G_{K/E}$. □

One *difference* between formation theory and galois theory proper is that in a formation one does *not* assume that the correspondence

$$G_K \leftrightarrow A_K = A^{G_K}$$

is one-to-one. It is perfectly possible, according to our definition, that the galois group of a formation operates trivially on the formation module, in which case all levels are the formation module itself. The point is that the one-to-oneness of the correspondence plays no role whatsoever in the cohomological considerations, so there is no point in assuming it.

2. Field Formations. The Brauer Groups

The main fact about the cohomology of a finite galois group operating on the multiplicative group of a normal field extension is that the 1-dimensional cohomology group is trivial. For the convenience of the reader we insert here a formal statement and proof of this fact:

PROPOSITION 3. *Let K be a field. Let A_K be the multiplicative group of K, and let $G_{K/F}$ be a finite group of automorphisms of K. Then $H^1(G_{K/F}, A_K) = 0$.*

PROOF. Let $f\colon G_{K/F} \to A_K$ be a crossed homomorphism of $G_{K/F}$, in A_K:

$$f(\sigma\tau) = (f(\tau))^{\sigma} f(\sigma) \qquad \sigma, \tau \in G_{K/F}.$$

Select $b \in K$ such that the element

$$a = \sum_{\tau \in G_{K/F}} b^{\tau} f(\tau)$$

is not zero, i.e. $a \in A_K$. The existence of such a b follows from the theorem on the independence of isomorphisms of a field (see, e.g., [**3**, Corollary in Section F], or [**4**,

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Chap. V, § 7, Theorem 3] or almost any modern textbook on Galois theory). Then the equations

$$a^\sigma f(\sigma) = \sum_\tau b^{\sigma\tau}(f(\tau))^\sigma f(\sigma) = \sum_\tau b^{\sigma\tau} f(\sigma\tau) = a$$

show that $f(\sigma) = a^{1-\sigma}$, i.e. f is a principal crossed homomorphism. □

It is the vanishing of the 1-dimensional groups which gives the ordinary galois cohomology theory its peculiar flavor. It is therefore of interest to study abstract formations whose normal layers have the corresponding property, i.e. which satisfy

Axiom I. $H^1(K/F) = 0$ for each normal layer K/F. We will call a formation a *field formation* if it satisfies this axiom. All the formations considered in class field theory are field formations, although in the case of the formation of idèle classes in the global theory this fact lies very deep.

In order to establish axiom I in the more difficult cases, it is necessary to know that it is a consequence of the seemingly weaker

Axiom I′. $H^1(K/F) = 0$ whenever K/F is a cyclic layer of prime degree.

The fact that Axiom I′ implies axiom I is a special case of the following lemma.

Lemma 1. *Let (G, A) be a formation in which all inflation-restriction sequences*

$$H^r(K/F) \xrightarrow{\text{inf}} H^r(L/F) \xrightarrow{\text{res}} H^r(L/K)$$

are exact for a certain positive dimension r. Then in order to prove that a divisibility of the form

$$(*) \qquad \text{order of } H^r(L/F) \mid [L:F]^\nu$$

holds for all normal layers L/F, it is enough to show that it holds for cyclic layers of prime degree.

Remark. From the case $r = 1$, $\nu = 0$ of this lemma, we see that Axiom I′ implies Axiom I, because the inflation-restriction sequence is always exact in dimension 1. We will use the case $r = 2$, $\nu = 1$ in the next section.

Proof. Let L/F be a given normal layer for which we wish to prove the divisibility $(*)$. By induction on the degree of the layers, we may assume $(*)$ holds for all layers of degree $< [L:F]$.

Case 1. $[L:F]$ is a prime power p^t. We may assume $t \geqslant 2$; otherwise $(*)$ is true by assumption. The galois group of the layer, $G_{L/F} = G_F/G_L$ is of order p^t, and therefore contains a proper normal subgroup different from identity, because p-groups are solvable. This subgroup is of the form $G_{L/K}$ where $F \subset K \subset L$, by Prop. 2 of §1. Using the exactness of the inflation-restriction sequence we find

$$h_r(L/F) \mid h_r(K/F) \cdot h_r(L/K),$$

where h_r denotes "order of H^r". Since the layers K/F and L/K are of lower degree than L/F, $(*)$ holds for them, by our inductive assumption, i.e.

$$h_r(K/F) \mid [K:F]^\nu \quad \text{and} \quad h_r(L/K) \mid [L:K]^\nu.$$

Combining these divisibilities with the above and using the obvious multiplicativity of the degrees, $[L:F] = [L:K][K:F]$, we obtain

$$h_r(L/F) \mid [L:F]^\nu$$

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which is the divisibility we are trying to prove.

Case 2. $[L : F]$ is not a prime power. For each prime p dividing $[L : F]$, let $G_{L/E}$ be a p-sylow subgroup of $G_{L/F}$. Then, using the fact that the restriction map from a group to a p-sylow subgroup is an isomorphism into on the p-primary part of the cohomology groups, we find

$$h_r(L/F) \mid \prod_p h_r(L/E_p).$$

Since $[L : F]$ is not a prime power, all the layers L/E_p corresponding to the Sylow subgroups are of lower degree than L/F; hence $(*)$ holds for them by our inductive assumption, i.e.

$$h_r(L/E_p) \mid [L : E_p]^\nu$$

multiplying over p and using $\prod_p [L : E_p] = [L : F]$ (the order of a group is the product of the orders of the p-sylow subgroups for the various p) we obtain the desired result. This completes the proof of Lemma 1. $\square$

In the proof of Lemma 1 the exactness of the inflation-restriction sequence is used only in the form of the consequence

$$h_r(L/F) \mid h_r(K/F)h_r(L/F).$$

This consequence could also be obtained from an exact sequence in the other direction such as the special 0-dimensional sequence

$$H^0(K/F) \leftarrow H^0(L/F) \leftarrow H^0(L/K)$$

mentioned at the beginning of §5 of Ch. IV, and from this we get a similar statement in dimension 0. However we only mention this in passing, since we will never require the result.

Throughout the remainder of this section we will assume that our formation is a field formation. The most important consequence of this assumption is the exactness of the inflation-restriction sequence in dimension 2. If $F \subset K \subset L$ with K/F and L/F normal, then the sequence

$$0 \to H^2(K/F) \xrightarrow{\text{inf}} H^2(L/F) \xrightarrow{\text{res}} H^2(L/K)$$

is exact. This follows either from the Hochschild–Serre spectral sequence, or by "dimensiona shifting", using the fact that the 1-dimensional cohomology group of the layer L/K corresponding to the subgroup $G_{L/K}$ of $G_{L/F}$, is trivial in our field formation.

One consequence of this exactness is that for $F \subset K \subset L$, the map

$$H^2(K/F) \xrightarrow{\text{inf}} H^2(L/F)$$

is an isomorphism into. We may therefore identify $H^2(K/F)$ with its image in $H^2(L/F)$, and view the inflation as an inclusion. If $F \subset K \subset L \subset M$, then, by the transitivity of inflation we see that the direct imbedding of $H^2(K/F)$ in $H^2(M/F)$ is the same as the imbedding via $H^2(L/F)$. Taking the injective limit we obtain a group which we may denote by $H^2(*/F) = \lim_K H^2(K/F)$, and which has the following properties:

1) For each normal layer K/F over the ground field $F, H^2(K/F)$ is a subgroup of $H^2(*/F)$.

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2) If $F \subset K \subset L$, then the subgroup $H^2(K/F)$ is contained in the subgroup $H^2(L/F)$, and the inclusion map is the inflation, $\inf_{(K/F),(L/F)}$.

3) $H^2(*/F)$ is the union of these subgroups.

The group $H^2(*/F) = \bigcup_K H^2(K/F)$ is called the *Brauer group over* F of the field formation (G, A). It was first constructed by Richard Brauer in the special case of the formations in which the formation module is the multiplicative group of the separable part of the algebraic closure of a ground field F. He viewed it as the group of classes of simple algebras with center F, the multiplication of algebra classes being induced by the Kronecker product of representative algebras. The connection with the 2-dimensional cohomology groups arises from the possibility of writing simple algebras explicitly as crossed products, in which the multiplication rules in the algebra appear in the form of a standard 2-cocycle. This procedure had been introduced by Wedderburn and Dickson in special cases, using non-standard 2-cocyles, and was first "standardized", and thereby systematized, by Brauer.

To review the definition of the Brauer group over F: Any element of it is equal to an element $\alpha \in H^2(K/F) = H^2(G_{K/F}, A_K)$ for some normal layer K/F. If $\alpha' \in H^2(K'/F)$ belongs to another such layer, then α' is equal to α in the Brauer group if and only if

$$\inf_{(K/F),(L/F)} \alpha = \inf_{(K'/F),(L/F)} \alpha'$$

$$\begin{tikzcd} & L \arrow[dl, dash] \arrow[dr, dash] & \\ K \arrow[dr, dash] & & K' \arrow[dl, dash] \\ & F & \end{tikzcd}$$

for a normal layer L/F such that $L \supset K$ and $L \supset K'$. (Of course if this equality holds for one such L, it holds for all.) The sum of α and α' in the Brauer group is equal to the element

$$\inf \alpha + \inf \alpha'$$

in $H^2(L/F)$. In this way, all computations in the Brauer group are referred to computations in the layers.

It is natural to denote the inclusion map of $H^2(K/F)$ into the Brauer group $H^2(*/F)$ by the symbol $\inf_{(K/F),(*/F)}$, and to view it as a symbolic inflation, since it is, so to speak, the limit of actual inflations, $\inf_{(K/F),(L/F)}$ as $L \to \infty$.

Our next task is to define a symbolic restriction map, $\mathrm{res}_{F,E}\colon H^2(*/F) \to H^2(*/E)$ between the respective Brauer groups of two fields $F \subset E$. Let $F \subset E \subset K \subset L$ with K/F and L/F normal. Then the following diagram is commutative:

$$\begin{tikzcd} H^2(K/F) \arrow[r, "\inf"] \arrow[d, "\mathrm{res}"'] & H^2(L/F) \arrow[d, "\mathrm{res}"'] \\ H^2(K/E) \arrow[r, "\inf"] & H^2(L/E). \end{tikzcd}$$

Therefore, if we view the two upper groups as subgroups of $H^2(*/F)$, the two lower ones as subgroups of $H^2(*/E)$, and the horizontal arrows as inclusions, we see that the effect of the vertical restriction maps is consistent—the one on the right extends the one on the left. Taken all together therefore, for all K, L etc. which contain E and are normal over F, these restriction maps $\mathrm{res}_{(K/F),(K/E)}$ constitute a map of $H^2(*/F)$ into $H^2(*/E)$ which we denote by $\mathrm{res}_{F,E}$, and view as a symbolic restriction.

In case E is normal over F, the kernels of both vertical restriction maps in our diagram are the same, namely they are $\inf\big(H^2(E/F)\big)$. This follows from the

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exactness of the horizontal inflation-restriction sequences

$$\begin{tikzcd} H^2(E/F) \arrow[r, "\inf"] \arrow[d, "1"'] & H^2(K/F) \arrow[r, "\mathrm{res}"] \arrow[d, "\inf"'] & H^2(K/E) \arrow[d, "\inf"'] \\ H^2(E/F) \arrow[r, "\inf"] & H^2(L/F) \arrow[r, "\mathrm{res}"] & H^2(L/E) \end{tikzcd}$$

and the commutativity of the diagram, in which we view the vertical inflations as inclusions. Consequently, the symbolic inflation-restriction sequence

$$0 \to H^2(E/F) \xrightarrow{\inf} H^2(*/F) \xrightarrow{\mathrm{res}} H^2(*/E)$$

is exact; the kernel of $\mathrm{res}_{F,E}$ is the subgroup $H^2(E/F)$ of $H^2(*/F)$ in case E/F is normal. This fact is of the utmost importance because it characterizes the subgroups of the Brauer group $H^2(*/F)$ which belong to the individual layers: For an element $\alpha \in H^2(*/F)$ to belong to $H^2(E/F)$, it is necessary and sufficient that $\mathrm{res}_{F,E}\,\alpha = 0$.

Let us review how the image of an element $\alpha \in H^2(*/F)$ under the symbolic restriction map is explicitly computed in terms of the layers. The given α will belong to some normal layer K/F, i.e. $\alpha \in H^2(K/F)$. Then we have

$$\mathrm{res}_{F,E}\,\alpha = \mathrm{res}_{(L,F),(L,E)}\,\inf_{(K/F),(L/F)}\,\alpha \in H^2(L/E) \subset H^2(*/E),$$

for any L containing E which is normal over F.

A symbolic transfer map can also be defined which maps $H^2(*/E)$ into $H^2(*/F)$ for any $F \subset E$. As in the case of restriction, it is induced by the ordinary transfers in finite layers, whose consistency is guaranteed by the fact that the following diagram is commutative:

$$\begin{tikzcd} H^2(K/F) \arrow[r, "\inf"] & H^2(L/F) \\ H^2(K/E) \arrow[r, "\inf"] \arrow[u, "V"] & H^2(L/E). \arrow[u, "V"'] \end{tikzcd}$$

We denote the resulting symbolic transfer by

$$V_{E,F}\colon H^2(*/E) \to H^2(*/F); \qquad F \subset E.$$

The important relation "transfer of restriction = multiplication by degree", i.e.

$$V_{E,F}\,\mathrm{res}_{F,E}\,\alpha = [E:F]\alpha,$$

holds for these symbolic maps because it holds in the finite layers.

Finally of course we have the isomorphisms between Brauer groups which are induced by conjugation with an element $\sigma \in G$:

$$\sigma_*\colon H^2(*/F) \to H^2(*/F^\sigma)$$

where F^σ is defined by $G_{(F^\sigma)} = (G_F)^\sigma$. We leave their construction to the reader.

3. Class Formations; Method of Establishing Axioms

In global class field theory the formation of idèle classes plays a central role, analogous to that of the field formation in local class field theory. If one analyzes the features common to these two formations one sees that they are both special cases of an abstract algebraic structure in which the constructions peculiar to class

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field theory can be carried out. We call this structure a *class formation*; it is a field formation which satisfies the following additional axiom

AXIOM II. For each field F, there is given an isomorphism $\alpha \to \operatorname{inv}_F \alpha$ of the Brauer group $H^2(*/F)$ into the group $\mathbb{Q}/\mathbb{Z}$ of rationals mod 1, which satisfies the following conditions:

a) If K/F is a normal layer of degree n, then inv_F, maps $H^2(K/F)$ onto the subgroup $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$ of $\mathbb{Q}/\mathbb{Z}$.

b) For any layer E/F of degree n we have

$$\operatorname{inv}_E \operatorname{res}_{F,E} = n \operatorname{inv}_F.$$

The rational number mod 1 which we denote by $\operatorname{inv}_F \alpha$ is called the *invariant* of the element $\alpha \in H^2(*/F)$. Since inv_F is by definition an isomorphism into, these invariants characterize the elements of the Brauer group uniquely.

The most difficult part of class field theory consists in proving that the formations of local fields and of global idèle classes are class formations. In the next few paragraphs, we shall outline the method of proof that is used, carrying out here the reduction steps which can be done abstractly. In the later sections of this chapter, we will assume given an abstract class formation and derive consequences from that assumption.

Notice that in the proof we are about to outline we do not assume at the beginning that our formation is a field formation. The proofs of Axioms I and II are intermingled, at least in the global theory. One first proves that the formation in question satisfies

AXIOM $0'$. In each cyclic layer of prime degree, the Herbrand quotient $h_{2/1}$ is defined and is equal to the degree.

The proof of Axiom $0'$ involves subtle arithmetical arguments, especially in the global case, but one has the advantage that the layers involved are of the simplest possible type, and that the Herbrand quotient has especially simple multiplicative properties with respect to changes in the coefficient module (cf. q.1, q.2, q.3 in the first chapter of the book).

Axiom $0'$ states that if K/F is a cyclic layer of prime degree, then

$$h_2(K/F) = [K : F]h_1(K/F),$$

where $h_r =$ "order of H^r". Thus, when it is proved, one has at one's disposal the "first inequality":

$$[K : F] \mid h_2(K/F)$$

in all cyclic layers of prime degree; and one knows that the necessary and sufficient condition for the equality

$$h_2(K/F) = [K : F]$$

to hold in these layers is given by Axiom I$'$ of the preceding section, i.e. $H^1(K/F) = 0$ for cyclic layers of prime degree. On the other hand, another necessary and sufficient condition is obviously given by

AXIOM I$''$. The "second inequality"

$$h_2(K/F) \mid [K : F]$$

holds in all cyclic layers of prime degree.

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Therefore one proves next *either* Axiom I′ *or* Axiom I″, whichever is more convenient. In the local theory Axiom I′ is immediate because the formation is that of the non-zero field elements and is therefore a field formation. But in the global theory, where one is dealing with the formation of idèle-classes, no direct proof of Axiom I′ is known. Instead, one proves Axiom I″, making heavy use of the first inequality and of Kummer theory. Here one again has the advantage that the layers involved are of the simplest possible type—cyclic of prime degree. Furthermore, in such a layer, the cohomology group in question, H^2, is isomorphic to H^0, which has a simple down-to-earth description: elements in the ground level modulo norms from the top level.

At this stage one knows that the formation is a field formation because, as we have seen in the preceding section, Axiom I′ implies Axiom I. Hence, the inflation-restriction sequence is exact in dimension 2, and we can use the case $r = 2$, $\nu = 1$ of Lemma 1 of §2 to show that the "second inequality"

$$\text{order of } H^2(K/F) \mid [K : F]$$

holds in *all* normal layers K/F, since we know by Axiom I″ that it holds in cyclic layers of prime degree.

Finally, by investigating the 2-dimensional cohomology groups of some special layers (corresponding to unramified extensions locally and cyclotomic extensions globally) one proves that the formation satisfies

Axiom II′. For each field F, there is a subgroup $\bar{H}^2(*/F)$ of the Brauer group $H^2(*/F)$, and an isomorphism $\overline{\mathrm{inv}}_F$ of this subgroup into $\mathbb{Q}/\mathbb{Z}$ such that:

a) For any layer E/F we have $\mathrm{res}_{F,E}\, \bar{H}^2(*/F) \subset \bar{H}^2(*/E)$, and

$$\overline{\mathrm{inv}}_E\, \mathrm{res}_{F,E}\, \alpha = [E : F]\, \overline{\mathrm{inv}}_F\, \alpha$$

for all $\alpha \in \bar{H}^2(*/F)$.

b) If there exists a layer E/F of degree n, then $\bar{H}^2(*/F)$ contains a subgroup which is cyclic of order n.

The point is that we now have enough information to prove that the subgroup $\bar{H}^2(*/F)$ must in fact be the whole Brauer group over F. To do this we have simply to show, for each normal layer K/F, that $H^2(K/F)$ is contained in $\bar{H}^2(*/F)$, because the Brauer group is the union of the groups $H^2(K/F)$. Let $n = [K : F]$ be the degree of the layer in question. According to II′b, there exists a subgroup T of $\bar{H}^2(*/F)$ which is cyclic of order n. Using II′a we find

$$\overline{\mathrm{inv}}_K\, \mathrm{res}_{F,K}\, T = n\, \overline{\mathrm{inv}}_F\, T = \overline{\mathrm{inv}}_F\, nT = \overline{\mathrm{inv}}_F\, 0 = 0.$$

Since $\overline{\mathrm{inv}}_K$ is an isomorphism into by hypothesis, it follows that $\mathrm{res}_{F,K}\, T = 0$, and the exactness of the symbolic inflation-restriction sequence

$$0 \to H^2(K/F) \to H^2(*/F) \xrightarrow{\mathrm{res}_{F,K}} H^2(*/K)$$

allows us to conclude $T \subset H^2(K/F)$. Recalling the second inequality, which states that the order of $H^2(K/F)$ divides the degree n, which is the order of T, we see that $H^2(K/F) = T$ and is therefore contained in $\bar{H}^2(*/F)$ as contended. Furthermore, its image under $\overline{\mathrm{inv}}_F$ is

$$\overline{\mathrm{inv}}_F\, H^2(K/F) = \overline{\mathrm{inv}}_F(T) = \frac{1}{n}\mathbb{Z}/\mathbb{Z}$$

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because $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$ is the only cyclic subgroup of order n of $\mathbb{Q}/\mathbb{Z}$. Thus the formation, together with the map $\overline{\mathrm{inv}}_F$, satisfies Axiom II. This concludes our discussion of how Axioms I and II follow from the seemingly weaker Axioms $0'$, I$'$ or I$''$, and II$'$.

For the remainder of this chapter we assume that our formation $(G, \{G_F\}, A)$ is a class-formation and discuss the rather remarkable consequences of this assumption. According to Axiom II a), the 2-dimensional cohomology group of any normal layer K/F of degree n is cyclic of order n, being in fact canonically isomorphic to the group $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$ under the isomorphism inv_F. Any rational number t which can be written with denominator n determines a unique element $\alpha \in H^2(K/F)$ such that $\mathrm{inv}_F\,\alpha \equiv t \pmod{\mathbb{Z}}$. This α is called the cohomology class with invariant t. If we are working with a complex X for the galois group $G_{K/F}$ of the layer, and $f\colon X_2 \to A_K$ is a cocycle in the class α, we also say that f is a cocycle with invariant t. The class with invariant $1/n$ has period n and generates $H^2(K/F)$. It is called the *canonical class*, or the *fundamental class* of the layer K/F. A cocycle f representing the fundamental class is called a fundamental 2-cocycle.

PROPOSITION 4. *Let $F \subset E \subset K$, with K/F normal. Then the restriction of the fundamental class of the layer K/F is the fundamental class of the layer K/E.*

PROOF. Let $n = [K : F]$ and $m = [K : E]$. Then $[E : F] = n/m$, and if α is the fundamental class of K/F we have by Axiom IIb:

$$\mathrm{inv}_E\,\mathrm{res}_{F,E}\,\alpha = \frac{m}{n}\,\mathrm{inv}_F\,\alpha \equiv \frac{n}{m}\cdot\frac{1}{n} \equiv \frac{1}{m} \pmod{\mathbb{Z}}. \qquad \square$$

Since the fundamental class of a layer generates its two dimensional cohomology group, the following corollary is immediate:

COROLLARY 1. *Let $F \subset E \subset K$, with K/F normal. Then the restriction maps $H^2(K/F)$ onto $H^2(K/E)$.*

Keeping F and E fixed, letting K vary, and remembering that $H^2(*/E) = \bigcup_K H^2(K/E)$ we see also

COROLLARY 2. *Let E/F be an arbitrary layer of a class formation. Then the symbolic restriction maps $H^2(*/F)$ onto $H^2(*/E)$.*

Corollary 2 is an important fact. For one thing, it shows that the isomorphisms inv_E for all fields $E \supset F$ are determined by the single isomorphism inv_F through the equation

$$\mathrm{inv}_E\,\mathrm{res}_{F,E} = [E : F]\,\mathrm{inv}_F\,.$$

Furthermore, the onto-ness of the symbolic restriction allows us to prove

PROPOSITION 5. *Let E/F be an arbitrary layer. Then*

1) *The symbolic transfer maps $H^2(*/E)$ isomorphically into $H^2(*/F)$, preserving invariants in the following sense*

$$\mathrm{inv}_F\,V_{E,F}\alpha = \mathrm{inv}_E\,\alpha, \qquad \alpha \in H^2(*/E).$$

2) *Conjugation by an element $\sigma \in G_F$ preserves invariants in the following sense*

$$\mathrm{inv}_{E^\sigma}(\sigma_*\alpha) = \mathrm{inv}_E\,\alpha, \qquad \alpha \in H^2(*/E).$$

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PROOF. 1) The homomorphism computation

$$\operatorname{inv}_F V_{E,F} \operatorname{res}_{F,E} = \operatorname{inv}_F [E:F] = [E:F] \operatorname{inv}_F = \operatorname{inv}_E \operatorname{res}_{F,E}$$

shows that

$$\operatorname{inv}_F V_{E,F} = \operatorname{inv}_E$$

because $\operatorname{res}_{F,E}$ is onto. Hence the transfer preserves invariants and is therefore an isomorphism into.

2) An element $\sigma \in G_F$ acts as identity on $H^2(K/F)$ for all normal K/F, hence also on $H^2(*/F)$. Using these facts we make the computation

$$\begin{aligned} \operatorname{inv}_{E^\sigma} \sigma_* \operatorname{res}_{F,E} &= \operatorname{inv}_{E^\sigma} \operatorname{res}_{F^\sigma,E^\sigma} \sigma_* \\ &= [E^\sigma : F^\sigma] \operatorname{inv}_{F^\sigma} \sigma_* \\ &= [E:F] \operatorname{inv}_F \\ &= \operatorname{inv}_E \operatorname{res}_{F,E} . \end{aligned}$$

This shows $\operatorname{inv}_{E^\sigma} \sigma_* = \operatorname{inv}_E$ because $\operatorname{res}_{F,E}$ is *onto.* □

4. The Main Theorem

The main fact about the cohomology of class formations is that the cohomology groups of a normal layer K/F depend only on the structure of the galois group $G_{K/F}$.

THEOREM 1 (Main Theorem). *Let K/F be any normal layer in a class formation. Let α be the fundamental class of the layer, i.e. the canonical generator of $H^2(K/F) = H^2(G_{K/F}, A_K)$. Consider the cup products associated with the natural pairing $A_K \times \mathbb{Z} \to A_K$. For each integer q, let α_q denote the map $\zeta \mapsto \alpha\zeta$ which is obtained by multiplying the variable element $\zeta \in H^q(G_{K/F}, \mathbb{Z})$ with α. We contend that these maps are* isomorphisms onto *for all q, $-\infty < q < \infty$:*

$$\alpha_q \colon H^q(G_{K/F}, \mathbb{Z}) \approx H^{q+2}(G_{K/F}, A_K) = H^{q+2}(K/F).$$

PROOF. We use an abstract cohomological theorem that was designed for just this purpose.[2] Each subgroup $G' \subset G_{K/F}$, is of the form $G' = G_{K/F'}$ for some field F', $F \subset F' \subset K$, by Proposition 2 of §1. If α' denotes the fundamental class of the layer K/F', then, by Proposition 4 of § 3, we have $\alpha' = \operatorname{res} \alpha$. Let $\alpha'_q \colon H^q(G_{K/F'}, \mathbb{Z}) \to H^{q+2}(G_{K/F'}, A_K)$ be the map obtained by multiplication with α'. Consider these maps in the three successive dimensions -1, 0, $+1$:

$$\begin{aligned} \alpha'_{-1} &\colon H^{-1}(G_{K/F'}, \mathbb{Z}) \to H^1(G_{K/F'}, A_K) = H^1(K/F') \\ \alpha'_0 &\colon H^0(G_{K/F'}, \mathbb{Z}) \to H^2(G_{K/F'}, A_K) = H^2(K/F') \\ \alpha'_1 &\colon H^1(G_{K/F'}, \mathbb{Z}) \to H^3(G_{K/F'}, A_K) = H^3(K/F'). \end{aligned}$$

In order to establish the hypotheses of the cohomological theorem, we must show that α'_{-1} is onto, α'_0 is an isomorphism onto, and α'_1 is an isomorphism into. Then we can conclude from that theorem that α'_q is an isomorphism onto for all q and our present theorem will be proved.

α'_{-1} is obviously onto, because $H^1(K/F') = 0$ by Axiom I. To show that α'_0 is an isomorphism onto we have only to point out that $H^0(G_{K/F'}, \mathbb{Z})$ is cyclic of

[2] For a very general version of this theorem, see [**21**, Theorem B in Chapter IXZ, § 8]. See also Section 2 of our chapter "Preliminaries", where there is a simple version, Theorem A, and sketch of the proof.

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order $(G_{K/F'} : 1) = [K : F']$ generated by $\varkappa 1$, $H^2(K/F')$ is cyclic of the same order generated by α', and $\alpha'_0(\varkappa 1) = \alpha'(\varkappa 1) = 1_*\alpha' = \alpha'$. Finally, α' is automatically an isomorphism into because $H^1(G_{K/F'}, \mathbb{Z}) = 0$. This concludes the proof of the main theorem. $\square$

The canonical isomorphisms which we have just established commute with restriction and transfer, according to

THEOREM 2. *Let $F \subset F' \subset K$, with K/F normal. Let α be the canonical class of the layer K/F, and α' that of K/F'. Then each pair of vertical arrows makes the following diagram commutative*:

$$\begin{tikzcd} H^q(G_{K/F}, \mathbb{Z}) \arrow[r, "\alpha"] \arrow[d, shift right, "\mathrm{res}"'] & H^{q+2}(K/F) \arrow[d, shift right, "\mathrm{res}"'] \\ H^q(G_{K/F'}, \mathbb{Z}) \arrow[r, "\alpha'"] \arrow[u, shift right, "V"'] & H^{q+2}(K/F'). \arrow[u, shift right, "V"'] \end{tikzcd}$$

PROOF. Recalling that $\alpha' = \mathrm{res}\,\alpha$, we see that this theorem is an immediate consequence of the identities relating products and restriction and transfer. Namely, for any $\zeta \in H^q(G_{K/F'}, \mathbb{Z})$

$$\alpha'(\mathrm{res}\,\zeta) = (\mathrm{res}\,\alpha)(\mathrm{res}\,\zeta) = \mathrm{res}(\alpha\zeta),$$

and for any $\zeta' \in H^q(G_{K/F'}, \mathbb{Z})$

$$\alpha(V\zeta') = V((\mathrm{res}\,\alpha)\zeta') = V(\alpha'\zeta'). \qquad \square$$

The isomorphisms of the main theorem commute with conjugation, but since this commutativity is a special case of their commutativity with the isomorphisms induced by isomorphisms between class formations, we digress for a moment to discuss this latter concept. If $(G, \{G_F\}, A)$ and $(G', G'_{F'}, A')$ are two formations, it is natural to define an isomorphism of one onto the other to be a pair of isomorphisms

$$\lambda\colon G' \approx G \qquad f\colon A \approx A'$$

such that $f((\lambda\sigma')a) = \sigma'(fa)$, and such that λ induces a one-to-one correspondence between the subgroups $\{G'_{F'}\}$ and the subgroups $\{G_F\}$. Then for each field F' we can define $\lambda F'$ by $G_{(\lambda F')} = \lambda(G'_{F'})$, and it is clear that $f(A_{\lambda F'}) = A'_{F'}$, $G_{\lambda K'/\lambda F'} = \lambda(G_{K'/F'})$, etc. Consequently, for each normal layer K'/F' there is an isomorphism

$$(\lambda, f)_*\colon H^r(\lambda K'/\lambda F') \approx H^r(K'/F')$$

induced by the isomorphism of pairs

$$\lambda\colon G'_{K'/F'} \approx G_{\lambda K'/\lambda F'} \qquad f\colon A_{\lambda K'} \approx A'_{K'}.$$

If one formation is a field formation, then so is the other, and the isomorphisms $(\lambda, f)_*$ on the 2-dimensional groups of the layers induce isomorphisms of the Brauer groups. If the formations are class formations, then one would also require that these isomorphisms between Brauer groups also preserve the invariants. When this is the case, it is clear that the two class formations are essentially the "same" algebraic structure and that any construction carried out in one could be carried out in

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the other with the "same" result. In particular, the isomorphisms constructed in the main theorem would correspond, i.e. the following diagram would be commutative

$$\begin{tikzcd} H^q(G_{\lambda K'/\lambda F'}, \mathbb{Z}) \arrow[r] \arrow[d, "(\lambda,1)_*"'] & H^{q+2}(\lambda K'/\lambda F') \arrow[d, "(\lambda,f)_*"'] \\ H^q(G'_{K'/F'}, \mathbb{Z}) \arrow[r] & H^{q+2}(K'/F'). \end{tikzcd}$$

If $(G, \{G_F\}, A)$ is any class formation, and $\tau \in G$, then the "inner automorphism"

$$\tau^{-1} \colon \sigma^\tau = \tau\sigma\tau^{-1} \to \tau \qquad \tau \colon a \to \tau a$$

is an automorphism of it in the sense we have described, because by Proposition 5 of §3 we know that conjugation preserves invariants. Thus we have proved

THEOREM 3. *Let $\tau \in G$, and let K/F be a normal layer. Let α be the fundamental class of the layer K/F. Then $\tau_*\alpha$ is the fundamental class of $\tau K/\tau F$ and the following diagram is commutative*:

$$\begin{tikzcd} H^q(G_{K/F}, \mathbb{Z}) \arrow[r, "\alpha"] \arrow[d, "\tau_*"'] & H^{q+2}(K/F) \arrow[d, "\tau_*"'] \\ H^q(G_{\tau K/\tau F}, \mathbb{Z}) \arrow[r, "\tau_*\alpha"] & H^{q+2}(\tau K/\tau F). \end{tikzcd}$$

The isomorphisms of the main theorem do *not* commute with inflation in positive dimensions. The correct rule is given by

THEOREM 4. *Let $F \subset K \subset L$ with K/F and L/F normal. Let $\alpha_{K/F}$ and $\alpha_{L/F}$ be the canonical classes of the respective layers. Then the following diagram is commutative for $q > 0$*:

$$\begin{tikzcd} H^q(G_{K/F}, \mathbb{Z}) \arrow[r, "\alpha_{K/F}"] \arrow[d, "{[L:K]\inf}"'] & H^{q+2}(K/F) \arrow[d, "\inf"'] \\ H^q(G_{L/F}, \mathbb{Z}) \arrow[r, "\alpha_{L/F}"] & H^{q+2}(L/F). \end{tikzcd}$$

PROOF. The theorem is an immediate consequence of the general formula $(\inf \alpha) \cup (\inf \beta) = \inf(\alpha \cup \beta)$. This rule can be proved by a dimension shifting induction, and it is also an immediate consequence of the well known formula for the cup product of standard cochains, in positive dimensions, namely

$$(f \cup g)(\sigma_1, \sigma_2, \ldots, \sigma_{p+q}) = f(\sigma_1, \sigma_2, \ldots, \sigma_p) \cup (\sigma_1, \ldots, \sigma_p) g(\sigma_{p+1}, \ldots, \sigma_{p+q}).$$

Granting this rule, the proof is immediate; we have only to observe that the inflation of $\alpha_{K/F}$ is $[L : K]\alpha_{L/F}$. Namely, these two classes have the same invariant, because

$$\frac{1}{[K : F]} = \frac{[L : K]}{[L : F]},$$

and inflation preserves invariants. (It is the inclusion in the Brauer Group!) Hence, for any $\zeta \in H^q(G_{K/F}, \mathbb{Z})$ we have

$$\begin{aligned} \inf(\alpha_{K/F}\zeta) &= (\inf \alpha_{K/F})(\inf \zeta) = ([L : K]\alpha_{L/F})(\inf \zeta) \\ &= \alpha_{L/F}([L : K] \inf \zeta) \end{aligned}$$

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and this was our contention. □

In dimensions $r > 2$, the inflation map in a class formation is very weak. For example we have

COROLLARY. *If $F \subset K \subset L$, and if the degree $[K : F]$ divides the degree $[L : K]$, then the inflation from K/F to L/F is the zero map in dimensions $r > 2$.*

PROOF. Since the horizontal arrows of the commutative diagram of the preceding theorem are isomorphisms onto, we need only show that the left hand vertical map, $[L : K]\inf$, is the zero map. But this follows from the fact that it is applied to the group $H^r(G_{K/F}, \mathbb{Z})$, in which every element has an order dividing $(G_{K/F} : 1) = [K : F]$. □

Combining the down-to-earth interpretations of the cohomology groups in low dimensions with the isomorphism of the main theorem we obtain the following special results:

$$\begin{aligned}
H^4(K/F) &\approx H^2(G_{K/F}, \mathbb{Z}) \approx \widehat{G}_{K/F} \\
H^3(K/F) &\approx H^1(G_{K/F}, \mathbb{Z}) = 0 \\
H^2(K/F) &\approx H^0(G_{K/F}, \mathbb{Z}) \approx \mathbb{Z}/n\mathbb{Z} \\
H^1(K/F) &\approx H^{-1}(G_{K/F}, \mathbb{Z}) = 0 \\
(A_F/N_{K/F}A_K) \approx H^0(K/F) &\approx H^{-2}(G_{K/F}, \mathbb{Z}) \approx (G_{K/F}/G^c_{K/F}) \\
((A_K)_{N_{K/F}}/IA_K) \approx H^{-1}(K/F) &\approx H^{-3}(G_{K/F}, \mathbb{Z}).
\end{aligned}$$

By far the most important of these special cases is the next to the last. It is the so-called reciprocity law isomorphism of class field theory, and the whole of the next section is devoted to a detailed study of its consequences. The following exercise, with which we close this section, concerns the case of $H^4(K/F)$.

Exercise

Using the isomorphism $\delta\colon H^1(\mathbb{Q}/\mathbb{Z}) \approx H^2(\mathbb{Z})$ we see that the elements of $H^4(K/F)$ are of the form $\alpha \cdot \delta\chi$, where $\chi \in H^1(G_{K/F}, \mathbb{Q}/\mathbb{Z})$ and α is the fundamental class of the layer K/F. If $\chi(\sigma)$ is the standard 1-cocycle representing χ, then the map $\sigma \to \chi(\sigma)$ is a character of G, and we may identify χ with this character as we have discussed. Thus the correspondence

$$\chi \leftrightarrow \alpha \cdot \delta\chi$$

is an isomorphism between $\widehat{G}_{K/F}$, the character group of the galois group of the layer K/F, and $H^4(K/F)$, the four dimensional cohomology group of the layer.

a) Let $F \subset E \subset K$ with K/F normal. Show that the restriction from $H^4(K/F)$ to $H^4(K/E)$ corresponds to restricting the character χ from $G_{K/F}$, to the subgroup $G_{K/E}$; and show that the transfer from $H^4(K/E)$ to $H^4(K/F)$ corresponds to the map of characters of the subgroup $G_{K/E}$ into characters of the big group $G_{K/F}$, obtained by composing them by the group theoretical transfer.

b) Notice that the restriction and transfer are weak maps in dimension 4; for example, if $F \subset E \subset K$, and the layer E/F contains the maximal abelian sublayer of K/F, then both maps are the zero map. (For the transfer statement, use the principal ideal theorem.)

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c) Let $F \subset K \subset L$ with K/F and L/F normal. Show that the inflation map from $H^4(K/F)$ to $H^4(L/F)$ corresponds to the procedure of viewing a character of the factor group $G_{K/F}$ as a character of the big group $G_{L/F}$, *and* raising it into the $[L:K]$-th power. (Use Theorem 4.)

d) Given a standard 4-cocycle $f = f(\sigma_1, \sigma_2, \sigma_3, \sigma_4)$ of $G_{K/F}$ in A_K, representing a class $\beta \in H^4(K/F)$, show that the corresponding character χ is given by

$$\chi(\tau) = \mathrm{inv}_F(f * \tau),$$

where $f * \tau$ is the standard 2-cocycle defined by

$$(f * \tau)(\sigma_1, \sigma_2) = \sum_\rho f(\sigma_1, \sigma_2, \rho, \tau).$$

5. The Reciprocity Law Isomorphism

In order that our notation correspond as closely as possible to that which will be used in applications, we will from now on write the formation module A multiplicatively. The effect of σ on a is then denoted by a^σ. If E/F is an arbitrary layer, the corresponding norm homomorphism $N_{E/F}\colon A_E \to A_F$ is defined by $N_{E/F}a = \prod_i a^{\sigma_i}$, where the elements σ_i are representatives of the left cosets of G_E in G_F: $G_F = \bigcup_i \sigma_i G_E$. It is the multiplicative analog of the trace; if $F \subset E \subset K$ and K/F is normal, then the map $N_{E/F}$ is what we have previously called the trace from the subgroup $G_{K/E}$ to the big group $G_{K/F}$.

For each normal layer K/F, the main theorem gives us a natural isomorphism

$$H^0(K/F) \approx H^{-2}(G_{K/F}, \mathbb{Z}).$$

Both of these groups have down-to-earth interpretations. $H^0(K/F)$ is isomorphic to $A_F/N_{K/F}A_K$, the factor group of elements in the ground level modulo norms from the top level; the isomorphism is induced by the map $\varkappa\colon A_F \to H^0(K/F)$ which is onto and has kernel $N_{K/F}A_K$. On the other hand, $H^{-2}(G_{K/F}, \mathbb{Z})$ is naturally isomorphic to the factor commutator group $G_{K/F}/G^c_{K/F}$ of the galois group of the layer K/F. The isomorphism is induced by the homomorphism[3] $\sigma \to \zeta_\sigma = \delta^{-1}\lambda\mathrm{K}(\sigma - 1)$ which is onto and has kernel $G^c_{K/F}$. The special case of the main theorem mentioned above may therefore be interpreted as an isomorphism

$$A_F/N_{K/F}A_K \approx G_{K/F}/G^c_{K/F}.$$

This is known as the reciprocity law isomorphism, for historical reasons. Being an isomorphism between a factor group of A_F and a factor group of $G_{K/F}$, it is induced by a multivalued correspondence

$$a \leftrightarrow \sigma, \qquad a \in A_F,\ \sigma \in G_{K/F}$$

between those two groups. Then, by definition, we have

$$a \leftrightarrow \sigma \quad \text{if and only if} \quad \varkappa a = \alpha \cdot \zeta_\sigma,$$

where α is the fundamental 2-dimensional class of the layer K/F. A dual description is given by

[3]Here $\delta : H^{-2}(G, \mathbb{Z}) \xrightarrow{\sim} H^{-1}(G, I)$is the connecting homomorphism of the exact sequence

$$0 \to I_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0.$$

We have $H^{-1}(G, T) \approx I_G/I_G^2$, and as is well known, the map $\sigma \mapsto \sigma - 1 \pmod{I}^2$ induces an isomorphism $G/G^c \approx I_G/I_G^2$.

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PROPOSITION 6. *Let $a \in A_F$ and $\sigma \in G_{K/F}$. Then $a \leftrightarrow \sigma$ if and only if*

$$(*) \qquad \mathrm{inv}_F(\varkappa a \cdot \delta\chi) = \chi(\sigma)$$

for all characters χ of $G_{K/F}$. (On the left hand side of this equation, the character χ is to be interpreted as an element of $H^1(G, \mathbb{Q}/\mathbb{Z})$ in the usual way, and $\delta\chi$ is the corresponding element of $H^2(G, \mathbb{Z})$.)

PROOF. Let α be the fundamental class for the layer K/F. If $\alpha \leftrightarrow \sigma$, then

$$\mathrm{inv}_F(ka \cdot \delta\chi) = \mathrm{inv}_F((\alpha \cdot \zeta_\sigma) \cdot \delta\chi) = \mathrm{inv}_F(\alpha \cdot (\zeta_\sigma \cdot \delta\chi)) = \frac{1}{n}\delta(\zeta_\sigma \cdot \chi) = \chi(\sigma).$$

This proves the proposition because the σ's to which a corresponds are characterized by the values $\chi(\sigma)$ for variable χ (see also pages 4–5). □

Our next theorem concerns the commutativity of the reciprocity law isomorphism with various natural mappings between different layers.

THEOREM 5. *Using the symbol $\updownarrow_{(K/F)}$ to denote the many valued correspondence inducing the reciprocity law isomorphism in the layer K/F, we assert that the following diagrams are commutative*:

a)
$$\begin{tikzcd}
A_F \arrow[r, "\text{inclusion}"] \arrow[d, leftrightarrow, "(K/F)"'] & A_E \arrow[d, leftrightarrow, "(K/E)"] \\
(G_{K/F}/G^c_{K/F}) \arrow[r, "\text{group theoretical transfer}"] & (G_{K/E}/G^c_{K/E})
\end{tikzcd} \qquad (F \subset E \subset K)$$

b)
$$\begin{tikzcd}
A_F \arrow[d, leftrightarrow, "(K/F)"'] & A_E \arrow[l, "N_{E/F}"'] \arrow[d, leftrightarrow, "(K/E)"] \\
G_{K/F} & G_{K/E} \arrow[l, "\text{inclusion}"']
\end{tikzcd} \qquad (F \subset E \subset K)$$

c)
$$\begin{tikzcd}
A_F \arrow[r, "\tau"] \arrow[d, leftrightarrow, "(K/F)"'] & A_{F^\tau} \arrow[d, leftrightarrow, "(K^\tau/F^\tau)"] \\
G_{K/F} \arrow[r, "\text{conjugation by } \tau"] & G_{K^\tau/F^\tau}
\end{tikzcd} \qquad (F \subset K,\ \tau \in G)$$

d)
$$\begin{tikzcd}
A_F \arrow[r, "\text{identity}"] \arrow[d, leftrightarrow, "(L/F)"'] & A_F \arrow[d, leftrightarrow, "(K/F)"] \\
G_{L/F} \arrow[r, "\text{natural hom. onto factor group}"] & G_{K/F}
\end{tikzcd} \qquad (F \subset K \subset L)$$

PROOF. To prove a) and b) we use Theorem 2 of §4 which states that the fundamental isomorphisms of the main theorem commute with restriction and transfer. We have only to recall how these homomorphisms are obtained explicitly in case of $H^0(A_K)$ and $H^{-2}(\mathbb{Z})$. The inclusion map $A_F \to A_E$ induces the restriction from $H^0(K/F)$ to $H^0(K/E)$; while the norm map from A_E to A_F induces the transfer from $H^0(K/E)$ to $H^0(K/F)$. On the other hand, the restriction from $H^{-2}(G_{K/F}, \mathbb{Z})$ to $H^{-2}(G_{K/E}, \mathbb{Z})$ is induced by the group theoretical transfer from $G_{K/F}$ to $G_{K/E}$, while the cohomological transfer in the other direction is induced by the inclusion of $G_{K/E}$ in $G_{K/F}$.

Statement c) follows similarly from Theorem 3 of § 4.

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For d) we must use another method because we have not introduced cohomology maps corresponding to the natural map of $A_F/N_{L/F}A_L$ onto $A_F/N_{K/F}A_K$, and to the natural map of $G_{L/F}$ onto the factor group $G_{K/F}$. We use the duality criterion of Proposition 6, which states that for $a \in A_F$ and $\sigma \in G_{L/F}$ we have $a \overset{(L/F)}{\longleftrightarrow} \sigma$ if and only if

$$(*) \qquad \chi(\sigma) = \mathrm{inv}_F(\varkappa a \cdot \delta\chi)$$

for all characters χ of $G_{L/F}$. We must therefore show that if $(*)$ holds for all χ, then

$$(**) \qquad \psi(\sigma G_{L/K}) = \mathrm{inv}_F(\varkappa a \cdot \delta\psi)$$

for each character ψ of $G_{K/F} = G_{L/F}/G_{L/K}$. To this effect, we let $\chi = \inf_{G_{K/F}, G_{L/F}} \psi$. Then χ is the character of $G_{L/F}$ defined by $\chi(\sigma) = \psi(\sigma G_{L/K})$. Of course we have $\inf \delta\psi = \delta\chi$ because inflation commutes with coboundaries. Hence

$$\psi(\sigma G_{L/K}) = \chi(\sigma) = \mathrm{inv}_F(\varkappa a \cdot \delta\chi) = \mathrm{inv}_F(\varkappa a \cdot \inf \delta\psi),$$

and since the cup product with the 0-dimensional class $\varkappa a$ is given by an induced map (Theorem 10, §3, Ch. V), and therefore commutes with inflation, we can write this as

$$\psi(\sigma G_{L/K}) = \mathrm{inv}_F(\inf(\varkappa a \cdot \delta\psi)) = \mathrm{inv}_F(\varkappa a \cdot \delta\psi).$$

(The last equality follows from the fact that we have $\inf(\varkappa a \cdot \delta\psi) = \varkappa a \cdot \delta\psi$ by the very definition of the Brauer group.) This concludes the proof of Theorem 5. $\square$

The reciprocity law isomorphism is of course induced by a homomorphism of A_F onto $G_{K/F}/G^c_{K/F}$ with kernel $N_{K/F}A_K$. This homomorphism is called the norm-residue map and is denoted by

$$a \to (a, K/F), \qquad a \in A_F.$$

The symbol $(a, K/F)$ is called the norm-residue symbol and is of historical origin. The reason for the name is that the value of $(a, K/F)$ determines the residue class of a modulo norms from K, i.e. modulo $N_{K/F}A_K$. Translating Proposition 6 and Theorem 6 into properties of this symbol we obtain:

Proposition 6′. *Let K/F be a normal layer and let $a \in A_F$. Then $(a, K/F)$ is the uniquely determined element of $G_{K/F}/G^c_{K/F}$ such that*

$$\mathrm{inv}_F(\varkappa a \cdot \delta\chi) = \chi(a, K/F)$$

for all characters χ of $G_{K/F}$.

Theorem 6. a) *If $F \subset E \subset K$, then for $a \in A_F$ we have*

$$(a, K/E) = V_{G_{K/F}, G_{K/E}} \qquad (a, K/F).$$

b) *If $F \subset E \subset K$, then for $a \in A_E$ we have*

$$(N_{E/F}a, K/F) = (a, K/E) \cdot G^c_{K/F}.$$

c) *If $F \subset K$ and $\tau \in G$, then for $a \in A_F$ we have*

$$(a^\tau, K^\tau/F^\tau) = (a, K/F)^\tau.$$

d) *If $F \subset K \subset L$, and $a \in A_F$, then*

$$(a, K/F) = (a, L/F) \cdot G_{L/K}.$$

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PROOF. These identities are simply restatements of the commutativities asserted in Theorem 5. □

The kernel of the norm residue homomorphism $a \to (a, K/F)$ is the subgroup $N_{K/F}A_K$ of A_F. We will call such a subgroup, and more generally, a subgroup of the form $N_{E/F}A_E$, for an arbitrary extension E/F, normal or not, a *norm subgroup* of A_F. Clearly, if $F \subset E_1 \subset E_2$, then $N_{E_2/F}A_{E_2} \subset N_{E_1/F}A_{E_1}$ because $N_{E_2/F} = N_{E_1/F}N_{E_2/E_1}$. Therefore, the map $E \to N_{E/F}A_E$ is a lattice-inverting map of the set of all extensions E of a fixed field F onto the set of all norm subgroups of A_F. We propose now to discuss the features of this correspondence in detail.

PROPOSITION 7. *Let $F \subset E \subset K$ with K/F normal. Then for $a \in A_F$ we have*

$$a \in N_{E/F}A_E \Leftrightarrow (a, K/F) \in G_{K/E}G^c_{K/F}.$$

PROOF. If $a \in N_{E/F}A_E$, say $a = N_{E/F}b$, with $b \in A_E$, then by Theorem 6b) we have

$$(a, K/F) = (b, K/E)G^c_{K/F} \in G_{K/E}G^c_{K/F}.$$

On the other hand, if $(a, K/F) \in G_{K/E}G^c_{K/F}$ then there exists an element $b \in A_E$ such that

$$(a, K/F) = (b, K/E)G^c_{K/F} = (N_{E/F}b, K/F)$$

because the map $b \to (b, K/E)$ maps A_E *onto* $G_{K/E}/G^c_{K/E}$. Since the kernel of the map $a \to (a, K/F)$ is $N_{K/F}A_K$ it follows that there exists an element $c \in A_K$ such that

$$a = N_{E/F}b \cdot N_{K/F}c = N_{E/F}(bN_{K/E}c).$$

Hence $a \in N_{E/F}A_E$ as contended. □

If E_1 and E_2 are any two fields of formation, we define their *compositum*, E_1E_2, to be the field which is defined by the property $G_{E_1E_2} = G_{E_1} \cap G_{E_2}$. Then $E_1 \cdot E_2$ contains both E_1 and E_2, and is contained in any field which contains both E_1 and E_2, because $G_{E_1} \cap G_{E_2}$ is contained in both G_{E_1} and G_{E_2}, and contains any subgroup of G which is contained in both G_{E_1} and G_{E_2}.

An extension M/F is called abelian if it is normal and its galois group $G_{M/F}$, is abelian; in other words, if $G_M \supset G^c_F$. From this latter criterion we see that if M_1/F and M_2/F are two abelian extensions, then their compositum M_1M_2/F is also an abelian extension. From this it follows immediately that any extension E/F, normal or not, contains a maximal abelian subextension M/F; that is, there exists a field M such that $F \subset M \subset E$, M/F is abelian, and any field between F and E which is abelian over F is contained in M.

The following theorem concerns the properties of the correspondence between extension fields of F and their norm subgroups.

THEOREM 7. *The norm group of an arbitrary extension E/F is the same as that of its maximal abelian subextension M/F, i.e. $N_{E/F}A_E = N_{M/F}A_M$. If $\{M\}$ is the set of all* abelian *extensions of F, then the correspondence*

$$M \leftrightarrow N_{M/F}A_M$$

is a one-to-one correspondence between $\{M\}$ and the set of all norm subgroups of A_F, having the following properties:

a) $M_1 \subset M_2 \Leftrightarrow N_{M_1/F}A_{M_1} \supset N_{M_2/F}A_{M_2}$;
b) $N_{M_1M_2/F}A_{M_1M_2} = (N_{M_1/F}A_{M_1}) \cap (N_{M_2/F}A_{M_2})$;

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c) $[M : F] = (A_F : N_{M/F}A_M)$.

Finally, every subgroup of A_F which contains a norm subgroup is itself a norm subgroup.

PROOF. Proposition 7 shows that for the extensions E/F which are contained in a given normal extension K/F, the norm group $N_{E/F}A_E$ depends only on the group $G_{K/E}G^c_{K/F}$. This group $G_{K/E}G^c_{K/F}$ is obviously the minimal normal subgroup of $G_{K/F}$ containing $G_{K/E}$ such that the factor group $G_{K/F}/G_{K/E}G^c_{K/F}$ is abelian. Consequently the corresponding intermediate field M (defined by $G_{K/M} = G_{K/E}G^c_{K/F}$) is the maximal abelian subextension of E. Thus, the norm group $N_{E/F}A_E$ depends only on the maximal abelian subextension of E/F; and since the maximal abelian subextension of the maximal abelian subextension is that extension itself, it follows that $N_{E/F}A_E = N_{M/F}A_M$. This proves the first part of the theorem, and shows in addition that every norm subgroup of A_F is a norm group of an abelian extension.

Let M/F be any abelian extension. The intermediate fields M_1, $F \subset M_1 \subset M$, are in one-to-one correspondence with the set of all subgroups G_{M/M_1} of the galois group $G_{M/F}$. Since the norm residue map $a \to (a, M/F)$ maps A_F onto $G_{M/F}$, and has kernel $N_{M/F}A_M$, it follows that the subgroups G_{M/M_1} are in one-to-one correspondence with their inverse images under the norm residue map, and that these inverse images are those subgroups of A_F which contain $N_{M/F}A_M$. On the other hand Proposition 7 shows that the inverse image of G_{M/M_1} is the norm subgroup $N_{M_1/F}A_{M_1}$. Since any two abelian extensions M_1/F and M_2/F are contained in a single abelian extension $M = M_1M_2$, it follows from what we have proved that $M_1 = M_2$ if and only if $N_{M_1/F}A_{M_1} = N_{M_2/F}A_{M_2}$; hence the correspondence between abelian extensions and their norm groups is one-to-one. Furthermore, since any norm subgroup of A_F is the norm group of an abelian extension M, we have shown that any subgroup of A_F which contains a norm subgroup is itself a norm subgroup.

All that remains is to establish properties a), b), and c). Concerning a) we know from the transitivity of the norm that the bigger the extension, the smaller its norm group; and in the case of abelian extensions we can say conversely that the smaller the norm group the bigger the extension, because of the one-one-ness of the correspondence. Now b) follows from a) by simply considering the lattice of norm subgroups of A_F as compared to the lattice of abelian extensions of F. Property c) is just a special consequence of the reciprocity law isomorphism $G_{M/F} \approx A_F/N_{M/F}A_M$. This concludes the proof of Theorem 7. □

If E/F is an arbitrary extension, the index $(A_F : N_{E/F}A_E)$ is called its *norm index*. From Theorem 7 it follows that the norm index of an extension is equal to the degree of its maximal abelian subextension. Hence, the norm index of an extension always divides its degree, and is equal to the degree if and only if the extension is abelian.

For historical reasons the top fields M of *abelian* extensions M/F are called *class fields* over F. According to Theorem 7, the class fields over F are in one-to-one correspondence with their norm groups in A_F. If B is a norm subgroup of A_F, the corresponding class field M over F such that $N_{M/F}A_M = B$ is called the *class field belonging to B*. The galois group $G_{M/F}$ is canonically isomorphic to the

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factor group A_F/B by the reciprocity law. If E/F is an arbitrary extension, then by Theorem 7 we have $M \subset E$ if and only if $B \supset N_{E/F}A_E$.

PROPOSITION 8. *Let B be a norm subgroup of A_F, and let M be the class field over F belonging to B. Then*

a) *For any $\tau \in G$, M^τ is the class field over F^τ belonging to B^τ.*

b) *If E is any field containing F, the group $C = N_{E/F}^{-1}(B)$ is a norm subgroup of A_E and the class field over E belonging to it is the compositum ME.*

PROOF. a) This follows from first principles: M^τ/F^τ is abelian because M/F is, and we have

$$N_{M^\tau/F^\tau}A_{M^\tau} = N_{M^\tau/F^\tau}(A_M^\tau) = (N_{M/F}A_M)^\tau = B^\tau.$$

Hence M^τ is the class field over F^τ belonging to B^τ.

b) Let L denote a variable extension of E. Then, by definition, ME is the smallest L such that $L \supset M$. Since M is the class field over F belonging to B, we have

$$L \supset M \Leftrightarrow N_{L/F}A_L \subset B \Leftrightarrow N_{L/E}A_L \subset C.$$

where $C = N_{E/F}^{-1}(B)$. Thus, ME is the smallest L such that $N_{L/E}A_L \subset C$. This shows in the first place that C is a norm subgroup of A_E, because C contains the norm group of ME; and now it follows that ME is the class field over E belonging to C because that is the smallest extension of E whose norm group is contained in C. □

6. The Abstract Existence Theorem

As we have seen (Theorem 7 of the preceding section), the lattice of abelian extensions M of a given field F is anti-isomorphic to the lattice of norm subgroups of the given ground level A_F. Thus, if we can in some way characterize those subgroups of A_F which are norm subgroups, then we will have gained an insight into the totality of all abelian extensions of F. In class field theory proper, both local and global, it turns out that a very simple characterization of the norm subgroups can be given, in terms of the natural topology in the levels A_F—the norm subgroups turn out to be the *open* subgroups of *finite index* in A_F. The aim of this section is to discuss a set of conditions on an abstract class formation, from which the above-mentioned characterization of norm subgroups follows, and which are relatively easy to check in the case of class field theory proper.

DEFINITION 2. *A* formation $(G, \{G_F\}, A)$ is a topological formation if each level A_F is a topological group and if:

a) In each layer E/F, the topology of the ground level A_F is that induced by the topology of the top level A_E; in other words, the inclusion map $A_F \to A_E$ is bicontinuous.

b) The galois group G acts continuously on the levels; if $\sigma \in G$, then the map $\sigma\colon A_F \to A_{(F^\sigma)}$ is continuous for each field F.

It follows that the map $\sigma\colon A_F \to A_{F^\sigma}$ is bicontinuous in a topological formation, because the inverse map, given by σ^{-1}, is also continuous. If E/F is an arbitrary layer of a topological formation, the norm map $N_{E/F}\colon A_E \to A_F$ is continuous, and furthermore the ground level A_F is a *closed* subgroup of the top level A_E. To prove these statements, we imbed E/F in a normal layer K/F, and make

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use of the fact that the topologies of A_E and of A_F are those they inherit as subsets of A_K. Obviously the norm map $a \to N_{E/F}a = \prod_i a^{\sigma_i}$ is continuous, because each of the isomorphisms $a \to a^{\sigma_i}$ is a continuous map of A_K into A_K. Furthermore, A_F, being the set of elements of A_K which are left fixed by the continuous operators $\sigma \in G_{K/F}$, is closed in A_K.

From now on in this section we assume that our class formation is a topological formation. In the course of studying the norm subgroups of a given level A_F, it will be useful to consider the intersection of all norm subgroups. We denote this intersection by D_F:

$$D_F = \bigcap_{E \supset F} N_{E/F} A_E.$$

Thus, D_F is the group of "universal norms", elements which are norms from every extension. For any layer E/F we have $N_{E/F}D_E \subset D_F$, i.e. a norm of a universal norm is a universal norm. This is obvious from the transitivity of the norm and from the fact that every extension of F is contained in an extension containing E.

Under certain topological assumptions one can prove the opposite inclusion, $D_F \subset N_{E/F}D_E$. These assumptions are:

Axiom IIIa). For each layer E/F the norm group $N_{E/F}A_E$ is a *closed* subgroup of A_F.

Axiom IIIb). For each layer E/F the kernel of the norm map, namely $N_{E/F}^{-1}(1)$, is a *compact* subgroup of A_E.

Proposition 9. *In a topological formation satisfying Axioms* IIIa) *and* b) *we have $D_F = N_{E/F}D_E$ for each layer E/F.*

Proof. According to the remarks above, we have only to show $D_F \subset N_{E/F}D_E$. Let $a \in D_F$. For each field L containing E, let

$$T_L = (N_{L/E}A_L) \cap (N_{E/F}^{-1}(a))$$

denote the set of elements of A_E which are norms from A_L and whose norm to F is a. Our task is to show that the intersection of the sets T_L, over all $L \supset E$, is non-empty, for an element of that intersection would be a universal norm in A_E whose norm to F is a. The individual sets T_L are not empty because, a being a universal norm in A_F, we have for each L,

$$a \in N_{L/F}A_L = N_{E/F}(N_{L/E}A_L).$$

It follows that the sets T_L have the finite intersection property, since $\bigcap_{i=1}^r T_{L_i} \supset T_L$ whenever $L \supset L_i$ for $i = 1, \ldots, r$. Therefore, in order to prove that the T_L have a common point it suffices to show they are compact. This is where Axioms IIIa) and b) come in; T_L is the intersection of the closed set $N_{L/E}A_L$ and the compact set $N_{E/F}^{-1}(a)$ and is therefore compact. This concludes the proof of Proposition 9. $\square$

In class field theory proper the formations satisfy

Axiom IIIc). In each level A_F the kernel of the map $a \to a^\ell$, namely $1^{\frac{1}{\ell}}$, is compact, for each prime number ℓ.

Axiom IIId). For each prime number ℓ we have $D_E \subset A_E^\ell$ for all sufficiently large fields E. (More precisely, there exists a field E_ℓ such that $D_E \subset A_E^\ell$ for all $E \supset E_\ell$. Note that the required "largeness" is allowed to depend on the prime ℓ.

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For example, in the application to ordinary class field theory, "sufficiently large" will mean "containing the ℓ-th roots of unity".)

PROPOSITION 10. *In a topological formation satisfying Axioms* IIIa), b), c) *and* d) *we have* $D_F = D_F^m$ *for all natural numbers* m *and all fields* F.

PROOF. It is clearly sufficient to prove $D_F = D_F^\ell$, for each prime ℓ. Using Axioms IIIa) and b) we know from the preceding proposition that $D_F = N_{E/F} D_E$ for each extension E/F. Taking E to be "sufficiently large" for ℓ in the sense of Axiom IIId) we have then

$$(*) \qquad D_F = N_{E/F} D_E \subset N_{E/F}(A_E^\ell) = (N_{E/F} A_E)^\ell.$$

For each $a \in D_F$, let $a^{1/\ell}$ denote the set of all elements *of* A_F whose ℓ^{th}-power is a. From $(*)$ we see that the sets

$$X_E = (N_{E/F} A_E) \cap (a^{1/\ell})$$

are non-empty. Therefore they have the finite intersection property. They are compact because $N_{E/F} A_E$ is closed (Axiom IIIa) and $a^{1/\ell}$ is compact (Axiom IIIc). Consequently their intersection is not empty. An element of their intersection is an element of D_F, whose ℓ^{th}-power is a. This concludes the proof of Proposition 10. $\square$

If our class formation satisfies one more axiom, then we can get information about the norm groups themselves rather than their intersection D_F. This final axiom is

AXIOM IIIe). For each field F there exists a *compact* subgroup U_F of A_F such that every open subgroup of finite index in A_F, which contains U_F, is a norm subgroup.

THEOREM 8. *In a topological class formation satisfying the five Axioms* IIIa)–e), *the norm subgroups of a level* A_F *are just the open subgroups of finite index in* A_F. *Their intersection* D_F *is given by* $D_F = \bigcap_{m=1}^\infty A_F^m$, *and we have* $D_F = D_F^m$ *for all natural numbers* m.

PROOF. The statements about D_F follow immediately from the preceding proposition. Indeed, for each field $E \supset F$ we have

$$\bigcap_{m=1}^\infty A_F^m \subset A_F^{[E:F]} = N_{E/F} A_F \subset N_{E/F} A_E$$

which proves that $\bigcap_{m=1}^\infty A_F^m \subset D_F$. Conversely, for each m we know from the preceding proposition that $D_F = D_F^m \subset A_F^m$, which proves $D_F \subset \bigcap_{m=1}^\infty A_F^m$.

In a class formation the norm indices $(A_F : N_{E/F} A_E)$ are *finite* (cf. Theorem 7 of the preceding section), and we have assumed (Axiom IIIa) that the norm subgroups $N_{E/F} A_E$ are *closed*. It follows that the norm subgroups are open. What remains to be shown is that conversely, each open subgroup B of finite index is a norm subgroup. Let N designate the general norm subgroup of A_F. Then B contains $D_F = \bigcap_N N$ because $B \supset A_F^{(A_F:B)} \supset D_F$. In other words, in terms of the complements, the open sets $A - N$ cover the closed set $A - B$. In particular, they cover the compact set $(A - B) \cap U_F$, where U_F is the compact subgroup of A_F mentioned in Axiom IIIe). Thus there is a finite set of norm subgroups $N_1, N_2, \ldots, N_h$ such that the sets $A - N_i$ cover $(A - B) \cap U_F$. The intersection $N = N_1 \cap N_2 \cap \cdots \cap N_h$ is a norm subgroup such that $(A - B) \cap U_F \cap N$ is empty,

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i.e. $U_F \cap N \subset B$. Now consider $N \cap B$. It is open and of finite index in A_F because both N and B are. Multiplying it by U_F we obtain a subgroup $(N \cap B)U_F$, which is open, of finite index, and contains U_F. Such a subgroup is a norm subgroup by Axiom IIIe). Since the intersection of two norm subgroups is a norm subgroup, it follows that $N \cap \big(U_F(N \cap B)\big)$ is a norm subgroup. This last subgroup is easily seen to be contained in B if one remembers that we have constructed N so that $N \cap U_F \subset B$. Thus B contains a norm subgroup and is therefore itself a norm subgroup. This concludes the proof of Theorem 8. $\square$

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CHAPTER XV

Weil Groups

In this section we shall apply the abstract theory of group extensions developed in Ch. XIII, §1, 2, 3 to the case of a group extension belonging to the fundamental class $\alpha_{K/F} \in H^2(K/F)$ of a normal layer K/F of our class formation. In doing so we will gain a new insight into the reciprocity law isomorphism. For the sake of efficiency and ultimate clarity, our discussion will be quite formal. We first define a certain type of mathematical structure called a Weil group of the normal layer K/F. We then prove the existence and essential uniqueness of such a structure. Finally we discuss various further properties of the structure.

Definition 1. Let K/F be a normal layer in a class formation. A *Weil group* $(U, g, \{f_E\})$ for the layer K/F consists of the following objects:

1) A group U (called the Weil group by abuse of language).
2) A homomorphism, g, of U *onto* the Galois group $G_{K/F}$.

Having g at our disposal, we can introduce for each intermediate field E between F and K the subgroup $U_E = g^{-1}(G_{K/E})$. U_E is the subgroup of U which is the inverse image, under g, of the subgroup $G_{K/E}$ of $G_{K/F}$. The final ingredient of the Weil group is:

3) A set of isomorphisms $f_E \colon A_E \approx U_E/U_E^c$ of the E-level A_E onto the factor commutator group of U_E, one for each intermediate field E.

In order to constitute a Weil group, these objects U, g, and $\{f_E\}$ must have the following four properties:

W1) For each intermediate layer $E'/E, F \subset E \subset E' \subset K$, the following diagram is commutative:

$$\begin{array}{ccc} A_E & \overset{f_E}{\approx} & U_E/U_E^c \\ \Big\downarrow{\scriptstyle \text{incl}} & & \Big\downarrow{\scriptstyle V_{E'/E}} \\ A_{E'} & \overset{f_{E'}}{\approx} & U_{E'}/U_{E'}^c \end{array}$$

where the left hand vertical arrow is the inclusion map between formation level, and the right hand vertical arrow $V_{E'/E}$ denotes the group theoretical transfer (Verlagerung) from U_E to $U_{E'}$.

W2) Let u be an element of U and put $\sigma = g(u) \in G_{K/F}$. Then it is clear that for each intermediate field E we have $U_E^u = U_{E^\sigma}$. Property W2) states that the

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following diagram is commutative:

$$\begin{array}{ccc} A_E & \overset{f_E}{\approx} & U_E/U_E^c \\ \Big\downarrow{\scriptstyle\sigma} & & \Big\downarrow{\scriptstyle u} \\ A_{E^\sigma} & \overset{f_{E^\sigma}}{\approx} & U_{E^\sigma}/U_{E^\sigma}^c \end{array}$$

where the left hand vertical arrow is the action of σ on the formation level A_E and the right hand vertical arrow is the map of the factor commutator groups induced by conjugation by $u\colon U_E \to uU_Eu^{-1} = U_{E^\sigma}$.

W3) Suppose L/E is a normal intermediate level, $F \subset E \subset L \subset K$. Then the map g induces an isomorphism

$$U_E/U_L \approx G_{K/E}/G_{K/L} = G_{L/E}$$

which we do not bother to name. Since A_L is isomorphic to U_L/U_L^c by f_L, we can view U_E/U_L^c as a group extension of A_L by $G_{L/E}$ as follows:

$$(*) \qquad (1) \to A_L \overset{f_L}{\approx} U_L/U_L^c \to U_E/U_L^c \to U_E/U_L \approx G_{L/E} \to (1).$$

The operation of $G_{L/E}$ on A_L associated with this extension is the natural one, as one sees by applying property W2) to an element $u \in U_E$ having a prescribed image in $G_{L/E}$ (replacing the field E mentioned in W2) by our present field L). Property W3) requires that the 2-dimensional class of our extension $(*)$ is the fundamental class $\alpha_{L/E}$ of the layer L/E.

W4) We finally require that $U_K^c = 1$. This concludes the definition of a Weil group.

Fortunately it is easier to prove the existence of Weil groups than it is to define them!

THEOREM 1. *Let K/F be a normal layer in a class formation. Then there exists a Weil group $(U, g, \{f_E\})$ for the layer K/F.*

PROOF. Let U be a group extension of A_K by $G_{K/F}$ belonging to the fundamental class $\alpha_{K/F}$ of the layer K/F (Cf. Ch. XIII, §1, especially Theorem 1). Thus, U is a group containing A_K as normal subgroup, together with a homomorphism g of U onto $G_{K/F}$, with kernel A_K. Choosing for each $\sigma \in G_{K/F}$ a preimage $u_\sigma \in U$ such that $\sigma = g(u_\sigma)$ we have then $a^{u_\sigma} = u_\sigma a u_\sigma^{-1} = a^\sigma$, for $a \in A_K$, and furthermore $a_{\sigma,\tau} = u_\sigma u_\tau u_{\sigma\tau}^{-1}$ is a fundamental standard 2-cocycle of $G_{K/F}$ in A_K.

For each intermediate field E between F and K we put $U_E = g^{-1}(G_{K/E})$. In the two extreme cases $E = F$ and $E = K$ we have $U_F = U$ and $U_K = A_K$. In general we have $A_K \subset U_E \subset U$ with $U_E/A_K \approx G_{K/E}$, the isomorphism being induced by g.

The 2-dimensional class of the group extension $U_E/A_K \approx G_{K/E}$ is the fundamental class $\alpha_{K/E}$ of the layer K/E, because it is the restriction to $G_{K/E}$ of the class $\alpha_{K/F}$, of our original extension, and we know that $\alpha_{K/E} = \operatorname{res} \alpha_{K/F}$.

Let us now consider the group theoretical transfer map V_{U_E,A_K}, designating it by $V_{K/E}$ for short. We have discussed this map for the case of an arbitrary group extension in §2 of Ch. XIII. It is shown there that $V_{K/E}$ carries U_E/U_E^c not only into A_K but into $A_K^{G_{K/E}} = A_E$. We therefore view $V_{K/E}$ as a homomorphism with

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values in A_E:

$$V_{K/E}\colon U_E/U_E^c \to A_E.$$

In the corollary of Theorem 4, §2, Ch. XIII the kernel and cokernel of this homomorphism is analyzed in terms of kernels and cokernels of the homomorphisms

$$\alpha_{-3}\colon H^{-3}(G_{K/E},\mathbb{Z}) \to H^{-1}(G_{K/E},A_K)$$

and

$$\alpha_{-2}\colon H^{-2}(G_{K/E},\mathbb{Z}) \to H^{0}(G_{K/E},A_K)$$

which are effected by cup product multiplication with the 2-dimensional class of the group extension involved. In our present case this 2-dimensional class is the fundamental class of the layer K/E, and by the main theorem of class field theory we know that the maps $\alpha_r\colon H^r(G_{K/E},\mathbb{Z}) \to H^{r+2}(K/E)$ are isomorphisms onto for all r. It follows that the transfer map is an isomorphism onto, $V_{K/E}\colon U_E/U_E^c \approx A_E$. We finish the construction of our Weil group by defining the isomorphism $f_E\colon A_E \approx U_E/U_E^c$ to be the inverse of $V_{K/E}$.

All that remains is to verify that properties W1)–W4) are satisfied, and this is not hard. Properties W1) and W2) concern the commutativity of diagrams involving the isomorphisms f_E. Replacing these isomorphisms by their inverses, $V_{K/E}$, we see that W1) amounts to the transitivity of the transfer, namely

$$V_{K/E}(a) = V_{K/E'}V_{E'/E}(a), \qquad a \in U_E/U_E^c,$$

and W2) amounts to the rule, for $\sigma = g(u)$,

$$(V_{K/E^{(a)}})^\sigma = uV_{K/E}(a)u^{-1} = V_{K/E^\sigma}(uau^{-1}), \qquad a \in U_E/U_E^c,$$

which follows from the naturality of the transfer.

To verify property W3) we refer to Ch. XIII, §3, where the map

$$v\colon H^2(G_{K/E},A_K) \to H^2(G_{L/E},A_L)$$

is defined. By the very definition of v we see that the class of the extension $(*)$ mentioned in W3) is the image under v of the class $\alpha_{K/E}$ of the extension $U_E/A_K \approx G_{K/E}$, because the extension

$$U_L/U_L^c \to U_E/U_L^c \to G_{L/E}$$

is a factor extension of the latter in the sense discussed in §3 of Ch. XIII, and the isomorphism $U_L/U_L^c \approx A_L$ is given by the transfer V_{U_L,A_K}. And since $\alpha_{L/E} = v\alpha_{K/E}$ it follows that $\alpha_{L/E}$ is indeed the class of the extension $(*)$.

Finally we see that W4) is satisfied because by our construction $U_K = A_K$ and is abelian. This concludes the proof of Theorem 1. □

Having defined the notion of Weil group, and shown the existence of Weil groups, it is natural to consider the question of isomorphisms of Weil group. It is clear how we should define isomorphisms, namely

Definition 2. Let K/F be a normal layer in a class formation. Let $(U, g, \{f_E\})$ and $(U', g', \{f'_E\})$ be two Weil groups for the layer K/F. Then a *Weil isomorphism* φ from one to the other is an isomorphism $\varphi\colon U \approx U'$ with the following two properties.

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WI 1) The following is commutative

$$\begin{tikzcd} U \arrow[r, "g"] \arrow[d, "\varphi"] & G_{K/F} \arrow[d, "\text{identity}"] \\ U' \arrow[r, "g'"] & G_{K/F}. \end{tikzcd}$$

WI 2) From WI 1) it is evident that $\varphi(U_E) = U'_E$, for each intermediate field E between F and K, and consequently φ induces an isomorphism $\varphi_E \colon U_E/U_E^c \approx U'_E/U'^c_E$. Property WI 2) requires the commutativity of

$$\begin{tikzcd} A_E \arrow[r, "f_E"] \arrow[d, "\text{identity}"] & U_E/U_E^c \arrow[d, "\varphi_E"] \\ A_E \arrow[r, "f'_E"] & U'_E/U'^c_E \end{tikzcd}$$

for each E.

THEOREM 2. *There exists a Weil isomorphism $\varphi \colon U \approx U'$ for any two Weil groups U and U' of a layer K/F. Furthermore, φ is unique up to an inner automorphism of U' effected by an element of U'_K.*

PROOF. If φ is a Weil isomorphism then the following diagram is commutative, by WI 1) and by WI 2) for the extreme case $E = K$:

$$\begin{tikzcd} (1) \arrow[r] & A_K \arrow[r, phantom, "\overset{f_K}{\approx}"] \arrow[d, "\text{id.}"] & U_K \arrow[r, "\text{incl}"] & U \arrow[r, "g"] \arrow[d, "\varphi"] & G_{K/F} \arrow[r] \arrow[d, "\text{id.}"] & (1) \\ (1) \arrow[r] & A_K \arrow[r, phantom, "\overset{f'_K}{\approx}"] & U'_K \arrow[r, "\text{incl}"] & U' \arrow[r, "g'"] & G_{K/F} \arrow[r] & (1) \end{tikzcd}$$

Conversely, we contend that any homomorphism $\varphi \colon U \to U'$ which makes this diagram commutative is a Weil isomorphism. Indeed, let φ be such a homomorphism. Then, from the exactness of the rows, it follows that φ is an isomorphism of U onto U', and by the commutativity of the right hand square we have $\varphi(U_E) = U'_E$ for each intermediate field E. Thus φ induces an isomorphism $\varphi_E \colon U_E/U_E^c \approx U'_E/U'^c_E$ and we can consider the following cube:

$$\begin{tikzcd} A_E \arrow[rr, "\text{incl}"] \arrow[dr, "f_E"] \arrow[dd, "\text{id.}"'] & & A_K \arrow[dr, "f_K"] \arrow[dd, "\text{id.}"] & \\ & U_E/U_E^c \arrow[rr, "V"] \arrow[dd, "\varphi_E"] & & U_K \arrow[dd, "\varphi_K"] \\ A_E \arrow[rr, "\text{incl}"] \arrow[dr, "f'_E"] & & A_K \arrow[dr, "f'_K"] & \\ & U'_E/U'^c_E \arrow[rr, "V'"] & & U'_K \end{tikzcd}$$

The top and bottom faces are commutative by property W1) of Weil groups. The back face is obviously commutative. The front face is commutative by the naturality of the transfer map V, because $\varphi : U \approx U'$ is an isomorphism mapping U_E on U'_E and U_K on U'_K. The right hand face is commutative because the left hand square in the preceding diagram is commutative. Since the horizontal arrows are

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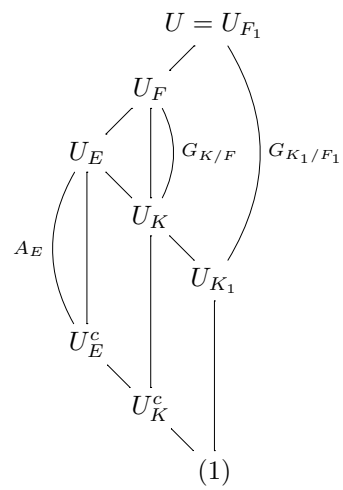
isomorphisms into, we conclude that the left hand face of our cube is commutative, and this shows that φ satisfies property WI 2) for all intermediate fields E and is therefore a Weil isomorphism.

The problem of Weil isomorphisms φ therefore boils down to the problem of middle arrows φ in the diagram at the beginning of this proof, that is, to the problem of homomorphisms between the two group extensions represented by the horizontal rows of that diagram. These two group extensions have the same 2-dimensional class, namely $\alpha_{K/F}$. Consequently, maps φ exist, by Theorem 2, §1 of Ch. XIII. Moreover, since the 1-dimensional cohomology group $H^1(G_{K/F}, A_K)$ is trivial, it follows from the uniqueness part of that same theorem that φ is determined uniquely up to an inner automorphism of U' effected by an element of $U'_K = f'_K(A_K)$. This concludes the proof of Theorem 2. $\square$

THEOREM 3. *Let $F_1 \subset F \subset K \subset K_1$ with K_1/F_1 and K/F normal. Let $(U, g, \{f_E\})$ be a Weil group for the big layer K_1/F_1. Then $(U_F/U_K^c, \bar{g}, \{f_E\}_{F\subset E\subset K})$ is a Weil group for the small layer K/F, where $\bar{g}$ denotes the homomorphism of U_F/U_K^c onto $G_{K/F}$, which is induced by g in the obvious way.*



PROOF. This theorem is evident from the definition of Weibl group. The lattice diagram at the right may help in visualizing the situation. $\square$

Now let K_1/F_1 be a fixed normal layer, and let $(U, g, \{f_E\})$ be a Weil group for it. In the next few paragraphs we suppose that all fields $F, E, K, \ldots$ under consideration are intermediate between K_1 and F_1. Clearly by choosing our fixed normal layer K_1/F_1 suitably large we can arrange that any prescribed finite set of fields $F, E, K, \ldots$ are contained between F_1 and K_1, and so are "under consideration".

THEOREM 4. *Let E/F be an arbitrary layer. Then the following diagram is commutative*:

$$\begin{array}{ccc} A_F & \overset{f_F}{\approx} & U_F/U_F^c \\ \uparrow{\scriptstyle N_{E/F}} & & \uparrow{\scriptstyle i_*} \quad \left(\begin{array}{l}\text{map induced by}\\ \text{the inclusion } U_E \subset U_F\end{array}\right) \\ A_E & \overset{f_E}{\approx} & U_E/U_E^c \end{array}$$

PROOF. This theorem follows from a certain simple property[1] of the group theoretical transfer when we analyze what it says. There is a minor technical difficulty arising from the fact that we do not assume E/F normal. Because of this non-normality, we must first choose a K (e.g. $K = K_1$) such that $E \subset K$ and K/F

[1]The general property of transfer which is essentially proved below is as follows: If $U \supset U_1 \supset A$ are groups with A abelian and normal of finite index in U, then, vieweing A as a (U/A)-module in the usual way, the transfer from U to A of an element $u_1 \in U_1$ is the norm from $A^{U_1/A}$ to $A^{U/A}$ of the transfer of u_1 from U_1 to A.

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is normal. We can then refer the things we are interested in to A_K and U_K/U_K^c by means of a cubic diagram of the following type:

$$\begin{tikzcd}
A_F \arrow[rr, "\mathrm{incl}"] \arrow[dr, "f_F"] & & A_K \arrow[dr, "f_K"] & \\
& U_F/U_F^c \arrow[rr, "V"] & & U_K/U_K^c \\
A_E \arrow[uu, "N_{E/F}"] \arrow[rr, "\mathrm{incl}"] \arrow[dr, "f_E"] & & A_K \arrow[uu, "N_1"] \arrow[dr, "f_K"] & \\
& U_E/U_E^c \arrow[uu, "i_*"] \arrow[rr, "V"] & & U_K/U_K^c \arrow[uu, "N_2"]
\end{tikzcd}$$

Here we must explain N_1 and N_2. We choose N_1 so that the back face of the cube is commutative, going back to the definition of $N_{E/F}$. Namely we write

$$G_{K/F} = \bigcup_i \sigma_i G_{K/E} \qquad \text{(disjoint union)}$$

and put

$$N_1(a) = \prod_i a^{\sigma_i}, \qquad a \in A_K.$$

Next, we choose N_2 so that the right face of the cube is commutative. According to property W2) of Weil groups, this can be done by choosing elements $v_i \in U_F$ such that $g(v_i) = \sigma_i$ and defining

$$N_2(u) = \prod_i u^{v_i} = \prod_i v_i u v_i^{-1} \qquad (\text{mod } U_K^c)$$

for $u \in U_K \pmod{U_K^c}$. Notice then that we have

$$U_F = \bigcup_i v_i U_E \qquad \text{(disjoint)}.$$

Now the top and bottom of the cube are commutative by property W1) of Weil groups. Since the horizontal arrows of the cube are isomorphisms into, the commutativity of the left side, which we want to show, will follow if we can show that the front face is commutative. This means that we must prove for $u \in U_E$ that

$$V_{U_F,U_K}(u) = \prod_i v_i (V_{U_E,U_K}(u)) v_i^{-1}.$$

To do this we write

$$U_E = \bigcup_j U_K w_j \qquad \text{(disjoint)}.$$

Then

$$U_F = \bigcup_i v_i U_E = \bigcup_{i,j} v_i U_K w_j = \bigcup_{i,j} U_K v_i w_j,$$

the last because U_K is normal in U_F. Now by the definition of the transfer we have

$$V_{U_F,U_K}(u) = \prod_{i,j} v_i w_j u w_{j_1}^{-1} v_{i_1}^{-1},$$

where $(i,j) \to (i_1, j_1)$ is the *unique* permutation of the pairs (i,j) such that each factor of this product lies in U_K. But since U_K is normal in U_F we see that this

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permutation is achieved by selecting first j_1 so that $w_j u w_{j_1}^{-1} \in U_K$ and then putting $i_1 = i$. And now we are through because

$$V_{U_E,U_K}(u) = \prod_j w_j u w_{j_1}. \qquad \square$$

COROLLARY. *f_F and f_E induce isomorphisms*

$$A_F/N_{E/F}A_E \approx U_F/U_E U_F^c$$
$$N_{E/F}^{-1}(1) \approx (U_E \cap U_F^c)/U_E^c.$$

THEOREM 5. *Let K/F be a normal layer. Then the reciprocity law isomorphism for that layer is given by*

$$A_F/N_{K/F}A_K \overset{f_F}{\approx} U_F/U_K U_F^c \approx G_{K/F}/G_{K/F}^c.$$

where the right hand isomorphism is that induced by g. In other words, if

$$g_*\colon U_F/U_F^c \to G_{K/F}/G_{K/F}^c$$

is the homomorphism induced by $g\colon U_F \to G_{K/F}$, then $g_* f\colon A_F \to G_{K/F}/G_{K/F}^c$ is the norm residue map.

PROOF. For each $\sigma \in G_{K/F}$, select a representative $u_\sigma \in U_F$ such that $\sigma = g(u_\sigma)$, and let $b_\sigma = f_F^{-1}(u_\sigma U_F^c) \in A_F$. We must then show that b_σ corresponds to σ under the reciprocity law map. From the commutativity of

$$\begin{array}{ccc} A_F & \xrightarrow{f_F} & U_F/U_F^c \\ \downarrow{\scriptstyle \text{incl}} & & \downarrow{\scriptstyle V} \\ A_K & \xrightarrow{f_K} & U_K/U_K^c \end{array}$$

we see that

$$f_K(b_\sigma) = V f_F(b_\sigma) = V(u_\sigma) = \prod_\tau u_\tau u_\sigma u_{\tau\sigma}^{-1} U_K^c = f_K\Big(\prod_\tau a_{\tau,\sigma}\Big),$$

where we have defined elements $a_{\tau,\sigma} \in A_K$ by $a_{\tau,\sigma} = f_K^{-1}(u_\tau u_\sigma u_{\tau\sigma}^{-1} U_K^c)$. Then $a_{\tau,\sigma}$ is a 2-cocycle belonging to the extension

$$A_K \xrightarrow{f_K} U_F/U_K^c \xrightarrow{g_*} G_{K/F}$$

and is therefore a fundamental 2-cocycle for the layer K/F, by property W 3) of Weil groups. Consequently, from

$$b_\sigma = \prod_\tau a_{\tau,\sigma} = \text{Image of } \sigma \text{ under Nakayama map}$$

we can conclude that b_σ and σ do correspond under the reciprocity law, as contended. $\square$

The theorem we have just proved shows that the entire theory of the reciprocity law is contained in the theory of Weil groups. The reader will easily check that all the results of Ch. XIV, § 5, can be recovered immediately from our present theory. The reciprocity relationship between levels A_F and galois groups $G_{K/F}$ becomes easy to visualize when one identifies A_F with U_F/U_F^c (by means of f_F) and identifies $G_{K/F}$ with U_F/U_K (by means of g). In this way, all the facts are wrapped up in

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one neat non-abelian bundle, namely a suitable Weil group U. From this point of view we get one additional dividend, the Shafarevič Theorem.[2].

THEOREM 6. *Let $F \subset K \subset L$ with K/F and L/F normal, and L/K abelian. Then we may view $G_{L/F}$ as a group extension of $G_{L/K}$ by $G_{K/F}$ so there is determined in a canonical way a 2-dimensional class $\beta \in H^2(G_{K/F}, G_{L/K})$, the class of this extension. By means of the reciprocity law isomorphism, $G_{L/K} \approx A_K/N_{L/K}A_L$ (which is a $G_{K/F}$ isomorphism), β determines then a class $\beta' \in H^2(G_{K/F}, A_K/N_{L/K}A_L)$. This class β' is the image of the fundamental class $\alpha_{K/F} \in H^2(G_{K/F}, A_K)$ under the natural projection of A_K onto $A_K/N_{L/K}A_L$.*

$$\begin{tikzcd}
U_F \arrow[d, dash, "G_{K/F}"] \arrow[dddd, dash, bend left, "G_{L/F}"] \\
U_K \arrow[d, dash, "G_{L/K}"] \arrow[dddd, dash, bend right, "A_K"'] \\
U_L \arrow[d, dash, "N_{L/K}A_L"] \\
U_K^c
\end{tikzcd}$$

PROOF. The proof is evident from the lattice diagram at the left. We identify the various galois groups with factor groups of subgroups of U, and we identify A_K with U_K/U_K^c by means of f_K. Then $N_{L/K}A_L$ is identified with U_L/U_K^c, and the reciprocity law isomorphism becomes the identity map of U_K/U_L^c, by the preceding theorem. Hence β' is the class of the extension

$$A_K/N_{L/K}A_L \to U_F/U_L \to G_{K/F}$$

and is therefore obviously the image of $\alpha_{K/F}$ because $\alpha_{K/F}$ is the class of the extension

$$A_K \to U_F/U_K^c \to G_{K/F}. \qquad \square$$

The theorem we have just proved shows that if K/F is a normal extension, and if B is a norm subgroup of A_K which is a $G_{K/F}$-submodule, and if L is the class field over K belonging to B (so that $B = N_{L/K}A_L$), then we can determine the structure of the galois group $G_{L/F}$ in terms of objects associated with the layer K/F. Indeed, $G_{L/F}$ is isomorphic to the group extension of A_K/B by $G_{K/F}$ belonging to the image of the fundamental class.

In the preceding paragraphs we have seen how a Weil group for a big normal layer K_1/F_1 contains information about all intermediate layers E/F, and in particular contains, as factor groups of subgroups, the Weil groups for all intermediate normal layers K/F. This suggests that we try to go to the limit and construct one universal group, a Weil group for the whole formation so to speak, which will have all the Weil groups of all finite normal layers as factor groups of subgroups of itself. This is the next step on our program. In order to carry it through we must assume that our formation is a topological formation (cf. Definition, Ch. XIV, §6) which satisfies a certain compactness condition, a condition which is satisfied in local and global class field theory. We shall also view the galois group, G, of our formation as a topological group, the neighborhoods of 1 in G being the subgroups G_F. Thus

[2]This theorem is due to Shafarevich. He observed that it is a consequence of a simple relation between the Akizuki–Witt map v and the description of the norm residue correspondence via the Nakayama map, about 10 years after those two things were discovered. If a is a 2-cocycle of G in A and H an abelian normal subgroup of G, then by (XIII, 3, Theorem 5), we have $(va)_{r,s} \equiv \prod_{h \in H} a_{h,\gamma_{r,s}} \pmod{N_H A}$. Here $\gamma_{r,s} = \overline{r}\ \overline{s}\ \overline{rs}^{-1}$ is a 2-cocycle defined by the extension $H \to G \to G/H$. The formula shows that the image of γ under the Nakayama map $H \to A^H/N_H A$ made with the restriction of a to H is equal to the image of va under the map $A^H \to A^H/N_H A$. In the class formation situation, if a is a fundamental 2-cocycle for G, then its restriction is one for H, and va is one for G/H, (by XIII, 3, Theorem 6), and the present theorem follows. This is what Shafarevich observed in the context of local class field theory (see [**24**])

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all groups and modules to be considered from now on are topological. As usual in the theory of topological groups we must distinguish between the notion of a representation and the notion of a homomorphism. A representation is a continuous map of one group into another which is algebraically a homomorphism. A homomorphism is a representation $f\colon G \to H$ such that f induces a homeomorphism between $G/\text{Kernel} f$ and Image f, in other words, such that $f(U)$ is open in $f(G)$ whenever U is open in G. These things being said, we can now define the type of object which we aim to construct:

Definition 3. Let $(G, \{G_F\}, A)$ be a topological class formation. A Weil group $(U, g, \{f_F\})$ for the formation consists of the following objects:

1) A topological group U.

2) A representation, g, of U onto an everywhere dense subgroup of the galois group G of the formation.

Having g at our disposal, we can introduce, for each field F of our formation the subgroup $U_F = g^{-1}(G_F)$. The U_F's will then be open subgroups of finite index in U whose lattice reflects exactly the lattice of the G_F's. We have $U_E \subset U_F \Leftrightarrow F \subset E$, and U_E is normal in U_F if and only if E/F is a normal layer. If this is the case, then g induces an isomorphism $U_F/U_E \approx G_{E/F}$. (These things are true because $g(U)$ is *dense* in G, and consequently, for any E, we have $g(U) \cdot G_E = G$, because G_E is open in G. In other words, every coset of G_E contains an element of the form $g(u)$.) The third ingredient of our Weil group is

3) For each field F of our formation, an isomorphism (topological and algebraic)

$$f_F\colon A_F \approx U_F/U_F^c,$$

where U_F^c denotes here, and from now on, the *closure* of the commutator subgroup U_F.

In order to constitute a Weil group for the formation, these objects U, g, and $\{f_F\}$ must have the following four properties:

W 1) For each layer E/F, the following diagram is commutative

$$\begin{array}{ccc} A_F & \overset{f_F}{\approx} & U_F/U_F^c \\ \Big\downarrow{\scriptstyle \text{incl}} & & \Big\downarrow{\scriptstyle V} \\ A_E & \overset{f_E}{\approx} & U_E/U_E^c \end{array}$$

where V is the transfer map. (It is easy to verify, in case of a topological group G and an open subgroup H of finite index, that the transfer map $V_{G,H}$ carries G^c, the *closure* of the commutator group of G into H^c, the *closure* of the commutator group of H, and consequently induces a map of G/G^c into H/H^c. It is this latter map which is meant by V here, and from now on.)

W 2) Let $u \in U$ and let $\sigma = g(u) \in G$. Then it is clear that $u(U_E)u^{-1} = U_{(E^\sigma)}$. Property W 2) states that the following diagram is commutative for each field E:

$$\begin{array}{ccc} A_E & \overset{f_E}{\approx} & U_E/U_E^c \\ \Big\downarrow{\scriptstyle \sigma} & & \Big\downarrow{\scriptstyle \text{conjugation by } u} \\ A_{E^\sigma} & \overset{f_{E^\sigma}}{\approx} & U_{E^\sigma}/U_{E^\sigma}^c \end{array}$$

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W 3) For each normal layer K/F, the class of the group extension

$$(1) \to A_K \approx U_K/U_K^c \to U_F/U_K^c \to U_F/U_K \approx G_{K/F} \to (1)$$

is the fundamental class of the layer K/F.

W 4) We finally require that

$$U \to \varprojlim U/U_K^c$$

be an isomorphism of topological groups. This concludes Definition 3.

Suppose for the moment that U is a Weil group for the formation. It is obvious from the definition that for each normal layer K/F, U_F/U_K^c is a Weil group for that layer. In particular, if k is the ground field of the formation, so that $U = U_k$, then the factor groups U/U_K^c for variable K normal over k are Weil groups for the various normal layers K/k. On the other hand, property W 4) above states that U is the projective (inverse) limit of these factor groups. This shows how we must go about constructing U; we must get it as the projective limit of Weil groups of finite layers K/k. Let us carry out this program.

From now on, unless specific mention is made to the contrary, all fields F, K, L, M, ... are understood to be normal over k. For each such field K, let $(U^K, g^K, \{f_E^K\})$ be a Weil group for the layer K/k. It will be convenient to suppose that the isomorphism $f_K^K \colon A_K \approx U_K^K$ is the identity map, in other words to identify A_K with its isomorphic image U_K^K, for each K. This being said, we topologize U^K by taking as fundamental system of neighborhoods of 1 in U^K a fundamental system of neighborhoods of 1 in A_K; in other words, we give U^K the unique topology for which A_K is an open subgroup of U_K, and for which the topology induced on A_K is the same as that which A_K gets as a level of our topological formation.

Lemma. *For each field E, $k \subset E \subset K$, normal or not, the commutator subgroup of U_E^K is closed; in other words, $(U_E^K)^c$ has the same meaning as before.*

Proof. Let for the moment $(U_E^K)^c$ denote just the commutator subgroup rather than its closure. Then $(U_E^K) \cap A_K$ is closed in A_K because it is the kernel of $N_{K/E}$, and the norm map $N_{K/E}$ is continuous in a topological formation. Being closed in A_K, it is closed in U^K. On the other hand $(U_E^K)^c \cap A_K$ is of finite index in $(U_E^K)^c$ because A_K is of finite index in U^K. Therefore, being the union of a finite number of closed cosets, $(U_E^K)^c$ is closed. □

Lemma. *The isomorphism $f_E^K \colon A_E \approx U_E^K/(U_E^K)^c$ is a homeomorphism if and only if $N_{K/E}$ is an open mapping.*

Proof. The following diagram is commutative

$$\begin{tikzcd}
A_E \arrow[r, phantom, "\overset{f_E}{\approx}"] & U_E^K/(U_E^K)^c \\
A_K \arrow[u, "N_{K/E}"] \arrow[r, phantom, "\overset{f_K=\mathrm{id.}}{\approx}"] & A_K \arrow[u, "i_*"'] \arrow[r, phantom, "\left(\begin{matrix}\text{map induced}\\ \text{by inclusion}\end{matrix}\right)"]
\end{tikzcd}$$

Since A_K is open in U^K, i_* is an open map. And since f_K is a homeomorphism, it follows that if f_E is a homeomorphism, then $N_{K/E}$ must be an open map. Conversely, if $N_{K/E}$ is an open map, then $N_{K/E}A_K$ is an open subgroup of A_E, and is

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homeomorphic to $A_K/N_{K/E}^{-1}(1)$. On the subgroup $N_{K/E}A_K$, f_E induces therefore the composition of two homeomorphisms, namely,

$$N_{K/E}A_K \approx A_K/N_{K/E}^{-1}(1) \approx i_*(A_K).$$

Since the left hand and right hand of these groups are open in A_E and in $U_E^K/(U_E^K)^c$ respectively, we conclude that f_E is a homeomorphism. □

Now to proceed to the construction of our projective limit we must find a collection of "reasonable" homomorphisms $\varphi_{L/K}\colon U^L \to U^K$, one for each pair (L, K) such that $L \supset K$, which are transitive in the sense that

$$\varphi_{M/K} = \varphi_{L/K}\varphi_{M/L}$$

whenever $M \supset L \supset K$. What is to be meant by "reasonable"? The answer is obvious once we recall that $U^L/(U_K^L)^c$ is a Weil group for K/k, just as U^K is. We shall require that our homomorphism $\varphi_{L/K}\colon U^L \to U^K$ has kernel $(U_K^L)^c$ and induces a Weil isomorphism (Definition 2) from $U^L/(U_K^L)^c$ to U^K. Let us give the name *Weil map* to a homomorphism $\varphi_{L/K}$ satisfying this condition.

Lemma. *Let $\varphi_{L/K}\colon U^L \to U^K$ be a Weil map. Then $\varphi_{L/K}$ carries $A_L(= U_L^L)$ into $A_K(= U_K^K)$ and the restriction of $\varphi_{L/K}$ to A_L is $N_{L/K}$. Furthermore, $\varphi_{L/K}$ is a homomorphism (in the sense of topological groups) if and only if $N_{L/K}\colon A_L \to A_K$ is an open map.*

Proof. In the following diagram

$$\begin{tikzcd}
A_K \arrow[rr, "\text{identity}"] \arrow[d, phantom, "\wr f_K^L"] & & A_K \arrow[d, phantom, "\wr \mathrm{id}"] \\
U_K^L/(U_K^L)^c \arrow[rr, "\text{restriction of the Weil isomorphism induced by } \varphi_{L/K}"'] & & U_K^K \\
A_L \arrow[u, "N_{L/K}"] & & \\
U_L^L \arrow[u, "i_*"] \arrow[uurr, bend right, "\text{restriction of } \varphi_{L/K}"'] \arrow[u, phantom, "\wr \mathrm{id}"] & &
\end{tikzcd}$$

the left side square is commutative, and the top square is commutative, by the definition of Weil isomorphism. Factoring the diagonal arrow (restriction of φ) into the product of i_* and the Weil isomorphism, we see that the restriction of φ does induce $N_{L/K}$ on A_L as contended.

Now since $\varphi_{L/K}$ maps U^L *onto* U^K, $\varphi_{L/K}$ will be a topological homomorphism if and only if it is an open map. Since the subgroups $A_L \subset U^L$ and $A_K \subset U^K$ are open subgroups, it is clear that $\varphi_{L/K}$ is an open map if and only if its restriction to A_L is open, i.e. if and only if $N_{L/K}$ is open. This concludes the proof of the lemma. □

According to Theorem 2, a Weil map $\varphi_{L/K}$ exists for each layer L/K, but is not unique, being determined only up to an inner automorphism of U^K by an element of $A_K = U_K^K$. This non-uniqueness will be the main obstacle we shall encounter in constructing our projective limit, since it forces us to make a selection. We must therefore discuss in some detail the inner automorphisms by which the Weil maps

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can be changed. In order to have a notation for them, let us denote by a_K the inner automorphism of U^K effected by an element $a \in A_K$; that is, we put $a_K(u) = aua^{-1}$ for $u \in U^K$. When is a_K the identity automorphism? Clearly, when, and only when, a commutes with every element of U^K. Selecting representatives $u_\sigma, \sigma \in G_{K/k}$, for the elements of $U^K/A_K \approx G_{K/k}$ we see that the condition is that a commute with each u_σ, because, A_K being abelian, it will then follow that a commutes with each element of U^K. Since $u_\sigma a u_\sigma^{-1} = a^\sigma$, this condition means that $a^\sigma = a$ for all $\sigma \in G_{K/k}$, in other words that $a \in A_k$. Thus we have shown that the group of inner automorphisms of U^K by elements of A_K is isomorphic to A_K/A_k.

For each layer $L \supset K$, let $X_{L,K}$ denote the set of all Weil maps $\varphi_{L/K} \colon U^L \to U^K$. If $\varphi^0_{L/K}$ is one fixed element of $X_{L,K}$, then the other elements are of the form $a_K \varphi^0_{L/K}$, with $a \in A_K$. Since $\varphi^0_{L/K}$ is an onto map we have $a_K \varphi^0_{L/K} = b_K \varphi^0_{L/K} \Leftrightarrow a_K = b_K \Leftrightarrow ab^{-1} \in A_k \Leftrightarrow aA_k = bA_k$. In other words, we have a one-one correspondence

$$aA_k \leftrightarrow a_k \varphi^0_{L/K}$$

between A_K/A_k and $X_{L,K}$. This allows us to topologize $X_{L,K}$ by the topology which is induced on it by the topology of A_K/A_k, i.e. such that the above one-one correspondence is a homeomorphism. This topologization of the set of Weil maps $X_{L,K}$ is independent of the choice of $\varphi^0_{L/K}$ because the topology of A_K/A_k is invariant under translation.

Lemma. *The composition of Weil maps is continuous. More precisely, if $M \supset L \supset K$, then the composed map $\varphi_{L/K}\varphi_{M/L} \in X_{M/K}$ is a continuous function of the two variables $\varphi_{L/K} \in X_{L/K}$ and $\varphi_{M/L} \in X_{M,L}$.*

Proof. Note first of all the rule

$$\varphi_{L/K} a_L = (N_{L/K} a)_K \varphi_{L/K}$$

for $a \in A_L$, $\varphi_{L/K} \in X_{L,K}$. Indeed, for $u \in U^L$ we have

$$\begin{aligned}\varphi_{L/K} a_L(u) = \varphi_{L/K}(aua^{-1}) = \varphi_{L/K}(a)\varphi_{L/K}(u)\big(\varphi_{L/K}(u)\big)^{-1} = \\ = (N_{L/K}a)\varphi_{L/K}(u)(N_{L/K}a)^{-1} = (N_{L/K}a)_K \varphi_{L/K}(u),\end{aligned}$$

because the restriction to A_L of $\varphi_{L/K}$ is $N_{L/K}$ by the preceding lemma.

For $a \in A_L$, $b \in A_K$ we have therefore

$$(b_K \varphi_{L/K})(a_L \varphi_{M/L}) = (bN_{L/K}a)_K \varphi_{L/K}\varphi_{M/L}.$$

Since $bN_{L/K}a$ is a continuous function of the variables $a \in A_L$ and $b \in A_K$, our contention follows. □

Lemma. *Suppose A_K/A_k is compact for each K. Then there exists a transitive collection of Weil maps $\varphi_{L/K} \colon U^L \to U^K$.*

Proof. A random choice of Weil maps may be thought of as an element

$$\varphi = (\varphi_{L/K})$$

of the cartesian product

$$X = \prod_{L \supset K} X_{L,K},$$

the product being taken over all pairs (L, K) such that $L \supset K$. Since we have assumed A_K/A_k is compact, we know each $X_{L,K}$ is compact and consequently their

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product X is compact in the product topology. Now for each triple $M \supset L \supset K$, let $X(M, L, K)$ be the subset of X consisting of the elements φ such that

$$\varphi_{M/K} = \varphi_{L/K}\varphi_{M/L}.$$

Our task is to show that the subsets $X(M, L, K)$ have a non-empty intersection, an element of their intersection being a choice of Weil maps which is transitive for each triple $M \supset L \supset K$. Each of our subsets $X(M, L, K)$ is closed, by the preceding lemma. And since X is compact, it will be enough if we show that the sets $X(M, L, K)$ have the finite intersection property.

To this effect, let (M_i, L_i, K_i), $1 \leqslant i \leqslant n$, be a finite set of triples. Select a field P such that $P \supset M_i$, for each i. We shall show that there exists an element $\varphi \in X$ such that

$$\varphi_{M/K} = \varphi_{L/K}\varphi_{M/L}$$

for *all* triples M, L, K such that $P \supset M \supset L \supset K \supset k$, and therefore in particular for our given finite set of triples. To do this we select, for each field F between k and P, a Weil isomorphism

$$\Theta_F \colon U^P/(U_F^P)^c \approx U^F$$

and then we put, for each couple $L \supset K$ between k and P:

$$\varphi_{L/K} = \Theta_K^{-1}\psi_{L/K}\Theta_L \colon U^L \to U^K$$

where $\psi_{L/K}$ denotes the natural map of $U^P/(U_L^P)^c$ onto $U^P/(U_K^P)^c$. These $\varphi_{L/K}$'s are obviously Weil maps, and their transitivity, in the levels between P and k, follows from the transitivity of the natural maps $\psi_{L/K}$. Finally, whenever $P \not\supset L$, we choose a $\varphi_{L/K}$ at random, obtaining then a $\varphi = (\varphi_{L/K})$ such that

$$\varphi \in \bigcap_{i=1}^{n} X(M_i, L_i, K_i).$$

This concludes the proof of the lemma. □

THEOREM 7. *Suppose $(G, \{G_F\}, A)$ is a topological class formation satisfying the following three conditions.*

WT 1) *The norm map $N_{E/F} \colon A_E \to A_F$ is an open map, for each layer E/F.*
WT 2) *The factor group A_E/A_F is compact for each layer E/F.*
WT 3) *The Galois group G is complete.*

Then there exists a Weil group $(U, g, \{f_F\})$ for the formation, and it is unique up to isomorphism.

PROOF. For the details of the proofs of the basic facts about projective limits which we shall use in our proof we refer the reader to Weil [**27**, section 5].

Choose a transitive collection of Weil maps $\varphi_{L/K} \colon U^L \to U^K$ (this is possible by hypothesis WT 2) and a lemma). Each $\varphi_{L/K}$ is a (topological) homomorphism of U^L onto U^K (by hypothesis WT 1)) and a lemma above. I contend that the kernel of $\varphi_{L/K}$ is compact. Since A_L is of finite index in U^L, $A_L \cap \operatorname{Ker}\varphi_{L/K}$ is of finite index in $\operatorname{Ker}\varphi_{L/K}$ so it suffices to prove that $A_L \cap \operatorname{Ker}\ \varphi_{L/K}$ is compact. Since the restriction of $\varphi_{L/K}$ to $A_{L/K}$ is $N_{L/K}$ we are reduced to proving that $\operatorname{Ker} N_{L/K}$ is compact. Now $I_{(G_{L/K})}A_L = \prod_{\sigma \in G_{L/K}} A_L^{(\sigma-1)}$ is of finite index in $\operatorname{Ker} N_{L/K}$ because the factor group is $H^{-1}(G_{L/K}, A_L)$. It will therefore suffice to prove $I_{G_{L/K}}A_L$ is compact. For each $\sigma \in G_{L/K}$, the map $a \to a^{\sigma-1}$ induces

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a representation of A_L/A_K onto $A_L^{\sigma-1}$, because $A_K^{\sigma-1} = 1$. Therefore, $A_L^{\sigma-1}$ is compact, being a continuous image of A_L/A_K which is compact by hypothesis WT 2). Consequently $I_{G_{L/K}} A_L$ is compact as contended.

We have now shown that the family of topological groups $\{U^K\}$ together with the family of homomorphisms $\{\varphi_{L/K}\}$ satisfies Weil's conditions LPI, LPII, and LPIII″. It follows that we can build a projective limit U with all the desirable properties one could wish for. We form the direct product

$$\prod_K U^K$$

of our Weil groups U_K, and in it we consider the subgroup U consisting of all elements

$$u = (u_K), \qquad u_K \in U^K$$

such that $u_K = \varphi_{L/K} u_L$ for all pairs $L \supset K$. We topologize U by giving it the topology which is induced by the product topology in the direct product. At first sight, this means that a neighborhood of 1 in U is given by a finite set of fields K_i, together with a neighborhood W_i of 1 in U^K for each i, the corresponding neighborhood in U consisting then of the elements $u = (u_K)$ such that $u_{K_i} \in W_i$ for each i. However, taking into account the "coherence" of the components of u, i.e. the fact that $u_{K_i} = \varphi_{L/K_i} u_L$ for a suitable $L \supset K_i$ all i, we see that it suffices to consider the neighborhoods of 1 in U which are given by a single field L together with a neighborhood W_L of 1 in U^L, these constituting a fundamental system.

For each L we have a map $\varphi_L \colon U \to U^L$ defined by

$$\varphi_L(u) = u_L$$

for $u = (u_K) \in U$. Using the compactness of the kernels of the maps $\varphi_{M/L}$ for fields $M \supset L$ one shows easily that φ_L is onto, and consequently is a topological homomorphism of U onto U^L.

We must now construct the representation $g \colon U \to G$, and to do so we must use the hypothesis WT 3) which states that G is complete in the topology for which the subgroups G_K are a fundamental system of neighborhoods of 1. This completeness assures us that G is the projective limit of its factor groups $G_{K/k} = G/G_K$, as follows: Let $\psi_{L/K} \colon G_{L/k} \to G_{K/k}$ be the natural map. Then, given any family of elements $\{\sigma_K\}$, with $\sigma_K \in G_{K/k}$, such that $\sigma_K = \psi_{L/K}\sigma_L$ for each pair $L \supset K$, there exists a unique element $\sigma \in G$ such that $\sigma_K = \sigma G_K$. (Namely, for each K, let $\bar{\sigma}_K \in G$ such that $\bar{\sigma}_K G_K = \sigma_K$, and put $\sigma = \lim \bar{\sigma}_K$, this limit existing because $\{\bar{\sigma}_K\}$ is a Cauchy directed set, and G is complete.) Conversely, given any $\sigma \in G$ we can put $\sigma_K = \sigma G_K$, obtaining a family of elements $\{\sigma_K\}$, with $\sigma_K \in G_{K/k}$, such that $\psi_{L/K}\sigma_L = \sigma_K$. Now for $u = (u_K) \in U$, we put

$$g(u) = \big(g^K(u_K)\big) \in G,$$

in other words, we define $g(u)$ to be the unique element of G such that

$$g(u)G_K = g^K(u_K)$$

for each K. From another point of view, if we introduce the natural maps $\psi_K \colon G \to G_{K/k}$, then we see that our map $g \colon U \to G$ is characterized by the property that $\psi_K g = g^K \varphi_K$ for each K.

Our g is obviously an algebraic homomorphism. Its image, $g(U)$, is everywhere dense in G. Indeed, for any neighborhood G_K of 1 in G, we have $g(U)G_K = G$,

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because g^K maps U^K *onto* $G_{K/k} = G/G_K$. Finally, one checks immediately that g is continuous.

Having g at our disposal, we can introduce the subgroup $U_E = g^{-1}(G_E)$ of U, for each field E (not necessarily normal over k).

LEMMA. *Let $K \supset E \supset k$, with K/k normal. Then $U_E = \varphi_K^{-1}(U_E^K)$ and $U_E^c = \varphi_K^{-1}\big((U_E^K)^c\big)$.*

PROOF. The first statement is almost evident. The formal proof depends on the rule $\psi_K g = g^K \varphi_K$ and runs as follows:

$$\begin{aligned}\varphi_K^{-1}(U_E^K) &= \varphi_K^{-1}(g^K)^{-1}(G_{K/E})\\ &= (g^K\varphi_K)^{-1}(G_{K/E})\\ &= (\psi_K g)^{-1}(G_{K/E})\\ &= g^{-1}\psi_K^{-1}(G_{K/E})\\ &= g^{-1}(G_E) = U_E.\end{aligned}$$

To prove the second statement, let U'_E denote the commutator group of U_E so that U_E^c is, by definition, the closure of U'_E in U. Let L be any field containing K such that L/k is normal. Then from what we already have proved, namely $U_E = \varphi_L^{-1}(U_E^L)$, and from the fact that φ_L is an onto mapping, we can conclude that

$$U'_E(\operatorname{Ker}\varphi_L) = \varphi_L^{-1}((U_E^L)^c).$$

(Recall that $(U_E^L)^c = (U_E^L)'$ by a lemma.) Now from the theory of Weil groups of finite layers and from the definition of Weil map we know that $(U_E^L)^c = \varphi_{L/K}^{-1}((U_E^L)^c)$. And since $\varphi_K = \varphi_{L/K}\varphi_L$, we conclude

$$U'_E(\operatorname{Ker}\varphi_L) = \varphi_K^{-1}((U_E^K)^c),$$

a formula in which the left side contains an arbitrary $L \supset K$, and the right side is independent of L. From the definition of the topology in U, we can make $\operatorname{Ker}\varphi_L$ an arbitrarily small subgroup by selecting L sufficiently large. Hence

$$U_E^c = \overline{U'_E} \supset \varphi_K^{-1}((U_E^K)^c).$$

The opposite inclusion is trivial since $\varphi_K^{-1}((U_E^K)^c)$ contains U'_E and is closed. This concludes the proof of the lemma and we now continue with our construction of the Weil group U. $\square$

Since $\varphi_K : U \to U^K$ is onto, we conclude from the lemma that φ_K induces an isomorphism

$$\varphi_K^E : U_E/U_E^c \approx U_E^K/(U_E^K)^c,$$

and this isomorphism is topological as well as algebraic because φ_K is a topological homomorphism. Thus there exists a unique isomorphism f_E such that $\varphi_K \circ f_E = f_E^K$, i.e. such that the diagram

$$\begin{tikzcd} A_E \arrow[rr, "f_E", "\sim"'] \arrow[dr, "\sim", "f_E^K"'] & & U_E/U_E^c \arrow[dl, "\sim"', "\varphi_K^{(E)}"] \\ & U_E^K/(U_E^K)^c & \end{tikzcd}$$

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is commutative. Moreover this isomorphism f_E is topological as well as algebraic because the same is true of f_E^K, by hypothesis WT1 and the second of the above lemmas. The properties W1–W4 for the Weil group $(U, g, \{f_E\})$ of the formation now follow readily from the corresponding properties for the Weil groups of finite layers K/E, and our existence proof is complete. □

Unicity was not discussed in the earlier editions. In these last several pages, added in 2008, we fix that gap and go on to discuss a few more things about Weil groups, including an account of Weil's original proof of their existence.

Concerning unicity, it is proved in [**25**, Proposition (1.3.1)] for the classical situations, and the proof there works in the present abstract context. If U and U' are two Weil groups for the same formation one shows, by the same type of compactness argument used to prove the lemma just before Theorem 4 of this chapter, that there exists a compatible family of isomorphisms $\theta_K : U/U_K^c \to U'/U_K'^c$ commuting with f_K and f'_K and with the maps induced by g and g', for normal extensions K of the ground field k. Then $\theta = \varprojlim \theta_K : U \to U'$ is an isomorphism of Weil groups. □

Theorem 8. (a) *Let $(U, g, \{f_F\})$ be a Weil group for a class formation (G, G_F, A). For each field F, the composed map*

$$A_F \xrightarrow{f_F} U_F^{\mathrm{ab}} \xrightarrow{g_F^{\mathrm{ab}}} G_F^{\mathrm{ab}} = G_F^{\mathrm{ab}} \tag{1}$$

where g_F^{ab} is the map induced by g, is the reciprocity mapping.

(b) *If for every normal layer K/k there is a cyclic layer L/k of the same degree, then in the definition of Weil group for a class formation* (*Definition* 3), *one can substitute statement* (a) *of this theorem for the condition* W 3) *involving the fundamental classes.*

Proof. For (a) it suffices to prove that for every abelian normal layer K/F, that the composed map

$$A_F \to U_F/U_K \to G_F/G_K = G_{K/F}$$

where the arrows are induced by f_F and G^{ab}, is the reciprocity map for that layer, and that is true by Theorem 5.

The fundamental class of a layer of degree n was defined in Chapter XIV as the class with invariant $1/n$, so to prove (b) we have only to show that in a class formation satisfying the hypothesis of (b) the reciprocity maps determine the invariant functions. That hypothesis implies that every two-dimensional class comes by inflation from a cyclic extension. For a cyclic extension K/F cut out by a character χ every class is of the form $a^{\delta\chi}$ and has invariant $\chi(\sigma)$, where σ is the image of $a \in A_F$ under the reciprocity map $A_F \to G_{K/F}$. □

If $(U, g, \{f_F\})$ is a Weil group for a class formation $(G, \{G_F\}, A)$ in which G is complete, it is clear from the definitions that we can recover the formation up to isomorphism from knowledge of only the topological group U and the collection of subgroups $\{U_F\}$, for we have isomorphisms

$$G = \varprojlim (G/G_K) = \varprojlim (U/U_K), \quad G_F = \varprojlim (G_F/G_K) = \varprojlim (U_F/U_K),$$

the projective limit being taken over all small U_K normal in U and the maps f_F give an isomorphism

$$A = \varinjlim A_F \xrightarrow{\varinjlim f_F} \varinjlim (U_F/U_F^c),$$

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the injective limit on the right being taken over all U_F's relative to the transfer homomorphisms $V_{E/F}$.

This suggests we define a new type of mathematical objects which we will call W-group.

DEFINITION 4. A W-*group* $(U, \{U_F\})$ is a topological group U, together with a family of subgroups $\{U_F\}$ such that

WG 1) The family $\{U_F\}$ consists of open subgroups of finite index in U, is closed under finite intersection and conjugation, and contains all subgroups containing any one of its members.

WG 2) For every pair $U_K \triangleleft U_F$, the transfer map $V_{K/F} : U_F/U_F^c \to U_K/U_K^c$ maps U_F/U_F^c isomorphically and homeomorphically onto $(U_K/U_K^c)^{U_F/U_K}$, and has compact cokernel. (Here U_F acts on U_K by conjugation and U_K acts trivially on U_K/U_K^c, so U_F/U_K acts on the latter.)

WG 3) The canonical map $U \to \varprojlim(U/U_K^c)$ is an isomorphism of topological groups.

It is clear that the Weil group of a class formation satisfying the conditions WT 1,2,3 of Theorem 7, gives us a W-group. In fact, the converse is also true.

THEOREM 9. *Up to isomorphism, every W-group comes from a class formation satisfying the three conditions of Theorem* 7.

PROOF. Let $(U, \{U_F\})$ be a W-group. As explained in a paragraph preceding Definition 4, we can construct from U a formation $(G, \{G_F\}, A)$ by putting

$$G = \varprojlim (G/G_K) = \varprojlim (U/U_K) \qquad \text{and} \qquad A = \varinjlim (U_F/U_F^c).$$

and taking for G_F the inverse image of U_F/U_K for all sufficiently small U_K normal in U. The level A_F, which by definition is A^{G_F} can be identified with the isomorphic image of U_F/U_F^c in A because the injective limit is made with the transfer maps, and for all small U_K normal in U we have $V_{K/F}(U_F/U_F^c) = (U_K/U_K^c)^{U_F/U_K} = (U_K/U_K^c)^{G_F/G_K}$. If $g : U \to G$ is the canonical map, and $f_F : A_F \to U_F/U_F^c$ is our identification map, then $(U, g, \{f_F\})$ is a Weil group for the formation.

This formation does satisfy the three conditions of Theorem 7. The norm map $N_{E/F}A_E \to A_F$ is reflected in the canonical map $U_E/U_E^c \to U_F/U_F^c$ which is open; the inclusion map $A_F \hookrightarrow A_E$ is reflected in the transfer map which has compact cokernel by the definition of W-group; and G is profinite, hence complete. The only question is whether our formation is a class formation. To see that it is, let K/F be a normal layer and consider the exact sequence

$$0 \to A_K \simeq U_K/U_K^c \to U_F/U_K^c \to U_F/U_K \simeq G_F/G_K \to 0$$

which to simplify notation we will denote by

$$0 \to A_K \to U_F^K \to G_{K/F} \to 0.$$

Let $\alpha \in \mathcal{H}^2(G_{K/F}, A_K)$ be the class of this group extension. Let $(U_F^K)'$ be the algebraic commutator subgroup of U_F^K.

LEMMA. *$(U_F^K)'$ is compact, hence $(U_F^K)' = (U_F^K)^c = U_F^c/U_K^c$, and U_F^c is compact.*

We postpone the proof of the lemma. Assume the lemma is true. Then the transfer map $U_F^K/(U_F^K)' \to A_K^{G_{K/F}}$ is bijective, by condition WG 2) of Definition 4,

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and it follows from (Chapter XIII, Section 2, Theorem 4) and its corollary, (in both of which U^c denotes U'), that α_{-3} is surjective and α_{-2} is bijective, where $\alpha_n : \mathcal{H}^n(G_{K/F}, \mathbb{Z}) \to \mathcal{H}^{n+2}(G_{K/F}, A_K)$ is the map given by cup product with α. Also, α_{-1} is injective because $\mathcal{H}^{-1}(G_{K/F}, \mathbb{Z}) = 0$. Hence, by the general cohomological theorem alluded to in the proof of Theorem 1 of Chapter XIV, §4, α_n is bijective for all $n \in \mathbb{Z}$. For $n = -1$ this implies our formation is a field formation, and for $n = 0$ it implies that $\mathcal{H}^2(G_{K/F}, \mathbb{Z})$ is cyclic of order $|G_{K/F}| = [K : F]$, generated by $\alpha = \alpha_0(1)$. We denote this class by $a_{K/F}$ when different layers are involved. For $K \subset L$ normal over F, (XIII, §3, Th.6) shows that $\mathrm{infl}_{L/K}(\alpha_{K/F}) = [L : K]\alpha_{L/F}$, because $\alpha_{K/F} = v(\alpha_{L/F})$, by definition of v and the α's. Also for $F \subset E \subset K$ we have $\alpha_{K/E} = \mathrm{res}_{E/F}\alpha_{K/F}$. These facts allow us to define "invariant maps" $\mathrm{inv}_F : \mathcal{H}^2(G_F, A) \to \mathbb{Q}/\mathbb{Z}$ which satisfy the class formation axiom, Axiom II of Chapter XIV, Section 3, by putting $\mathrm{inv}_F(\alpha_{K/F}) = \frac{1}{[K:F]}$.

Proof of the lemma. Consider the inclusions

$$I_{K/F}A_K := \prod_{\sigma \in G_{K/F}} A_K^{1-\sigma} \subset (U_F^K)' \cap A_K \subset (U_F^K)'.$$

The index of the first inclusion is finite, because, by Theorem 3 in Chapter XIII, Section 2, the quotient is a homomorphic image of $\mathcal{H}^{-3}(G_{K/F}, \mathbb{Z})$. The index of the second inclusion is finite because A_K is of finite index in U_F^K. Each $A_K^{1-\sigma}$ is compact, as continuous image of A_K/A_F. Thus $I_{K/F}A_K$ is compact and so also is $(U_F^K)'$, as finite union of compacts. Hence $(U_F^K)'$ is closed, i.e, equal to its closure $(U_F^K)^c = U_F^c/U_K^c$. Finally, U_F^c is compact as projective limit of the compacts U_F^c/U_K^c. This proves the lemma and the theorem. □

Theorem 10. *The existence theorem holds for the class formation associated to a W-group $(U, \{U_F\})$ if and only if the family $\{U_F\}$ consists of all open subgroups of finite index in U.*

Proof. The existence theorem for a "field" F means that every open subgroup of finite index in U_F which contains U_F^c is of the form U_K for some "field" K, which is then the "class field over F to the subgroup U_K/U_F^c of $U_F/U_F^c = A_F$". This is certainly true if every open subgroup of finite index in U is a U_K, because finite index in U_F implies finite index in U. Conversely, suppose the existence theorem holds. Let W be open of finite index in $U = \varprojlim_K (U/U_K^c)$. A neighborhood of 1 in the projective limit topology contains the inverse image in U of a neighborhood of 1 in U/U_K^c for some K. Thus, $W \supset U_K^c$ for some K. By the existence theorem for K, there exists a "classfield L over K belonging to the subgroup $(W \cap U_K)/U_K^c$ of U_K/U_K^c", that is, a field L such that $U_L = W \cap U_K$, and since W contains U_L, it is of the form U_F for some F. □

It follows that a topological class formation satisfying the existence theorem and the three conditions of Theorem 7 is mathematically equivalent to a special type of topological group U, one which satisfies the two conditions WG 2) and WG 3) of Definition 4, if we take the collection of all open subgroups of finite index as the family denoted there by $\{U_F\}$.

In class formations in which the reciprocity maps $A_F \to G_F/G_F^c$ are injective, the map $U \to G$ is injective, and one can "find" a Weil group U for the formation

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inside the Galois group G. This is the case for nonarchimedean local fields and global function fields. We discuss the latter case as an example.

Let k be a global function field, p its characteristic, k_{sep} a separable algebraic closure, k_0 the (finite) constant field, and $\bar{k}_0$ the algebraic closure of k_0 in k_{sep}. Let $G = \text{Gal}(k_{\text{sep}}/k)$ and let U be the subgroup of G consisting of the elements which act on $\bar{k}_0$ as a positive or negative power of the Frobenius automorphism $x \mapsto x^p$. Let $U_1 = G_1 = \text{Gal}(k_{\text{sep}}/\bar{k}_0 k)$ be the subgroup of U consisting of the elements of G which act trivially on $\bar{k}_0$. Topologize U by declaring U_1 to be open in U and giving U_1 the profinite Krull topology of G_1. Thus $U/U_1 = \mathbb{Z}$ with the discrete topology, whereas $G/G_1 = \hat{\mathbb{Z}}$.

Let $g : U \to G$ be the inclusion map. For each finite extension F of k in k_{sep}, let $G_F = \text{Gal}(k_{\text{sep}}/F)$, and $U_F = U \cap G_F = g^{-1}(G_F)$. Let $r_F : C_F \to G_F/G_F^c$ be the reciprocity map (denoted by ω in Chapter VIII). The image of r_F is U_F/G_F^c. We have $G_F^c = U_F^c$, because U_F is dense in G_F and $G_F^c \subset G_1 = U_1$. Let $f_F : C_F \to U_F/U_F^c$ be the bijection induced by r_F. Our choice of the topology of U is such that the f_F's are isomorphisms of topological groups (cf. Ch. VIII, §3). Thus $(U, g, \{f_F\})$ is a Weil group for the formation $(G, \{G_F\}, C_{k_{\text{sep}}})$. This should be clear from the preceding discussion, except perhaps for the fact that for each normal layer K/F the class of the group extension

$$0 \to C_K = U_K/U_K^c \to U_F/U_K^c \to U_F/U_K = G_{K/F} \to 0$$

is the fundamental class. But that is true by part (b) of Theorem 8, because part (a) holds, by our construction of U.

It was Weil who first focused on the fact that the Takagi–Artin class field theory, as expressed for infinite extensions by Chevalley with idèles, could be interpreted, in the case of function fields, in terms of the group U we have just described. His belief in the deep analogy between function fields and number fields and his hope to find a non-abelian analog of Hecke's L-functions with Grössencharacters led him to expect that a similar group U might exist in the case of number fields, a group in which the quotients U_F/U_F^c are isomorphic to the idele class groups C_F, so that one would have a Galois-like interpretation of the full group C_F, as a refinement of Chevalley's interpretation of the group of connected components C_F/D_F as G_F/G_F^c. Weil believed that to find a theory in which such a group U appears as naturally for number fields as it does for function fields is a very important problem, perhaps holding the key to the Riemann hypothesis. He was at least able to show that this idea was not a pipe dream, by proving in [**28**] by an artificial construction that such a group U, with all the properties one would expect, does indeed exist for number fields, and is characterized up to isomorphism by those properties. To find it occurring naturally is still, 60 years later, an open problem.

How did Weil construct the group U using only the classical Takagi-Artin theory, without the theory of global fundamental classes which was the basis of our method? Although our way is more general, and perhaps more natural, once one has the theory of fundamental classes, Weil's is a natural direct attack on the specific problem of global number fields, and is certainly of intrinsic and historic interest. We finish this chapter by describing Weil's method in an abstract situation briefly, in a series of exercises.

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To begin, we must define what Weil constructed. For that we have only to modify our Definition 1 at the beginning of this chapter as follows:

(i) Replace "class formation" by "field formation with reciprocity maps $A_F \to G_F/G_F^c$ satisfying (Chapter XIV, Section 5, Theorem 6)"

(ii) Have g be a map of U into G_F/G_K^c rather than into $G_{K/F} = G_F/G_K$.

(iii) Replace condition W 3) involving the fundamental class by:

W 3′) For each intermediate field $F \subset E \subset K$, let $g_E^{\mathrm{ab}} : U_E^{\mathrm{ab}} \to G_E^{\mathrm{ab}}$ denote the map induced by g. Then the composed map

$$A_E \xrightarrow{f_E} U_E^{\mathrm{ab}} \xrightarrow{g_E^{\mathrm{ab}}} G_e^{ab}$$

is the reciprocity map r_E.

Call a triple $(U, g, \{f_E\})$ satisfying this modified definition a *Weil group in Weil's sense* for the layer K/F. Note that it has a key feature, the map g in (ii) above, which is missing from our definition of Weil group of a layer. We recover the existence of g only after passing to the inverse limit to construct a Weil group for the whole formation.

In [**28**], Weil showed for the formation of idèle classes of global number fields that for normal K/F a Weil group in Weil's sense exists and is unique up to isomorphism.The key special properties of number fields which he used are:

(1) For each intermediate field $F \subset E \subset K$ the reciprocity map $r_E : C_E \to G_E/G_E^c$ is surjective.

(2) The $G_{K/F}$-module $D_K := \mathrm{Ker}(r_K)$ is isomorphic to a direct sum of modules induced from subgroups of order 1 or 2 of $G_{K/F}$.

(3) For each intermediate E, $\mathcal{H}^1(G_{K/E}, D_K) = 0$ and $N_{K/E} D_K = D_E$.

THEOREM 11. *Let $(G, \{G_F\}, A)$ be a field formation with reciprocity laws satisfying Theorem* 6 *in Chapter* XIV, §5, *properties* (1) *and* (3) *just above and, instead of* (2), *the slightly more liberal condition*

(2′) *The $G_{K/F}$-module $D_K := Ker(r_K)$ is isomorphic to a direct sum of modules induced from cyclic subgroups of $G_{K/F}$.*

Assume also that G is complete so that we have reciprocity maps $r_F : A_F \to G_F^{ab}$. Then a Weil group in Weil's sense exists for a normal layer K/F of the formation, and is unique up to isomorphism.

PROOF. (Sketch, in a series of exercises. The method is essentially Weil's, except for the additional technical difficulty posed by assuming (2′) instead of (2).)

1. Define a *W-diagram for K/F* to be an exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_K & \xrightarrow{i} & U_F & \xrightarrow{j} & G_{K/F} & \longrightarrow & 0 \\ & & \downarrow r_K & & \downarrow g & & \downarrow \mathrm{id.} & & \\ 0 & \longrightarrow & G_K/G_K^c & \xrightarrow{\mathrm{incl.}} & G_F/G_K^c & \xrightarrow{\mathrm{canon.}} & G_{K/F} & \longrightarrow 0. & \end{array}$$

Using the long exact cohomology sequence associated with the short exact sequence of $G_{K/F}$-modules

$$0 \longrightarrow D_K \hookrightarrow A_K \xrightarrow{r_K} G_K/G_K^c = G_K^{ab} \longrightarrow 0,$$

show that a W-diagram exists.

2. The group $\mathcal{H}^2(G_{K/F}, D_K)$ acts simply and transitively on the set of isomorphism classes of W-diagrams for K/F. To see this, fix a section $\sigma \mapsto \gamma_\sigma$ of the

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canonical map $G_F/G_K^c \to G_{K/F}$ which occurs in the bottom row of W-diagrams for K/F. Let $a_{\sigma,\tau}^{\mathrm{gal}} = \gamma_\sigma\gamma_\tau\gamma_{\sigma\tau}^{-1} \in \mathcal{Z}^2(G_{K/F}, G_K/G_K^c)$ denote the corresponding 2-cocycle, which represents the class of the Galois group extension in the bottom row of each such diagram. The set of lifts of this cocycle to A_K is a coset of $\mathcal{Z}^2(G_{K/F}, D_K)$ in $\mathcal{Z}^2(G_{K/F}, A_K)$, where $\mathcal{Z}$ denotes cocycle. Each lift $a_{\sigma,\tau}$ defines a W-diagram $(U, g, \dots)$ with elements $u_\sigma \in U$ such that

$$U = \coprod_{\sigma\in G_{K/F}} A_K u_\sigma, \qquad u_\sigma u_\tau u_{\sigma\tau}^{-1} = a_{\sigma,\tau}, \qquad g(u_\sigma) = \gamma_\sigma.$$

Show that the product action of $\mathcal{Z}^2(G_{K/F}, D_K)$ on the coset of lifts induces a simply transitive action of $\mathcal{H}^2(G_{K/F}, D_K)$ on the set of isomorphism classes of W-diagrams for K/F. Call this action "twisting".

3. Suppose (U_F, g_F, i, j) is a W-diagram for K/F. For each intermediate field E, $F \subset E \subset K$, let $U_E = j^{-1}(G_{K/E}) = g^{-1}(G_E/G_K^c)$ and note that by replacing F's by E's in the diagram in (1) we obtain a W-diagram for K/E as "subdiagram" of the one there for K/F. The transfer map $U_E^{\mathrm{ab}} \to U_K^{\mathrm{ab}} = A_K$ factors through the inclusion $A_E \hookrightarrow A_K$. Let $V_{K/E} : U_E^{\mathrm{ab}} \to A_E$ denote the map it induces. Show that a Weil group $(U, g, \{f_E\})$ for K/F in Weil's sense is the "same" as a W-diagram $(U, g, \dots)$ in which, for each intermediate E, the map $V_{K/E} = f_E^{-1}$ is an isomorphism and the composed map $r_E \circ V_{K/E}$ is equal to the map $g_E^{\mathrm{ab}} : U_E^{\mathrm{ab}} \to G_E^{\mathrm{ab}}$ induced by g. Our first goal is to show that there exists a W-diagram in which $r_E \circ V_{K/E} = g_E^{\mathrm{ab}}$ for all E, and that it is unique up to isomorphism.Then we shall prove that for such a diagram the transfer maps $V_{K/E}$ are isomorphisms.

4. Suppose $(U_F, g_F, \dots)$ is a W-diagram for K/F. Show that the maps $r_F \circ V_{K/F}$ and g_F^{ab} coincide on the image of $A_K = U_K^{\mathrm{ab}}$ in U_F^{ab} and that their compositions with $V_{K/F}^{\mathrm{gal}} : G_F^{\mathrm{ab}} \to G_K^{\mathrm{ab}}$ are equal. (Use the "translation" and "transfer" theorems, i.e., parts (b) and (a) of Theorem 6 in Chapter XIV, Section 5).

5. Note that the quotient of U_F^{ab} by the image of A_K is $G_{K/F}^{\mathrm{ab}} = \mathcal{H}^{-2}(G_{K/F}, \mathbb{Z})$. Also, show that r_F maps $\mathcal{H}^0(G_{K/F}, D_K) = D_K^{G_{K/F}}/N_{K/F}D_K = (D_K \cap A_F)/D_F$ isomorphically onto $\mathrm{Ker}(V_{K/F}^{\mathrm{gal}})$.

6. By 4 and 5, the ratio $\frac{r_F \circ V_{K/F}}{g_F^{\mathrm{ab}}} : U_F^{\mathrm{ab}} \to G_F^{\mathrm{ab}}$ can be factored as follows

$$U_F^{\mathrm{ab}} \to G_{K/F}^{\mathrm{ab}} = \mathcal{H}^{-2}(G_{K/F}, \mathbb{Z}) \xrightarrow{\psi_F} \mathcal{H}^0(G_{K/F}, D_K) = (D_K \cap A_F)/D_F \xrightarrow{r_F} G_F^{\mathrm{ab}}.$$

The same consideration applies to each subdiagram $(U_E, g_E, ..)$. In a this way, a W-diagram determines a collection of homomorphisms $\psi_E : \mathcal{H}^{-2}(G_{K/E}, \mathbb{Z}) \to \mathcal{H}^0(G_{K/E}, D_K)$, one for each intermediate field E. Our first goal is to show the existence and uniqueness up to isomorphism of a W-diagram for which $\psi_E = 0$, i.e., $r_E \circ V_{K/E} = g_E^{\mathrm{ab}}$, for all E.

7. As notation, for a group G, a G-module M and $\alpha \in \mathcal{H}^2(G, M)$, let $\bullet\alpha : \mathcal{H}^{-2}(G, \mathbb{Z}) \to \mathcal{H}^0(G, M)$ denote the Nakayama map $\zeta \mapsto \zeta \cdot \alpha$ given by cup product with α. With notation as in 2., show that $V_{K/F}(u_\sigma) = \prod_{\rho\in G_{K/F}} a_{\rho,\sigma}$ and $g_F^{\mathrm{ab}}(u_\sigma) = \gamma_\sigma$. Conclude that twisting a W-diagram for K/F by a class $\delta \in \mathcal{H}^2(G_{K/F}, D_K)$ multiplies its ψ_E map by the map $\bullet\mathrm{res}_{E/F}\delta$, for each intermediate field E.

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8. For finite cyclic G and G-module M, show that $\alpha \mapsto \bullet\alpha$ is an isomorphism from $\mathcal{H}^2(G,M)$ to $\mathrm{Hom}(\mathcal{H}^{-2}(G,\mathbb{Z}),\mathcal{H}^0(G,M))$. Thus a W-diagram defines, for each intermediate field E such that K/E is cyclic, a unique class $\delta_E \in \mathcal{H}^2(G_{K/E},D_K)$ such that $\psi_E = \bullet\delta_E$.

9. Let $(U,g,\dots)$ be a W-diagram for K/F. let $E \subset E'$ be intermediate fields corresponding to subgroups $H \supset H'$ of $G_{K/F}$. Let $\delta \in \mathcal{H}^2(H,D_K)$. Check that the following diagram is commutative with the middle horizontal arrows being either ψ_E and $\psi_{E'}$, as in the diagram, or being $\bullet\delta$ and $\bullet\mathrm{res}\,\delta$ (V denotes the transfer map and $\widetilde{\mathrm{id}}$ the map induced by the identity on D_K):

$$\begin{array}{ccccccc} H^{\mathrm{ab}} & = & \mathcal{H}^{-2}(H,\mathbb{Z}) & \xrightarrow{\psi_E} & \mathcal{H}^0(H,D_K) & = & D_K^H/N_H D_K \\ \downarrow V & & \downarrow \mathrm{res} & & \downarrow \mathrm{res} & & \downarrow \widetilde{\mathrm{id}} \\ H'^{\mathrm{ab}} & = & \mathcal{H}^{-2}(H',\mathbb{Z}) & \xrightarrow{\psi'_E} & \mathcal{H}^0(H',D_K) & = & D_K^{H'}/N_{H'}D_K. \end{array}$$

Prove that if $\mathcal{S}$ is the set of cyclic subgroups of $G_{K/F}$, then the family $(\delta_E)_{G_{K/E}\in\mathcal{S}}$ defined in 8 is coherent in the sense of the following

Proposition. *Let G be a finite group and M a G-module. Let $\mathcal{S}$ be a set of subgroups of G such that if $H \in \mathcal{S}$, then each subgroup of H is in $\mathcal{S}$ and each conjugate $H^\sigma = \sigma H \sigma^{-1}$ is in S. Call a family of elements $\alpha_{\mathcal{S}} = (\alpha_H)_{H\in\mathcal{S}} \in \prod_{H\in\mathcal{S}} \mathcal{H}^r(H,M)$* coherent *if $res\,\alpha_H = \alpha_{H'}$ for all pairs $H' \subset H \in \mathcal{S}$, and $\alpha_{(H^\sigma)} = \sigma\alpha_H$ for all pairs $H \in \mathcal{S}$ and $\sigma \in G$. Suppose that M is a finite direct sum of modules induced from subgroups $H \in \mathcal{S}$. Then the map $\mathcal{H}^r(G,M) \to \prod_{H\in\mathcal{S}} \mathcal{H}^r(H,M)$ defined by $\alpha \to (res_H^G\alpha)_{H\in\mathcal{S}}$ is an isomorphism of $\mathcal{H}^r(G,M)$ onto the set of coherent families $(\alpha_H)_{H\in\mathcal{S}}$.*

10. To prove the above proposition, suppose H and G are in $\mathcal{S}$. Let

$$G = \coprod_{\tau\in T} H\tau G_0 \quad \text{and} \quad H\tau G_0 = \coprod_{\sigma\in R_\tau} \sigma\tau G_0.$$

Then $H = \coprod_{\sigma\in R_\tau} \sigma(H\cap\tau G_0\tau^{-1})$ and M as H-module is a direct sum, $M = \sum M_\tau$, where $M_\tau = \sum_{\sigma\in R_\tau} \sigma\tau M_0$ is an H-module induced from the $(H\cap\tau G_0\tau^{-1})$-module τM_0.

Let $\alpha_H \in \mathcal{H}^r(H,M)$ be the H-component of a coherent family $\alpha_{\mathcal{S}}$. Let $\alpha_H = \sum_{\tau\in T} \alpha_{H,\tau}$ with $\alpha_{H,\tau} \in \mathcal{H}^r(H,M_\tau)$. Let $\alpha^0_{M,\tau} = \mathrm{pr}(\mathrm{res}\,\alpha_{H,\tau}) \in \mathcal{H}^r(H\cap\tau G_0\tau^{-1},\tau M_0)$ be the class corresponding to $\alpha_{H,\tau}$ by the semilocal theory. Show that $\alpha^0_{H,\tau} = \tau(\mathrm{res}^{G_0}_{(\tau^{-1}H\tau)\cap G_0}\alpha_{G_0,1})$. Conclude that $\alpha_H = \mathrm{res}^G_H\alpha$, where $\alpha \in \mathcal{H}^r(H,M)$ is the unique class such that $\alpha_{G_0,1} \in \mathcal{H}^r(G_0,M_0)$ is the class corresponding to α by the semilocal theory. Since H was arbitrary in $\mathcal{S}$, this proves the theorem.

11. Taking S to be the set of cyclic subgroups of $G = G_{K/F}$, show that 8, 9, and 10 imply that any W-diagram for K/F can be twisted by a unique $\delta \in \mathcal{H}^2(G_{K/F},D_K)$ to a W-diagram for K/F in which $\psi_E = 0$ for all intermediate E.

12. To finish we must show that in a W-diagram (U,g,i,j) for K/F in which $r_E \circ V_{K/E} = g_E^{\mathrm{ab}}$ for all intermediate E, the transfer maps $V_{K/E}$ are isomorphisms. It suffices to do this for $E = F$. We write $U = U_F$, $g = g_F$, and $V = V_{K/F}$.

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Surjectivity: Let $a \in A_F$. Since g^{ab} is surjective, there exists $x \in U^{\mathrm{ab}}$ such that $g^{\mathrm{ab}}(x) = r_F(a)$. Then

$$\frac{V(x)}{a} \in \mathrm{Ker}(r_F) = D_F = N_{K/F} D_K.$$

Hence $V(x) = aN_{K/F}d = aV(i(d))$ for some $d \in D_K$, and $a = V(\frac{x}{i(d)})$.

Injectivity: Let $u \in U$ such that $V(uU^c) = 0$. Then $g^{\mathrm{ab}}(uU^c) = r_F(V(uU^c)) = 0$. Hence there is a $d \in D_K$ such that $u \in i(d)U^c$, because the kernel of g in a W-diagram is $i(D_K)$. We must show $i(d) \in U^c$. We know that $1 = V(uU^c) = V(i(d)U^c) = N_{K/F}d$. By our hypotheses on D_K, $\mathcal{H}^1(G_{K/F}, D_K) = 0$ and the cohomology of D_K has period 2. Thus, $\mathcal{H}^{-1}(G_{K/F}, D_K) = 0$ and consequently, $d = \prod_\nu d_\nu^{\sigma_\nu - 1}$ for some finite set of pairs $(d_\nu, \sigma_\nu) \in D_K \times G_{K/F}$. Let $i(d_\nu) = y_\nu$ and $j(u_\nu) = \sigma_\nu$. Then $i(d) = \prod_\nu u_\nu y_\nu u_\nu^{-1} y_\nu^{-1} \in U^c$ as was to be shown. $\square$

That finishes our sketch of the existence and uniqueness of a Weil groups in Weil's sense for layers in the special type of field formation we are considering, following Weil's proof for the formation of idele class groups of number fields. If, as in the case of number fields, the formation is a class formation with cyclic layers over the ground field of arbitrary degree, then the class of the group extension of $G_{K/F}$ by A_K given by the Weil group, is the fundamental class. This was proved for number fields by Nakayama in a paper in the same Takagi memorial volume of the Journal of the Japanese Mathematical Society as [**28**], by the same method we used to prove Theorem 8(b). The fundamental class was discovered at almost the same time by Nakayama and Weil, in completely different ways, Weil as a byproduct of his discovery of the Weil group and Nakayama by a systematic study of the Galois cohomology of class field theory, partly in collaboration with G. Hochschild, leading to most of the cohomological results we have presented in Chapter XIV (cf. [**11**]).

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Bibliography

[1] Y. Akizuki, *Eine homomorphe Zuordnung der Elemente der galoisschen Gruppe zu den Elemente einer Untergruppe der Normklassengruppe*, Math. Annalen, **112** (1935–1936), 566–571.

[2] E. Artin, *Algebraic Numbers and Algebraic Functions*, AMS Chelsea, Providence, RI, 1967.

[3] ———, *Galois Theory*, Notre Dame Mathematical Lectures, no. 2, Notre Dame, IN, 1942; Reprinted in *Expositions by Emil Artin: A Selection*, M. Rosen (ed.), History of Math., vol. 30,, Amer. Math. Soc., Providence, RI, 2007, pp. 61–107.

[4] N. Bourbaki, *Algèbre, Chapitre 5: Corps commutatifs*. Hermann, Paris, 1959.

[5] J. W. S. Cassels and A. Fröhlich (eds.), *Algebraic Number Theory*, Academic Press, London, 1976.

[6] C. Chevalley, *La théorie du corps de classes*, Ann. of Math. (2) **41** (1940), no. 2, 391–418.

[7] B. Dwork, *Norm residue symbol in local number fields*, Abh. Math. Sem. Hamburg, **22** (1958), 180–190.

[8] I. V. Fesenko and S. V. Vostokov, *Local Fields and Their Extensions: Second Edition*, Amer. Math. Soc., Providence, RI, 2002.

[9] E. S. Golod and I. R. Shafarevich, *On class number towers*, Amer. Math. Soc. Transl., Ser. 2, vol. 48, Amer. Math. Soc., Provience, RI, 1964, pp. 91–102.

[10] W. Grunwald, *Ein allgemeines Existenztheorem für algebraische Zahlkörper*, J. Reine Angew. Math. **169** (1933), 103–107.

[11] G. Hochschild and T. Nakayama, *Cohomology in class field theory*. Ann. Math. (2) **55** (1952), 348–366.

[12] K. Iwasawa, *Local Class Field Theory*, Oxford Univ. Press, 1986.

[13] ———, *An explicit formula for the norm residue symbol*, J. Math. Soc. Japan **20** (1968), 151–165.

[14] S. Iyanaga, *Zum Beweis des Hauptidealsatzes*, Abh. Math. Sem. Hamburg, **10** (1934), 349–357.

[15] S. Lang, *On quasi algebraic closure*, Ann. of Math. (2) **55** (1952), 373–390.

[16] ———, *Algebraic Number Theory*. Second edition. Springer-Verlag, New York, 1994.

[17] J. Lubin and J. Tate, *Formal complex multiplication in local fields*, Ann. Math. (2) **80** (1964), 464–484.

[18] K. Miyake (ed.), *Class Field Theory: Its Centenary and Prospect* Advanced Studies Pure Math., vol. 30, Math. Soc. Japan, Tokyo, 2001.

[19] T. Nakayama, *Über die Beziehungen zwischen den Faktoren Systemen und der Normclassengruppe eines galoisschen Erweiterungs Körpers*, Math. Annalen, **112** (1935–1936), 85–91.

[20] J. Neukirch, A. Schmidt, and K. Wingberg, *Cohomology of number fields*, Springer-Verlag, Heidelberg, 2000.

[21] J-P. Serre, *Local Fields*, Springer-Verlag, New York–Berlin, 1979.

[22] ———, *Galois Cohomology*, Springer-Verlag, Berlin, 1997.

[23] ———, *Cohomologie et arithmétic*, Séminaire Bourbaki, Volume 2, Exposé 77, Soc. Math. France, Paris, 1995, pp. 263–269.

[24] I. R. Shafarevich, *On the Galois groups of* $\mathfrak{p}$*-adic fields*, Dokl. Akad. Nauk SSSR **53** (1946), no. 1, 15–16; see also C. R. Acad. Sci. Paris **53** (1946), 15–16 and Collected Mathematical Papers of Shafarevich, Springer-Verlag, Heidelberg, 1989, pp. 5–6.

[25] J. Tate, *Number theoretic background*, Automorphic Forms, Representations and L functions, A. Borel and W. Casselman (eds.), Proc. Symp. Pure Math., vol. 33, Part 2, Amer. Math. Soc., Providence, RI, 1979, pp. 3–26.

[26] Shianghaw Wang, *On Grunwald's theorem*, Ann. of Math. (2) **51** (1950), 471–484 no. 2, 471–484.

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[27] A. Weil, *L'intégration dans les groupes topologiques et ses applications*, Hermann, Paris, 1938.

[28] ———, *Sur la théorie du corps de classes.* J. Math. Soc. Japan **3** (1951), 1–35; see also A. Weil, *Collected Papers*, Volume 1, [1951b], Springer-Verlag, New York–Heidelberg, 1979, pp. 487–581.

[29] ———, *Basic Number Theory*, Springer-Verlag, Berlin–Heidelberg–New York, 1967.

[30] G. Whaples, *Non-analytic class field theory and Grunwald's theorem*, Duke Math. J. **9** (1942), 455-473.

[31] E. Witt, *Zwei Regeln über verschränkte Produkte*, J. Reine Angew. Math. **173** (1935), 191–192.

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ERRATA, as of 1/23/09

p.154, footnote: Theorem 13, Ch. IX
p.191, reference [23]: arithmétique
p.191, reference [19]: Factorensystemen, Erweiterungskörpers, Normklassen
p.63, line 1: $\mathbb{Q}$ is in wrong font.
p.50, third line of Theorem 7: Axiom II

The following errata have to do only with references to theorems in the missing chapters which should have been eliminated in this new edition, but were overlooked.

p.131, line 11: replace 'According to the corollary of Theorem 16 of Ch. V, section 4' by 'It is well known that'.
p.131, lines 18,19: Delete ', by Theorem 16, Ch. V, section 4' .
p.131, lines -8,-9: replace 'corollary of .... which states' by 'the fact that'

[the fact alluded to here is implied by the duality $\hat{H}^{-r-1}(G,A)' = H^r(G,A')$, where $M' = \mathrm{Hom}(M, \mathbb{Q}/\mathbb{Z})$. If we identify $\hat{H}^{-r-1}$ with H_r, as explained for example in Ch. 4 of [5], then cup product becomes cap product, and the duality in question becomes the well known $H_r(G,A)' = H^r(G,A')$. This can be proved by applying the identity $\mathrm{Hom}_Z(A \otimes_G X, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}_G(X, \mathrm{Hom}_Z(A, \mathbb{Q}/\mathbb{Z})$ to a projective resolution of the G-module $\mathbb{Z}$, and using the fact that $M \mapsto M'$ is an exact functor.]

p.132, last three lines: Delete 'which we have ..... alluded there'
p.160, line 16: Delete '(Theorem ... Ch. V)'
p.148, line 20: Delete 'mentioned at ... Ch. IV'
p.128, line 11: Delete '(cf. the ... Ch. II)'